



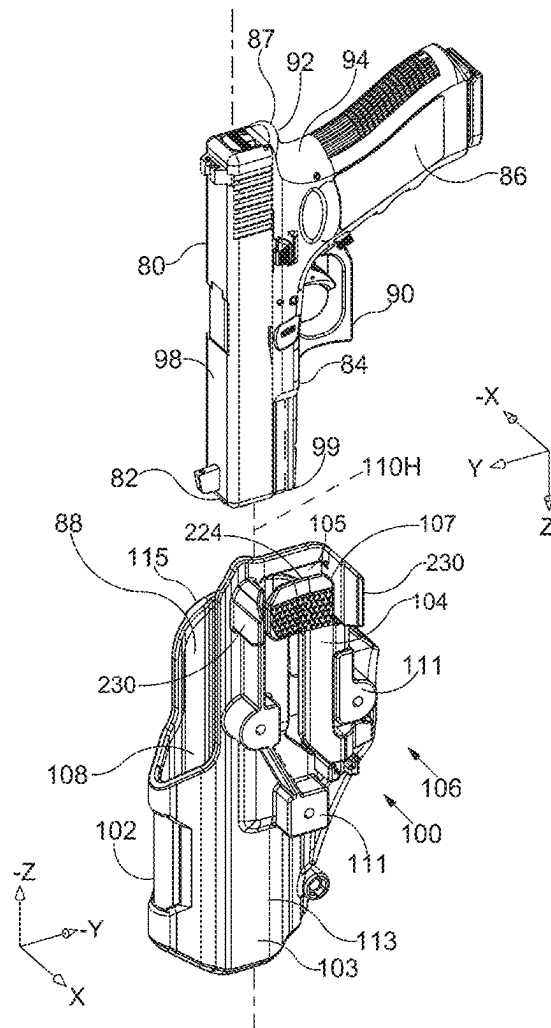
US 20250277647A1

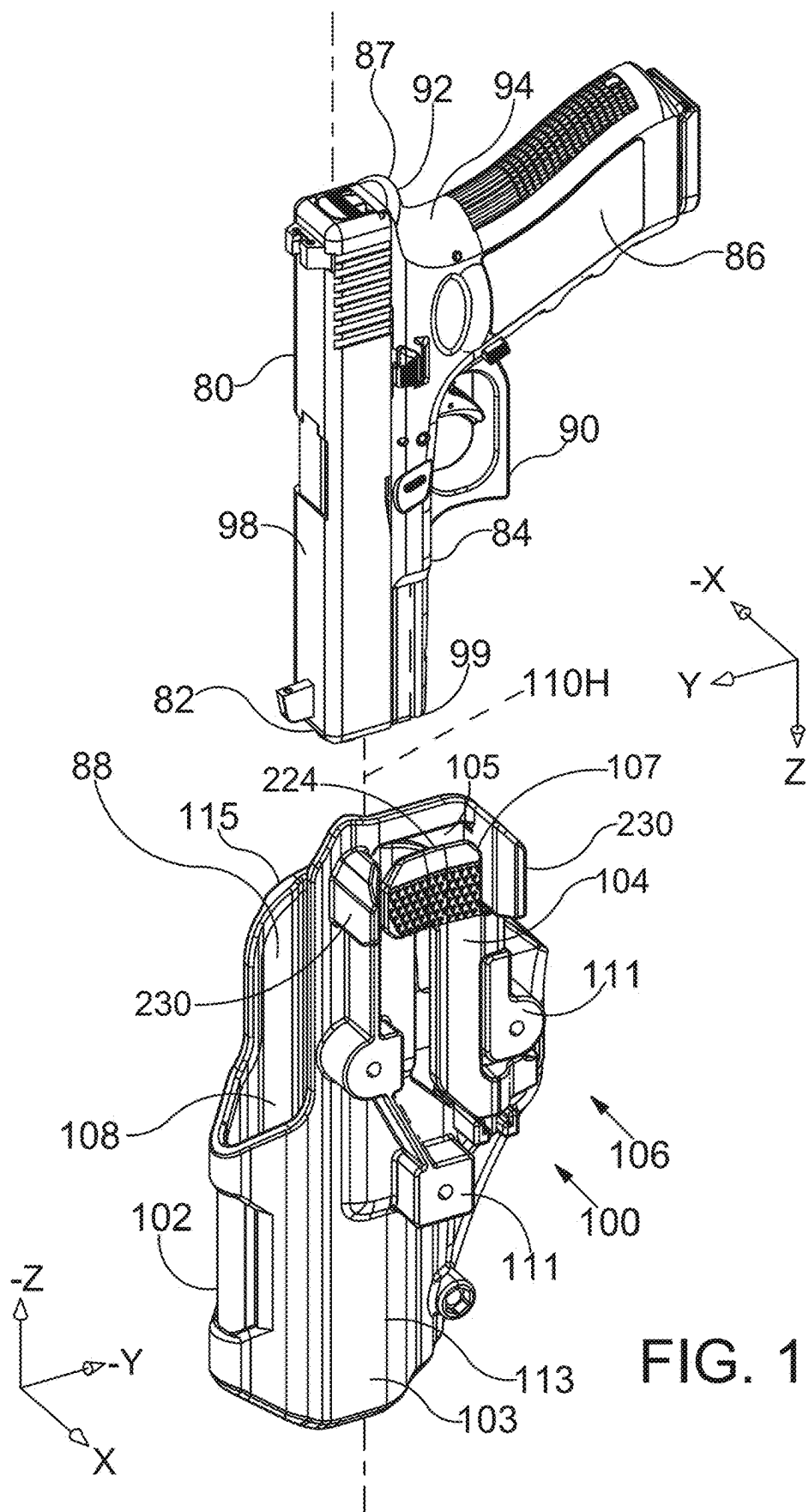
(19) **United States**(12) **Patent Application Publication**
Smith et al.(10) **Pub. No.: US 2025/0277647 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ROTATING LOCK LEVER**(71) Applicant: **Revelyst Operations LLC**, Anoka, MN
(US)(72) Inventors: **Paul N. Smith**, Bozeman, MT (US);
Christopher J. Michael, Belgrade, MT
(US); **Daniel D. Sereday**, Belgrade,
MT (US)(21) Appl. No.: **19/049,447**(22) Filed: **Feb. 10, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/116,806, filed on
Mar. 2, 2023, now Pat. No. 12,247,805.(60) Provisional application No. 63/315,971, filed on Mar.
2, 2022.**Publication Classification**(51) **Int. Cl.**
F41C 33/02 (2006.01)(52) **U.S. Cl.**CPC **F41C 33/0263** (2013.01); **F41C 33/0227**
(2013.01); **F41C 33/0272** (2013.01)

(57)

ABSTRACT

A rotating lock lever assembly configured as an additional security feature for a holster and may be useable in combination with a first user actuatable handgun retention mechanism and a second user actuatable handgun retention mechanism. The rotating lock lever assembly may include a housing, a pivot portion pivotally mounted to the housing and an elongate member extending from the pivot portion. The elongate member may include a lever button portion and a blocking projection extending from the pivot portion. The blocking projection may have an engagement surface configured to engage with a protrusion portion of a thumb button of the second user actuatable handgun retention mechanism when in the protrusion portion blocking position. The rotating lock lever assembly is configured for manual movement of the rotating lock lever assembly between a protrusion portion blocking position and a protrusion portion non-blocking position.





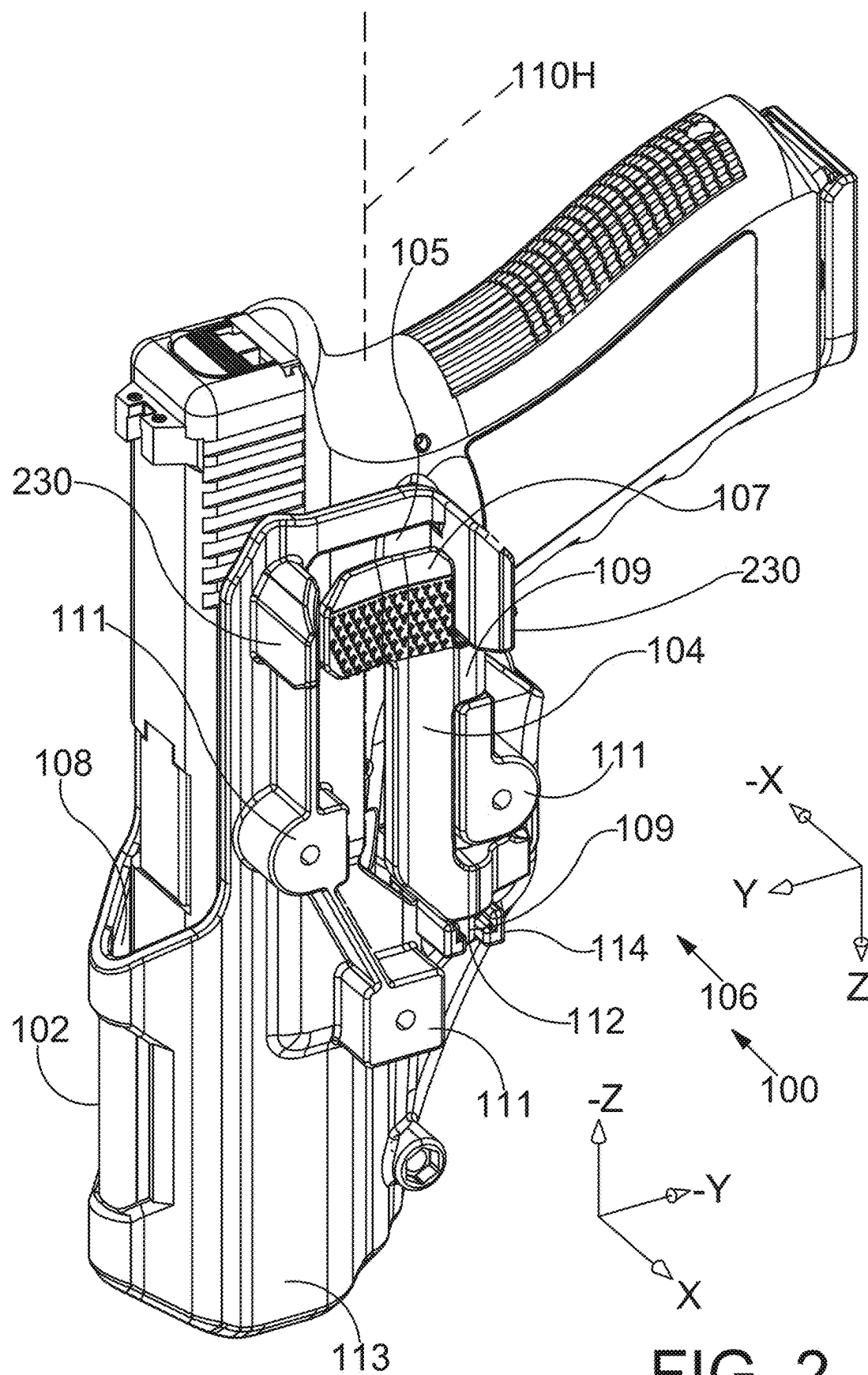


FIG. 2

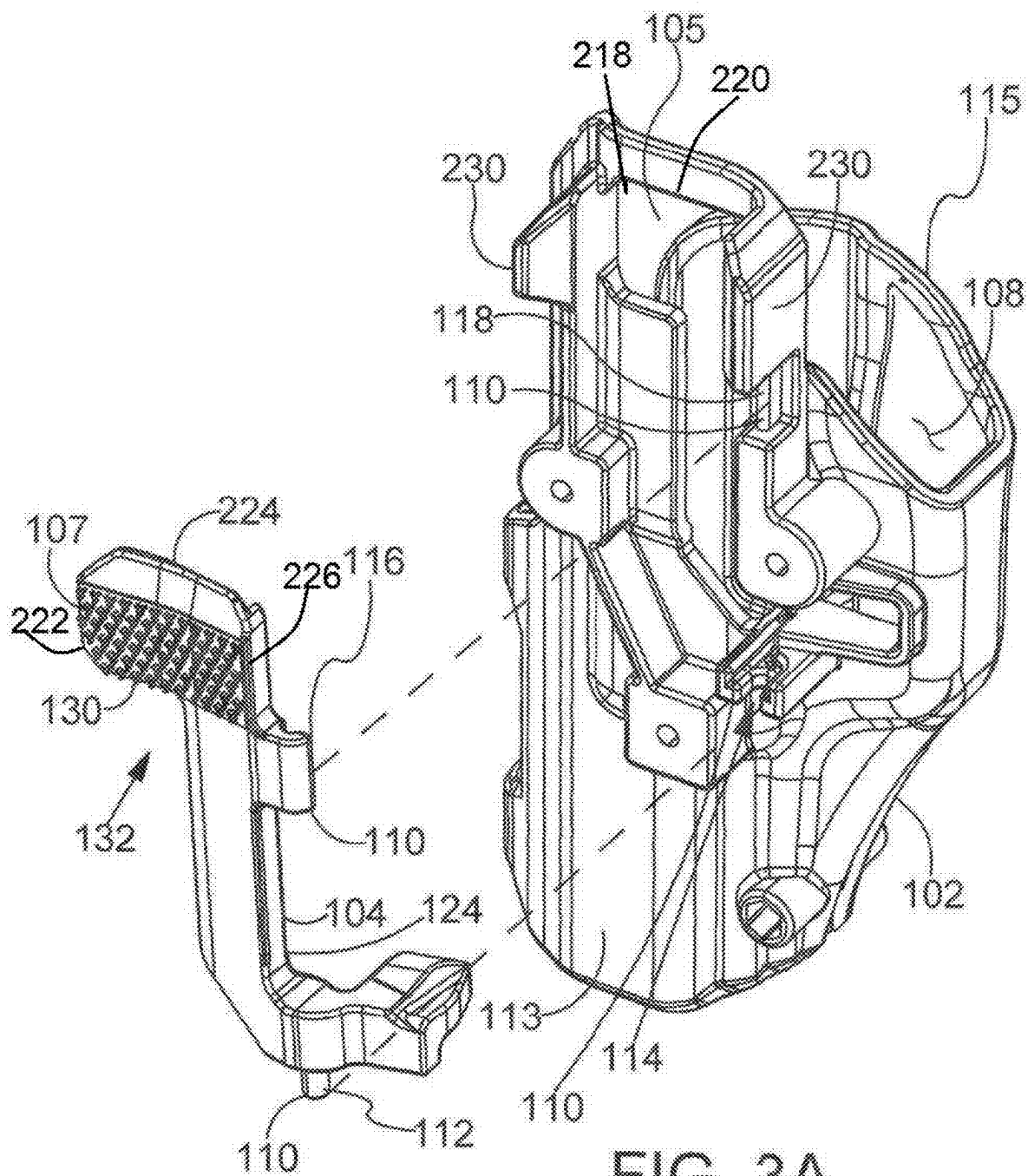
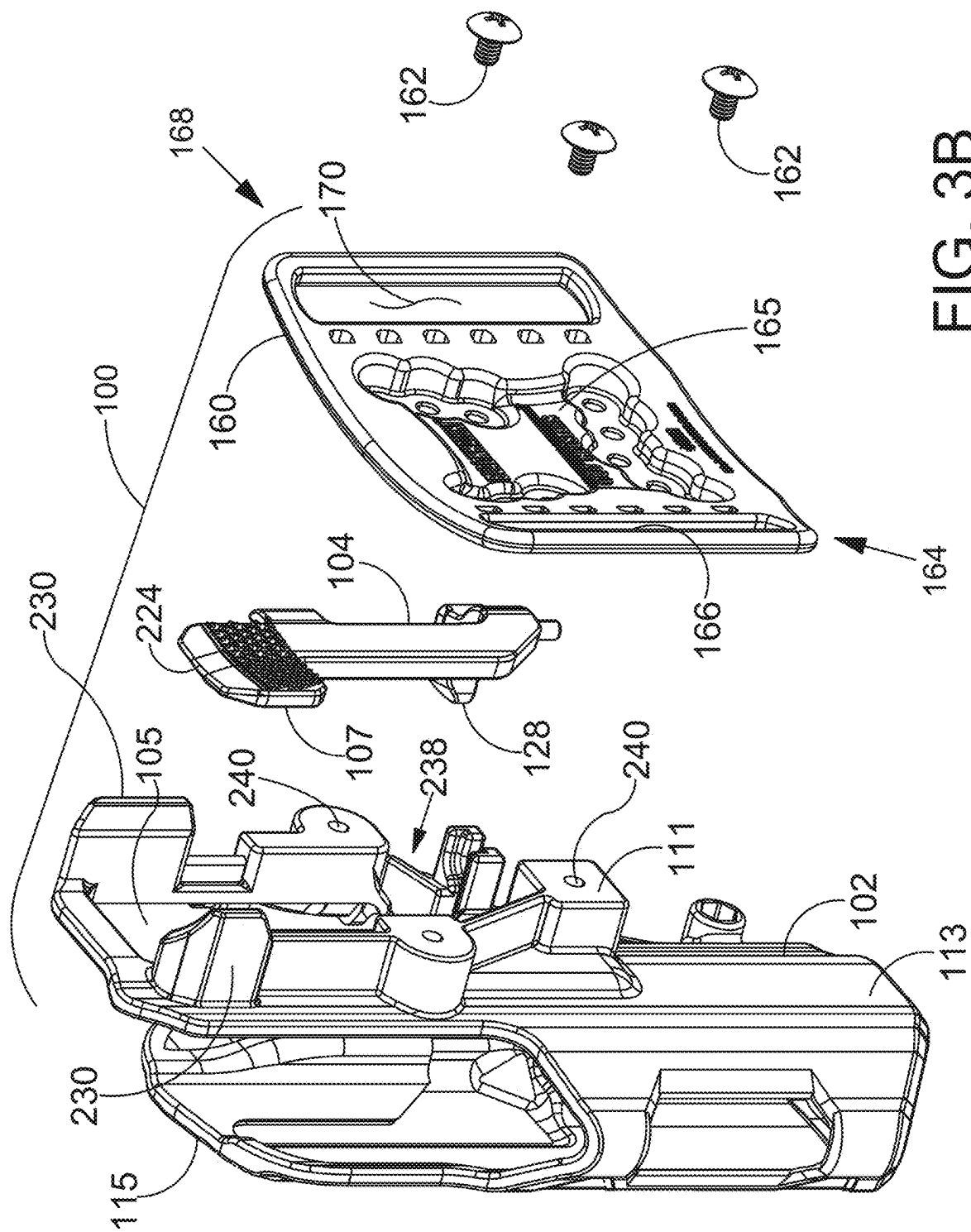
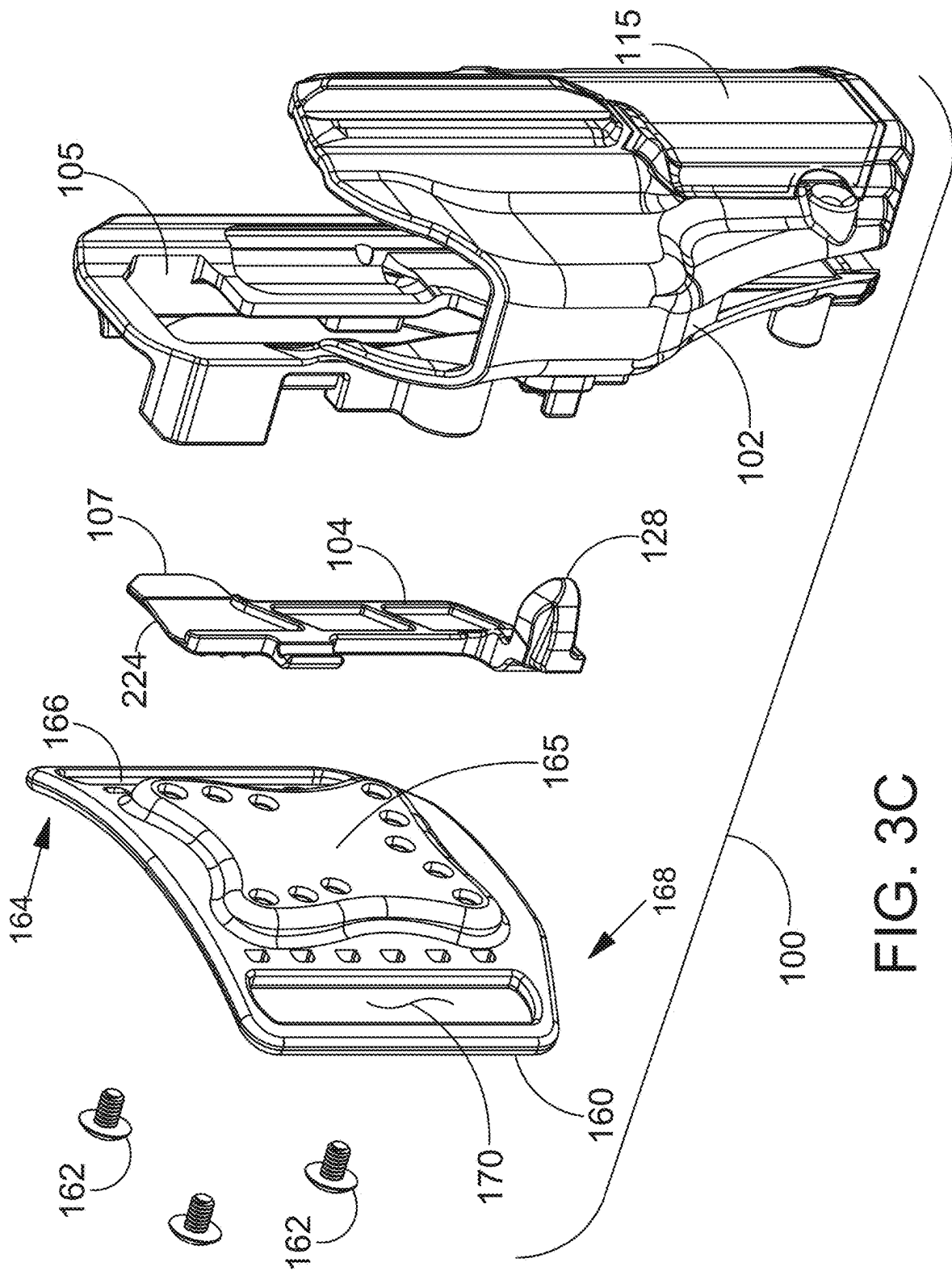


FIG. 3A



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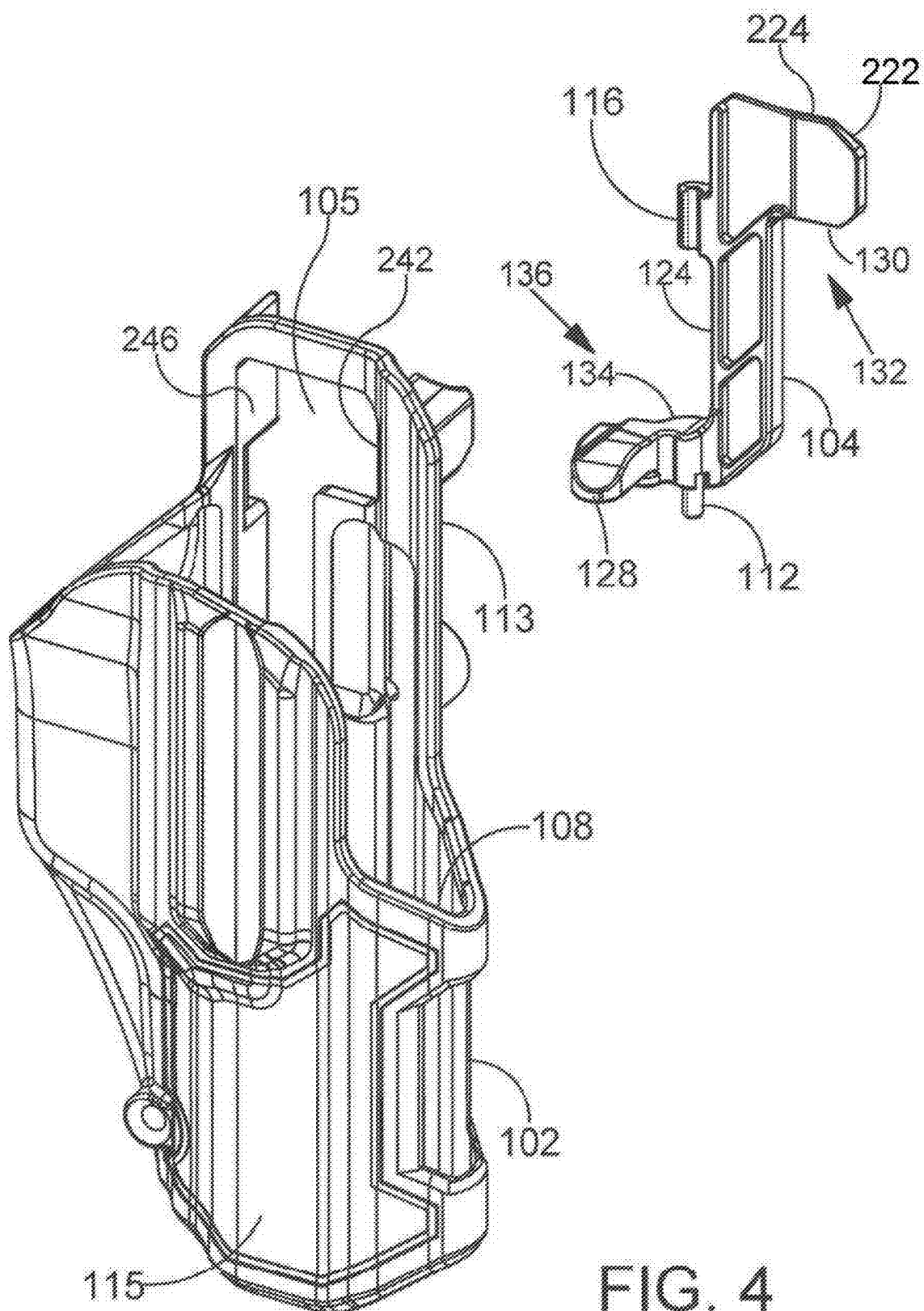
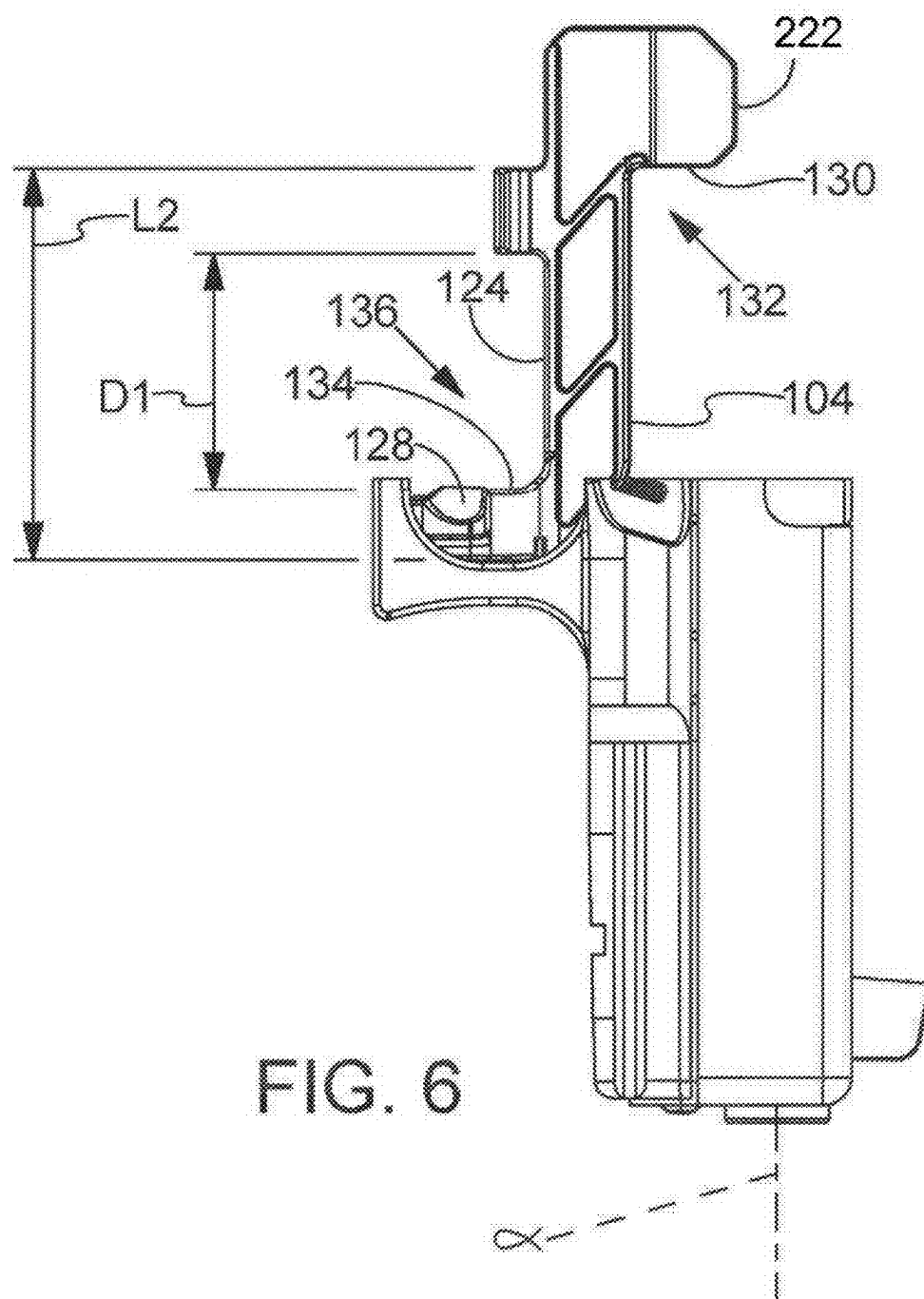
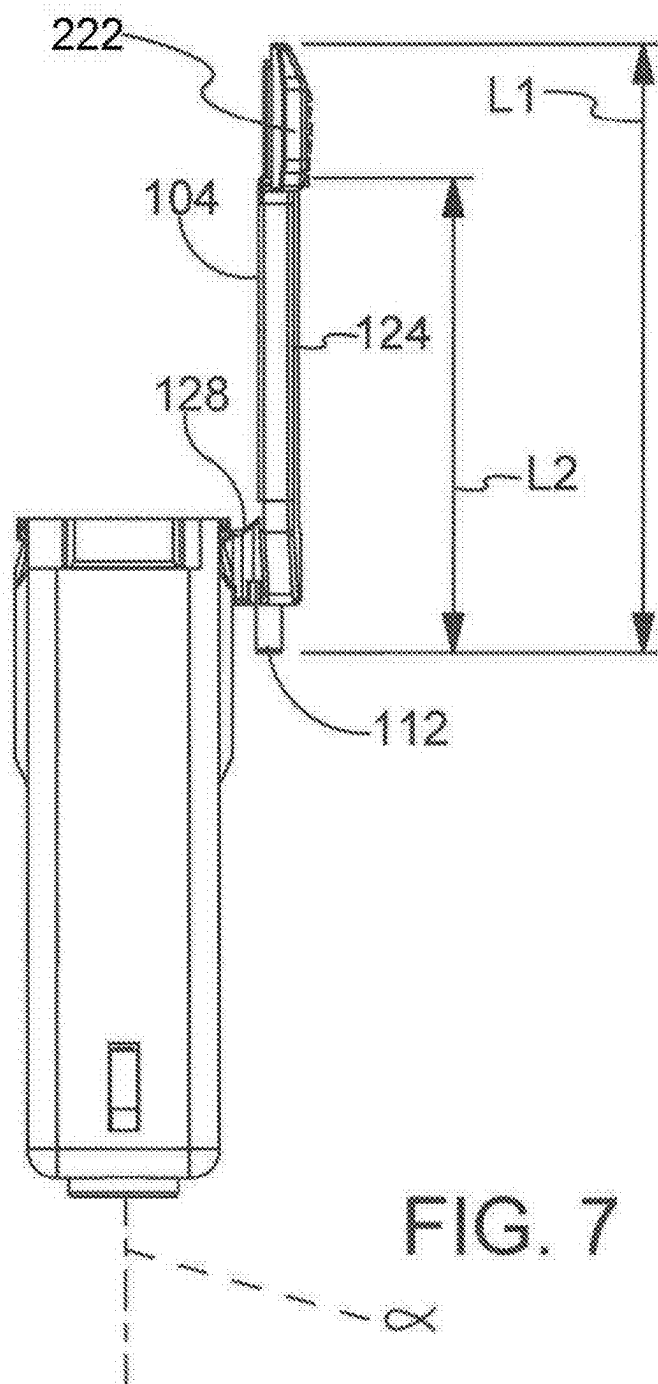


FIG. 5





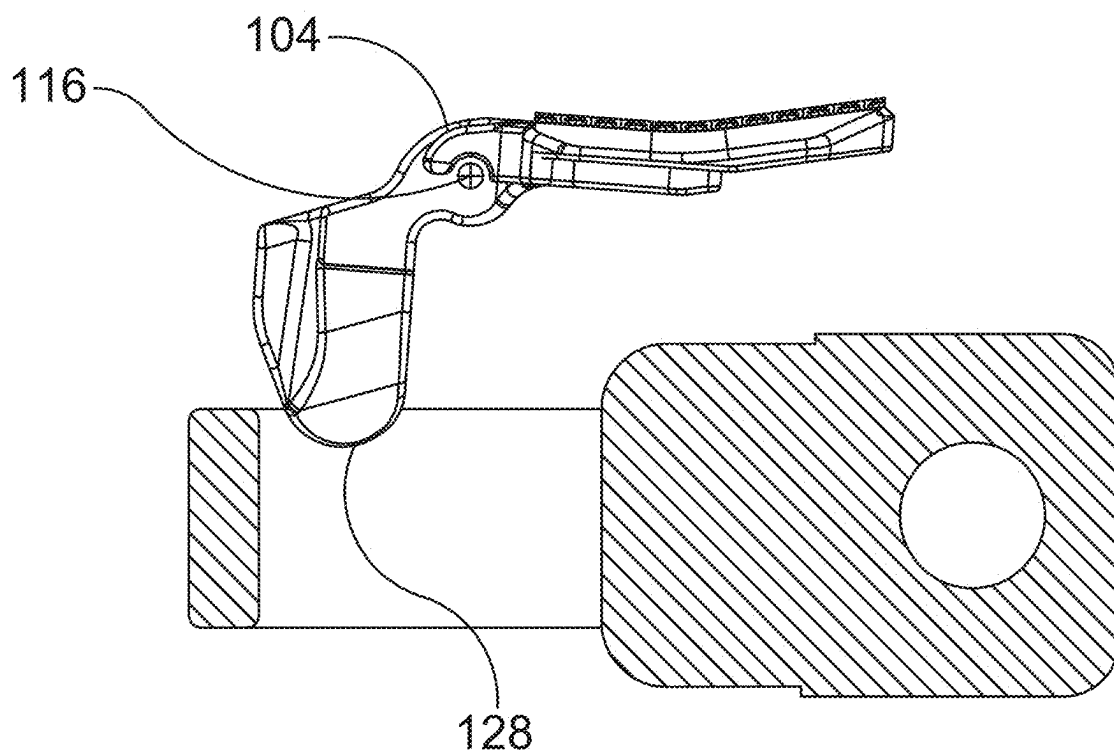


FIG. 8A

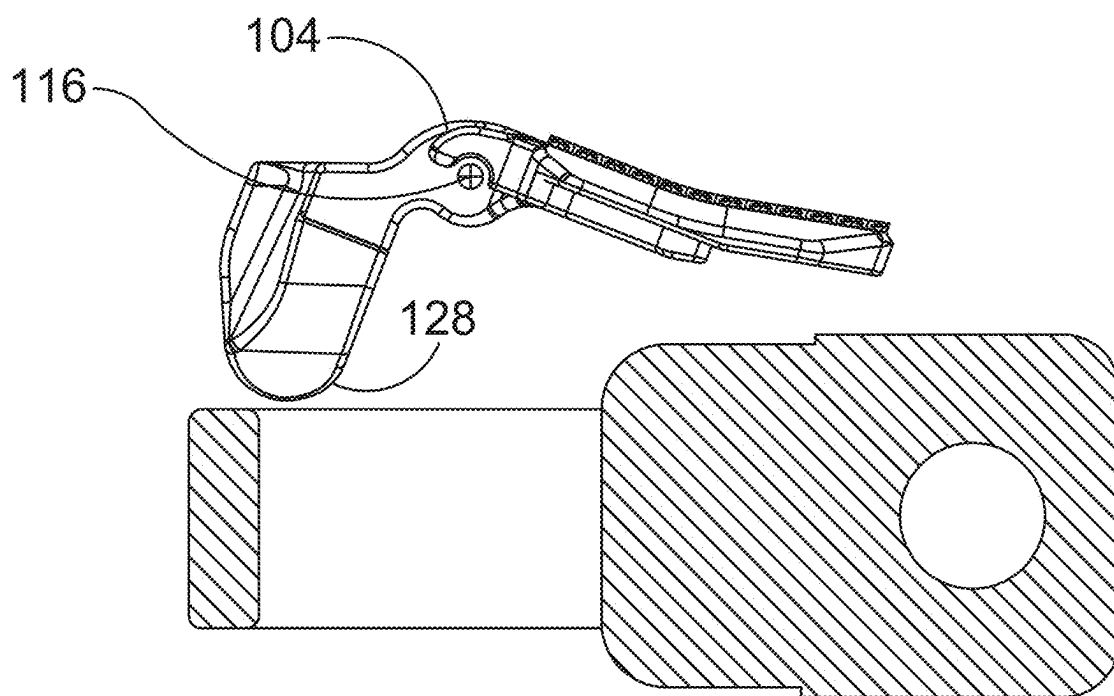
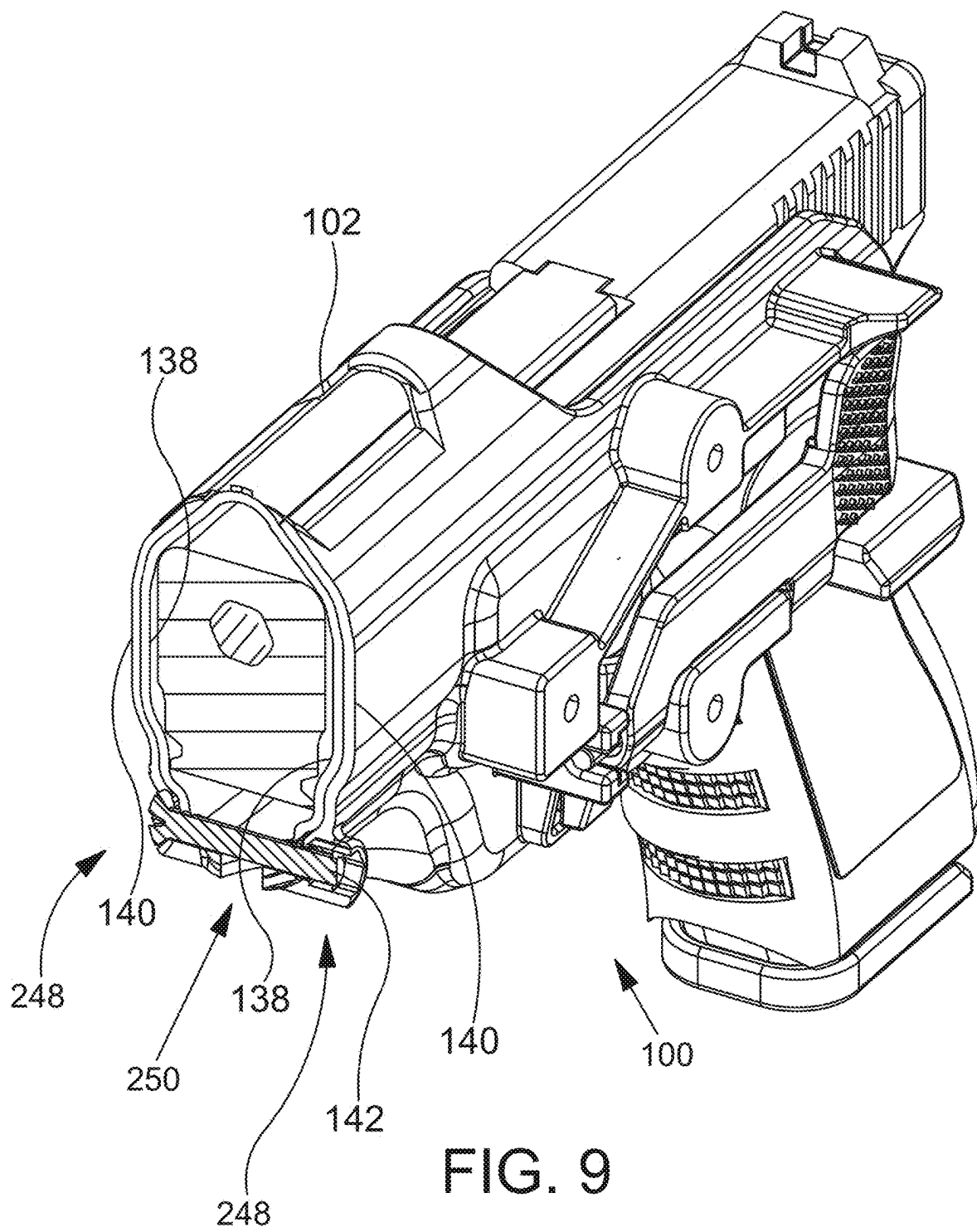


FIG. 8B



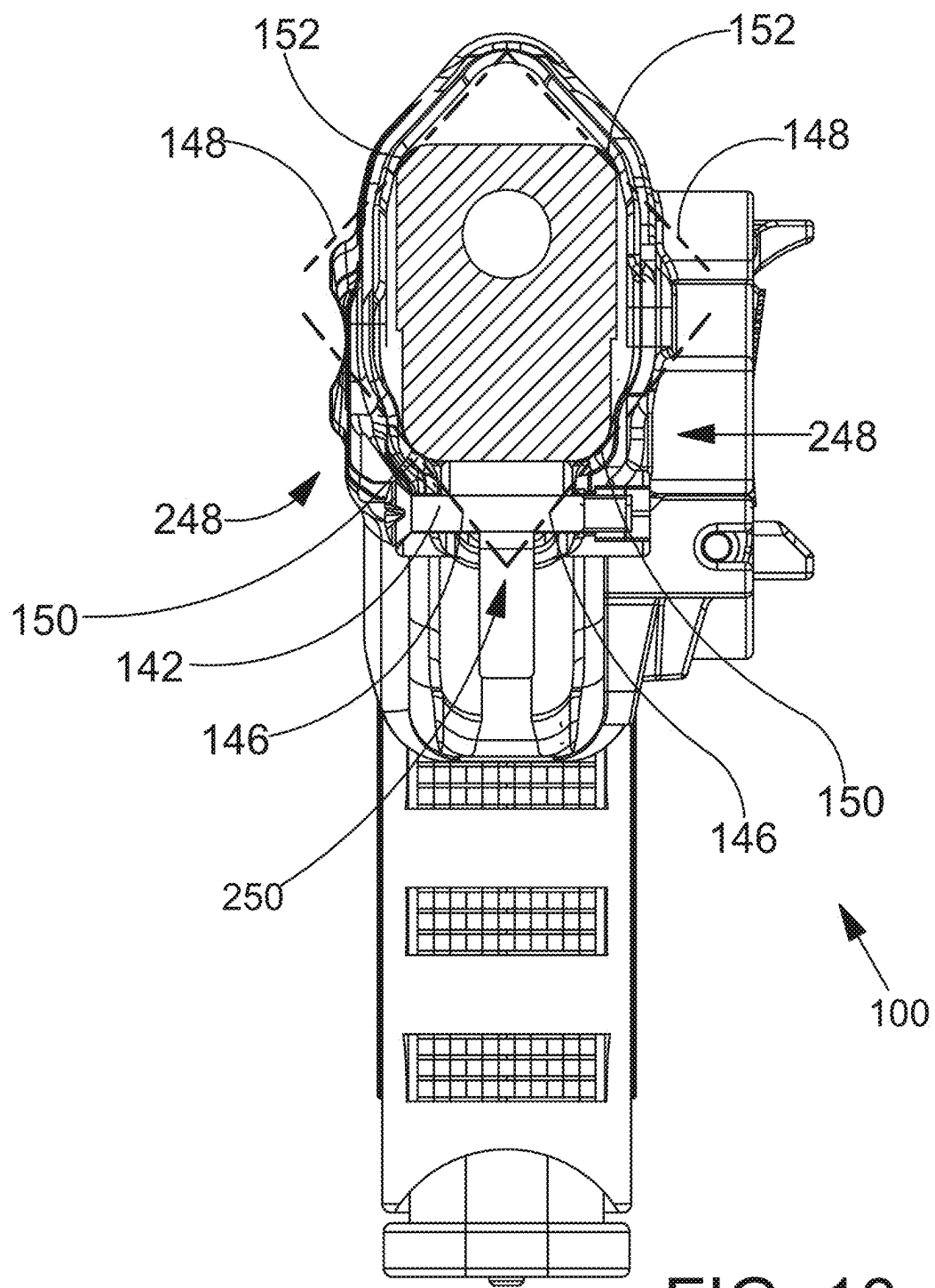


FIG. 10

FIG. 11A

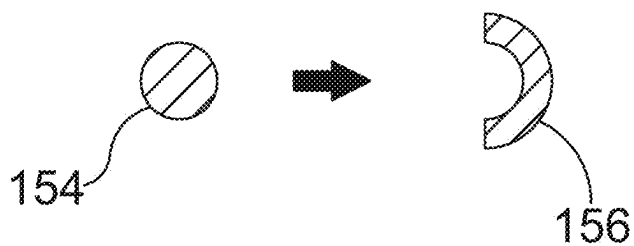


FIG. 11B

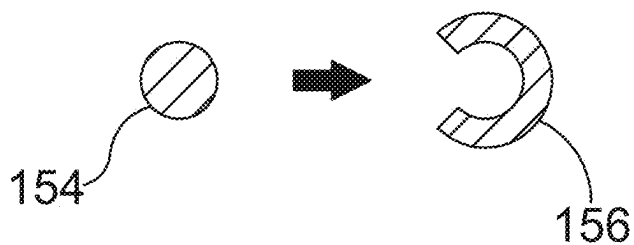


FIG. 11C

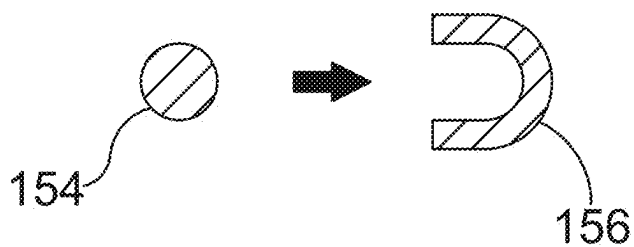
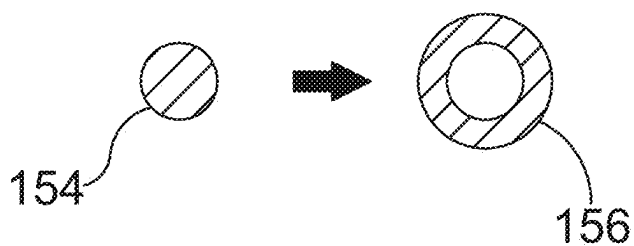


FIG. 12



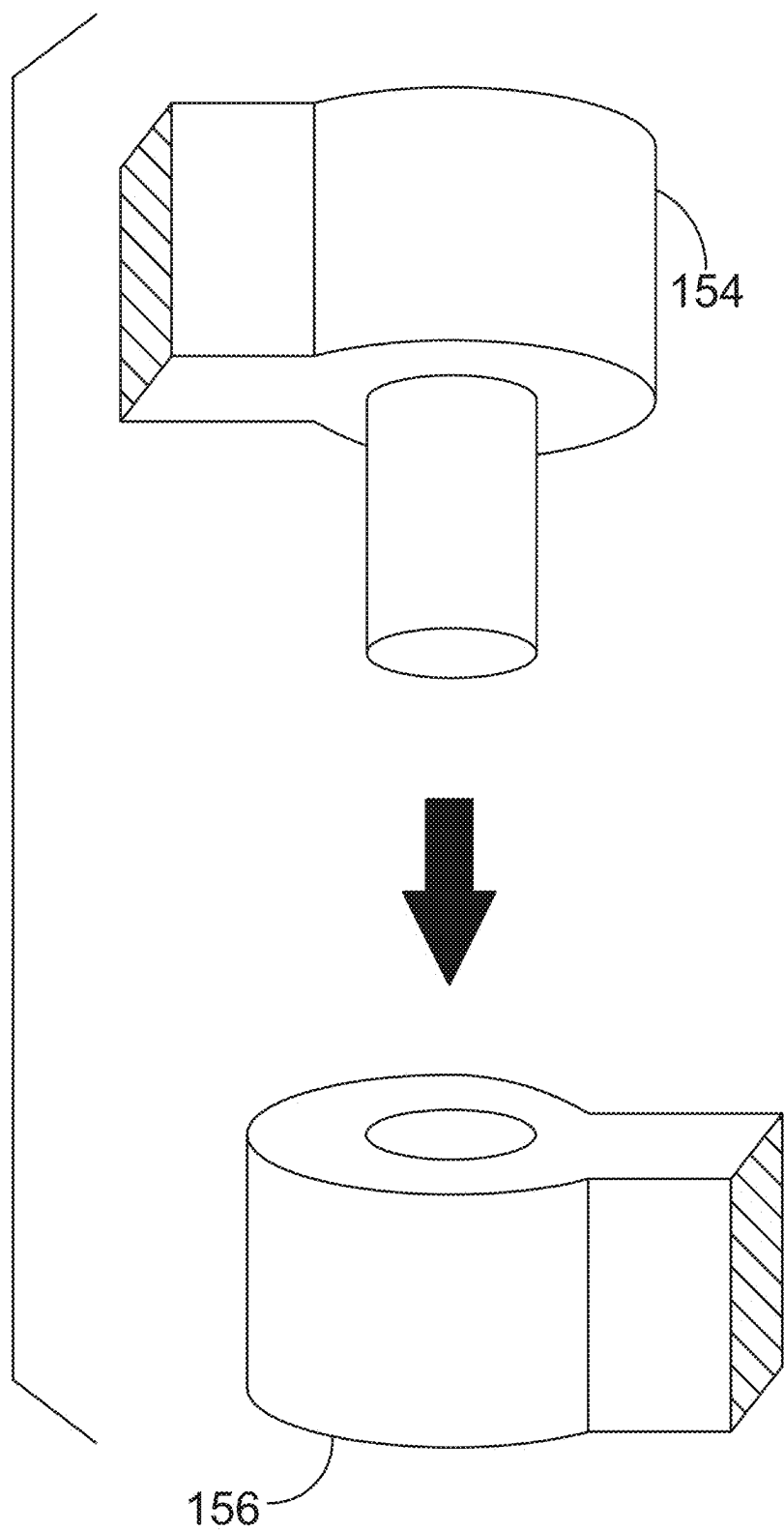


FIG. 13

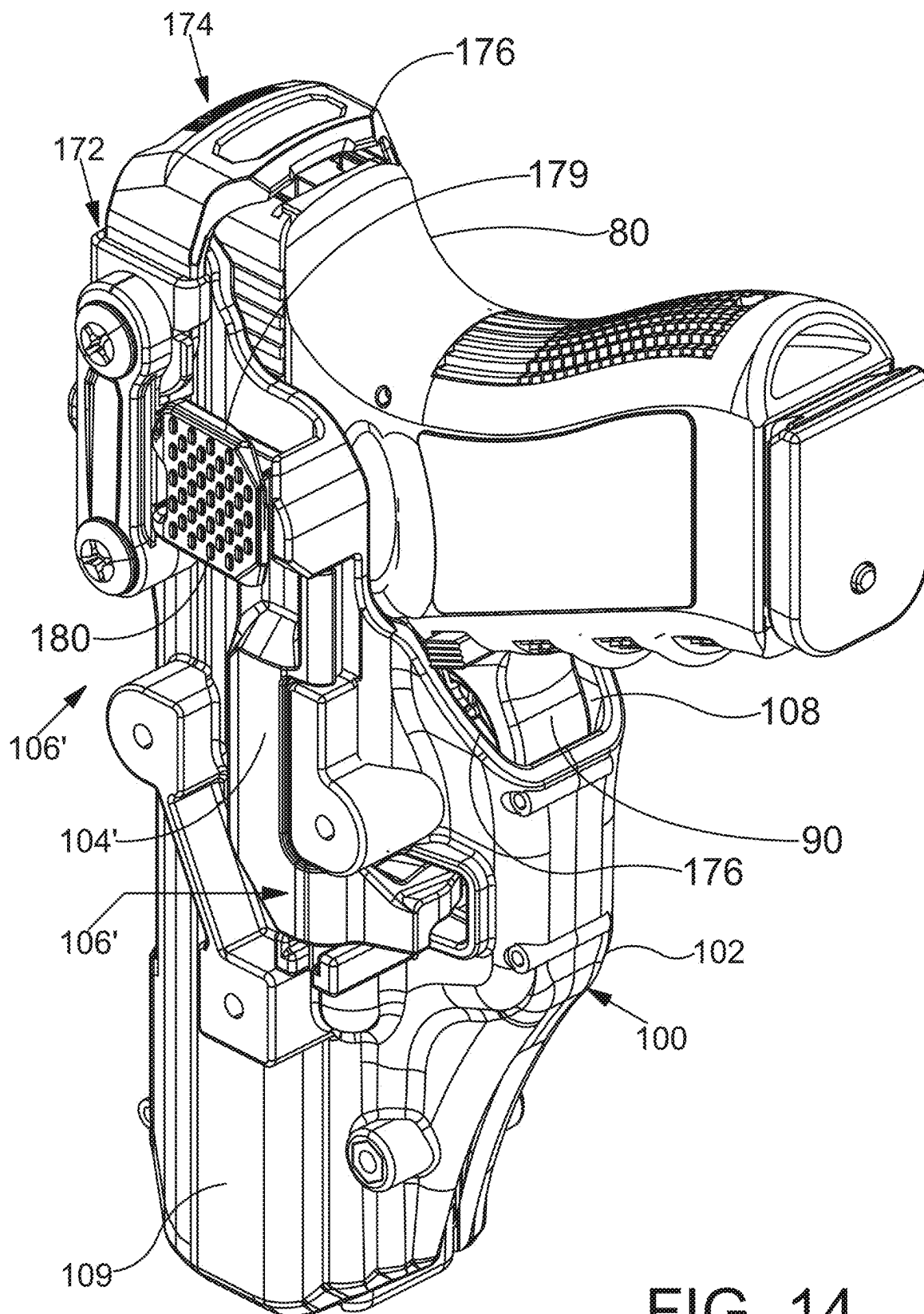
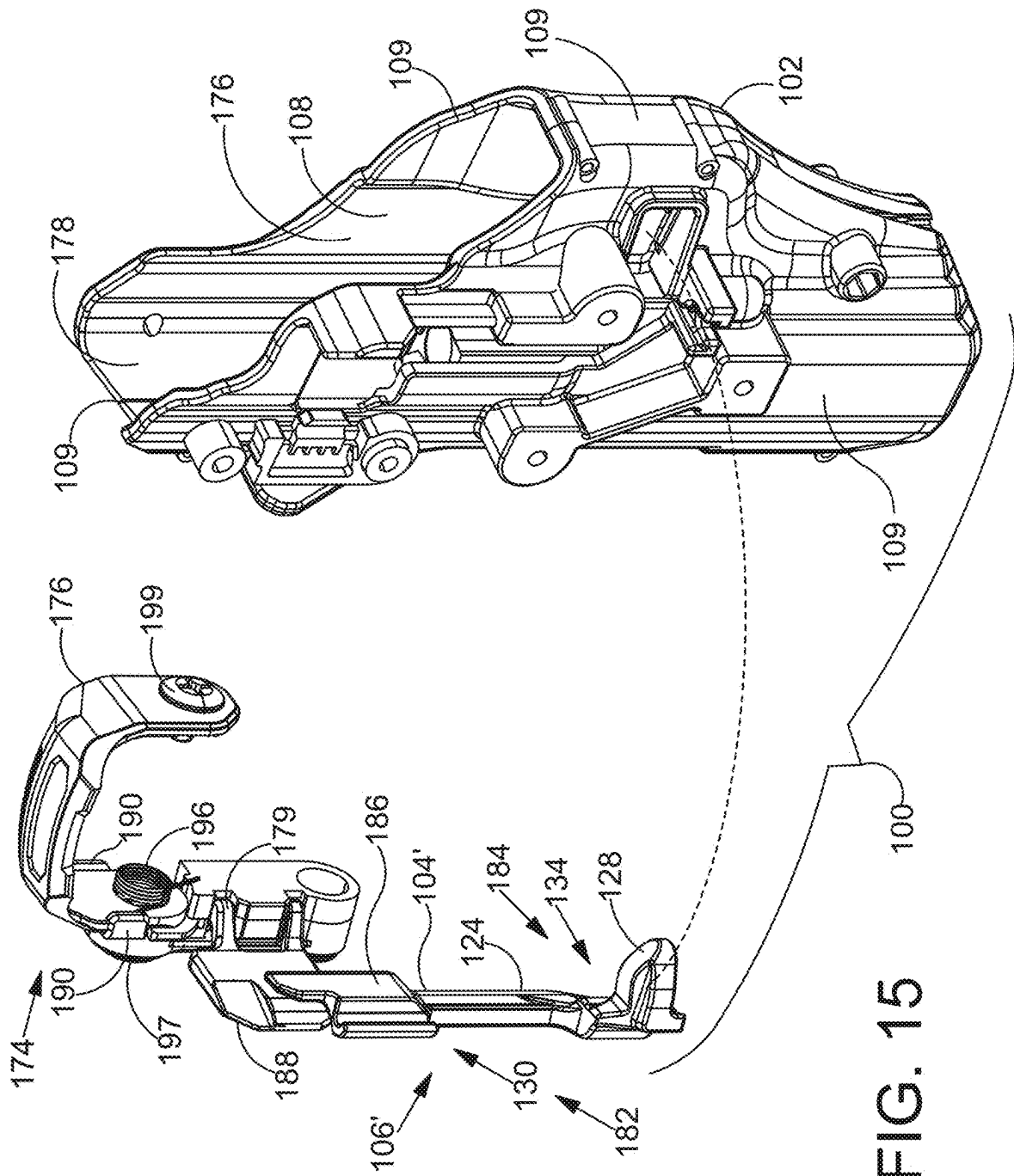
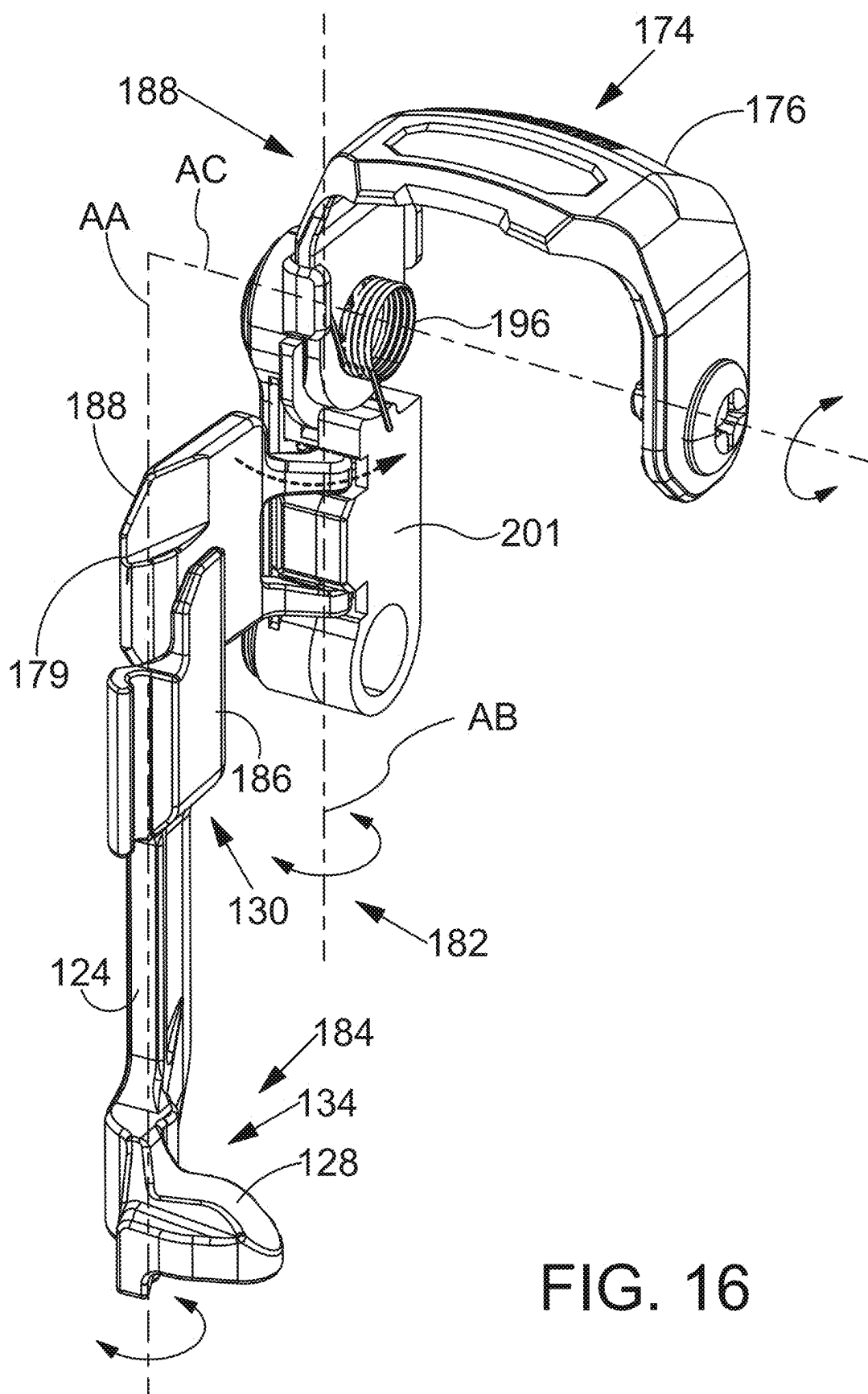
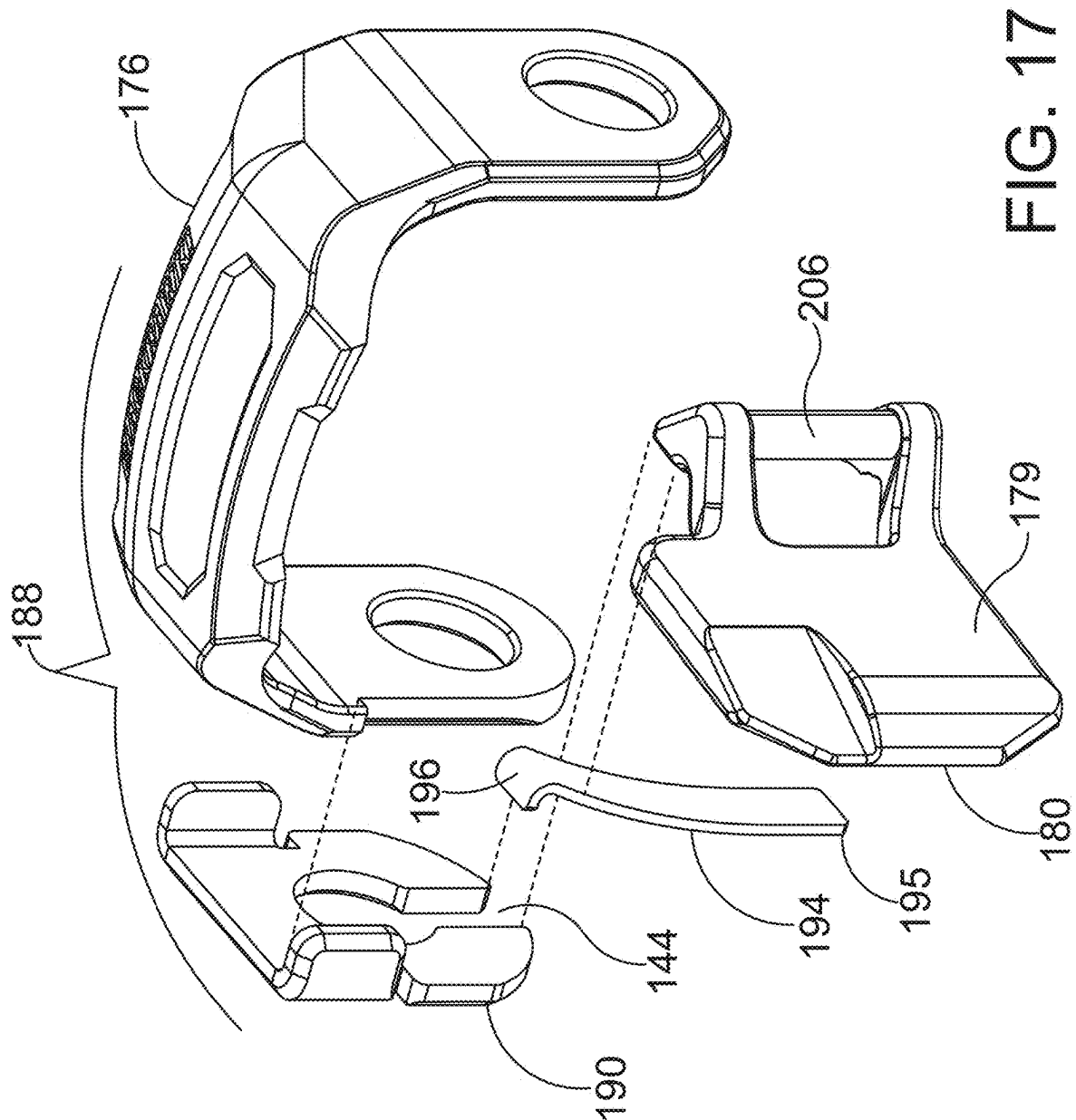


FIG. 14







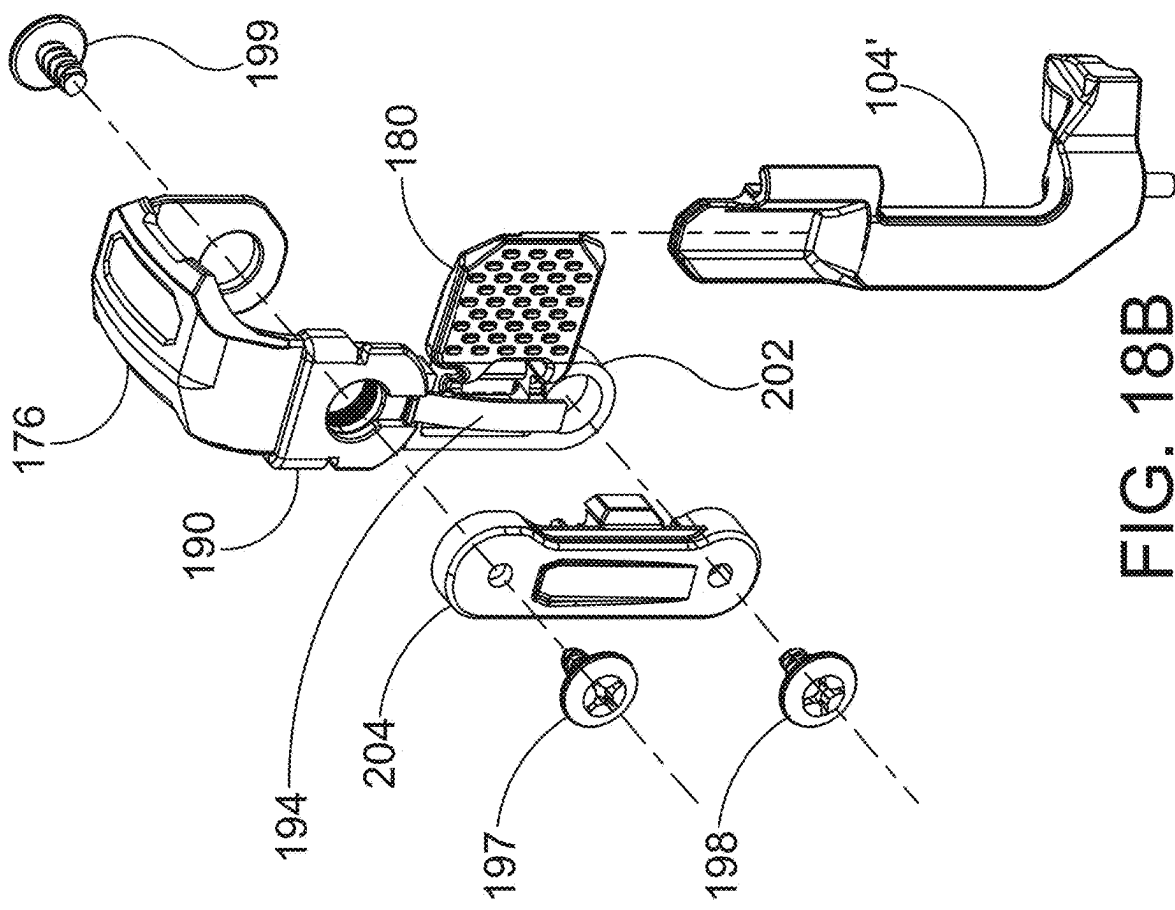


FIG. 18B

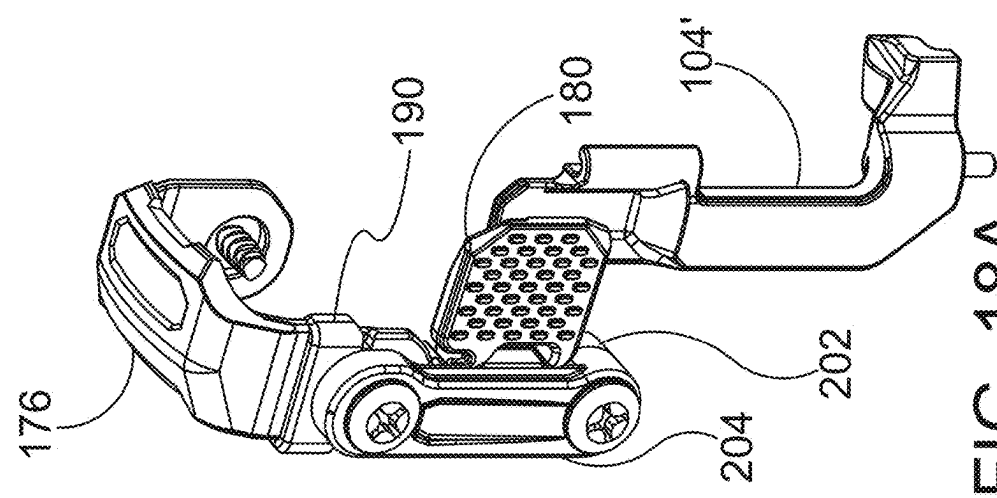


FIG. 18A

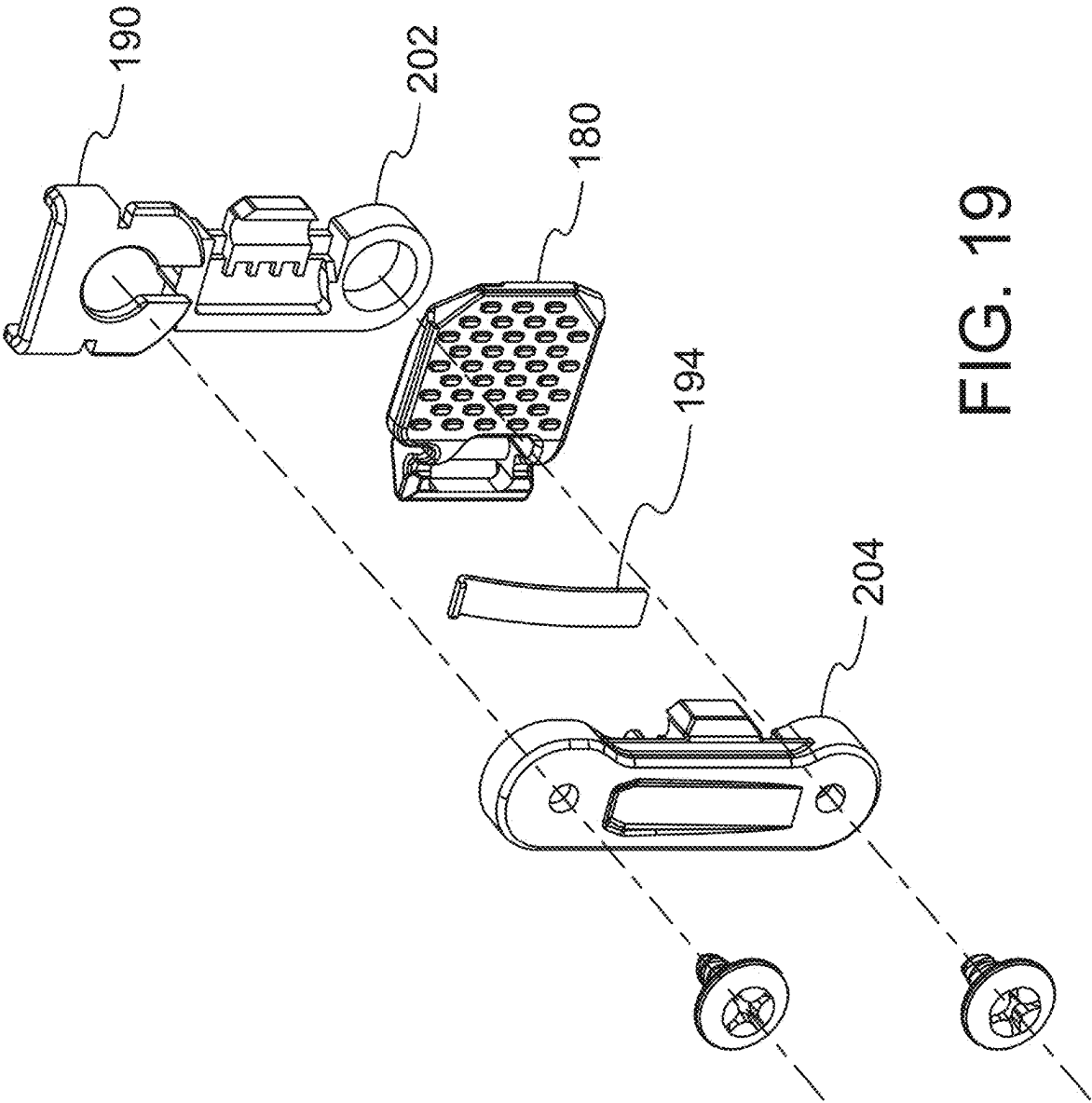


FIG. 19

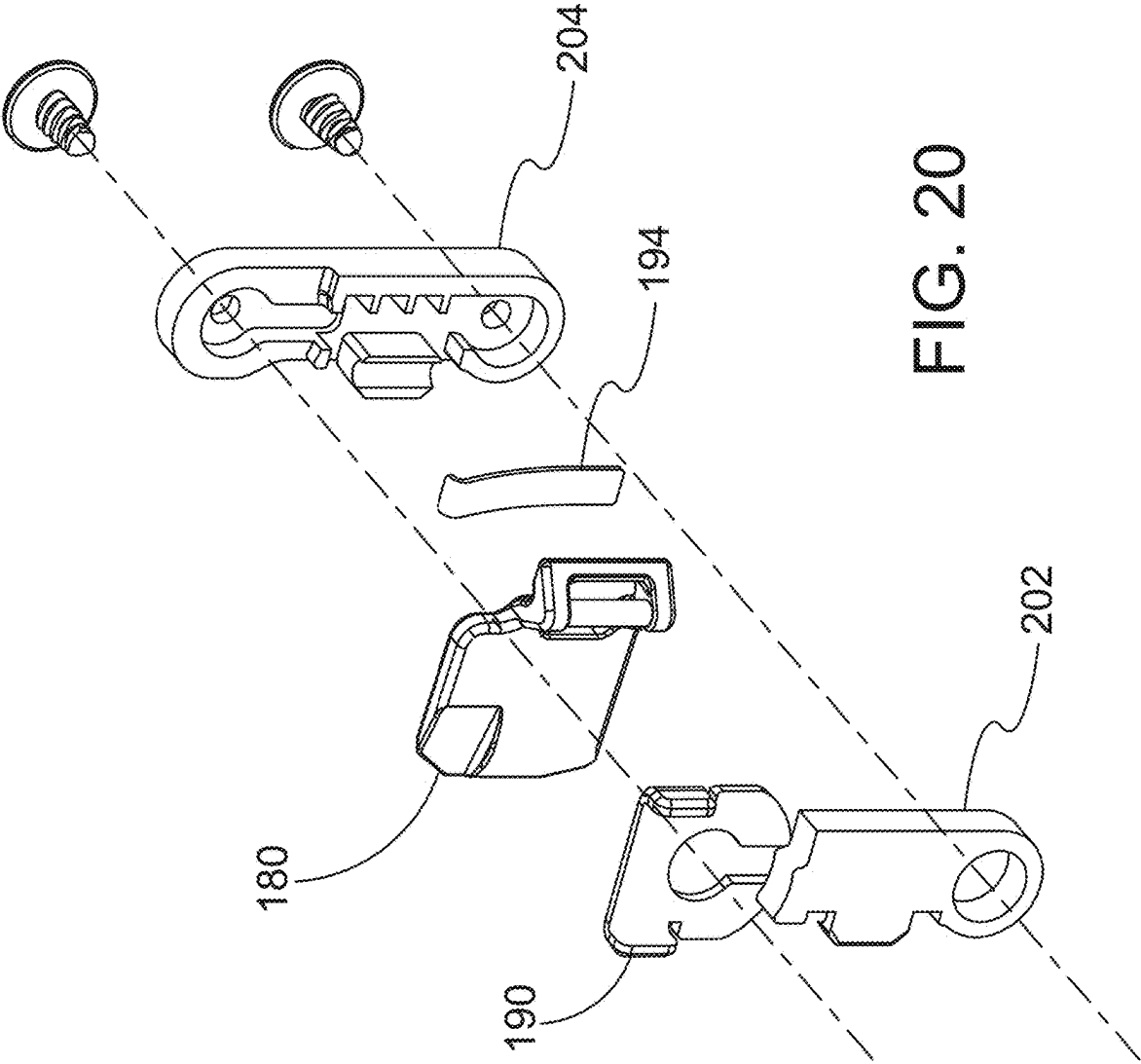


FIG. 20

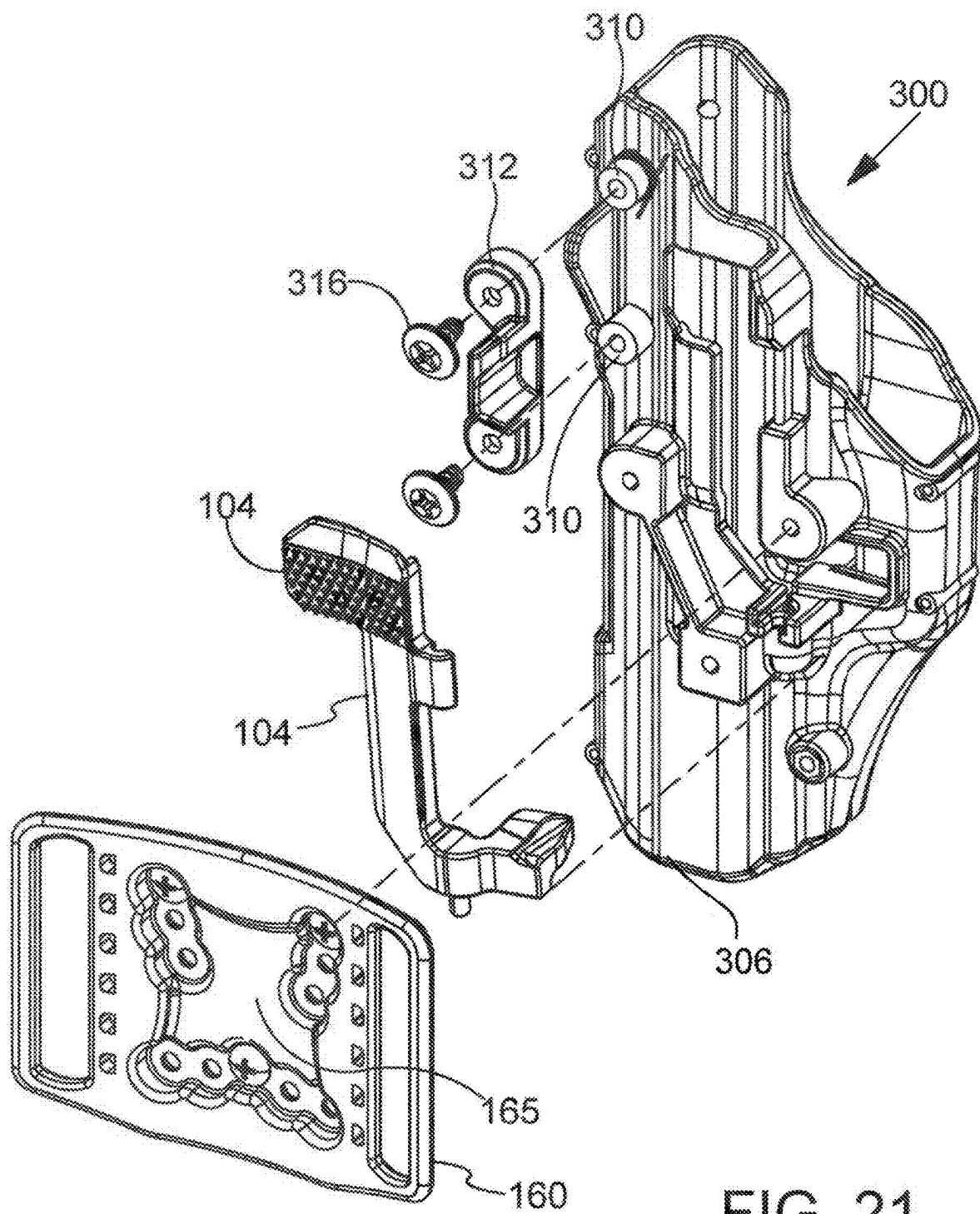


FIG. 21

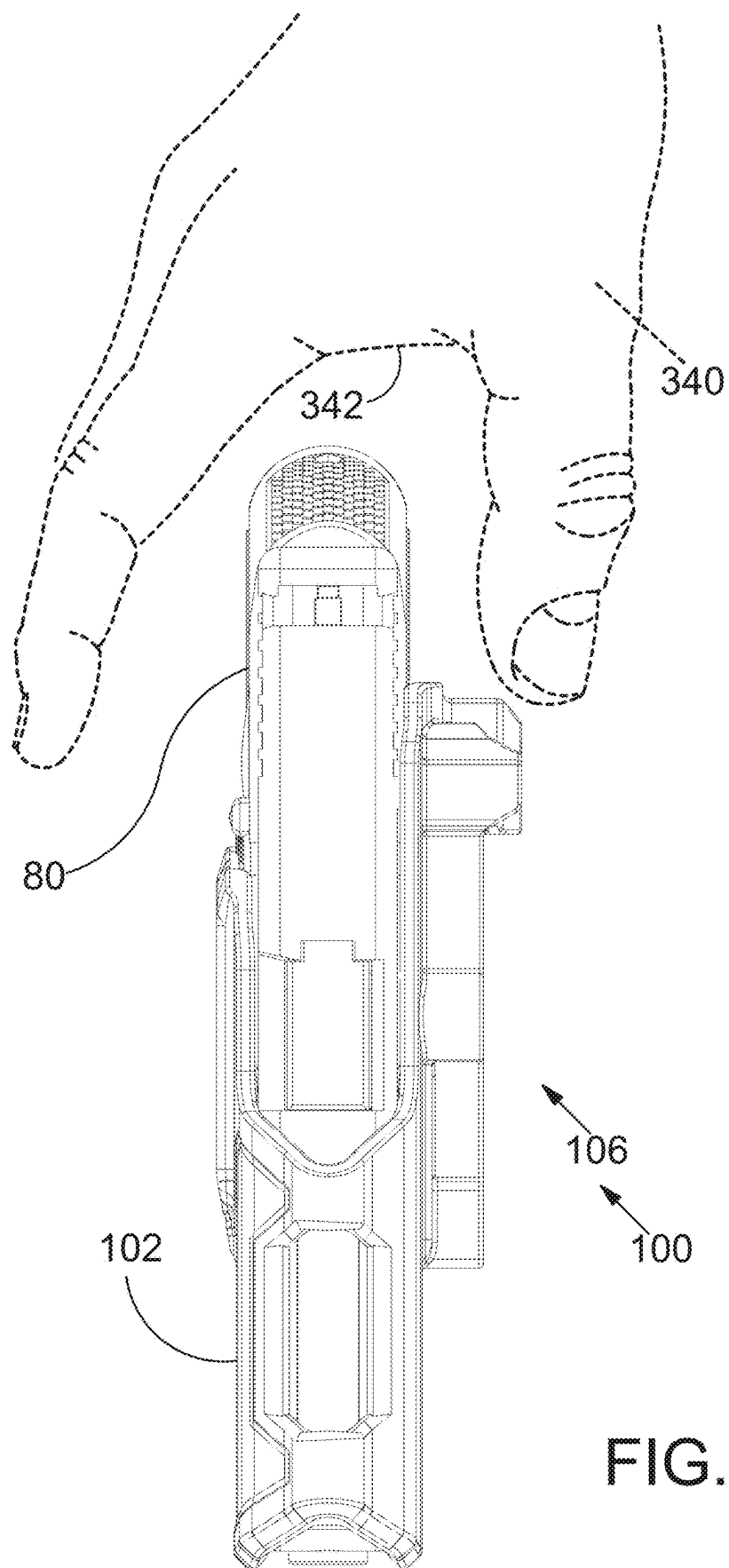


FIG. 22

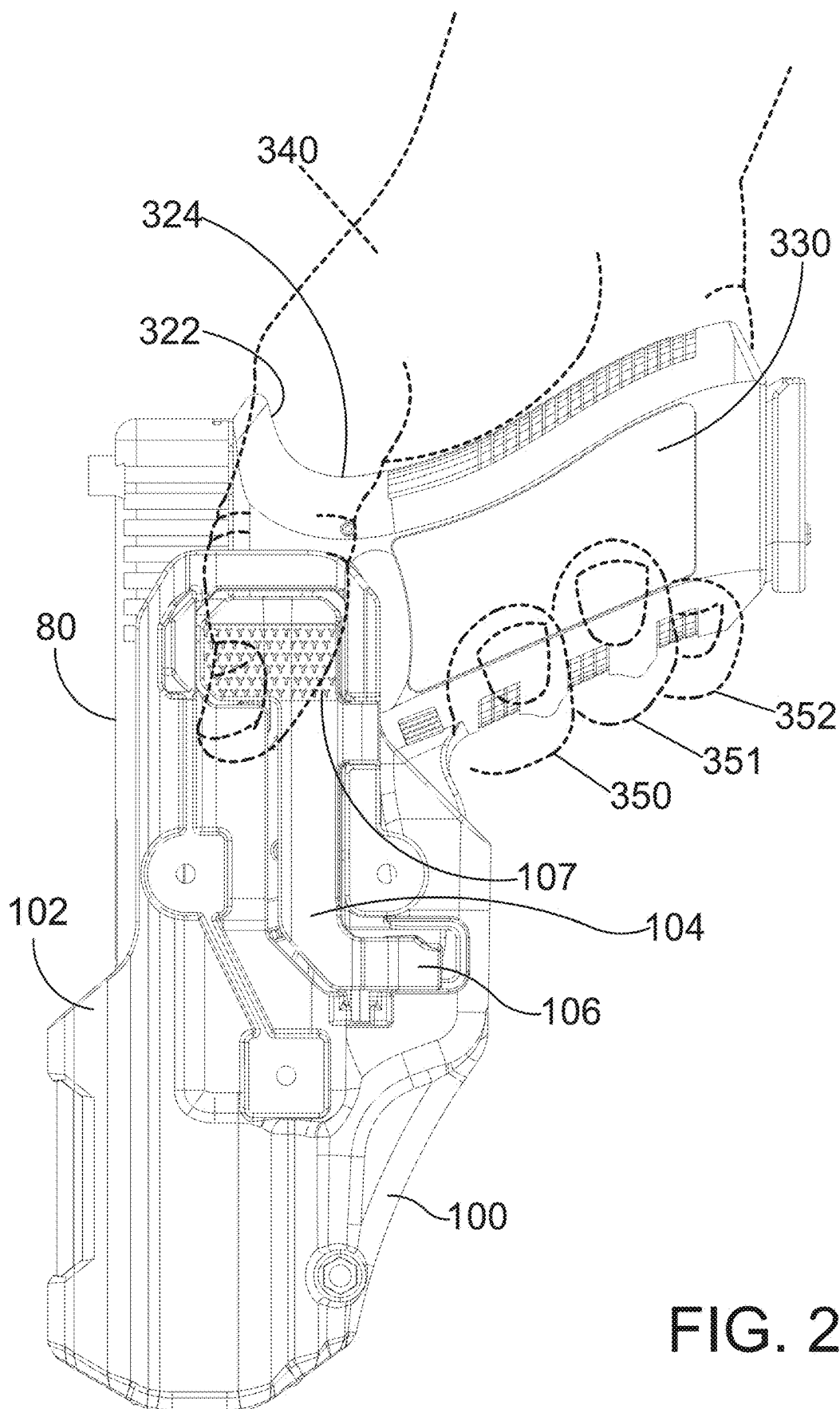


FIG. 23

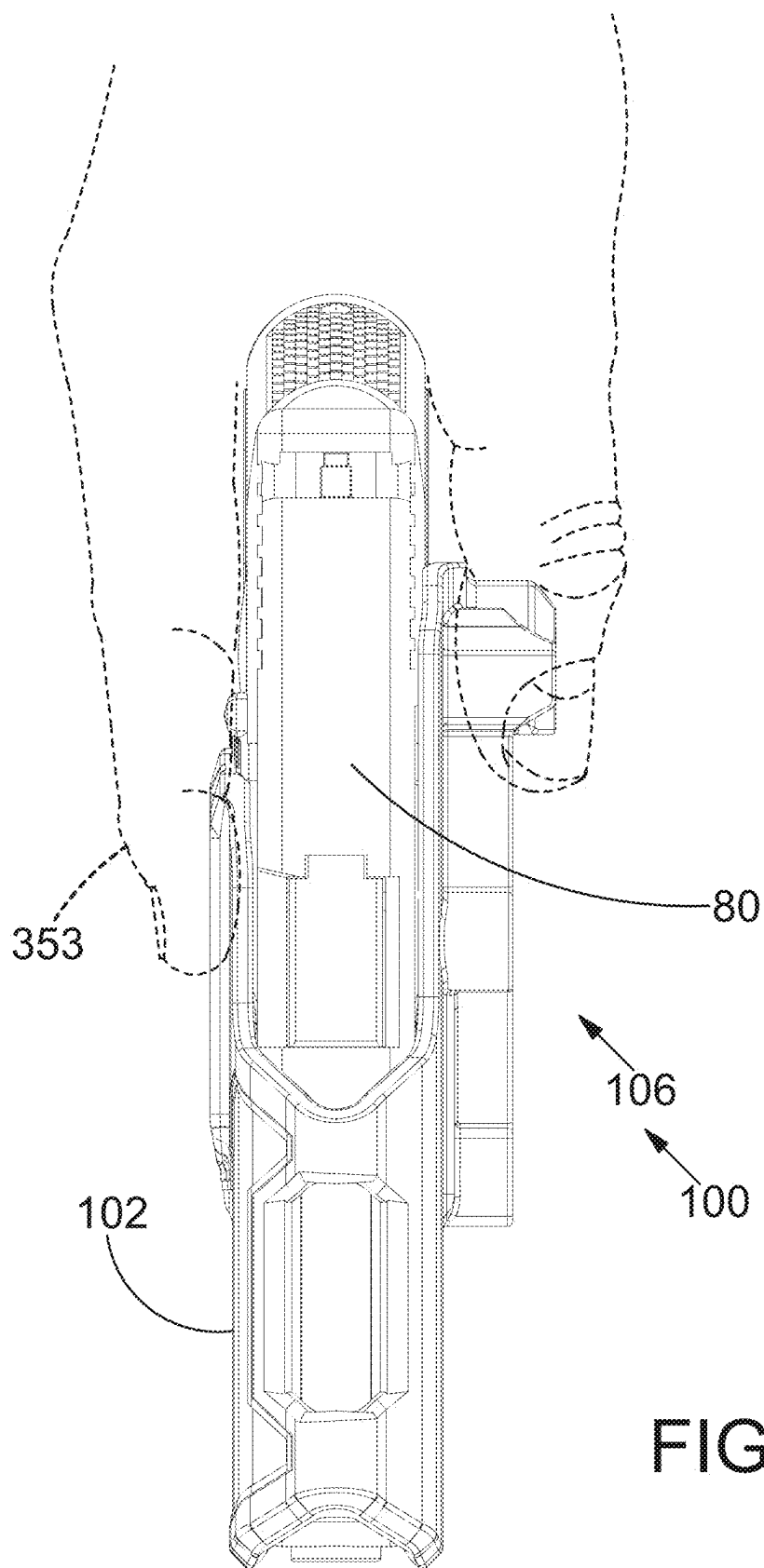


FIG. 24

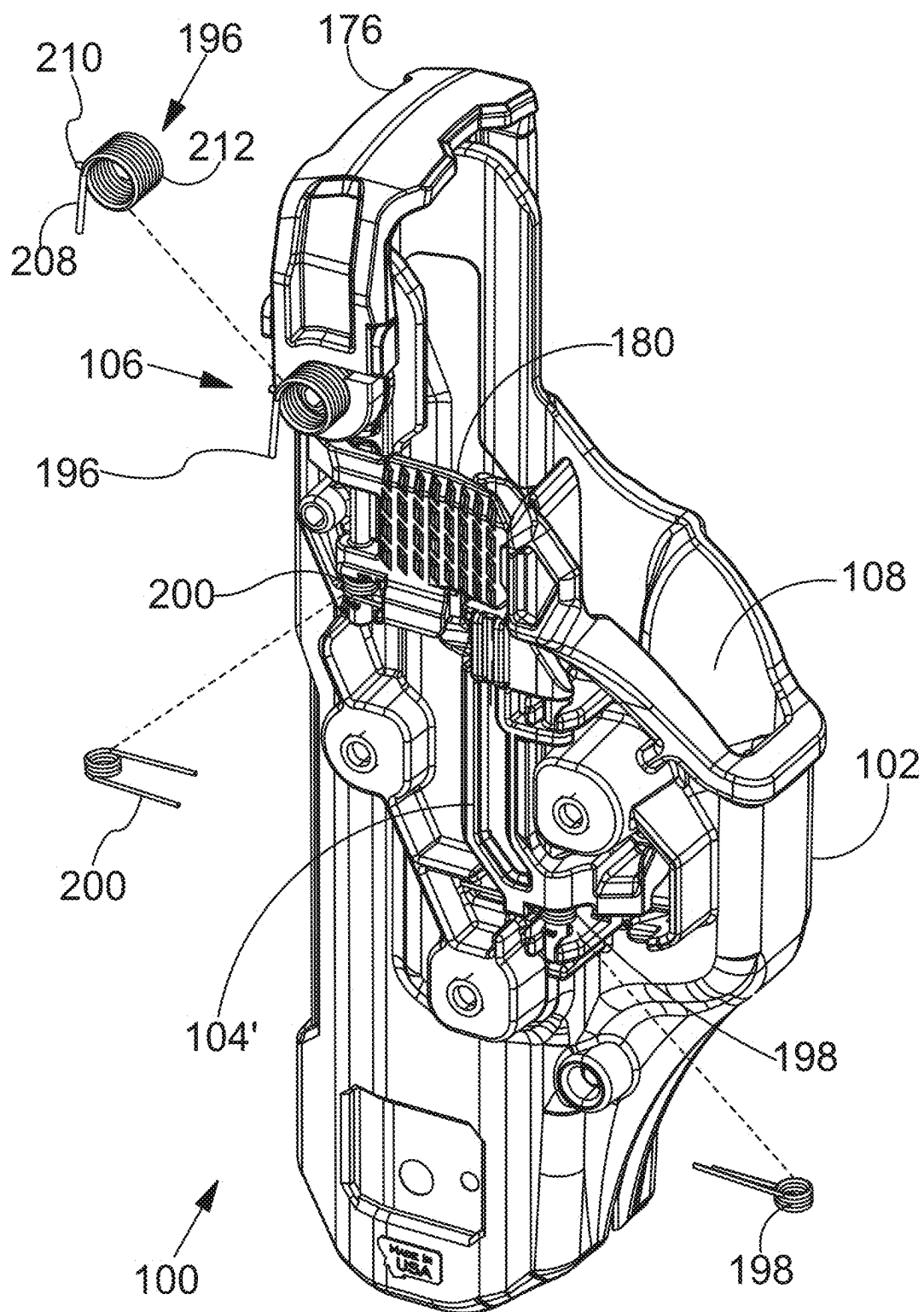
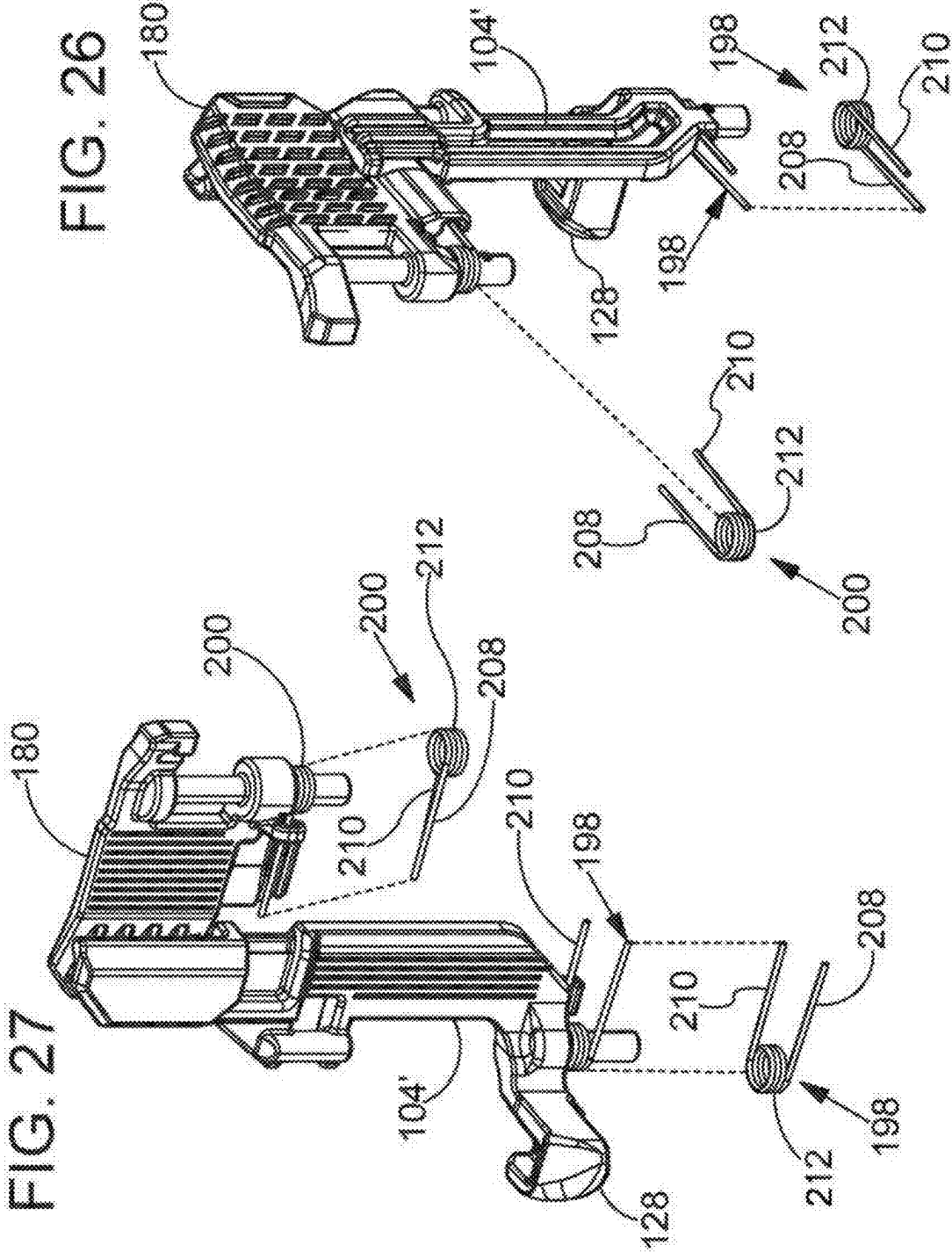
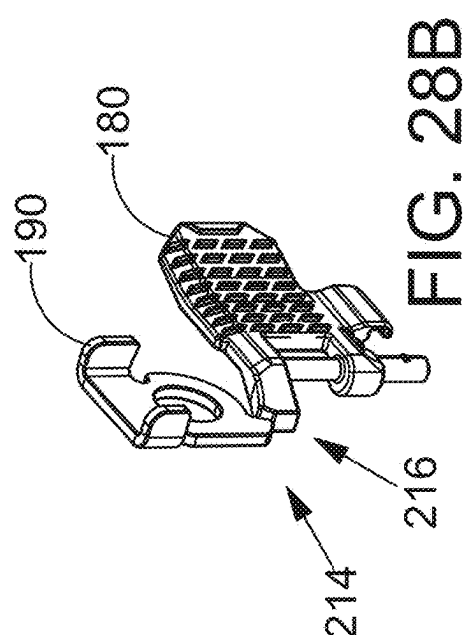
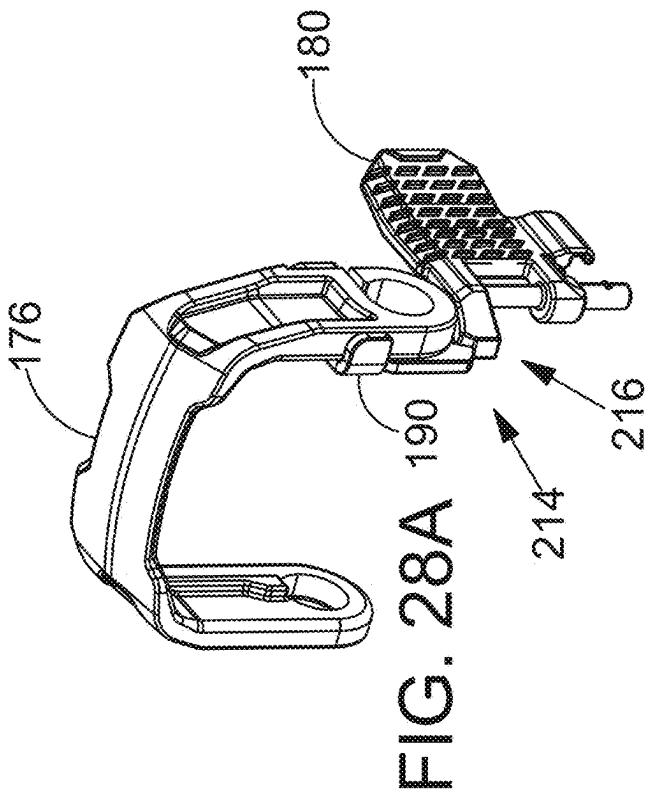
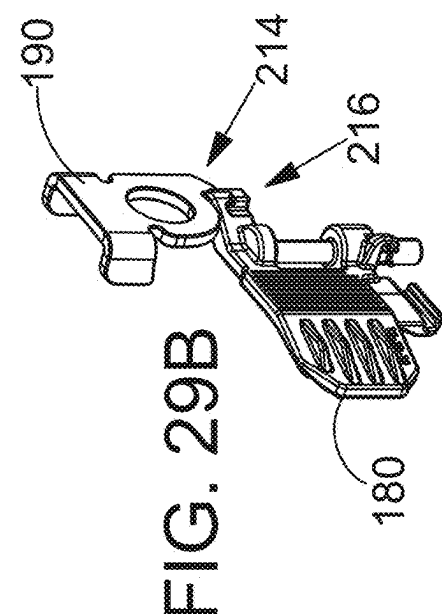
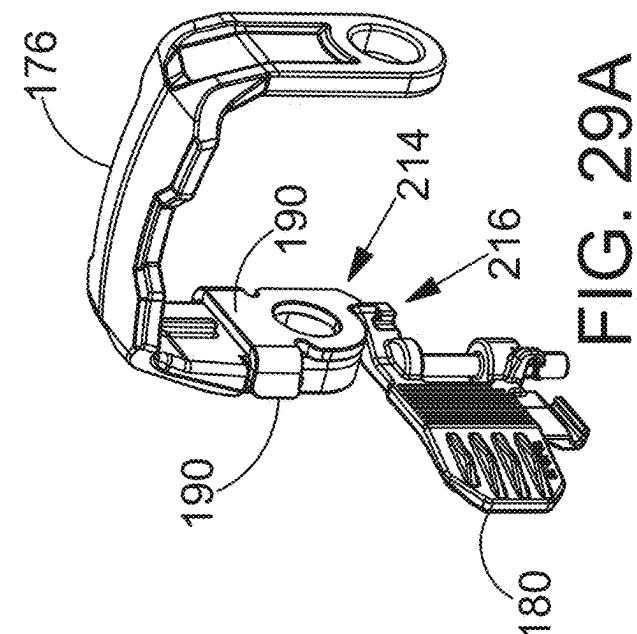
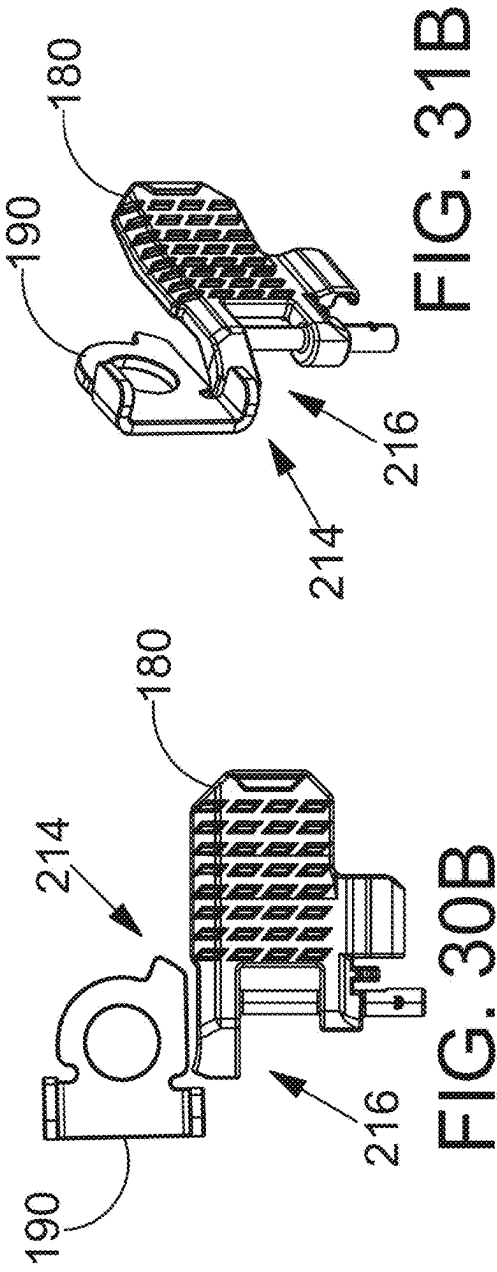
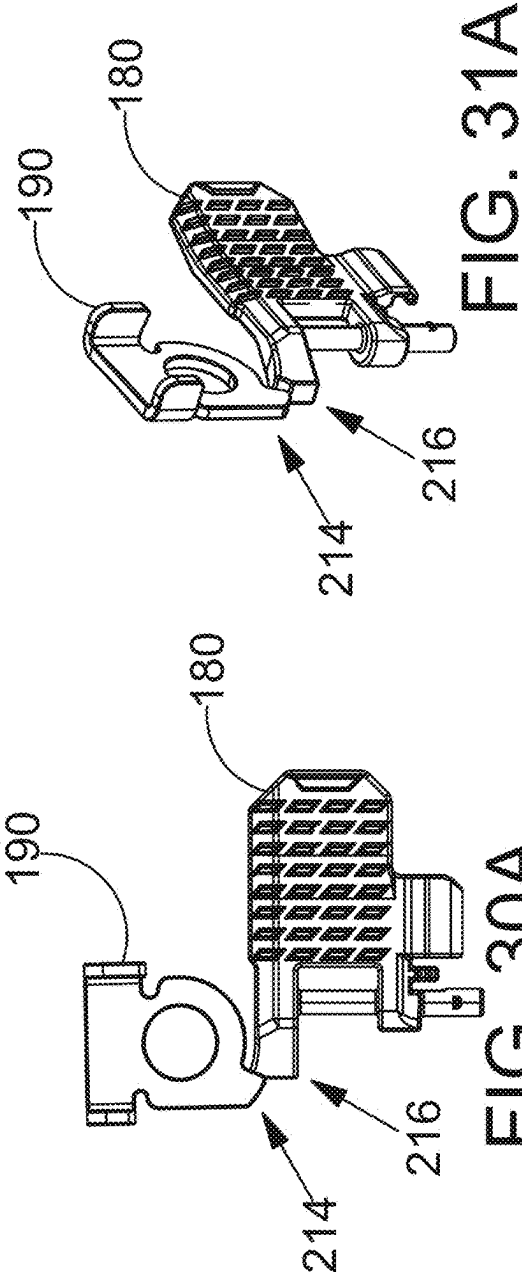
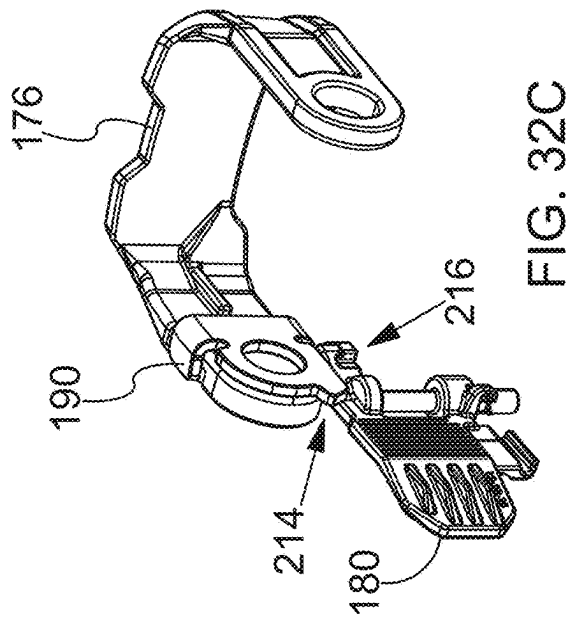
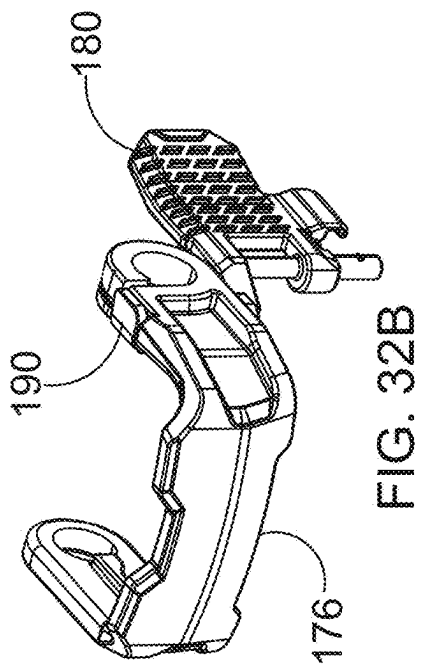
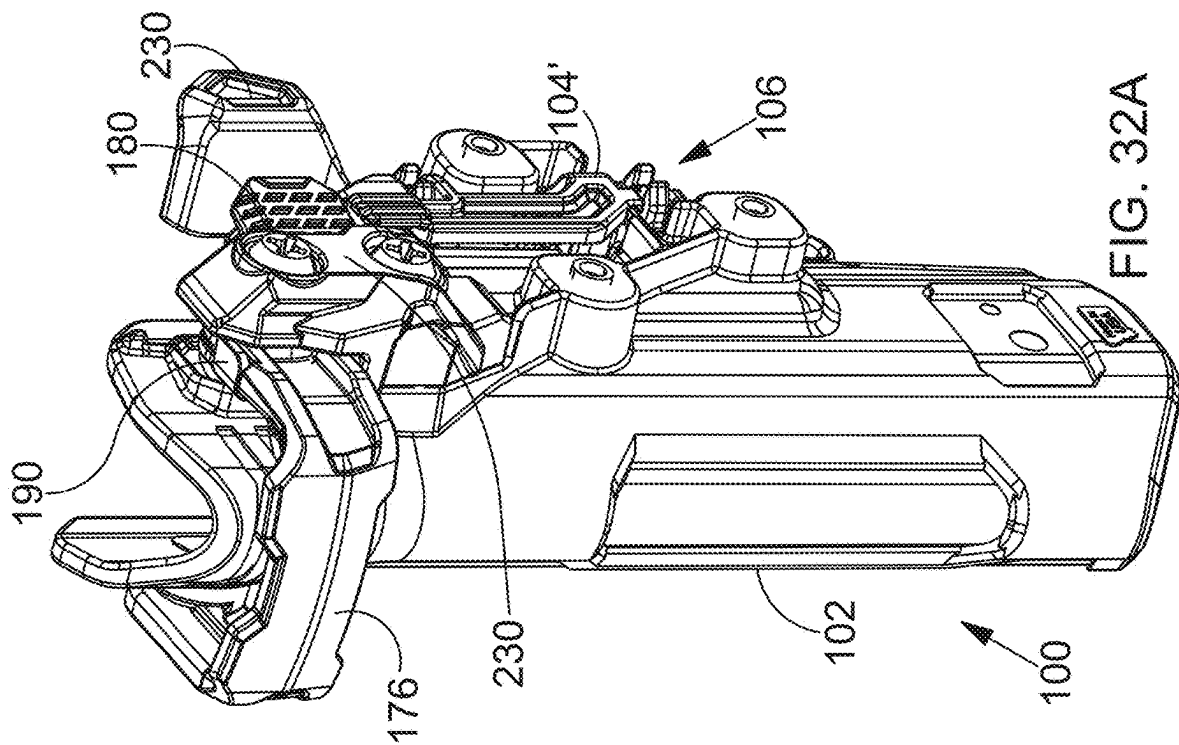


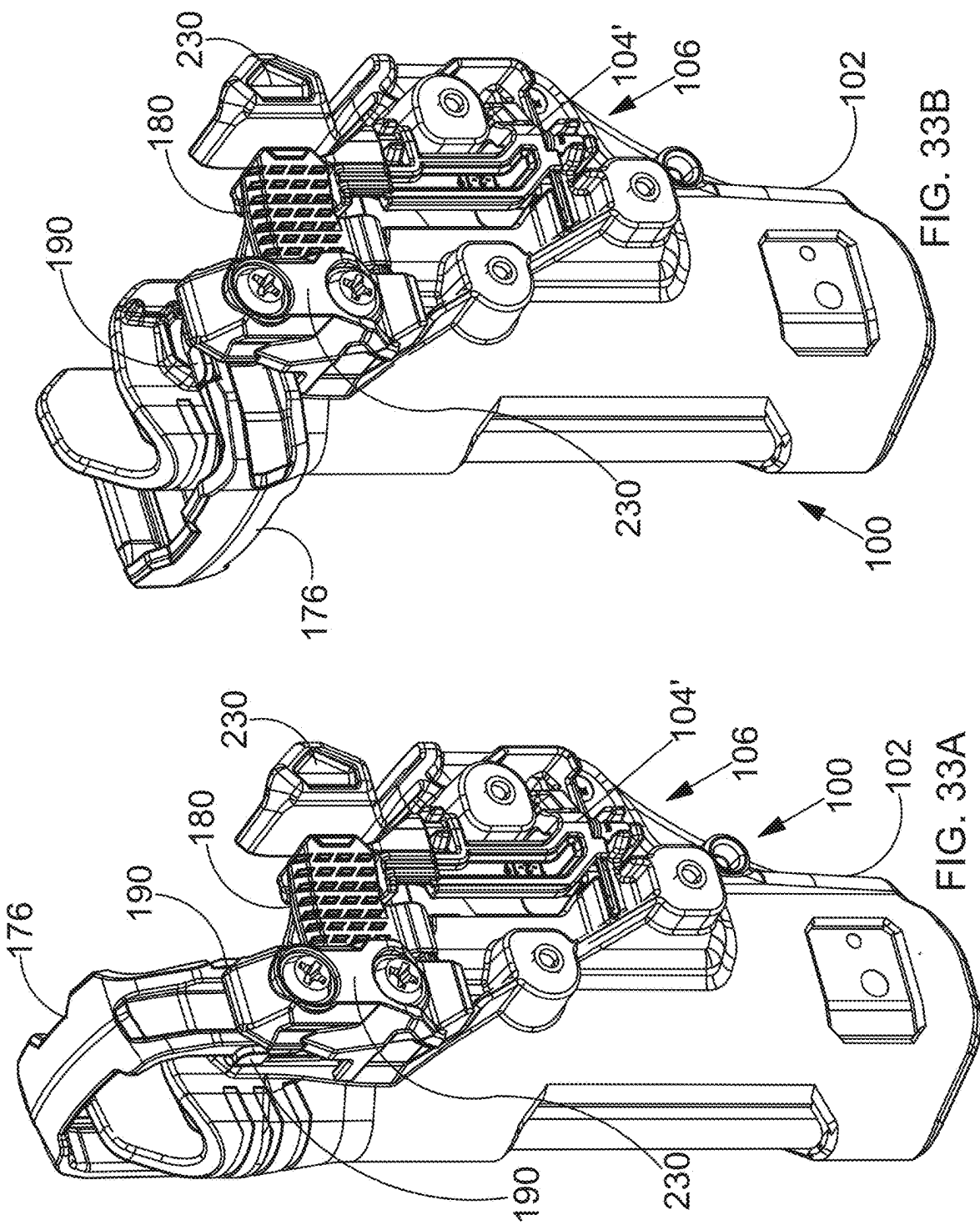
FIG. 25

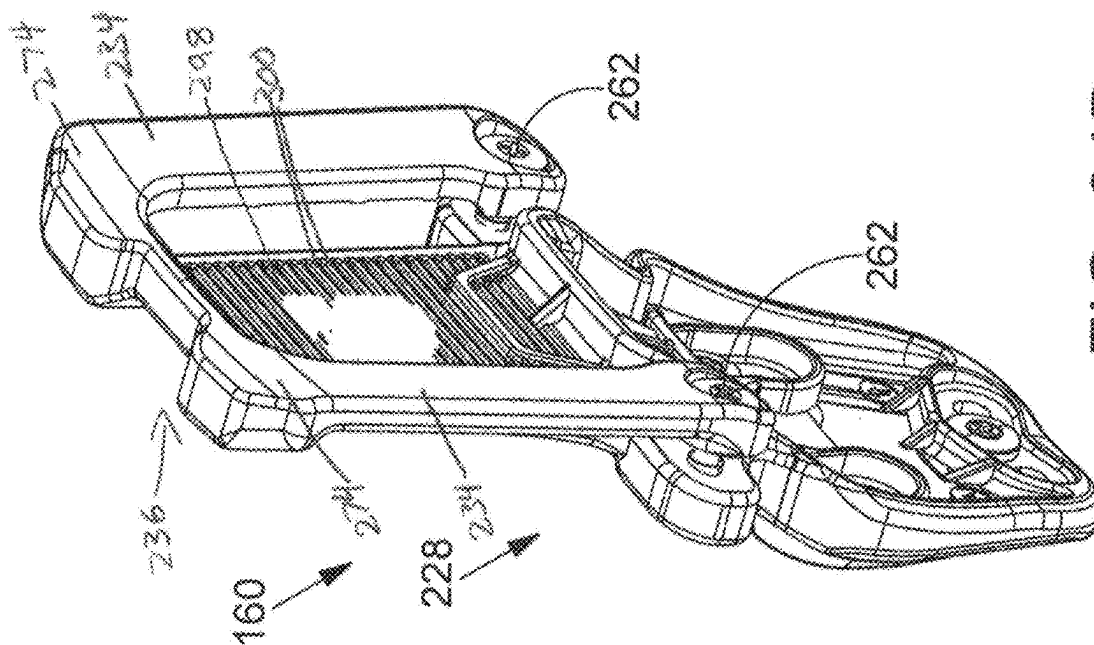
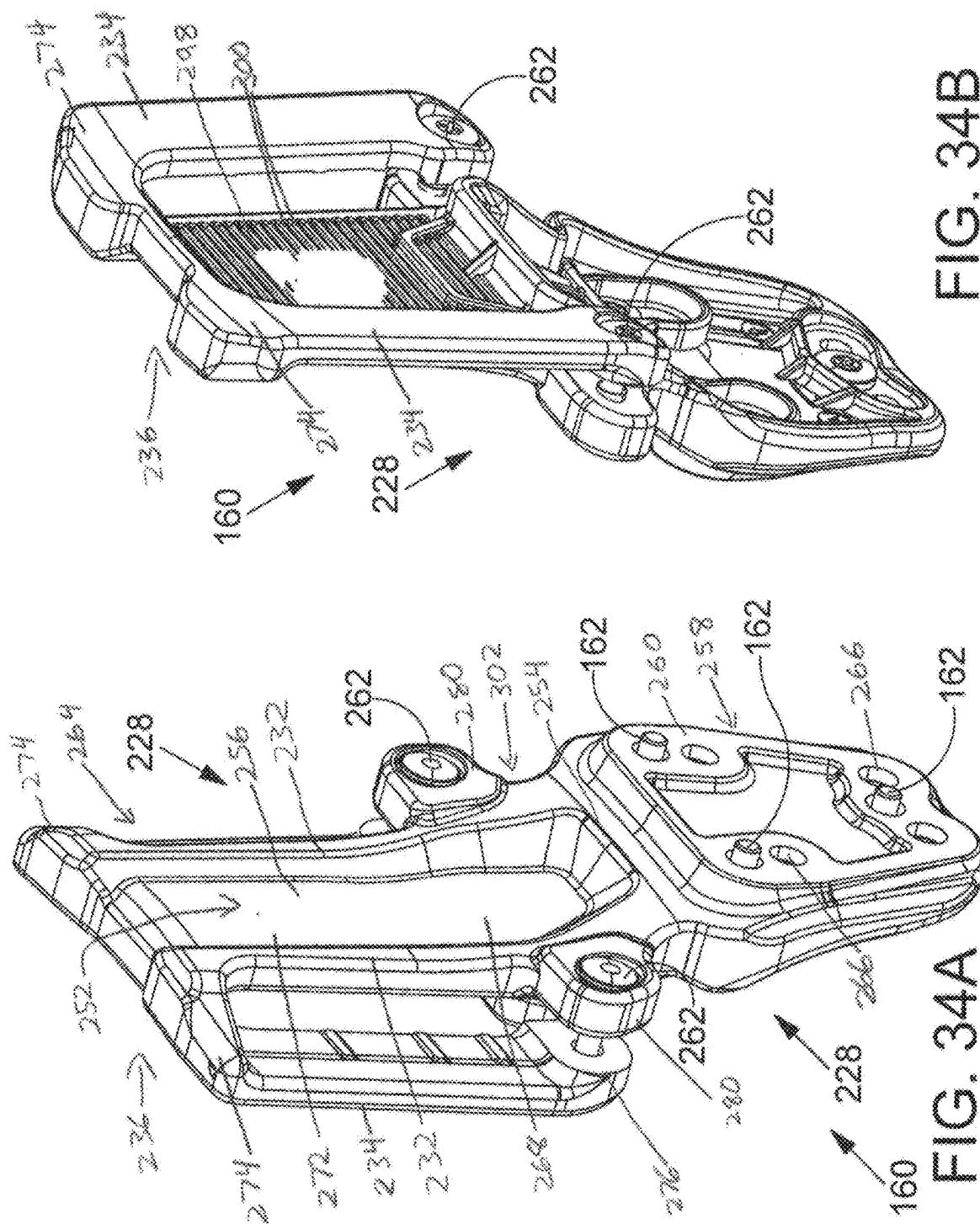












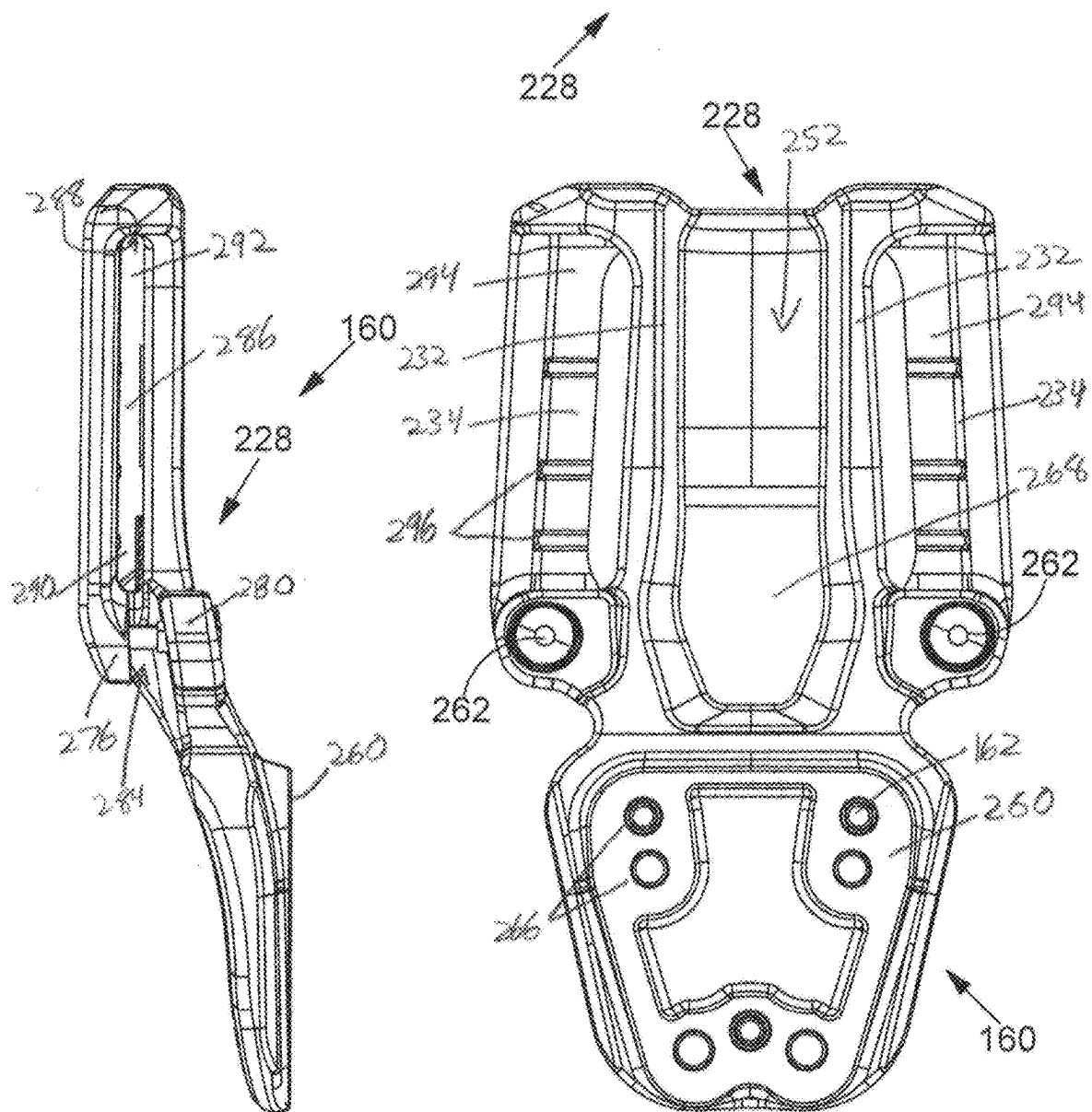
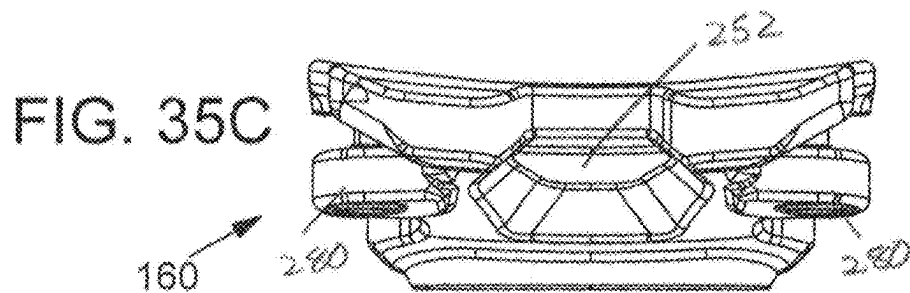


FIG. 35A

FIG. 35B

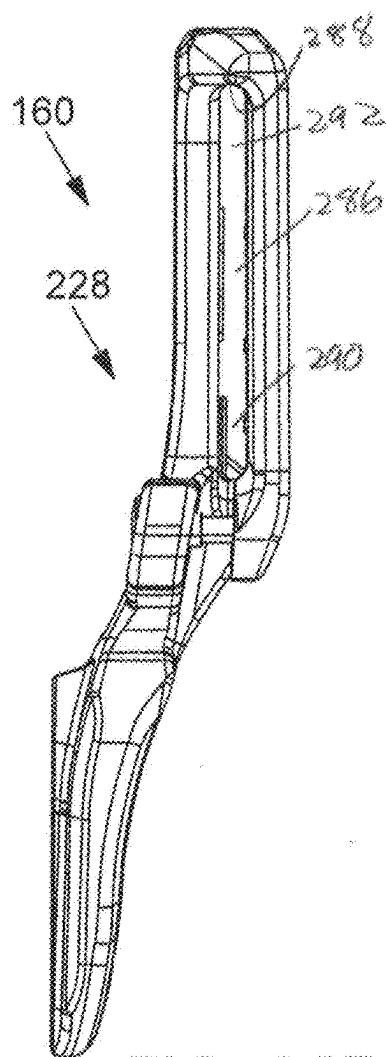


FIG. 35D

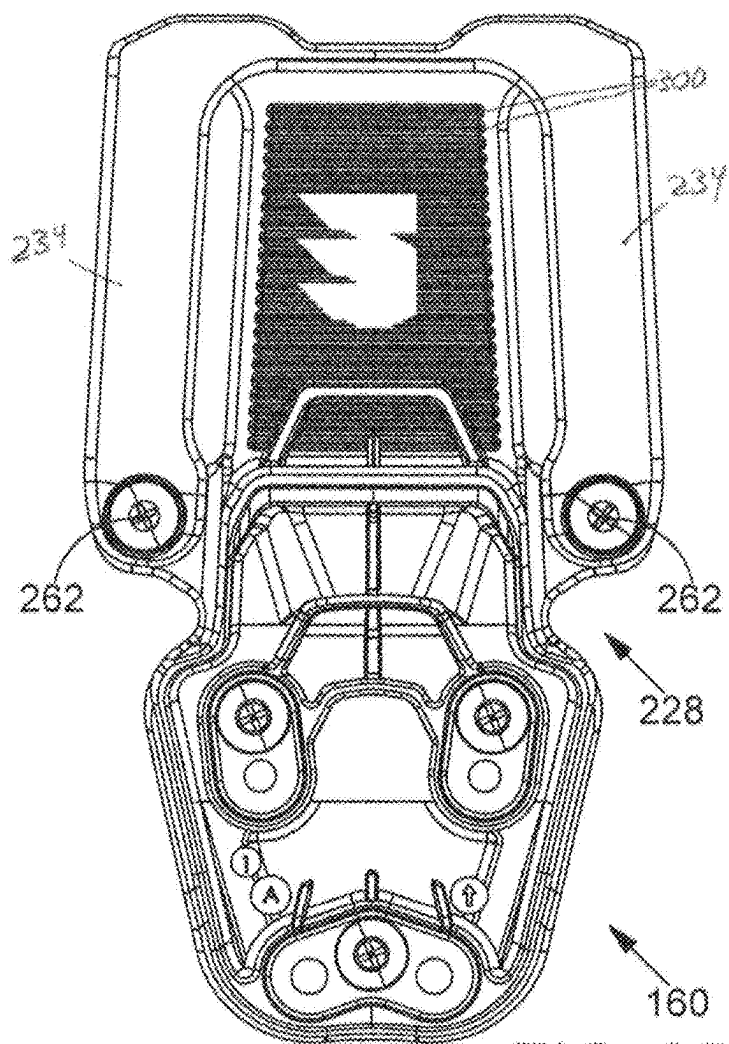


FIG. 35E

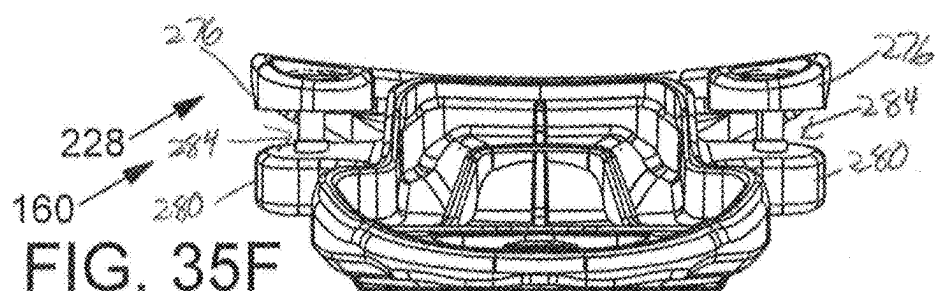
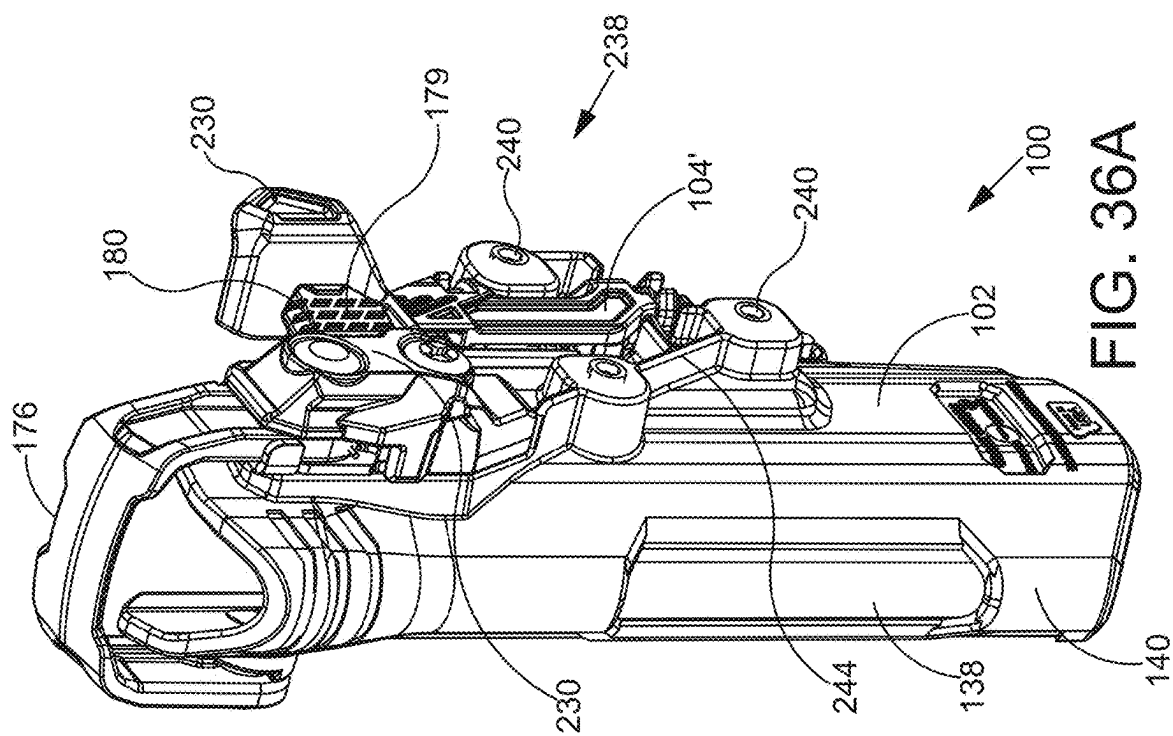
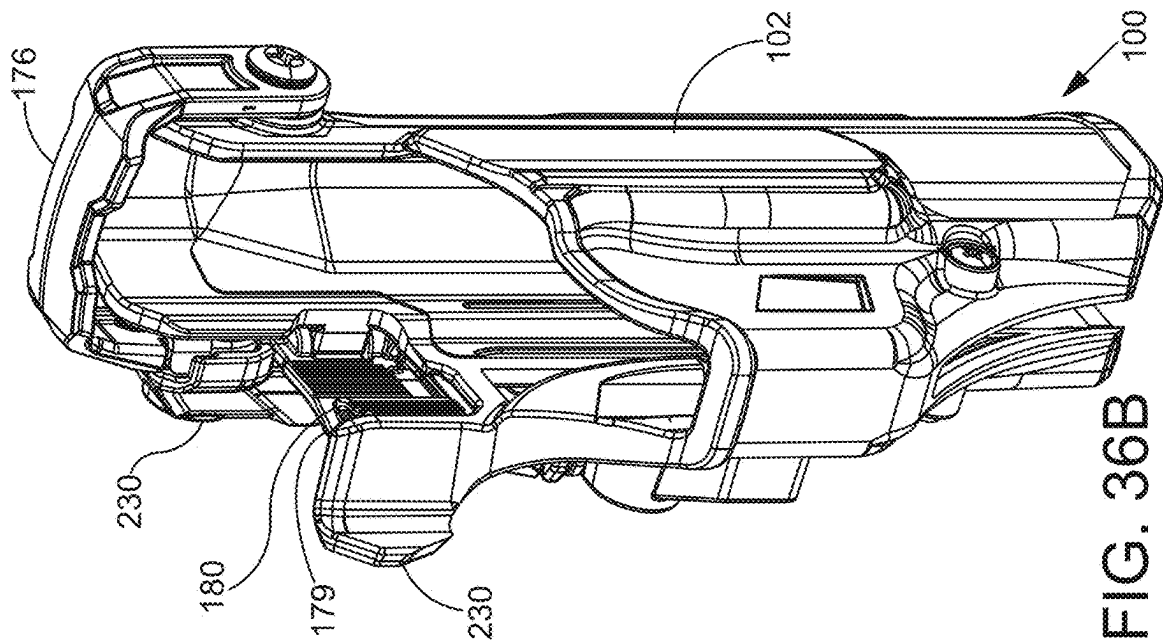
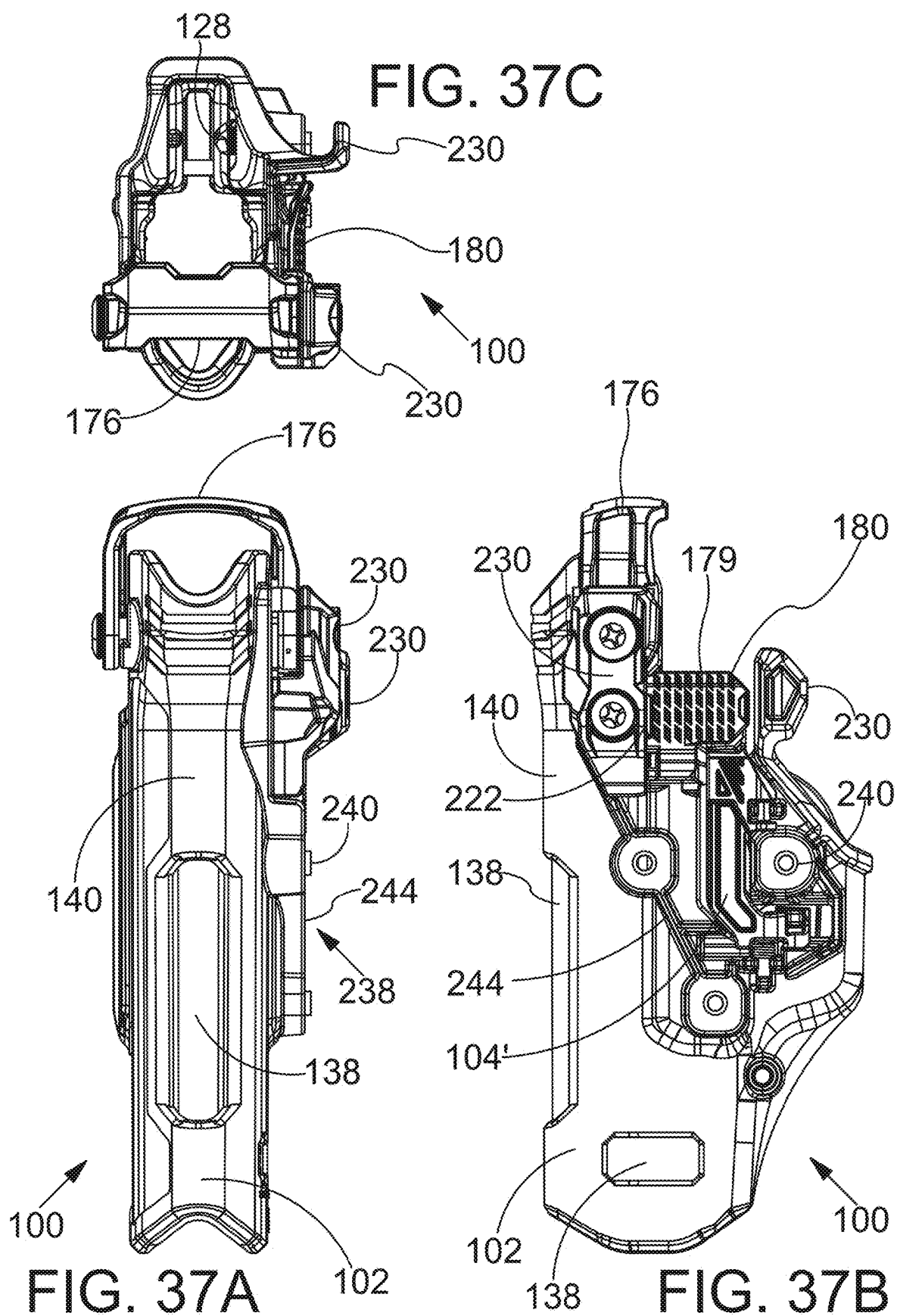


FIG. 35F





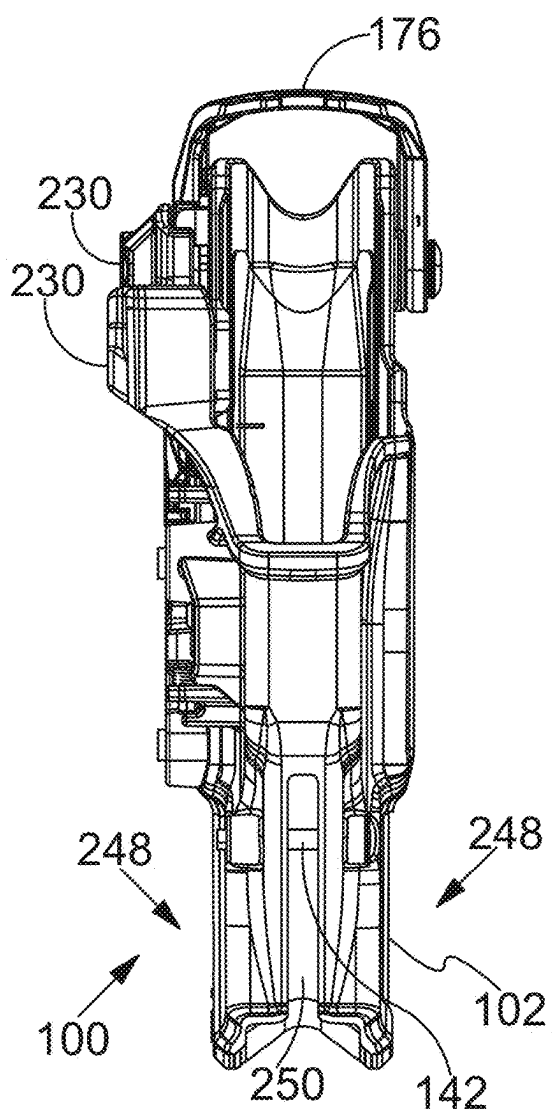


FIG. 37D

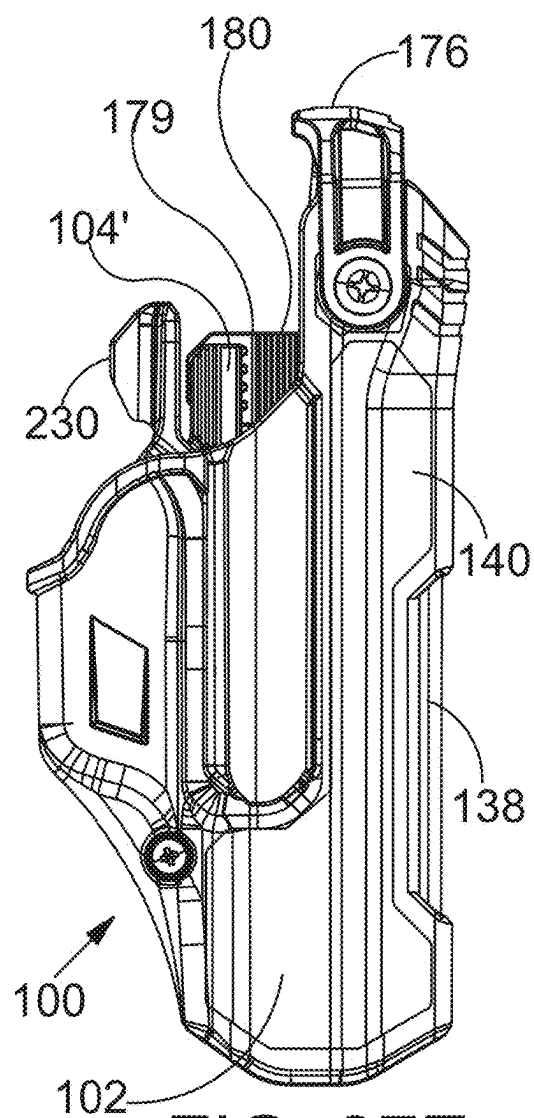


FIG. 37E

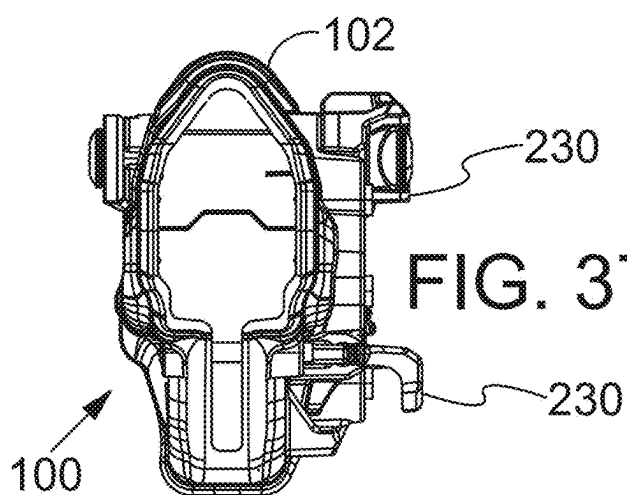


FIG. 37F

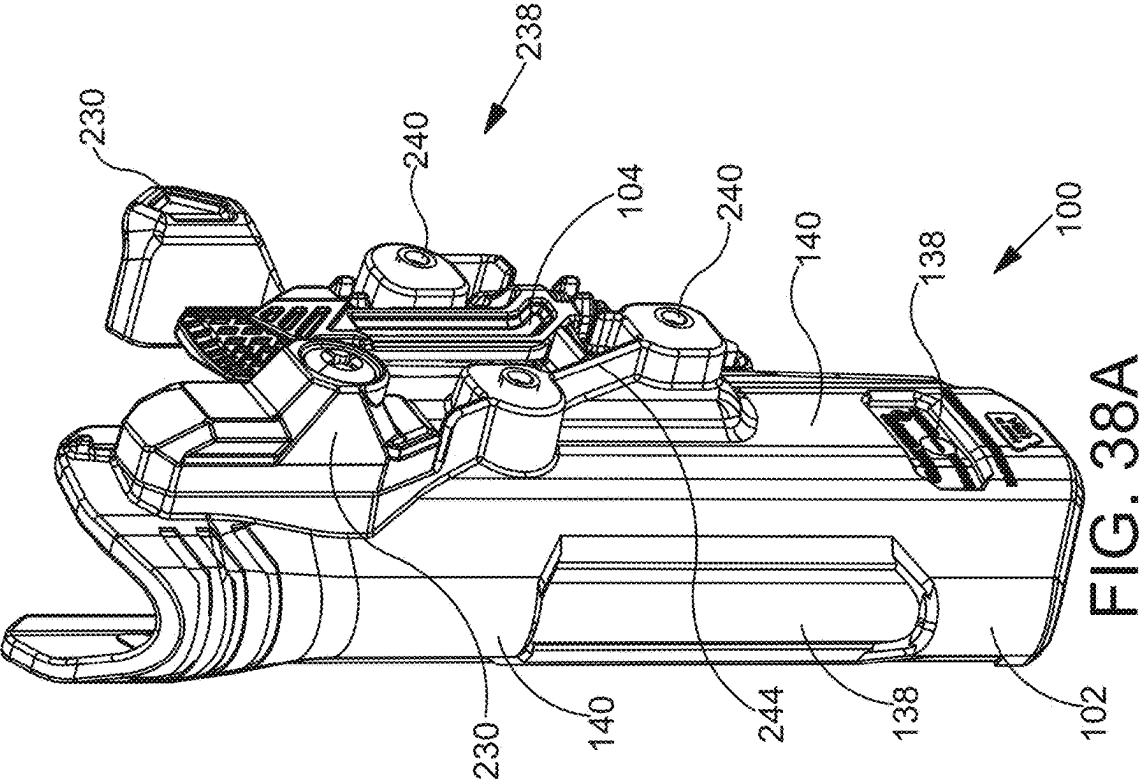
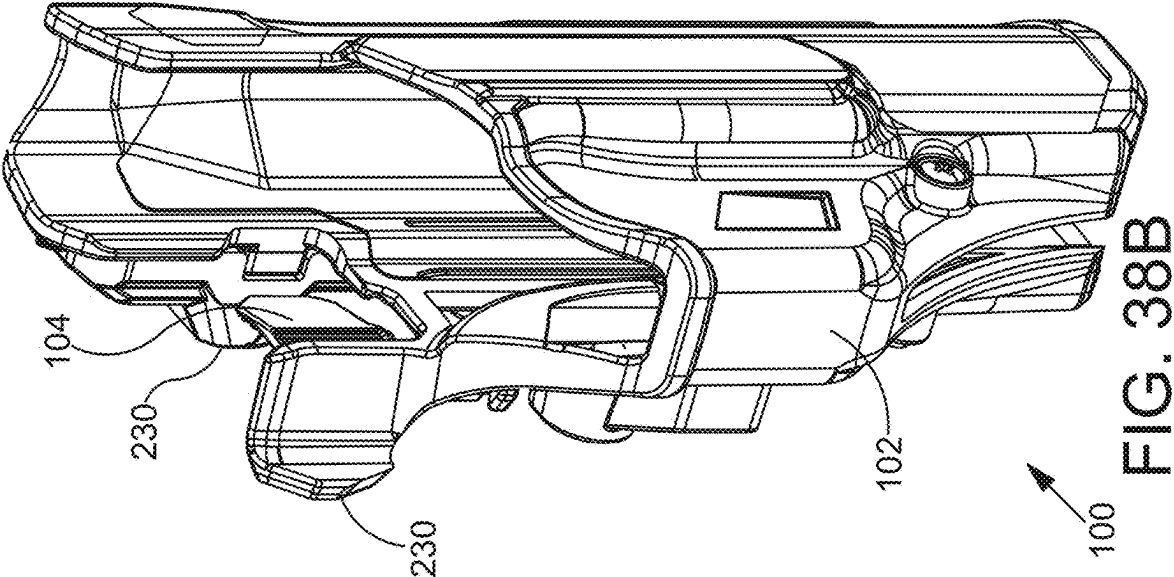


FIG. 39C

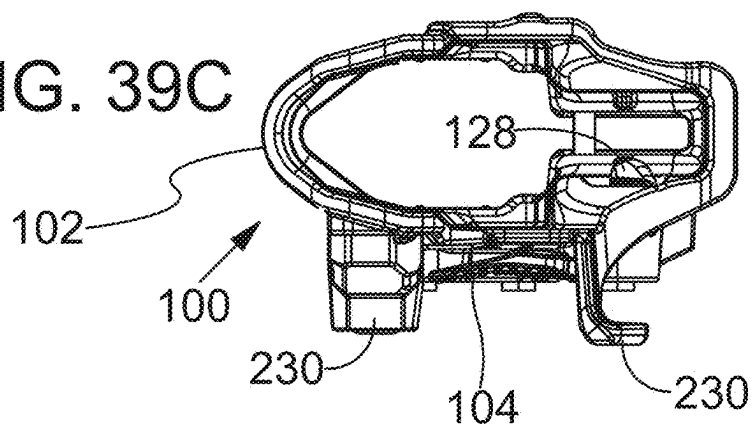


FIG. 39A

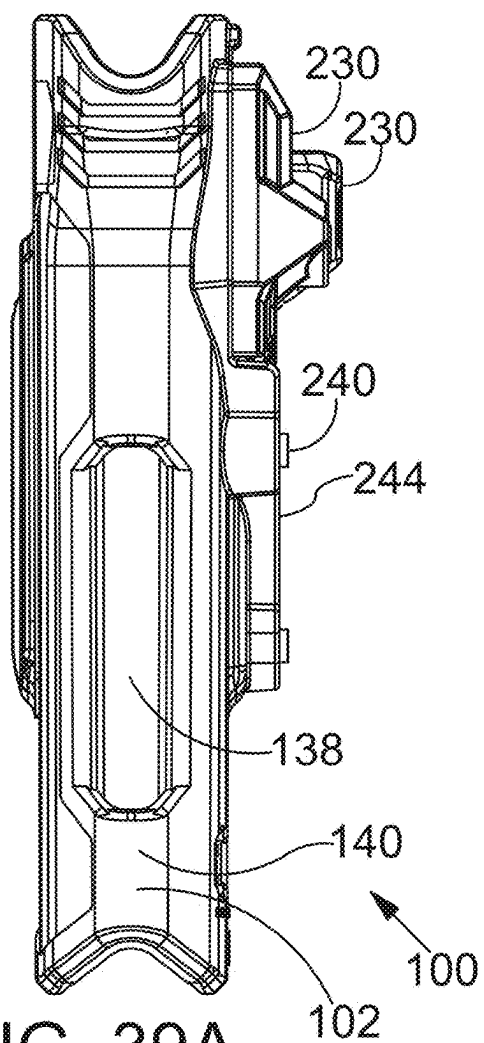
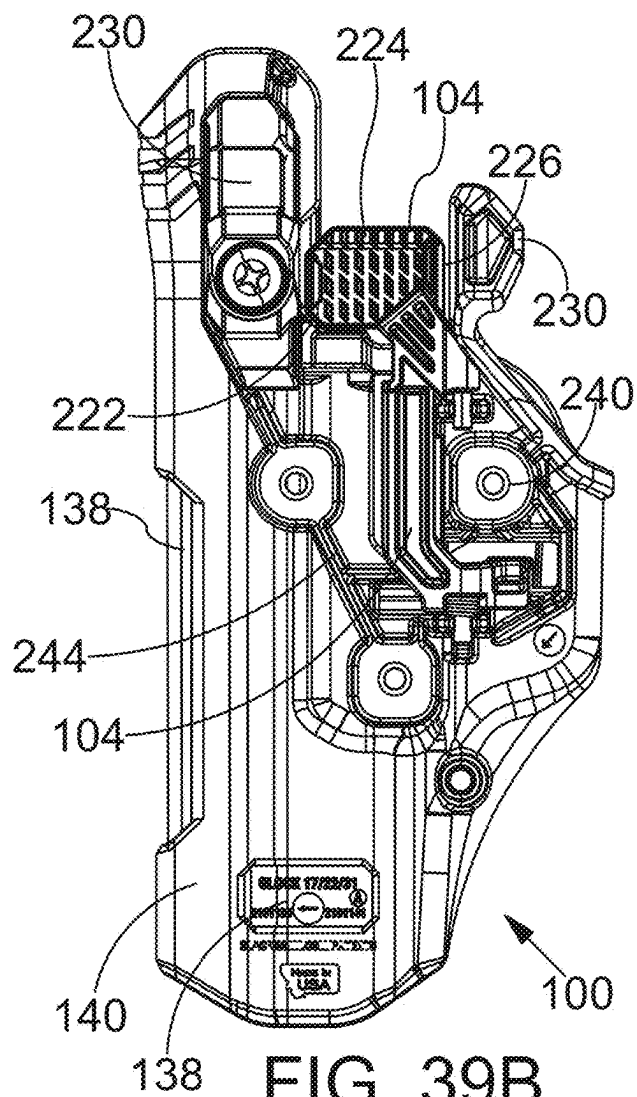


FIG. 39B



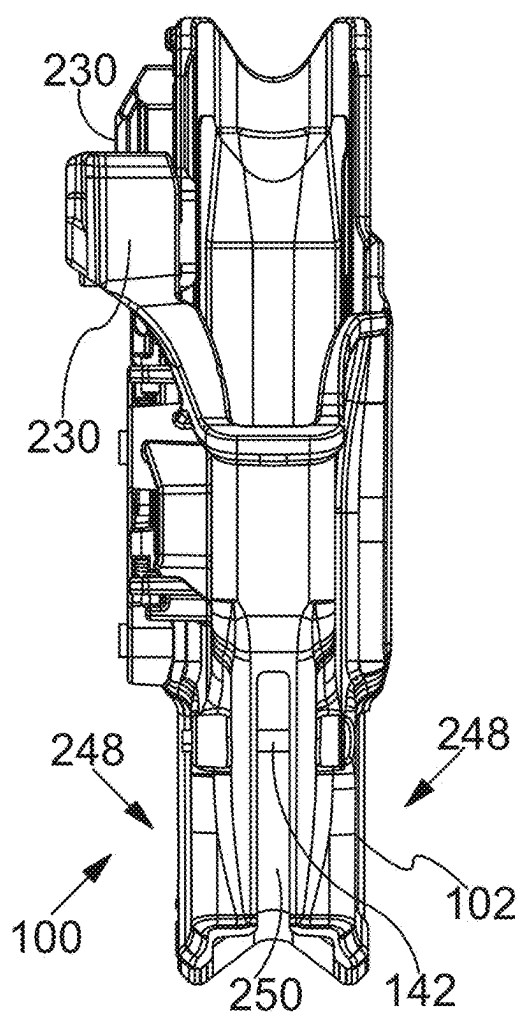


FIG. 39D

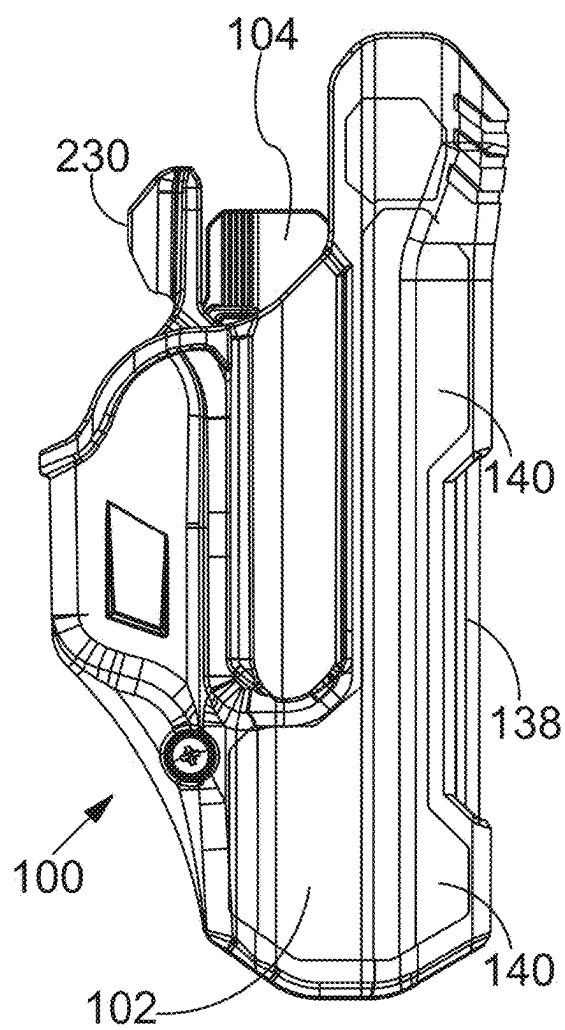


FIG. 39E

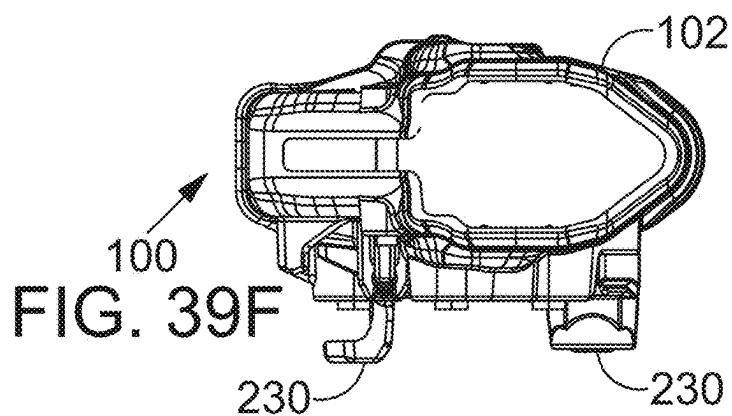
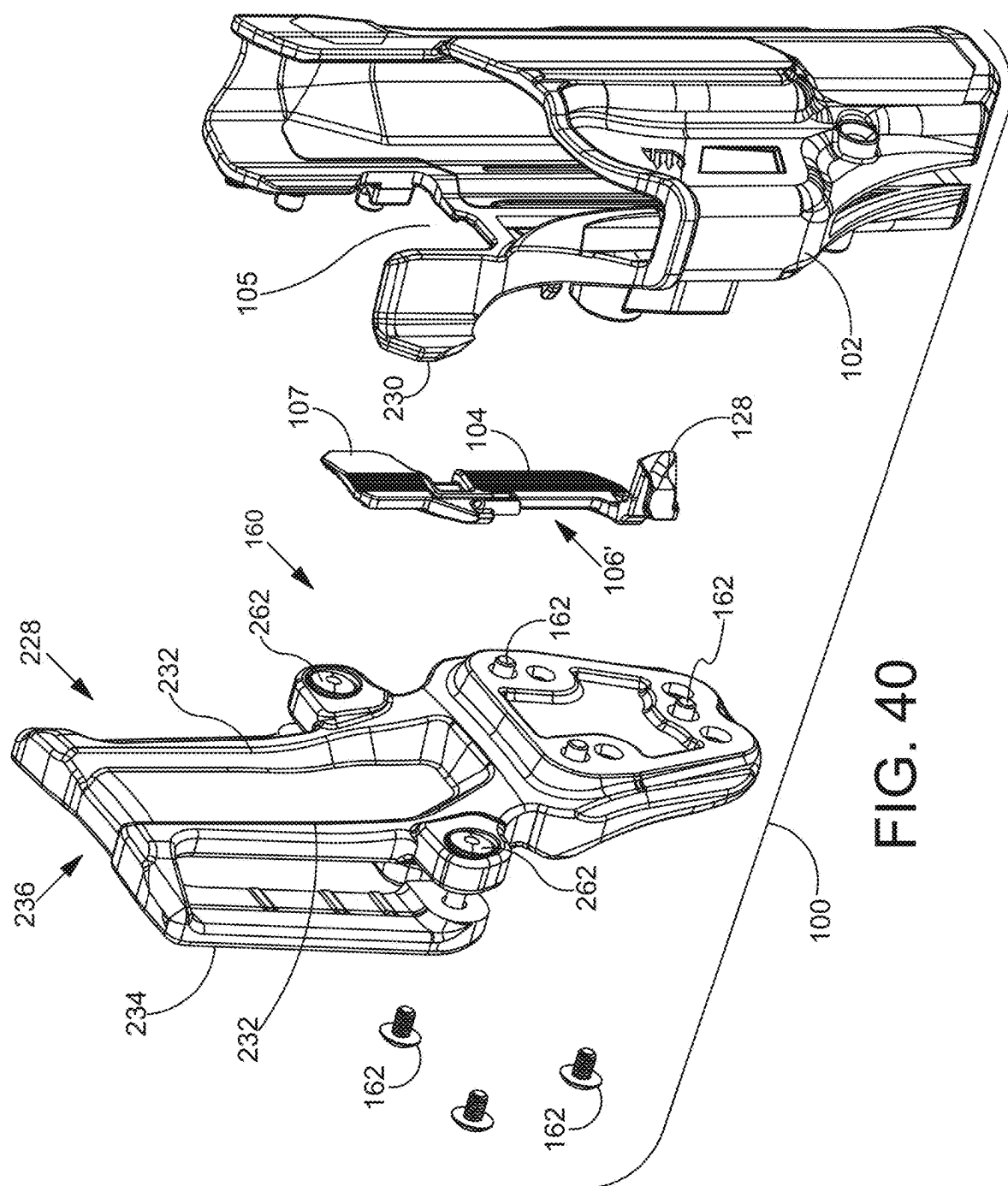


FIG. 39F



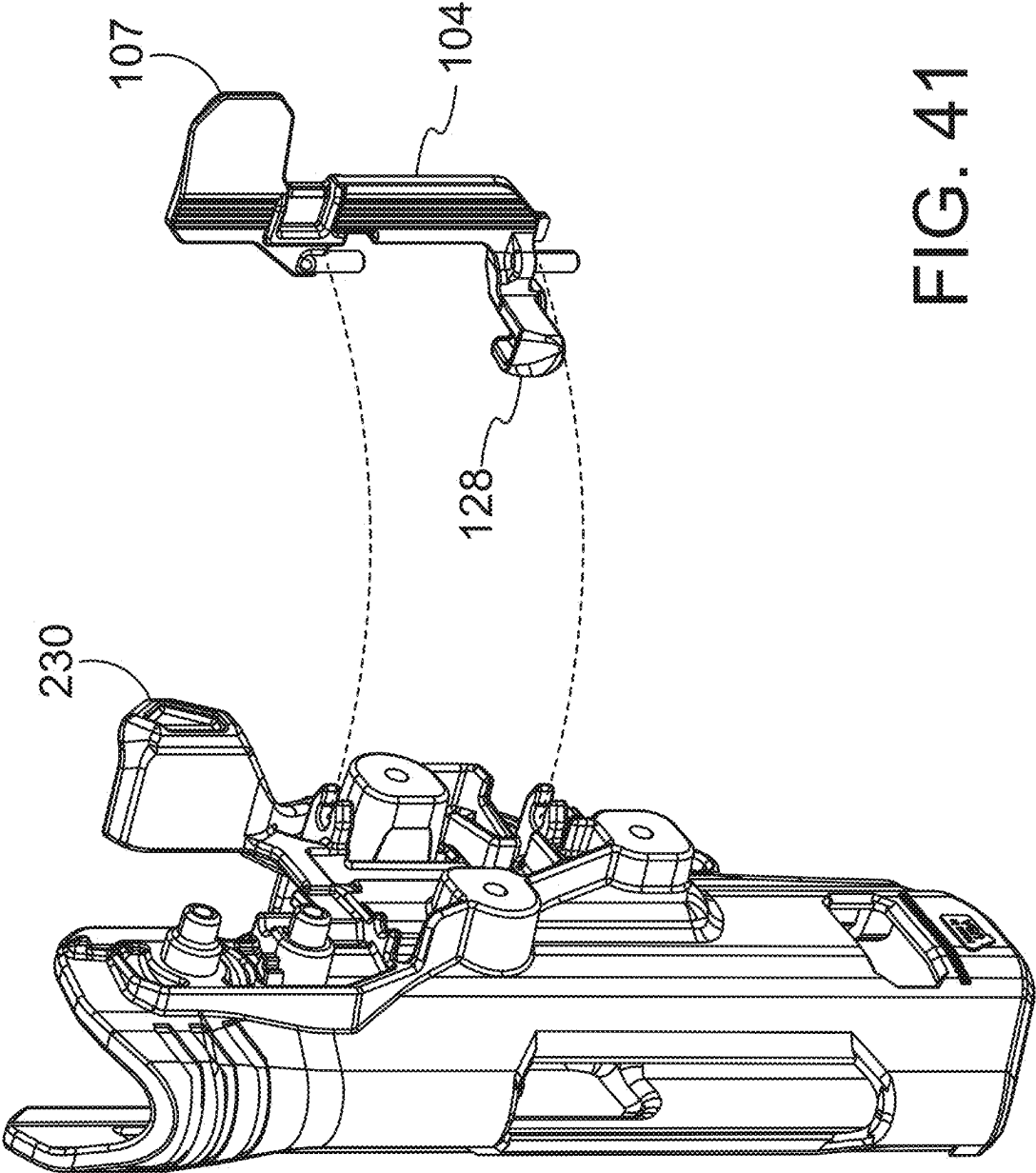
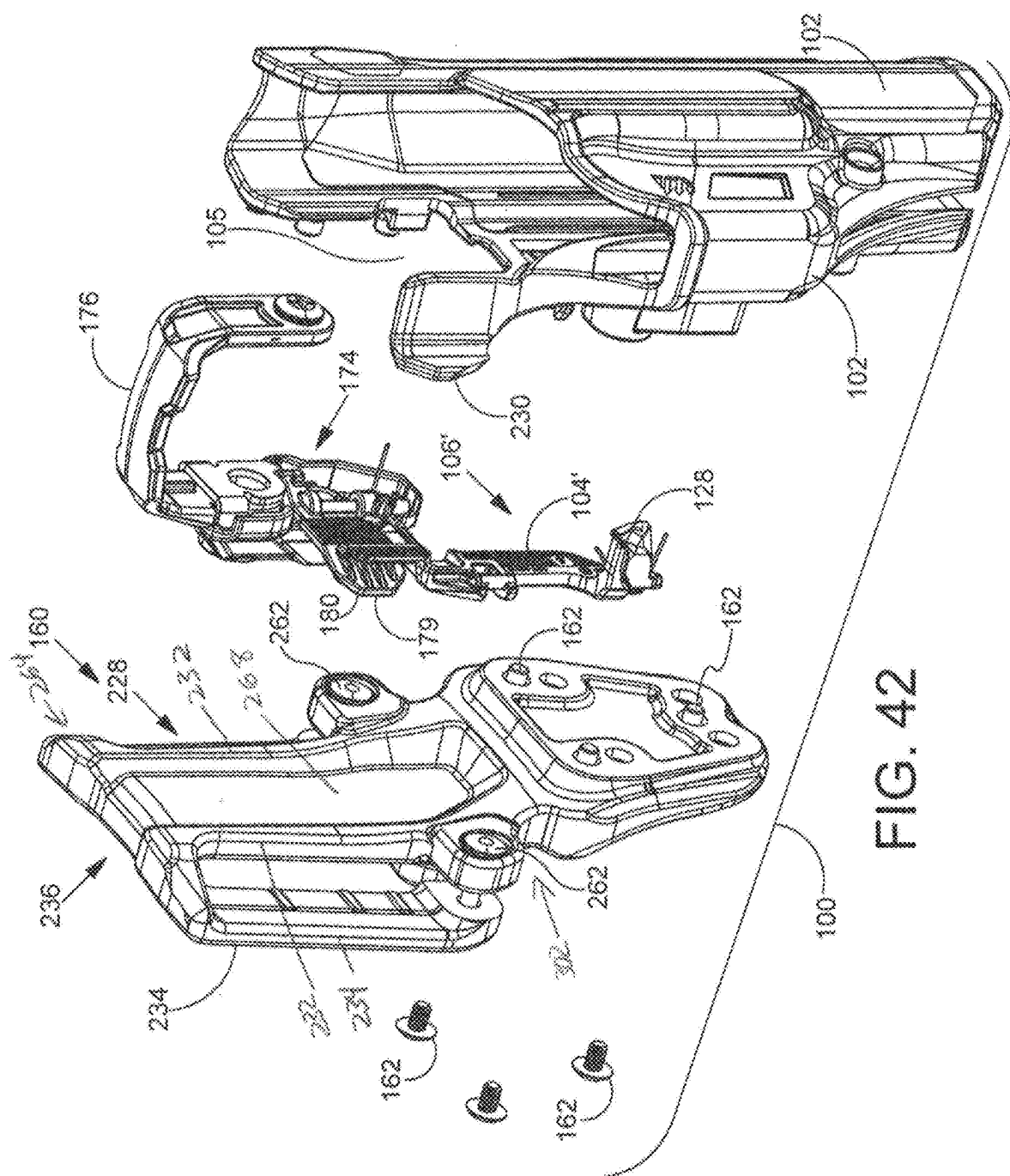


FIG. 41



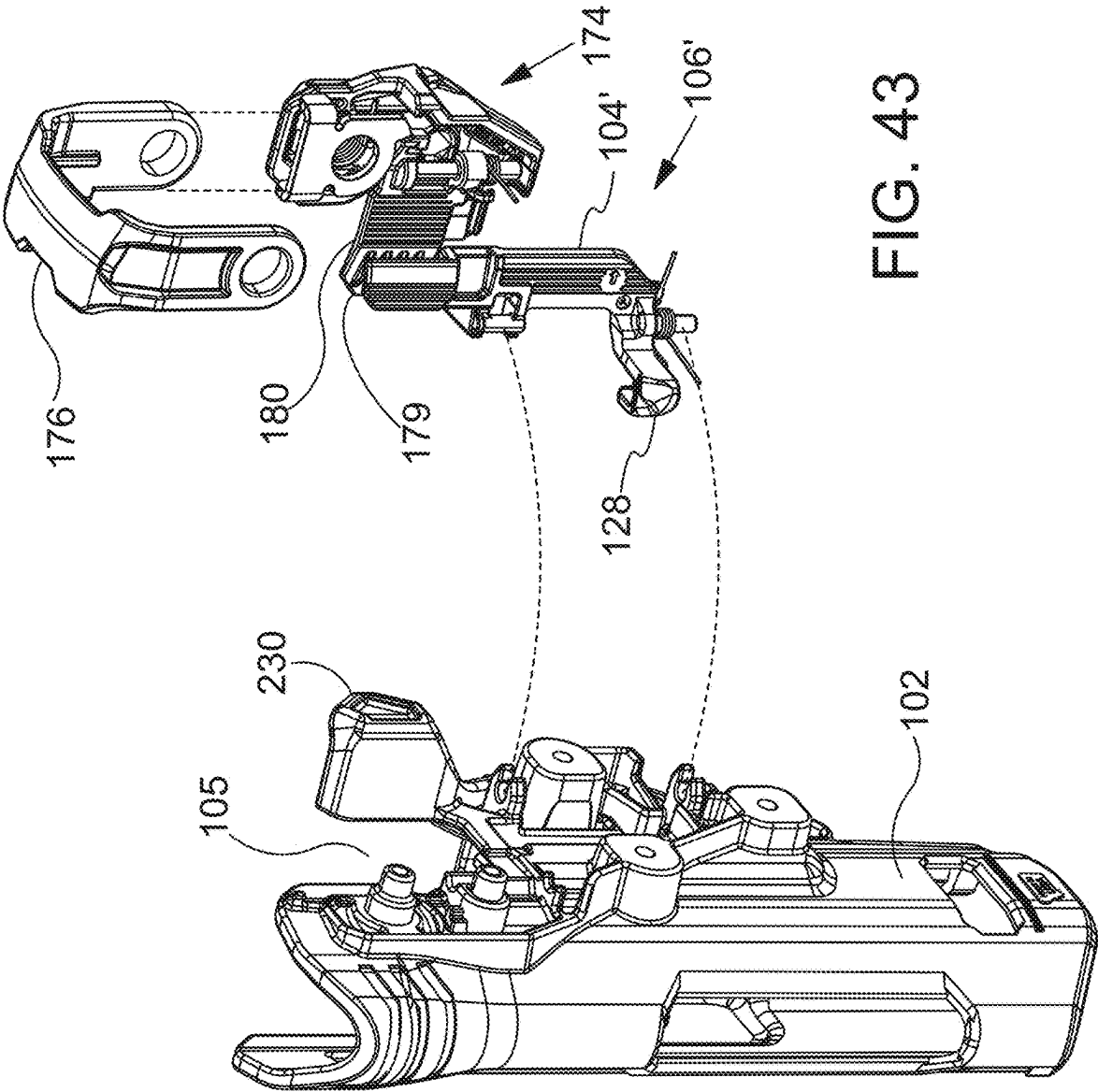
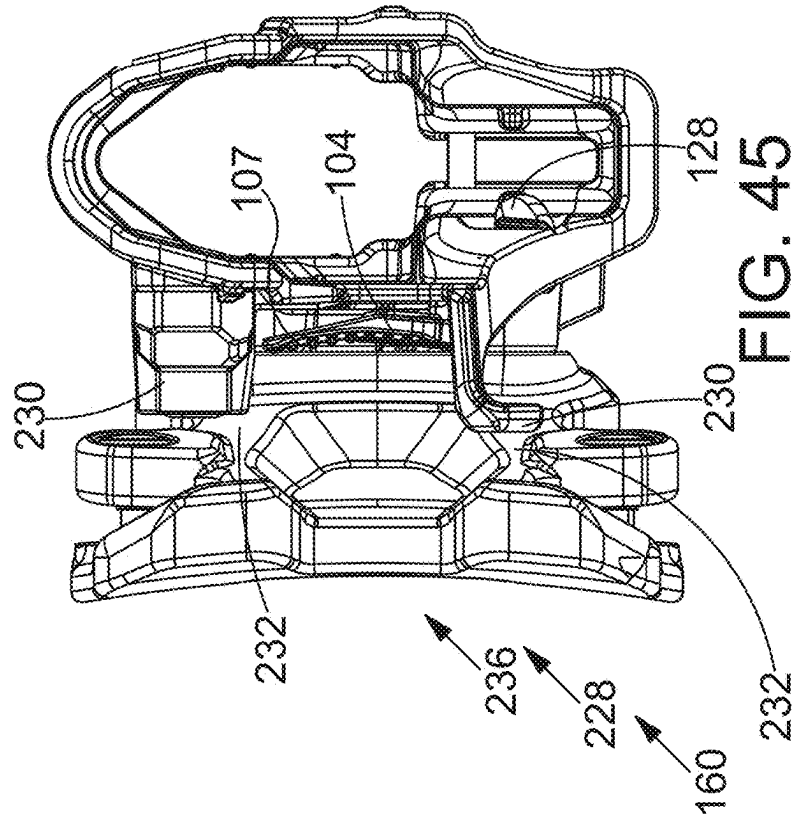
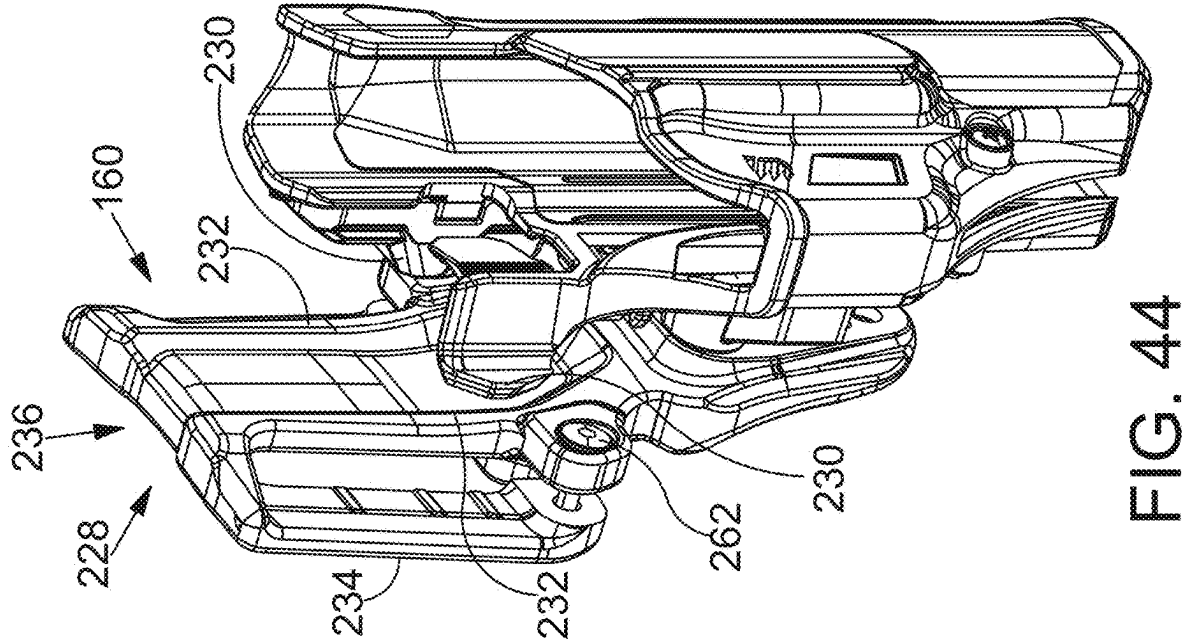
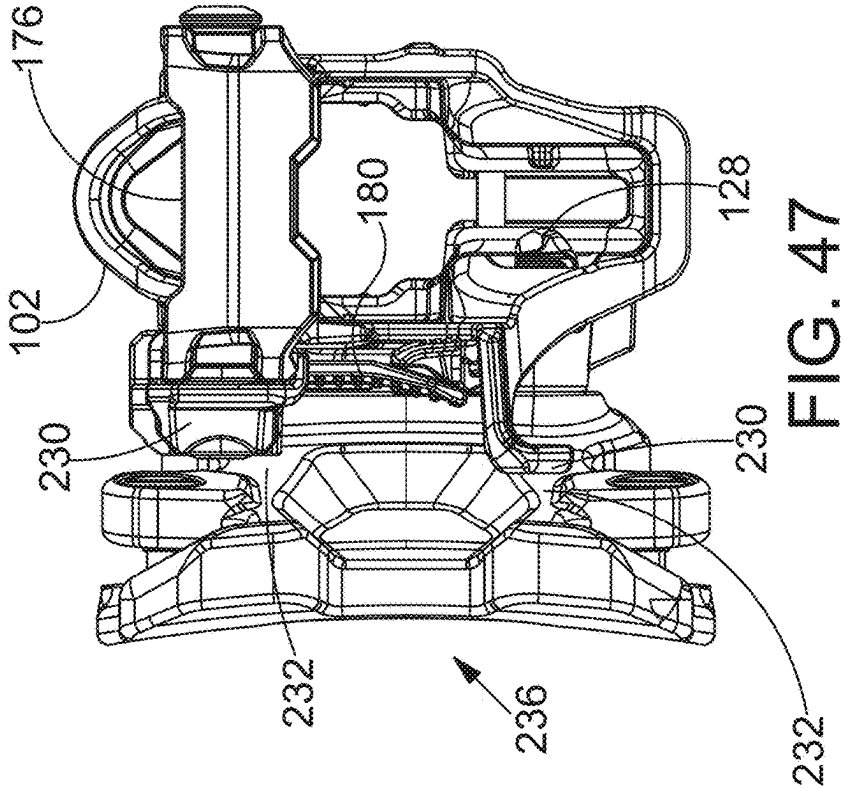
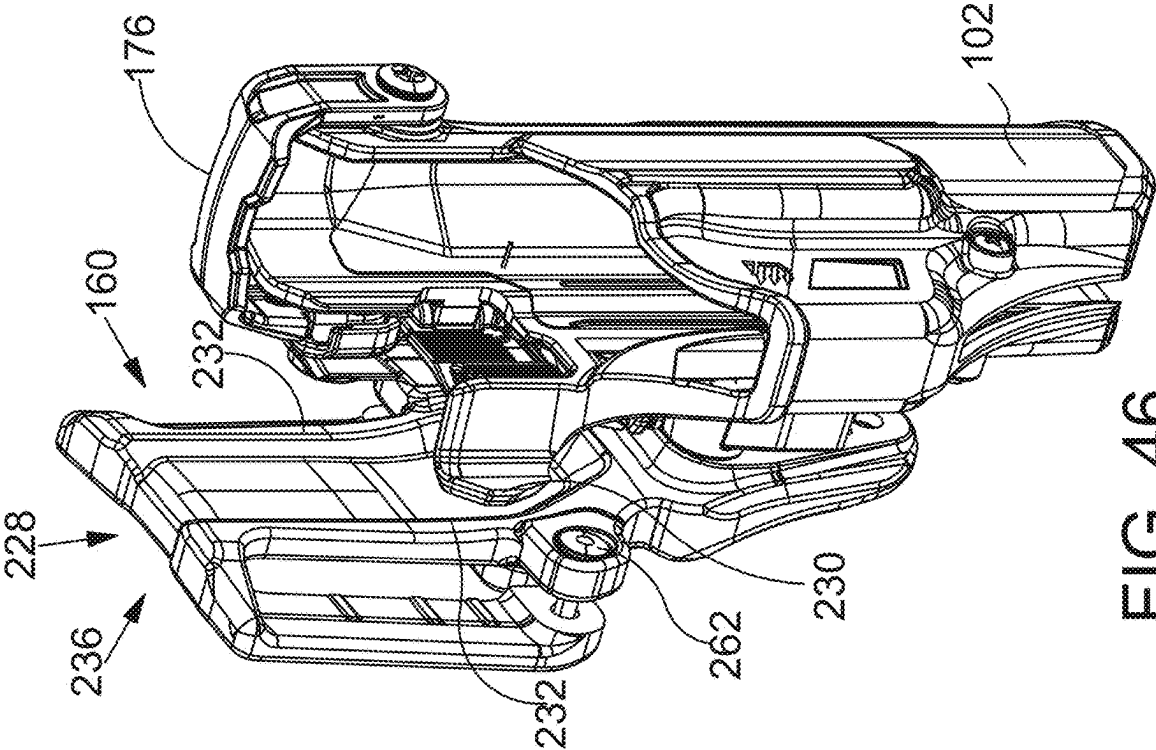


FIG. 43





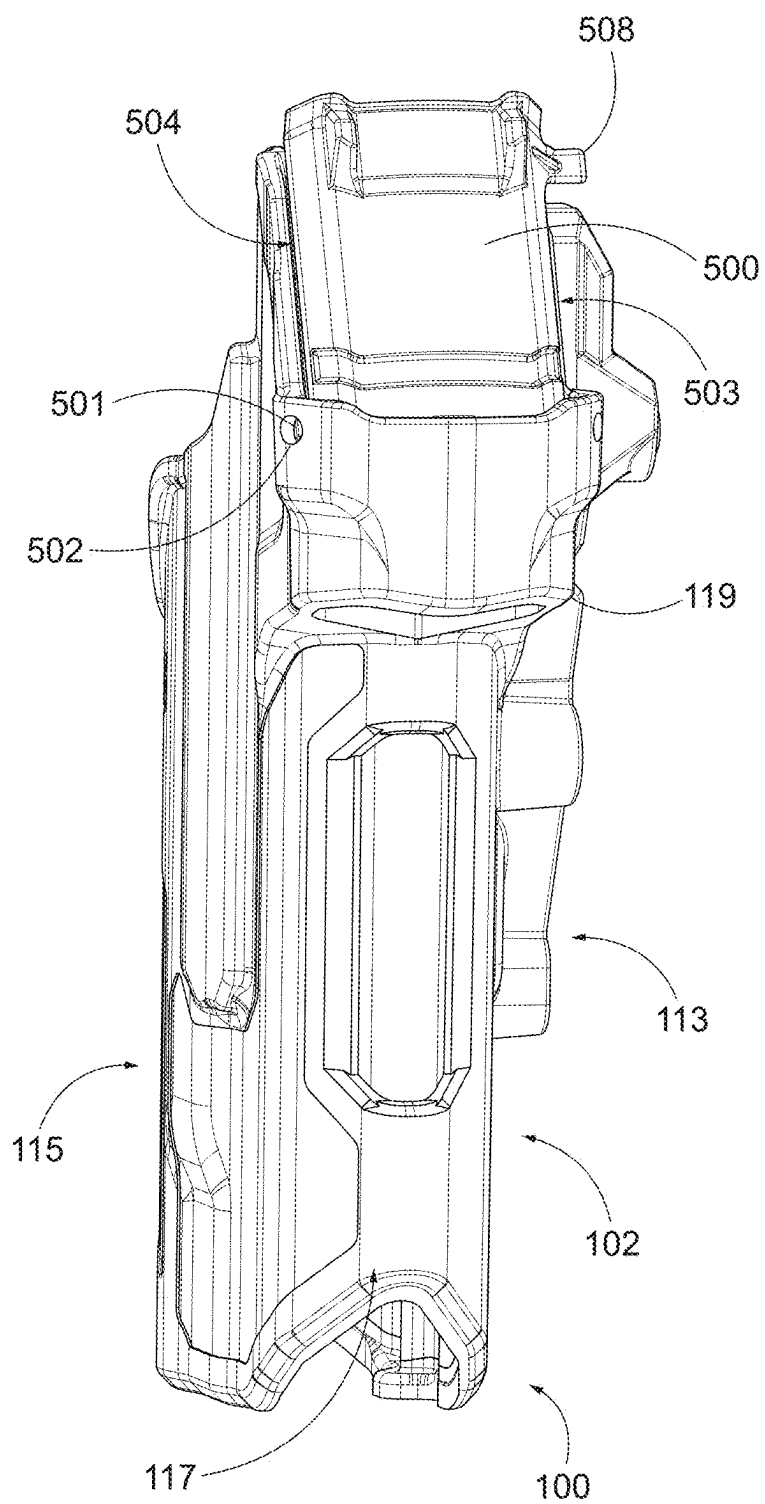


FIG. 48A

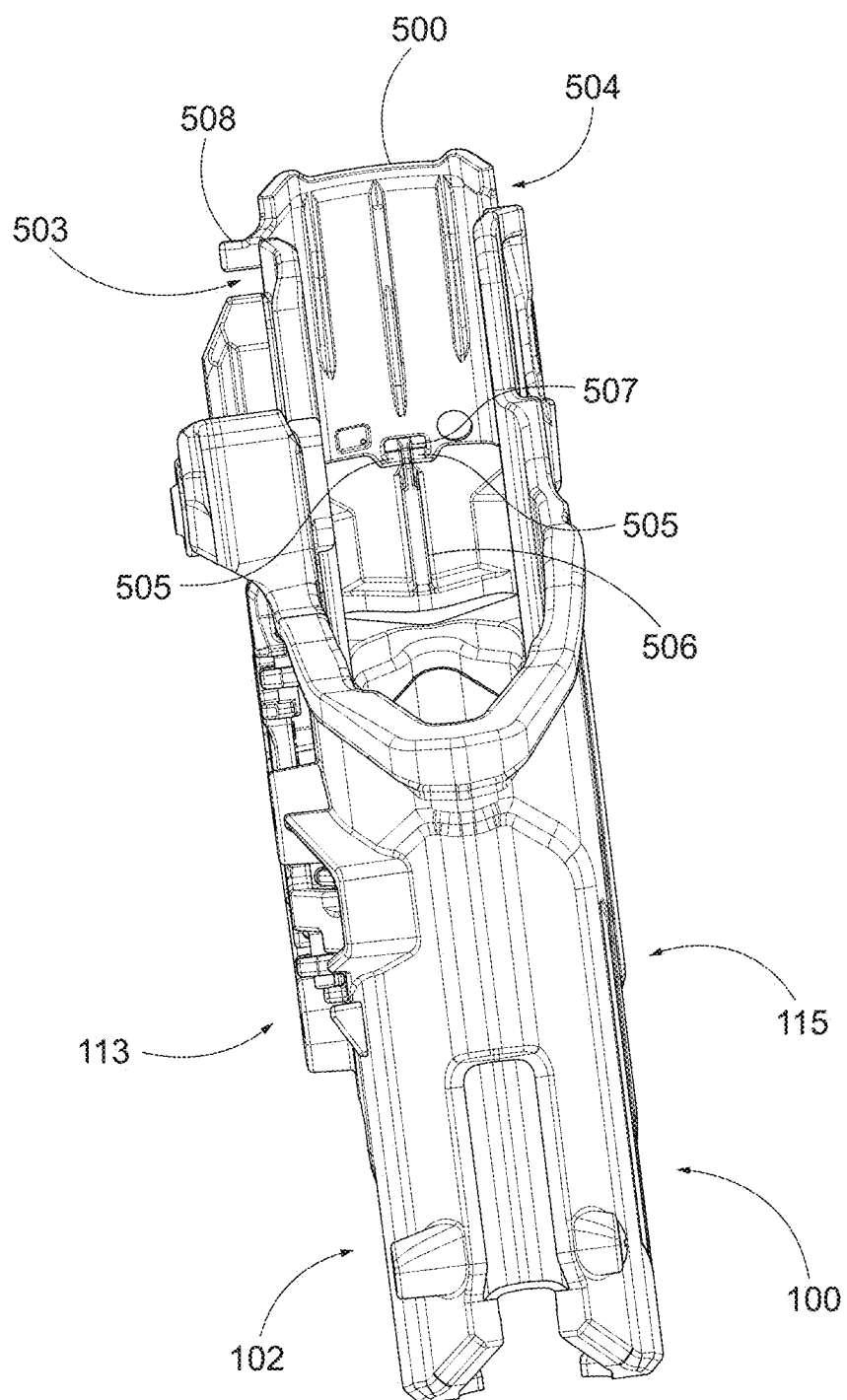


FIG. 48B

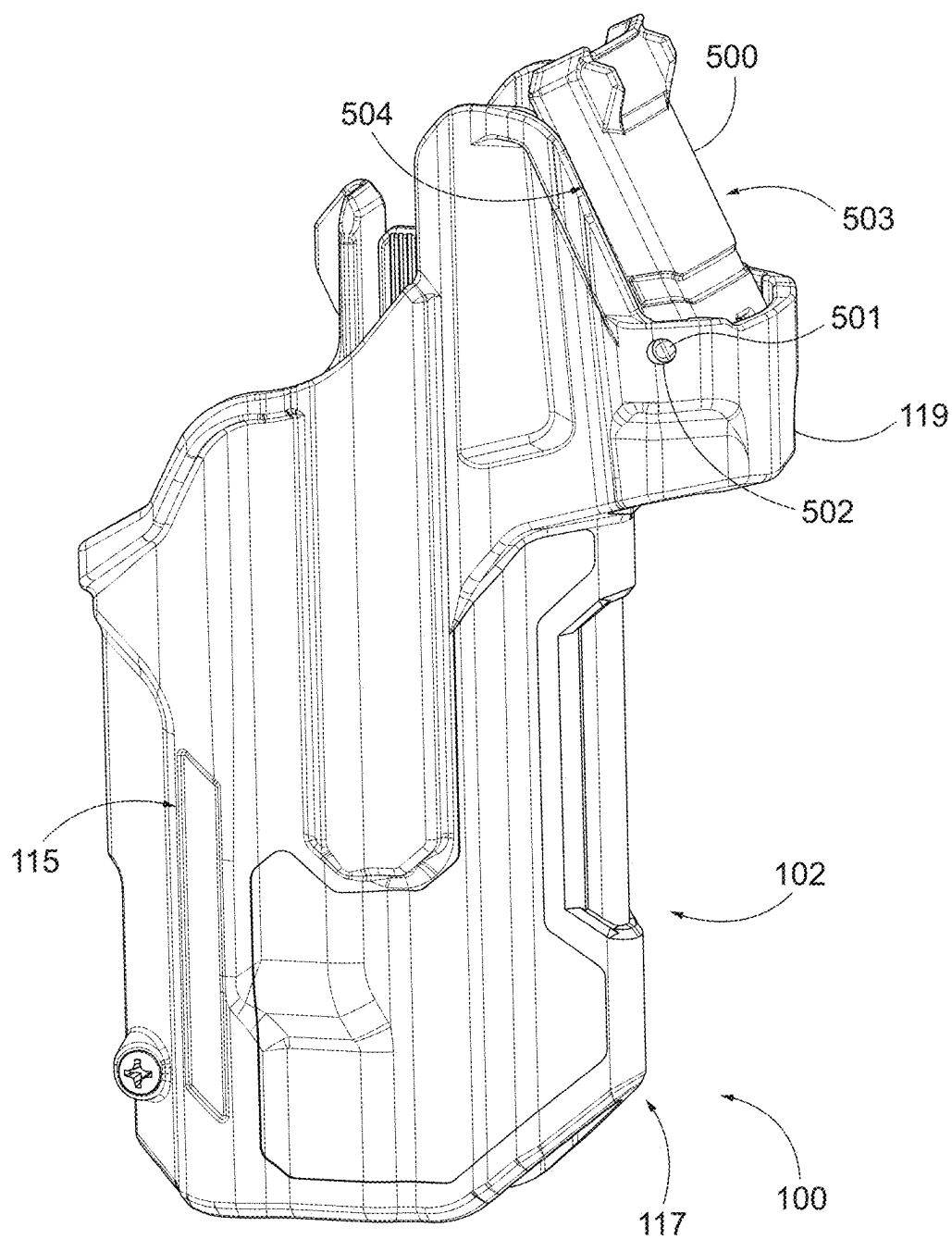


FIG. 48C

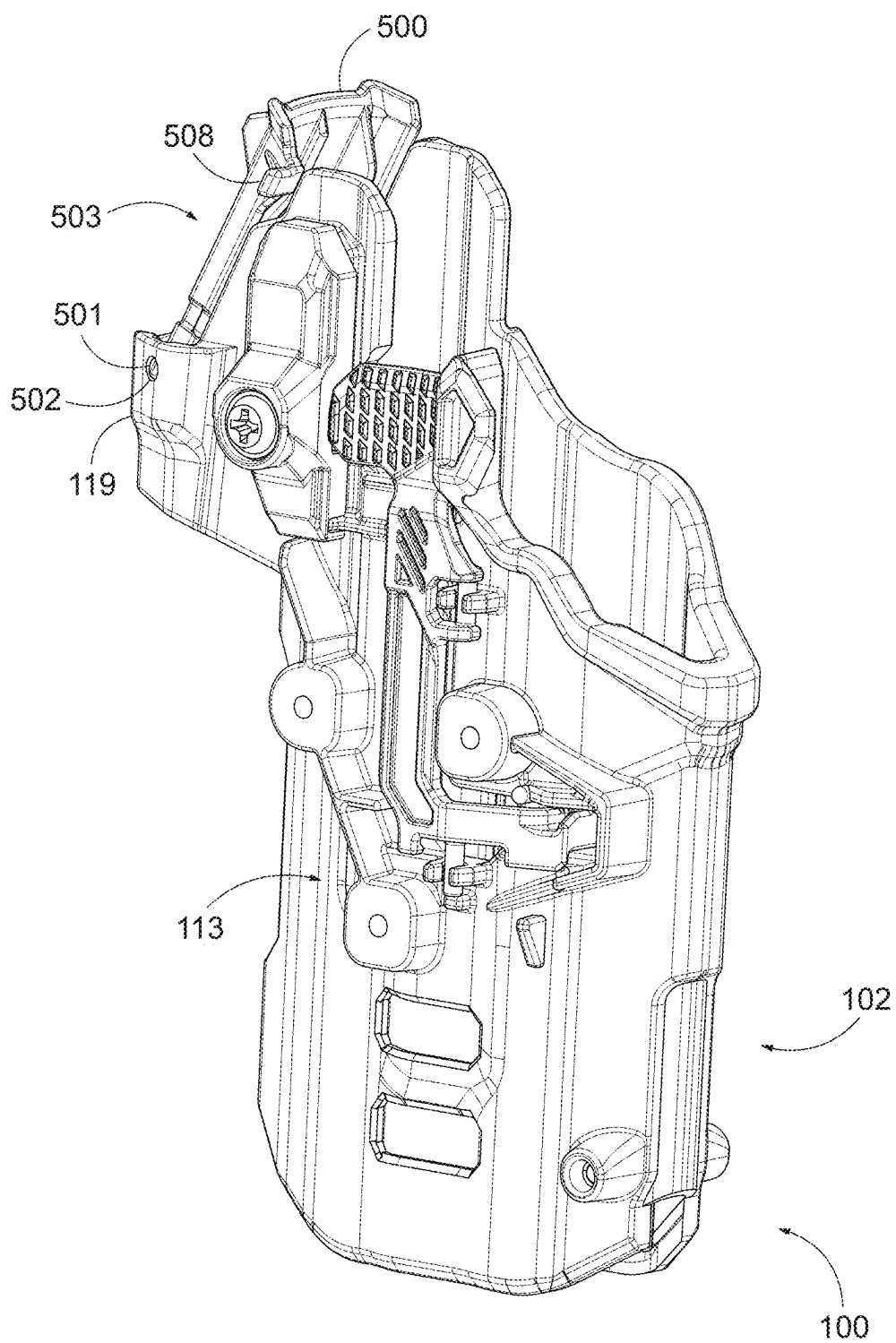


FIG. 48D

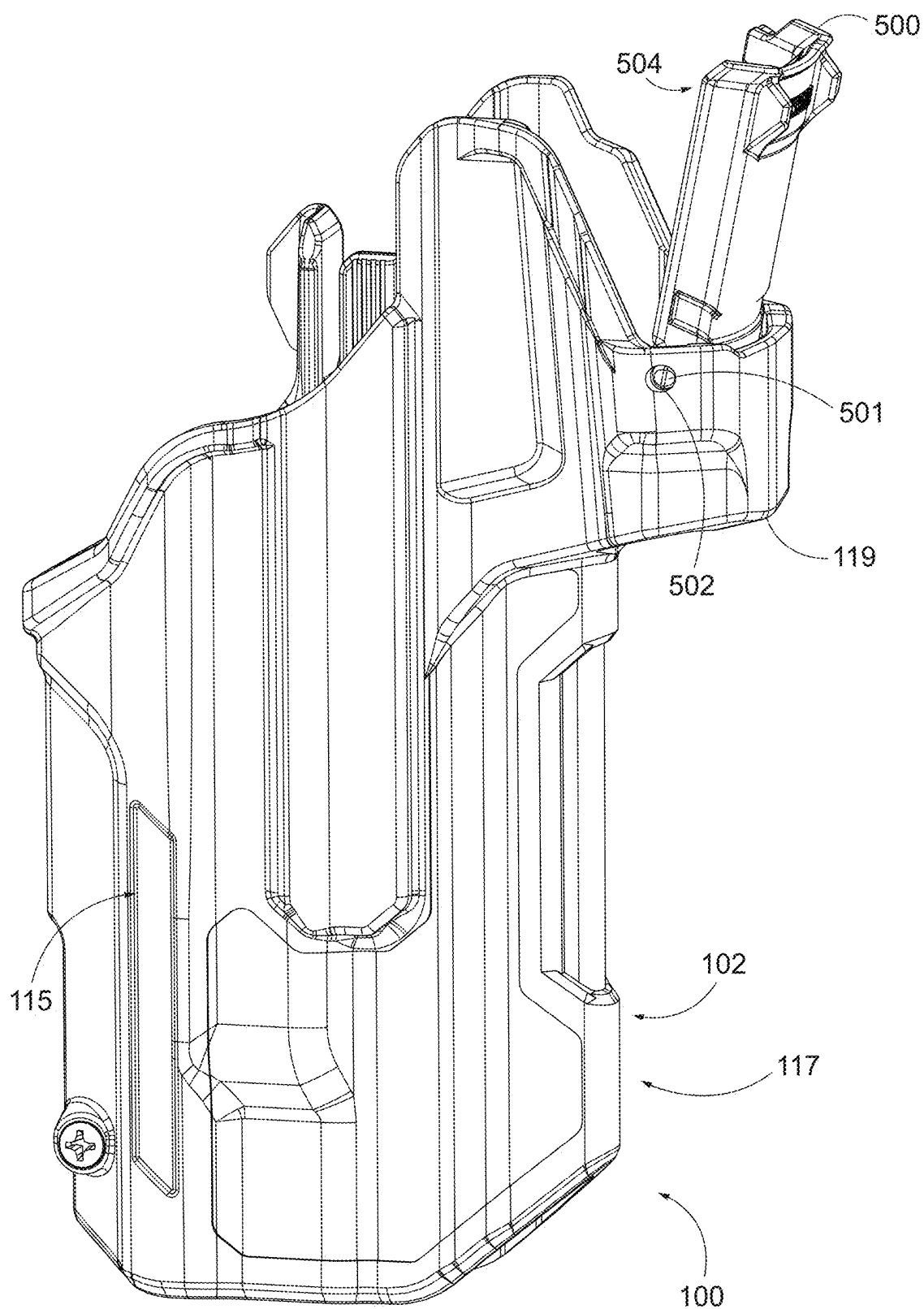


FIG. 48E

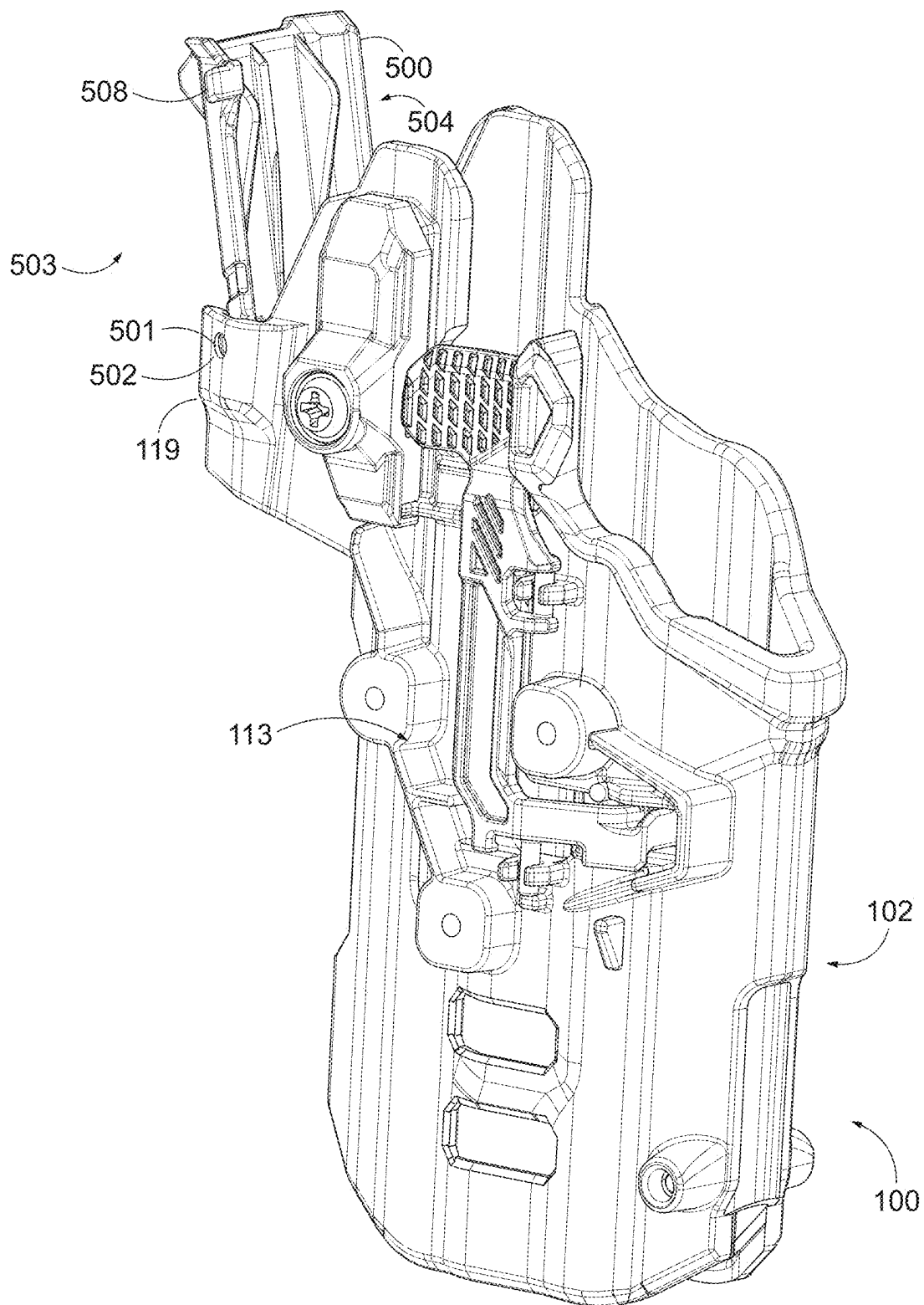


FIG. 48F

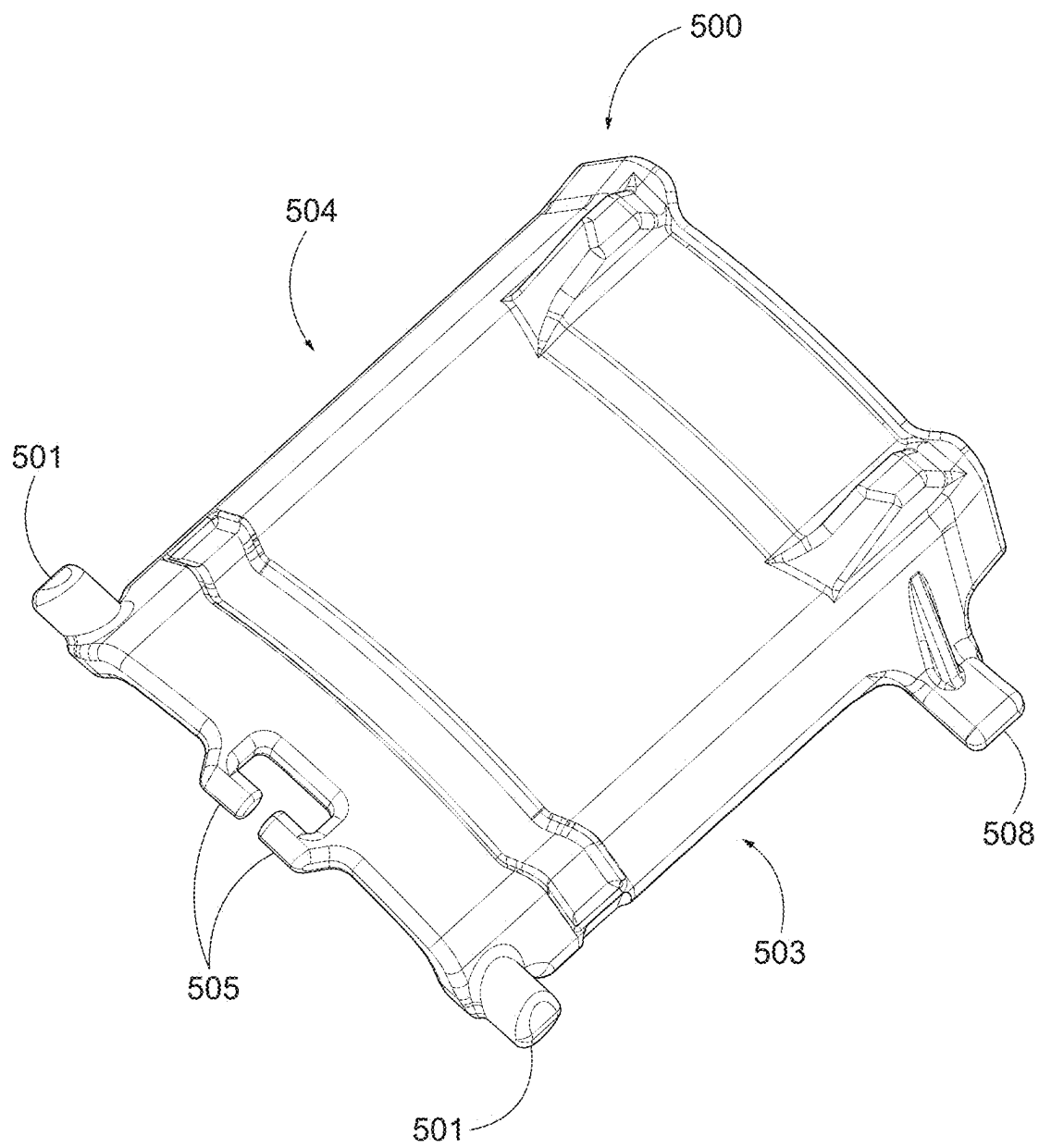


FIG. 49

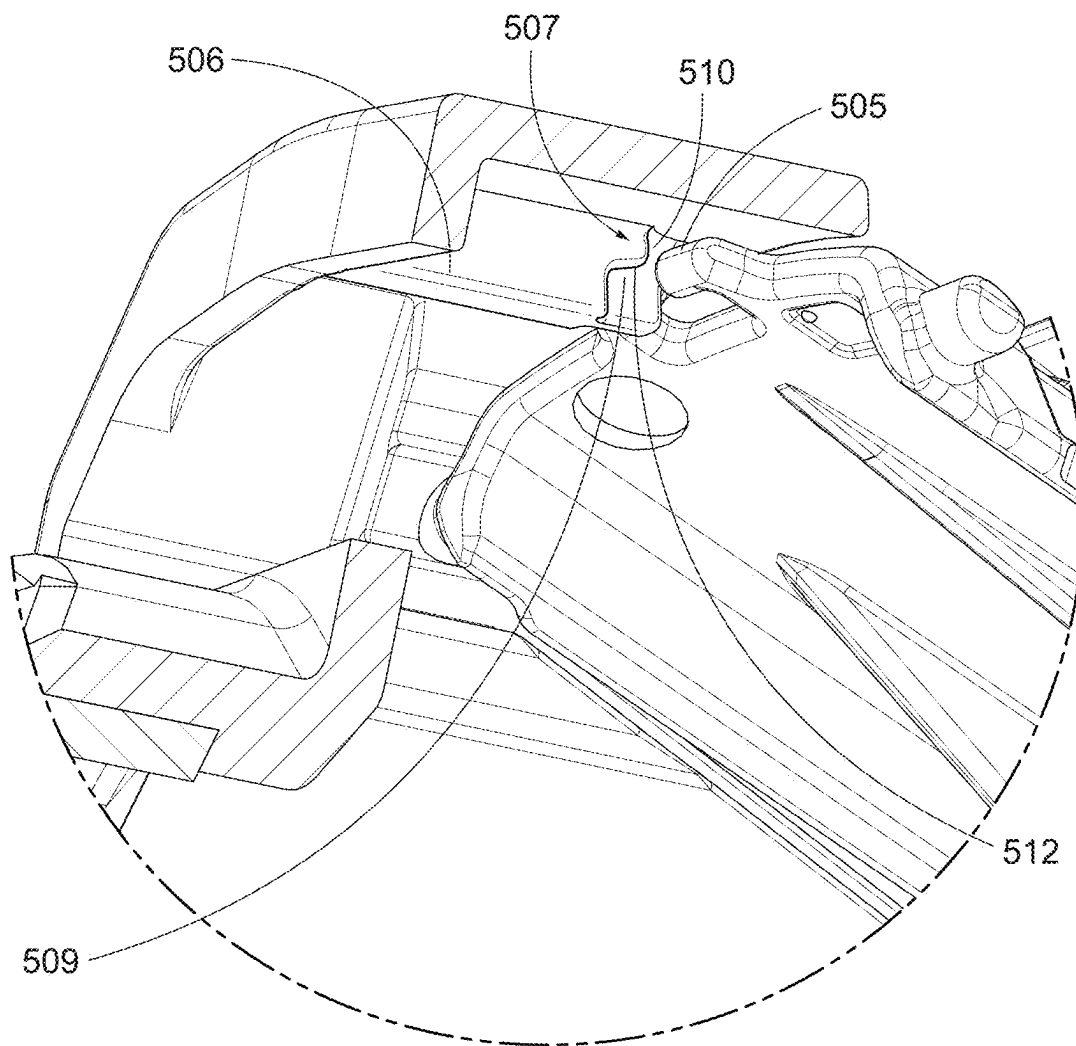


FIG. 50A

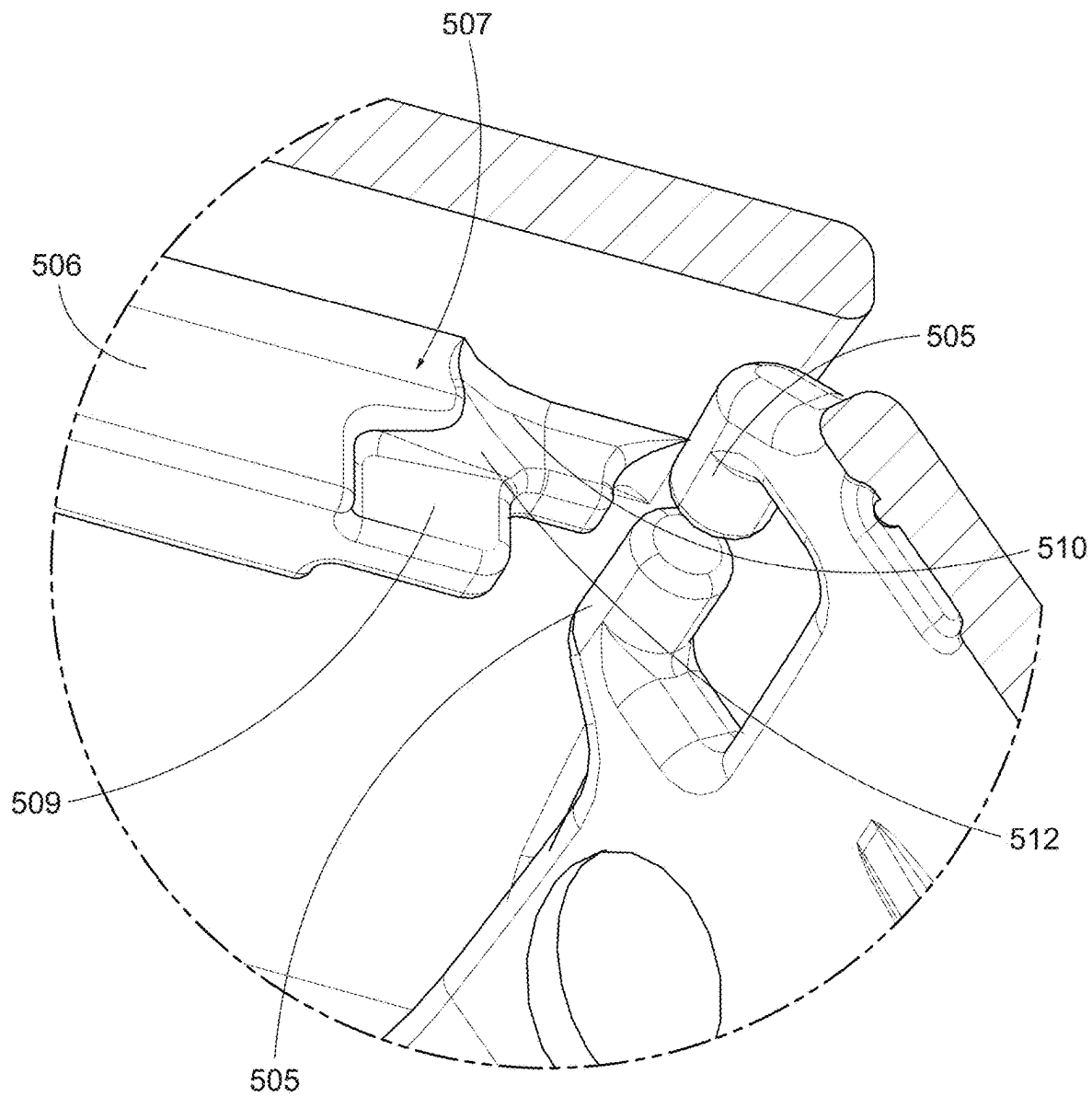


FIG. 50B

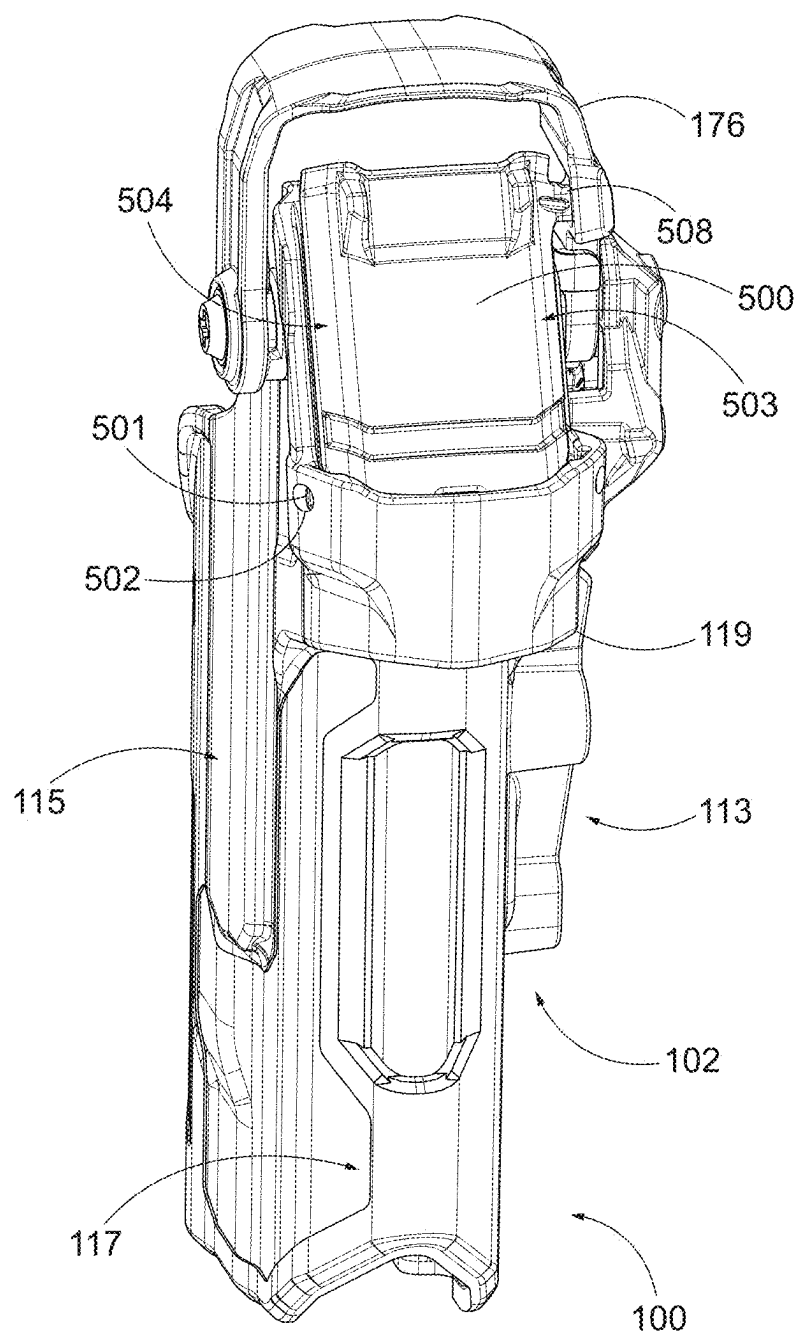


FIG. 51A

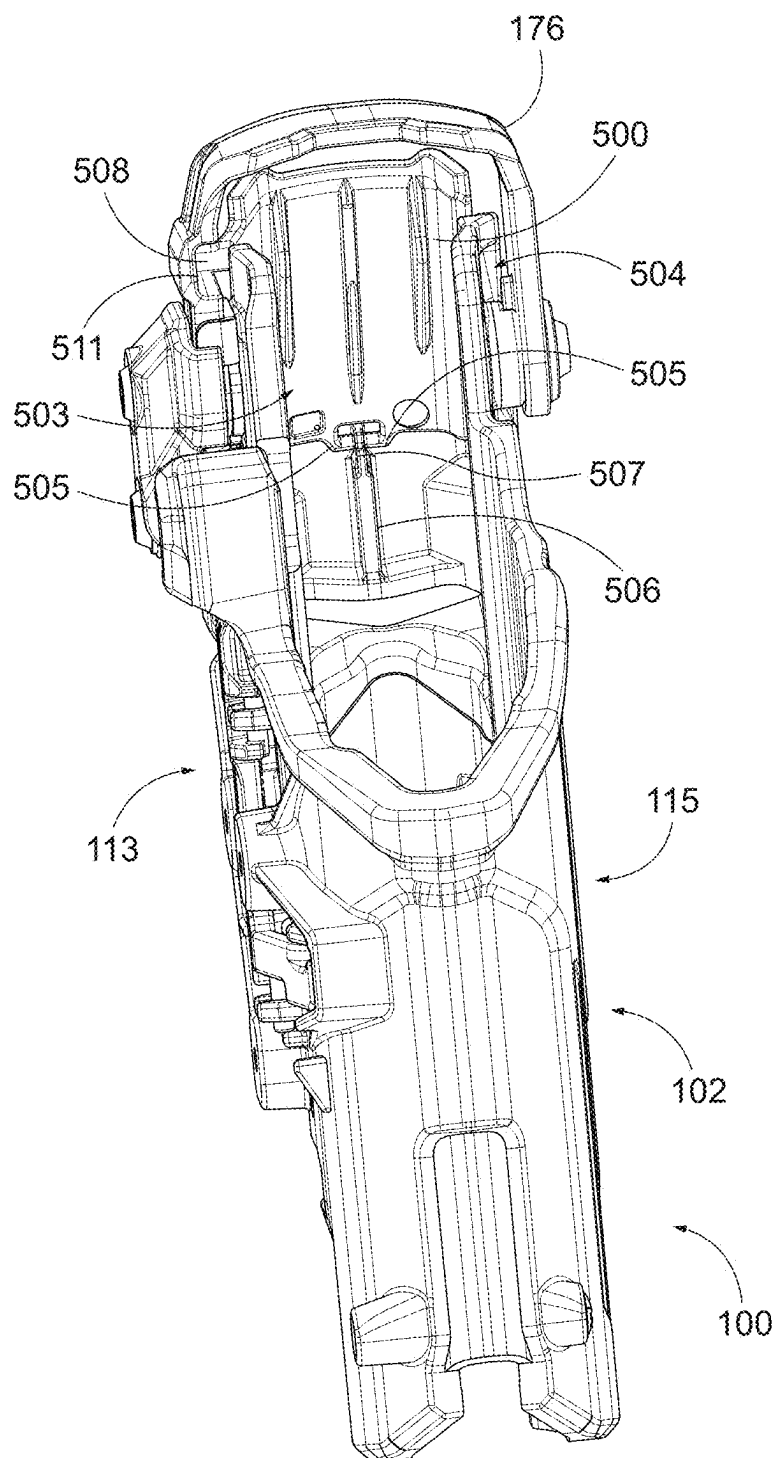


FIG. 51B

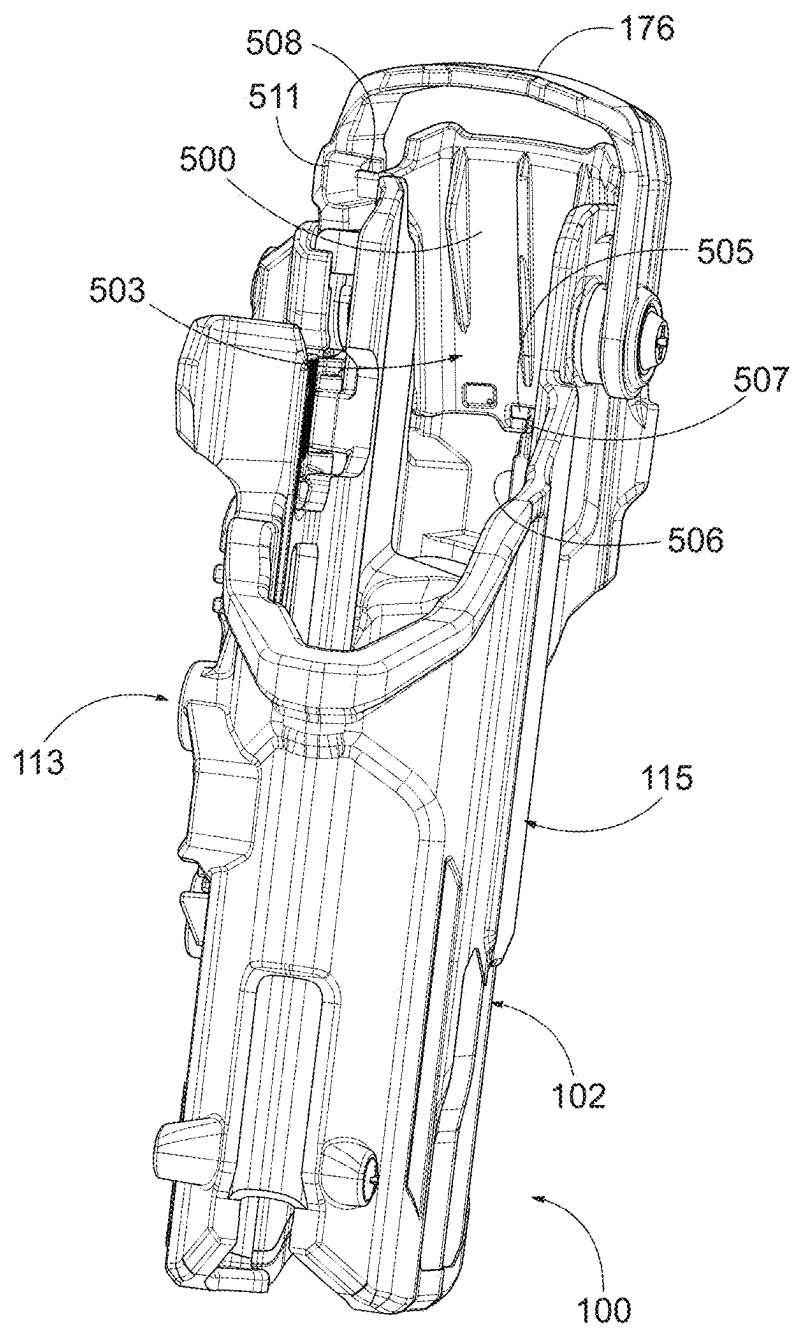


FIG. 51C

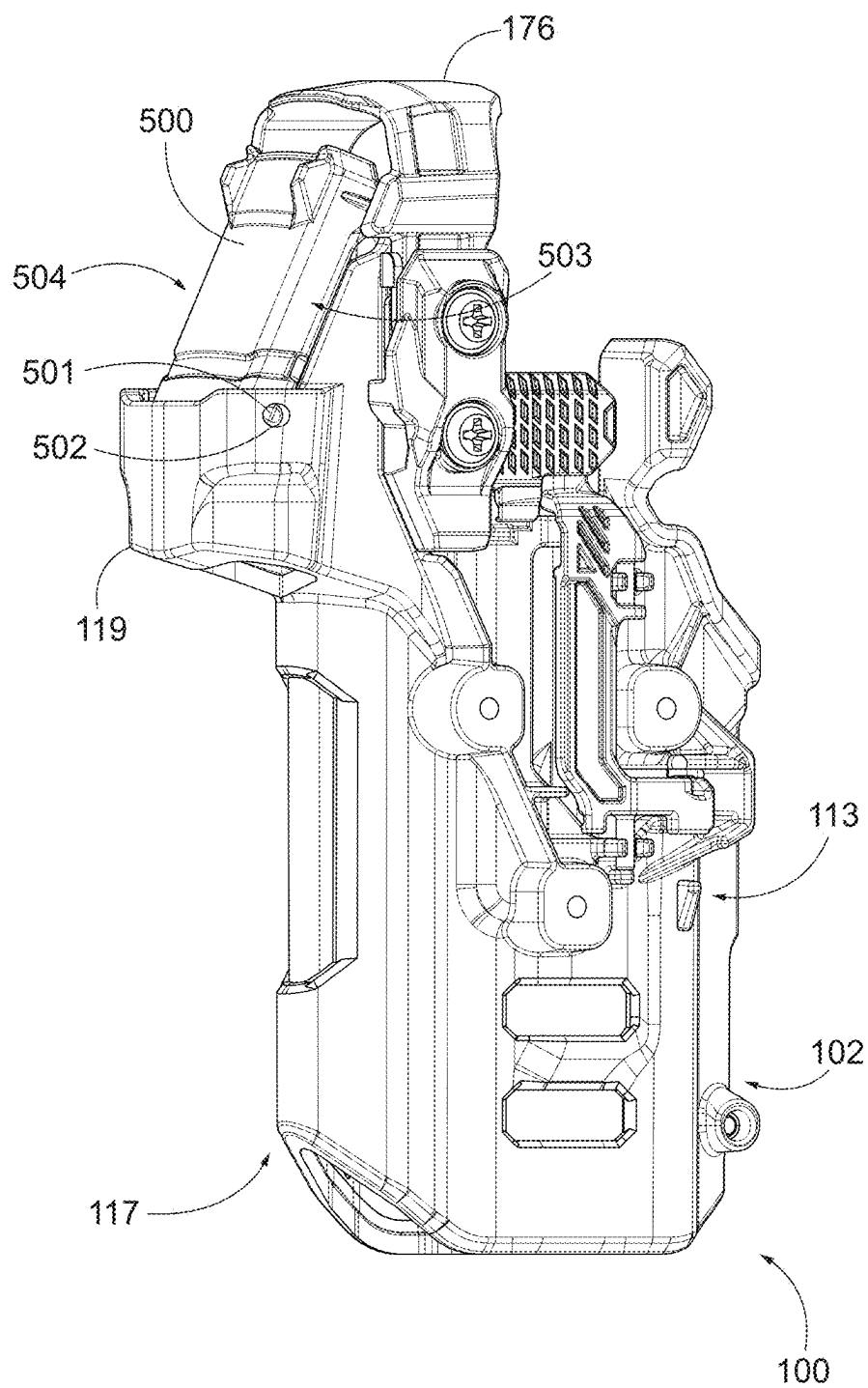


FIG. 51D

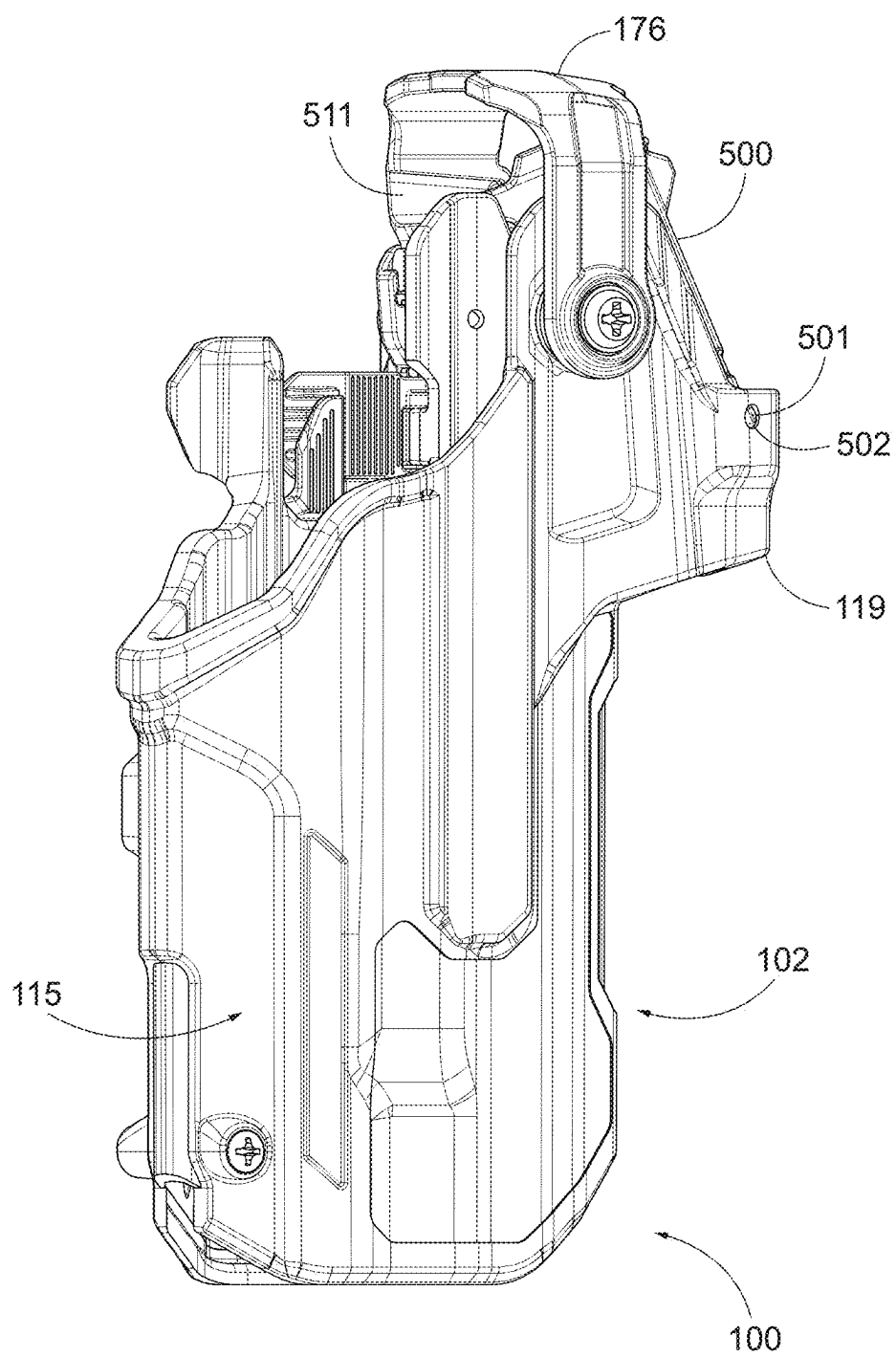


FIG. 51E

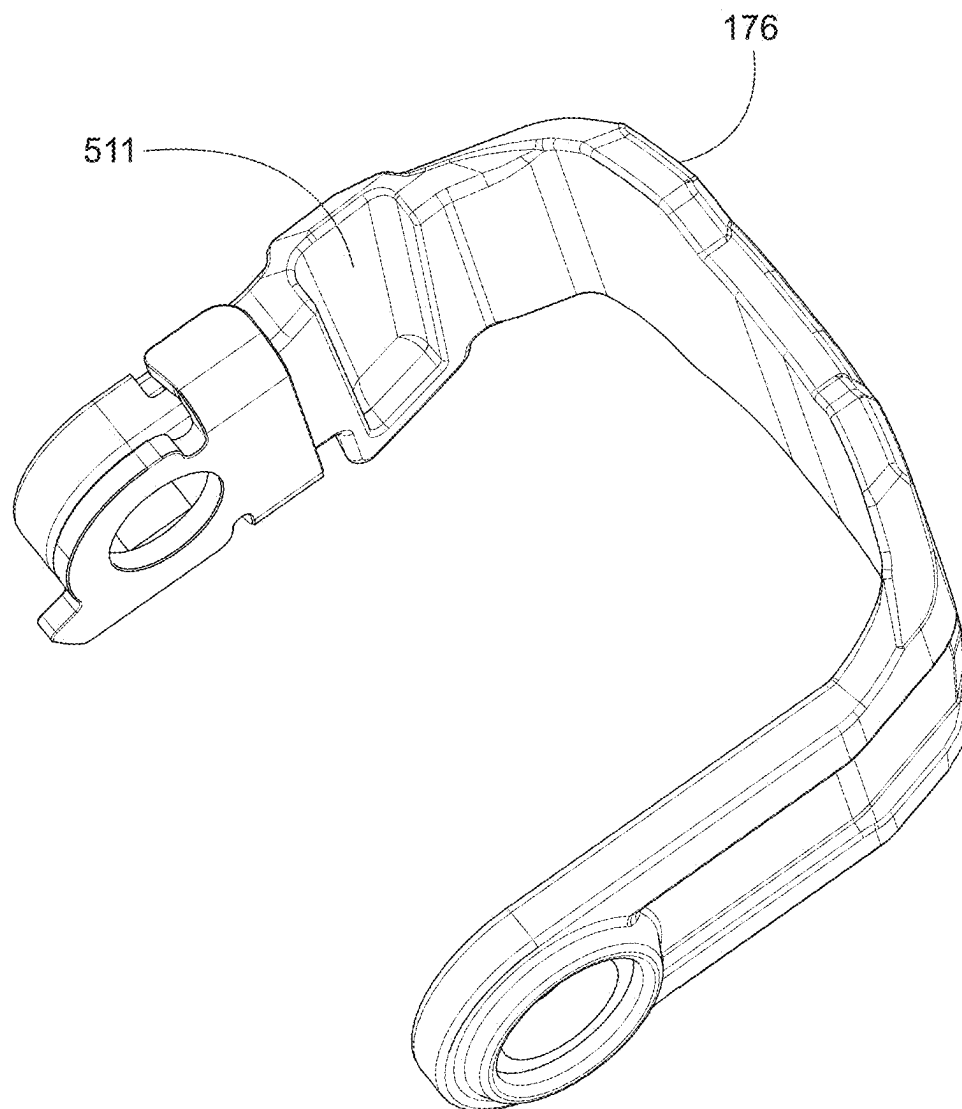


FIG. 51F

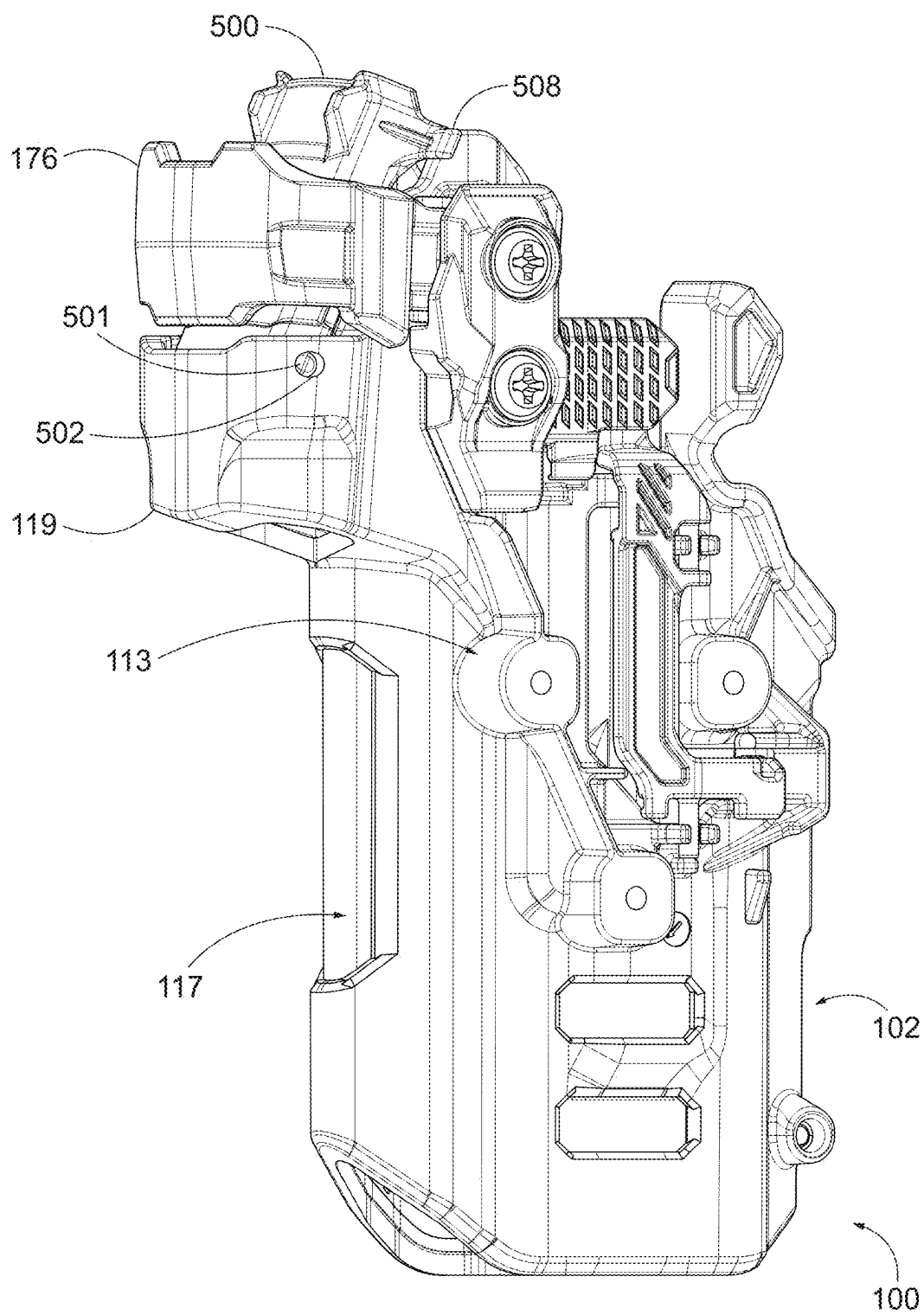


FIG. 51G

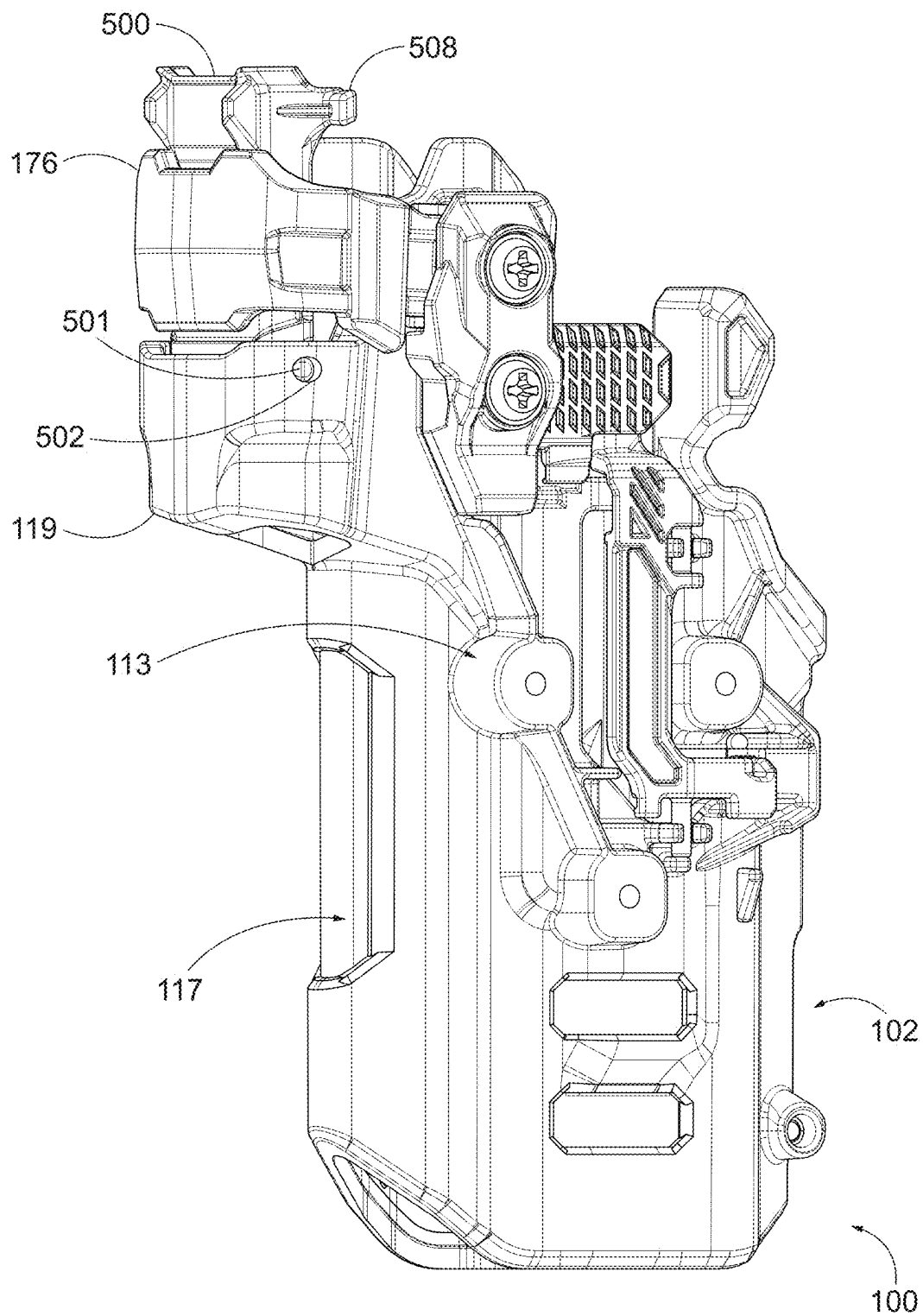


FIG. 51H

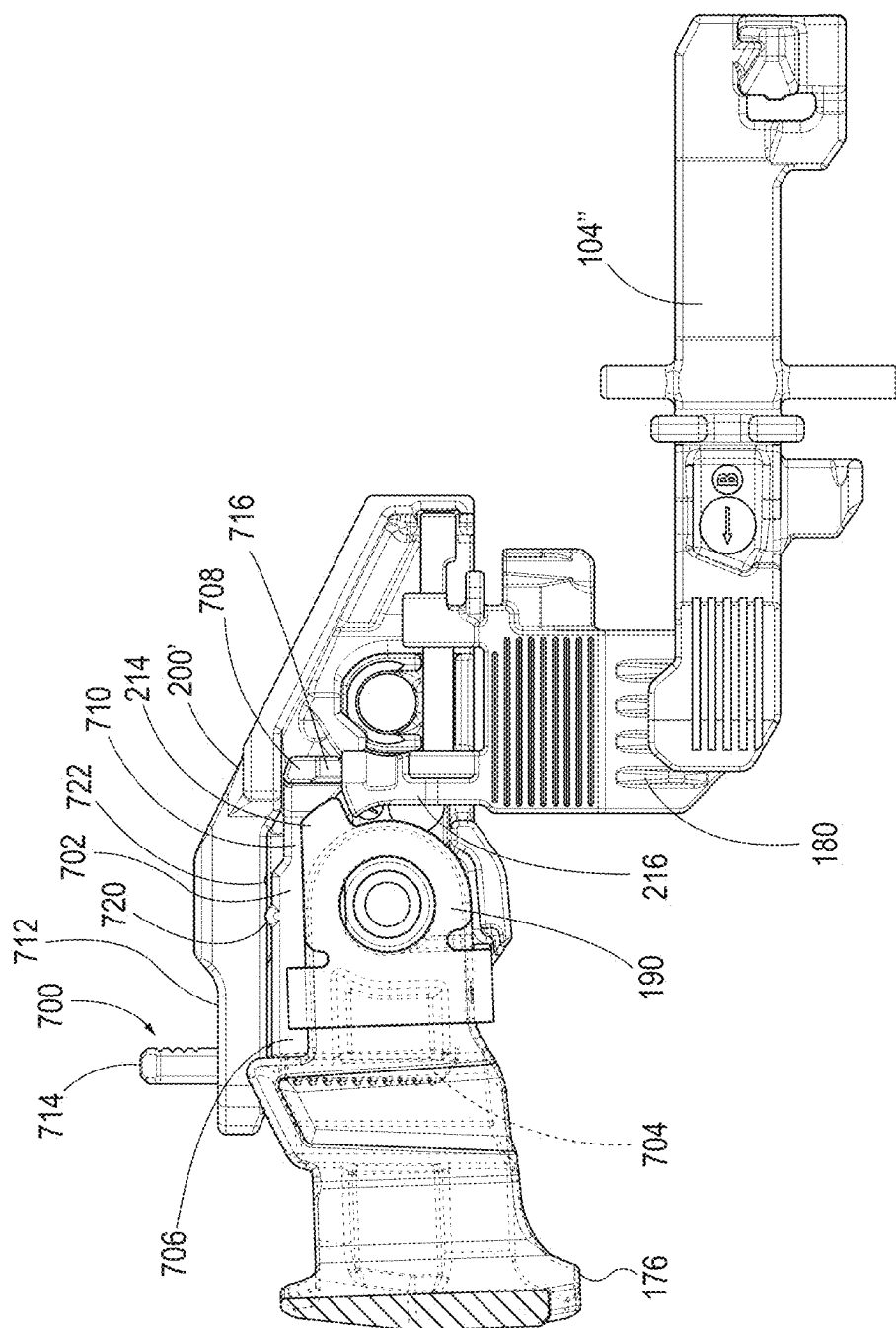


FIG. 52A

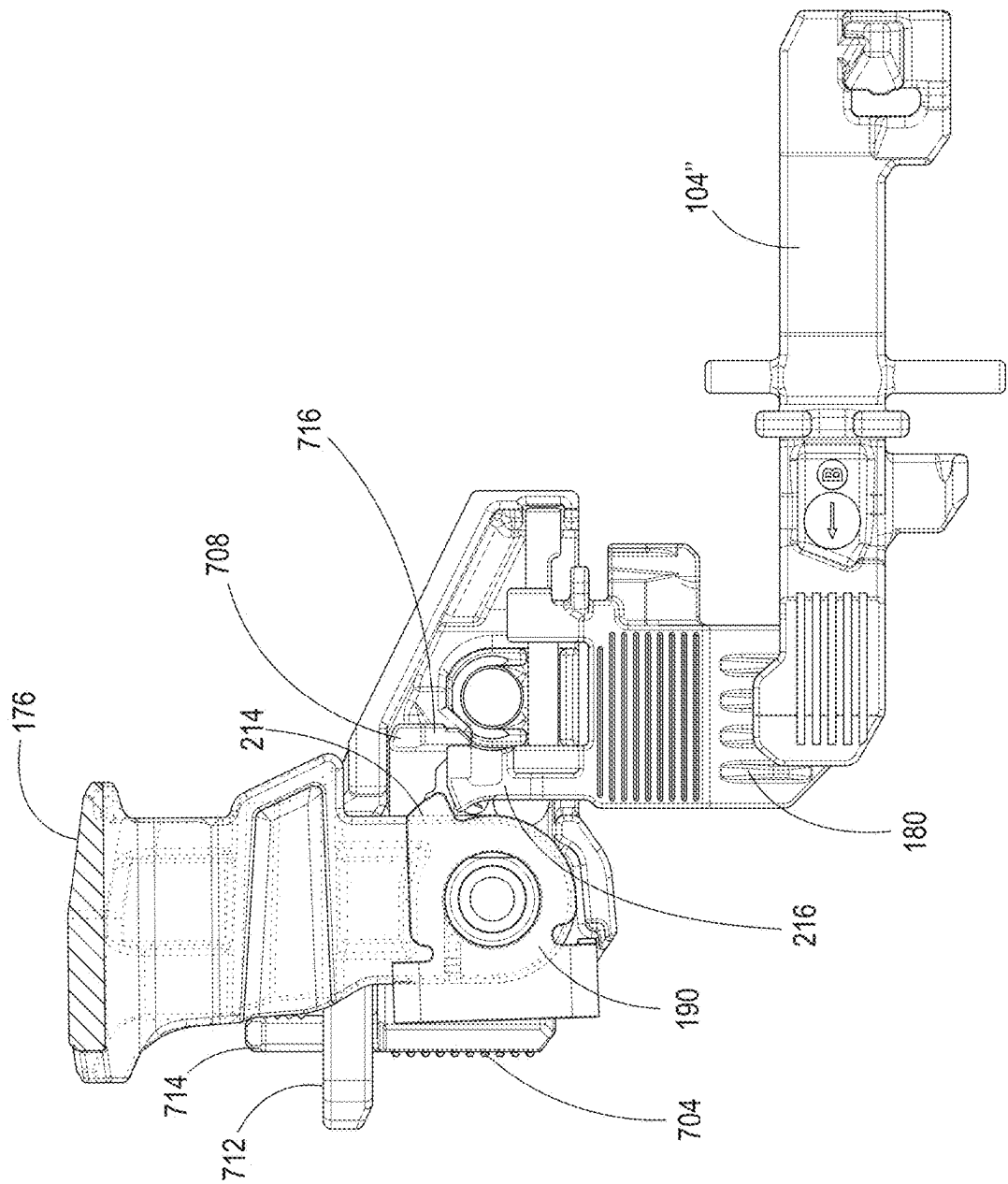


FIG. 52B

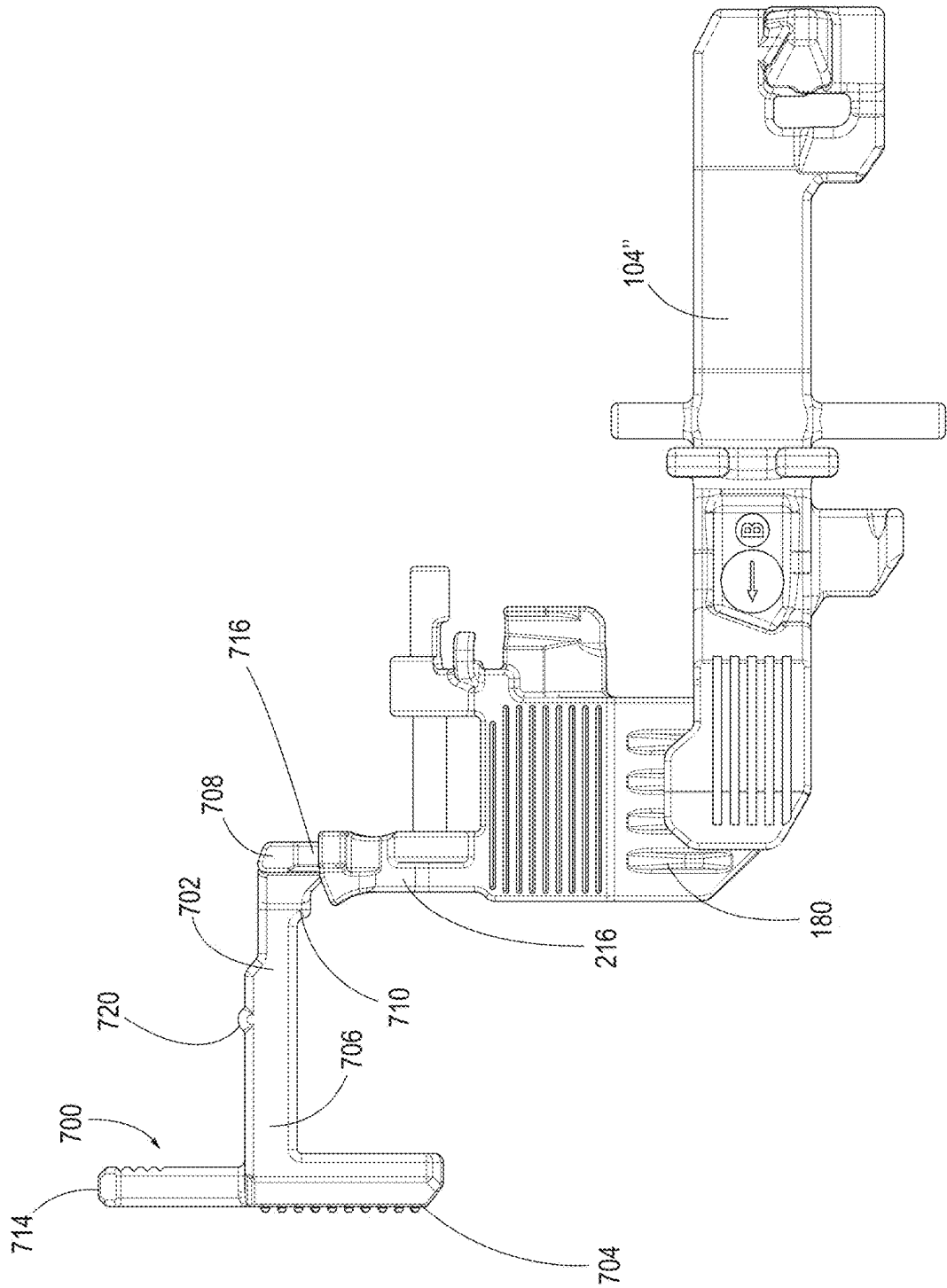


FIG. 52C

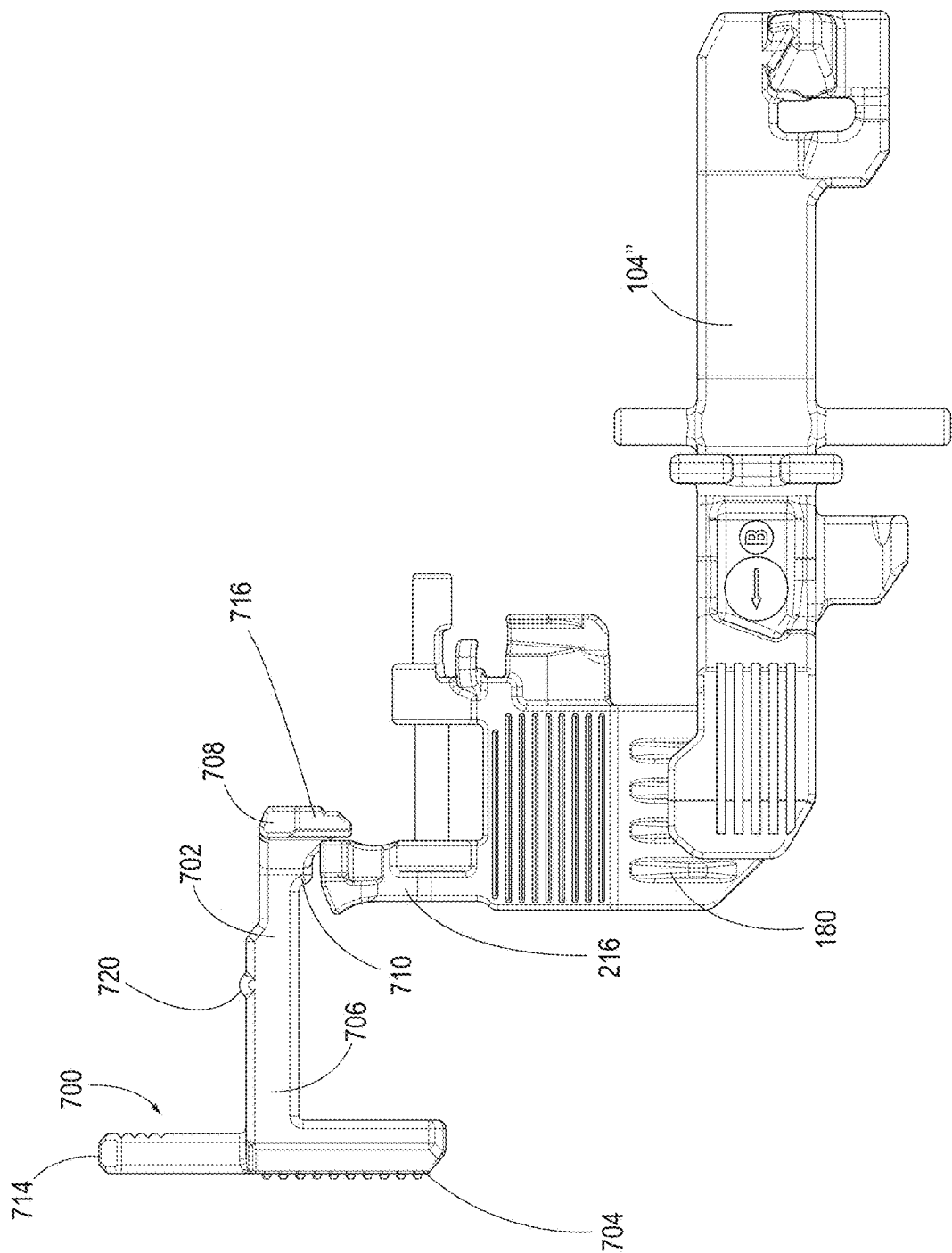


FIG. 52D

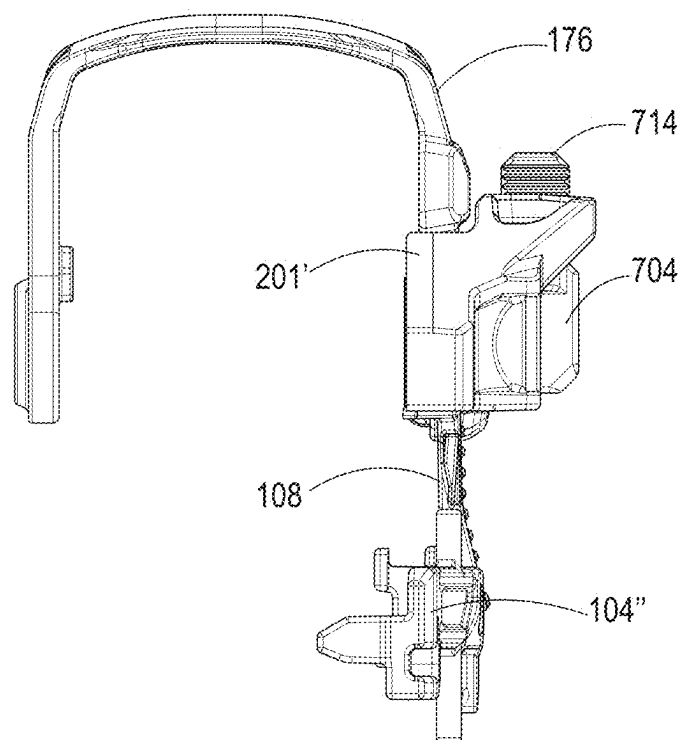


FIG. 53A

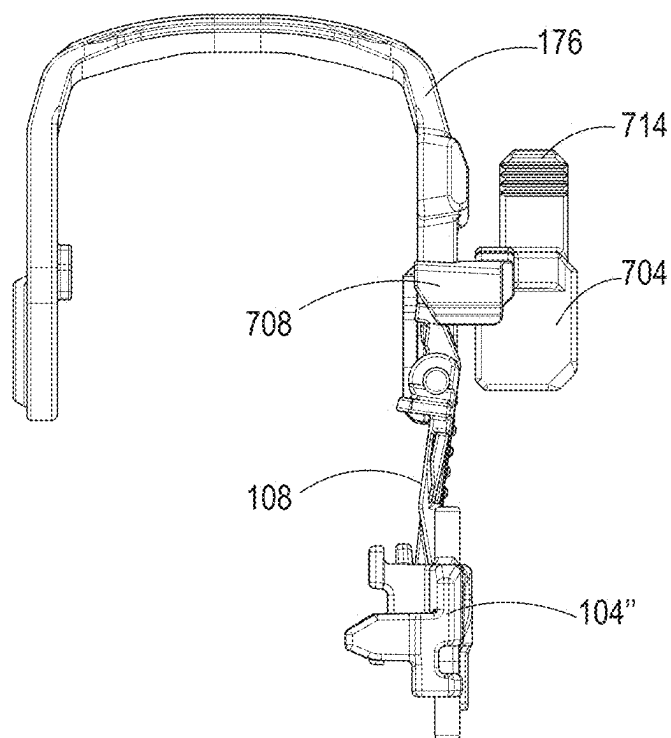


FIG. 53B

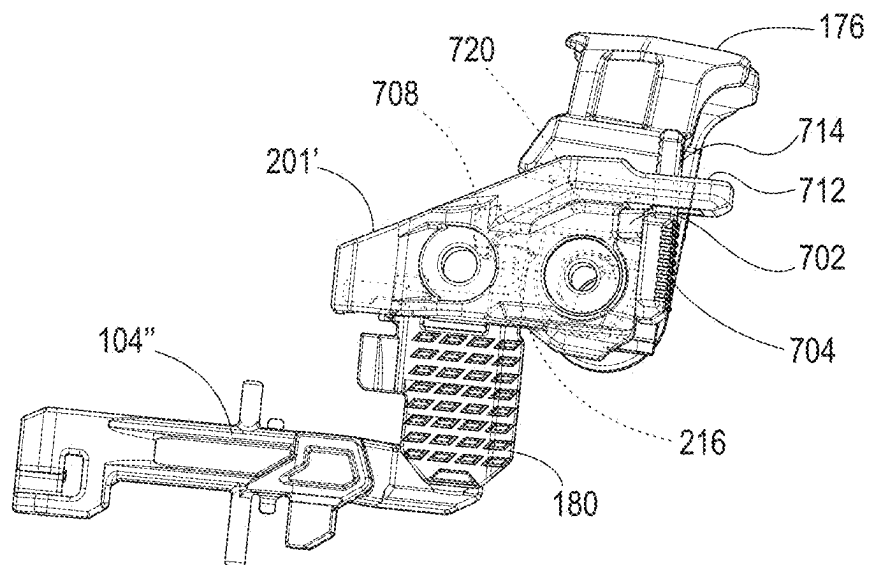


FIG. 54A

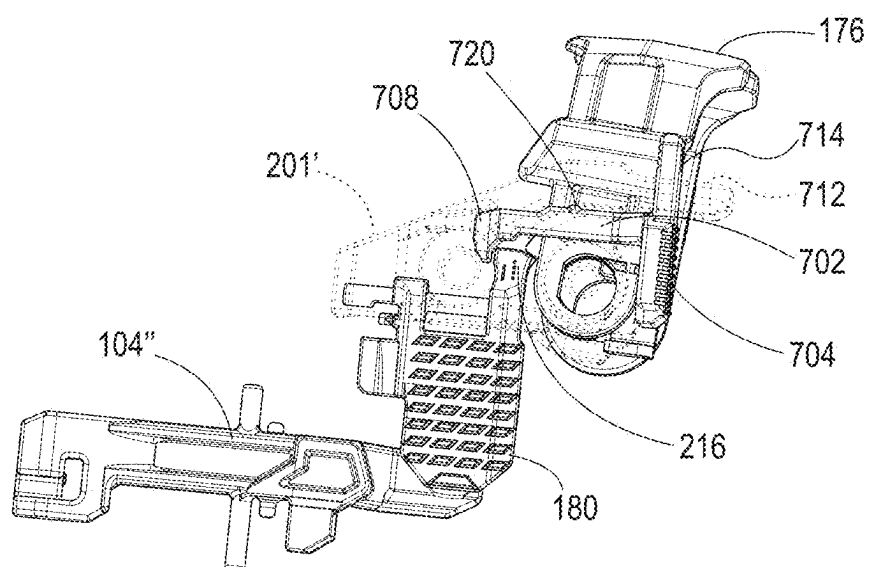


FIG. 54B

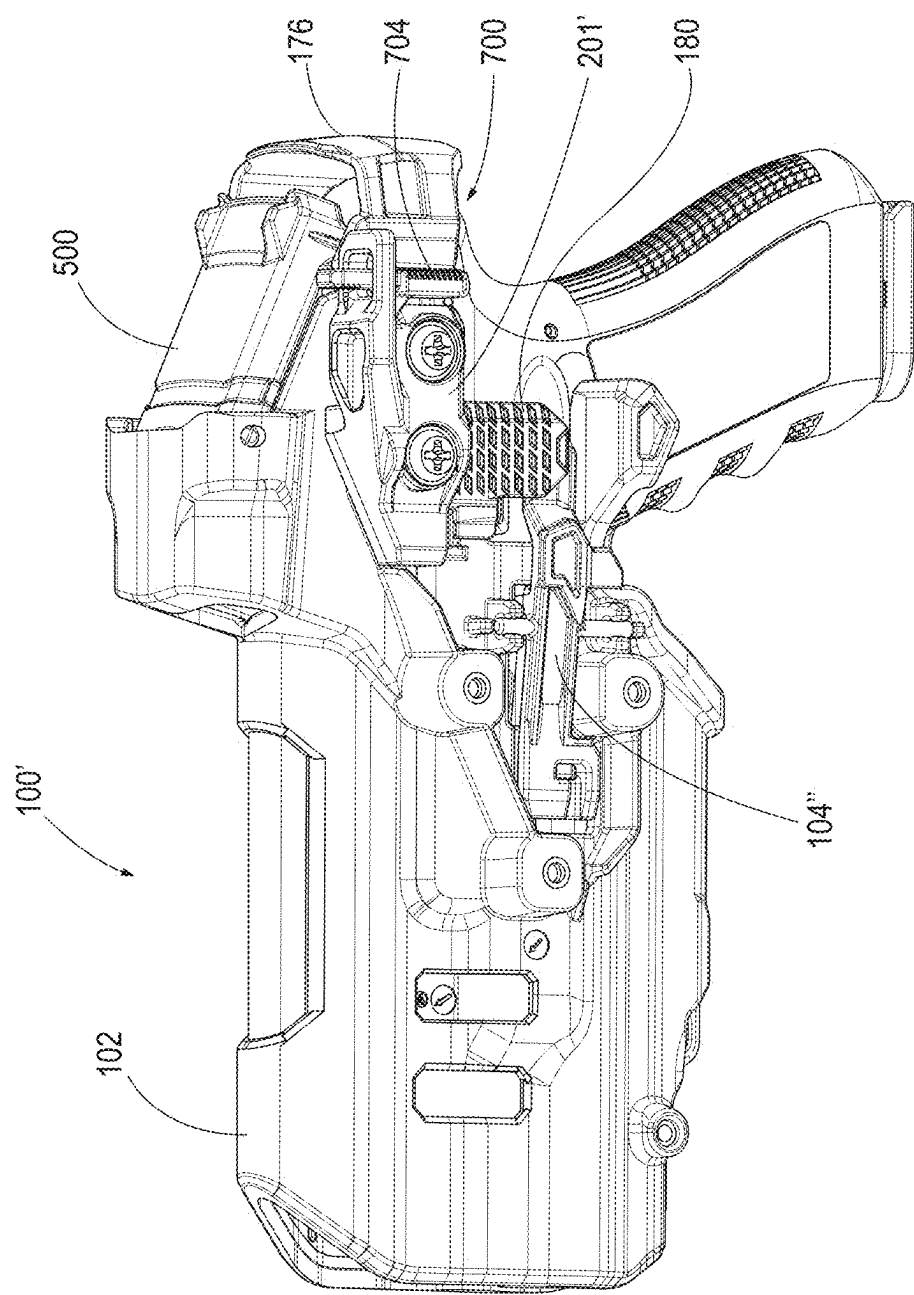


FIG. 55

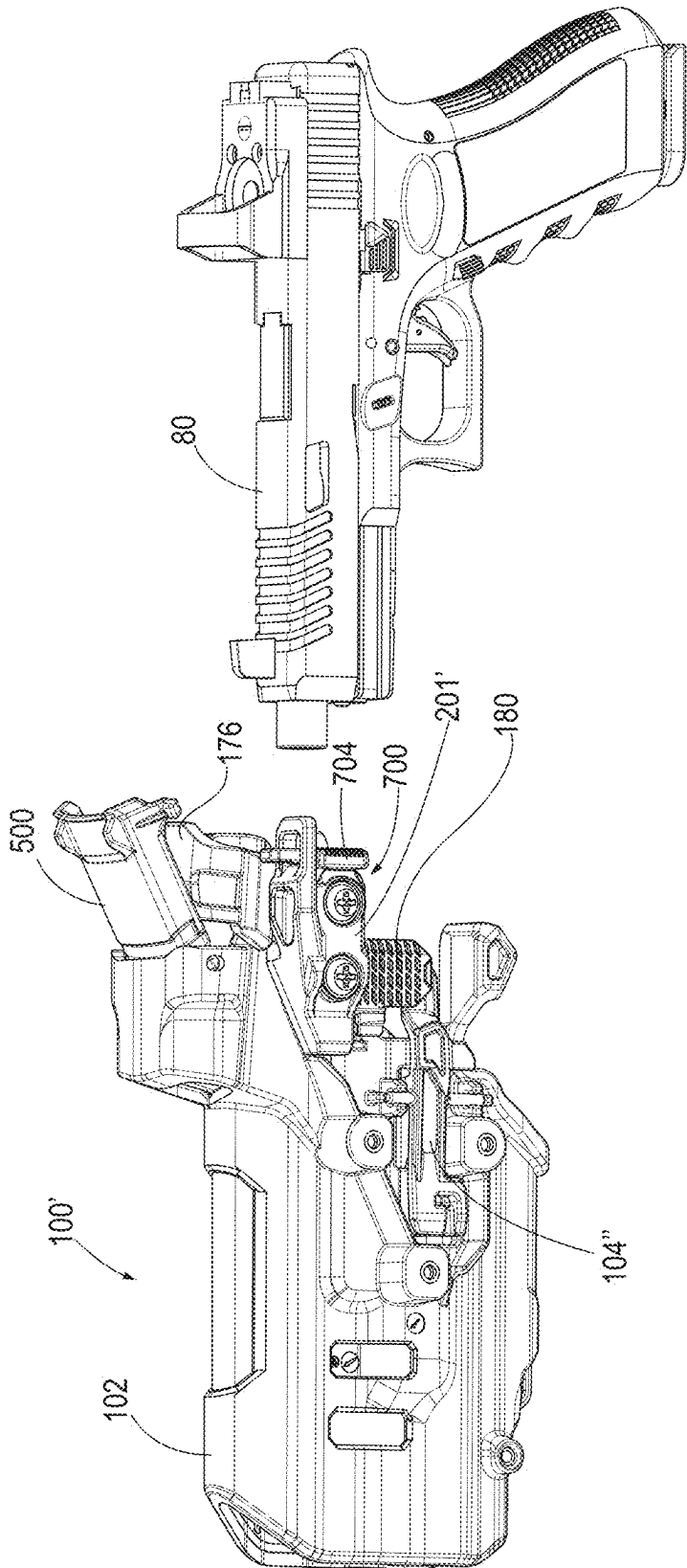


FIG. 56

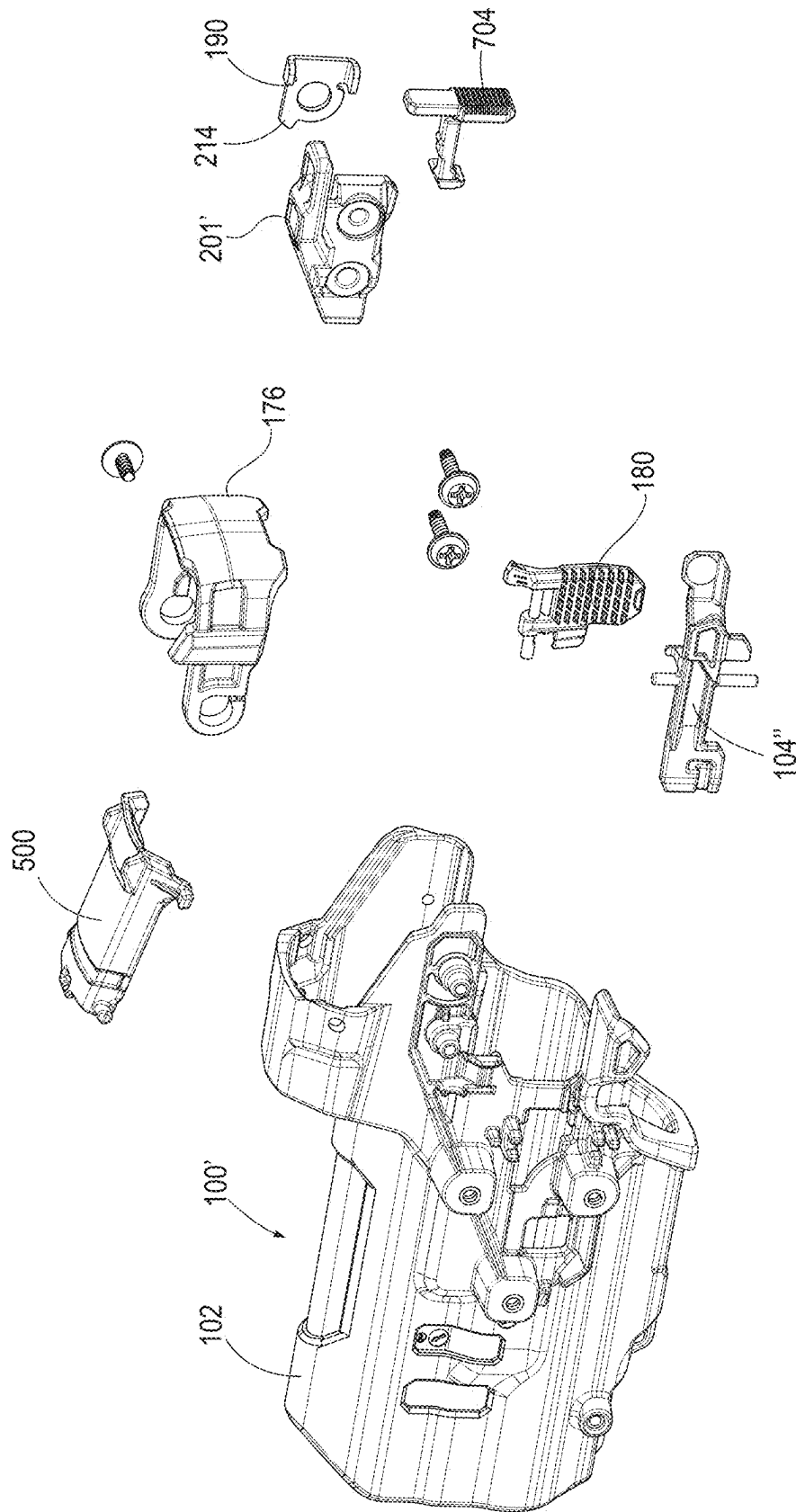


FIG. 57

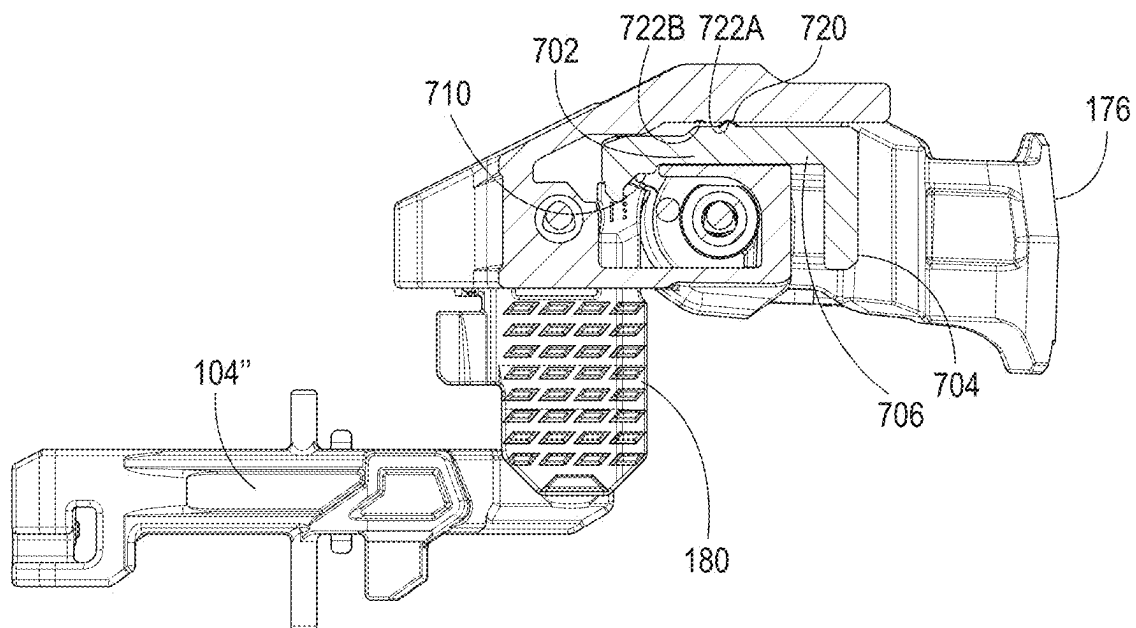


FIG. 58A

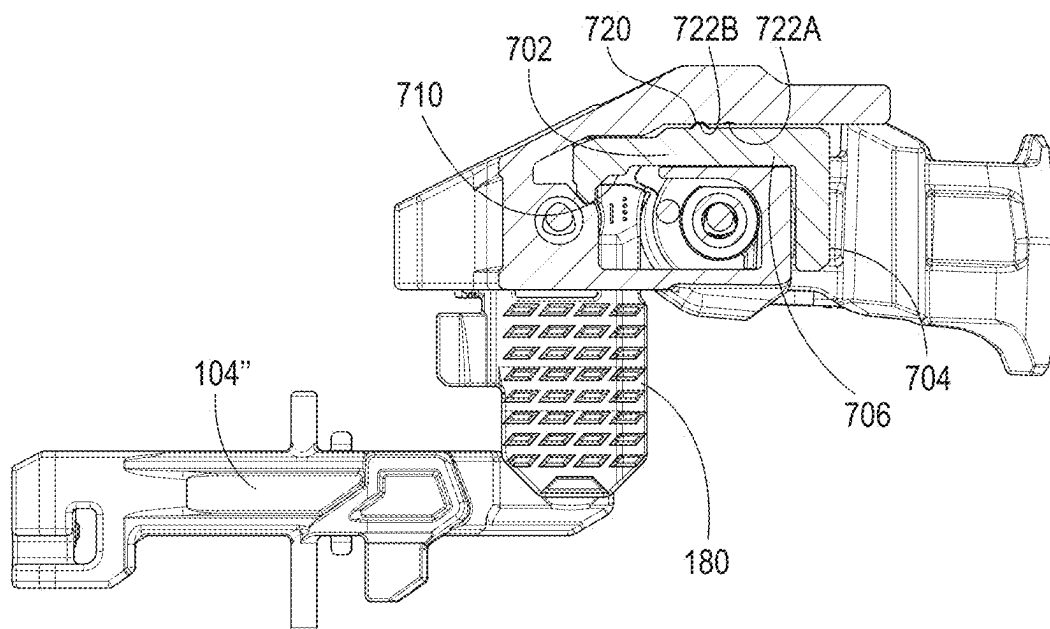


FIG. 58B

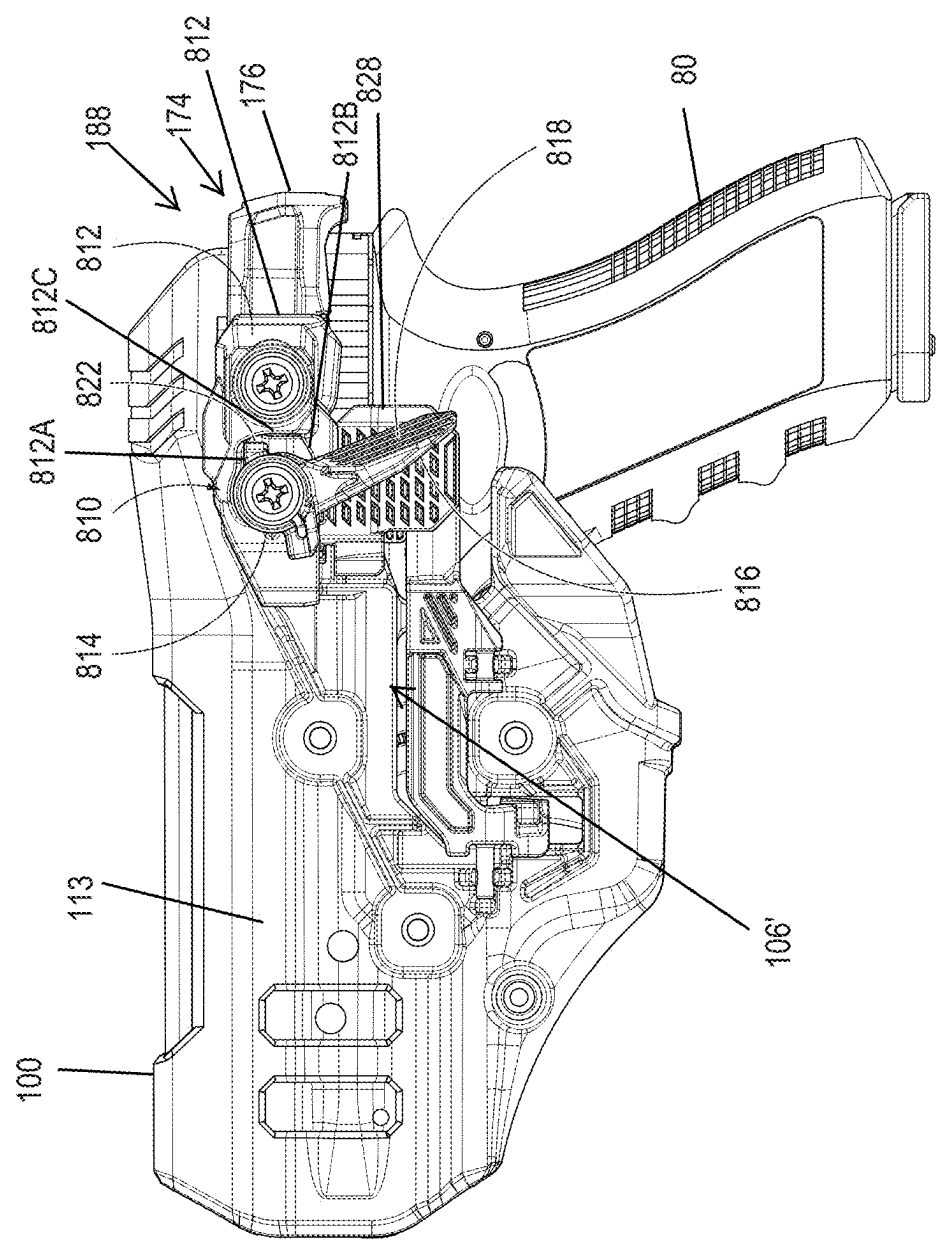


FIG. 59

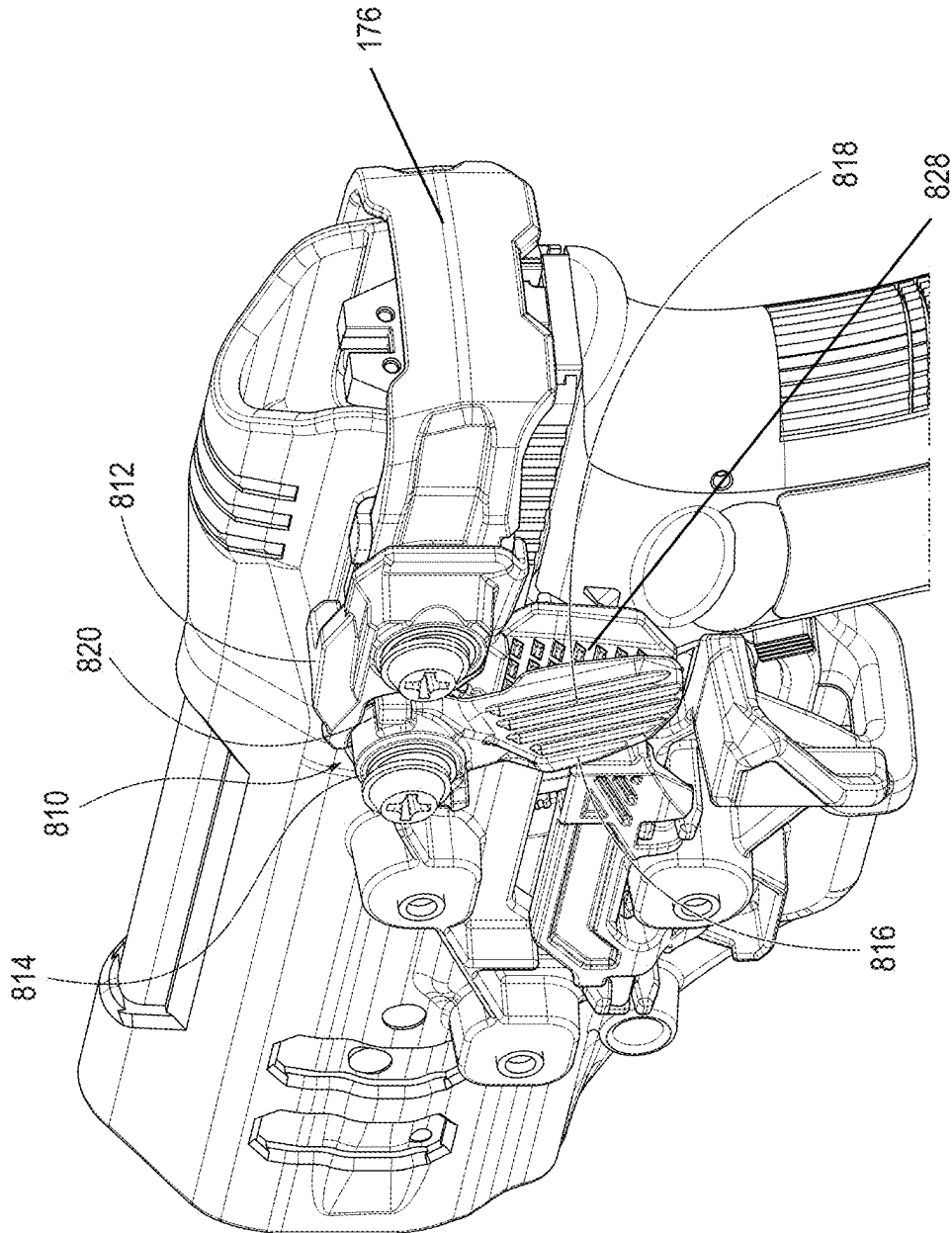


FIG. 60

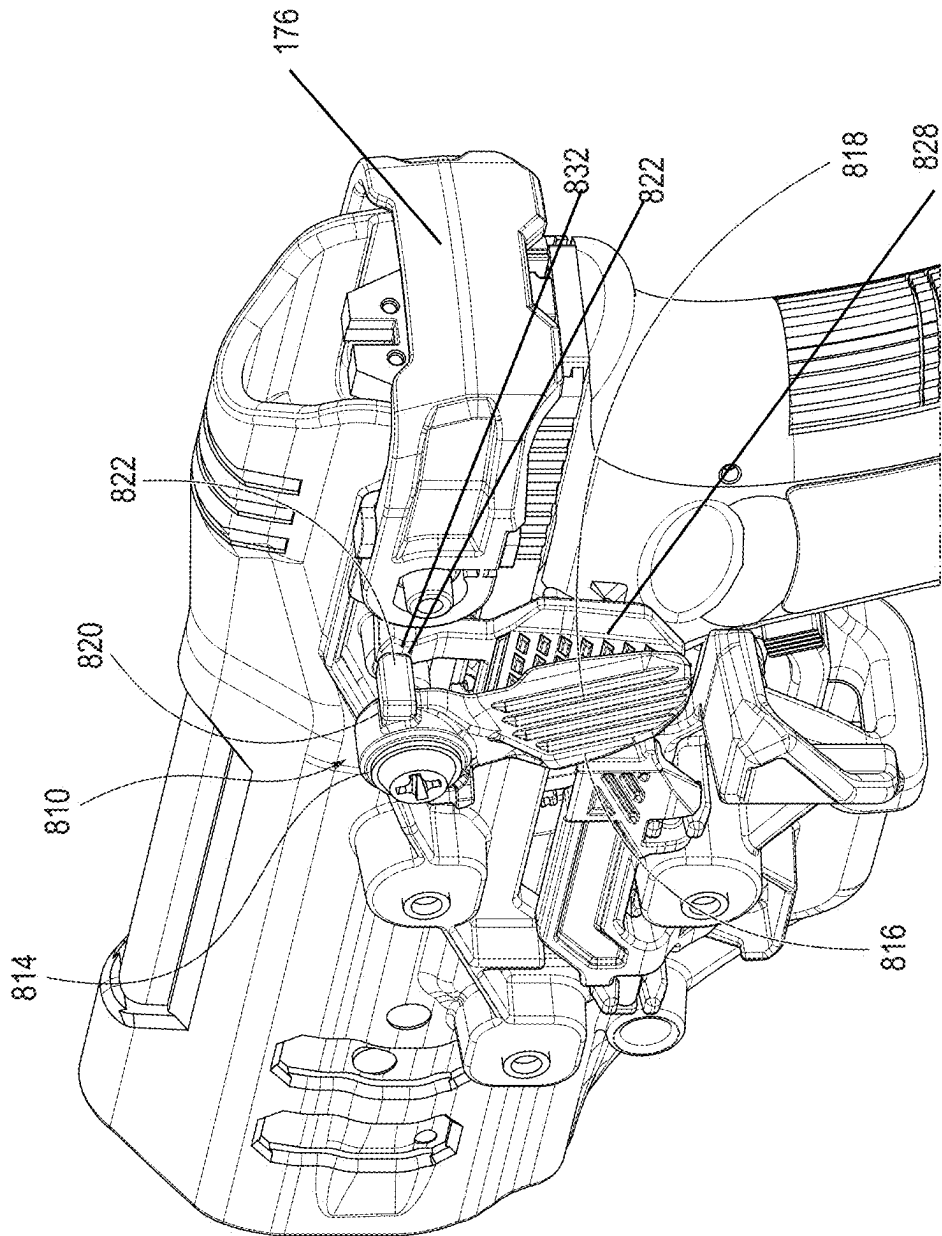


FIG. 61

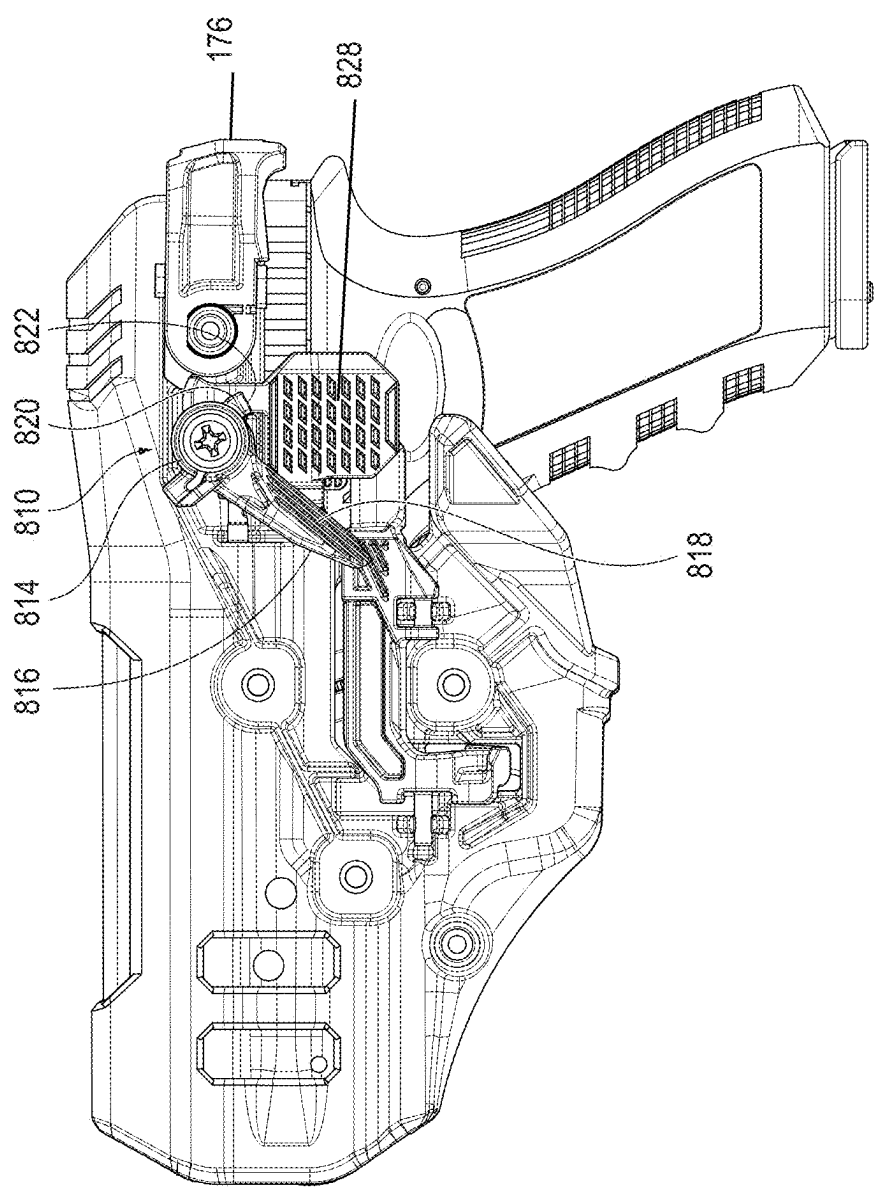


FIG. 62

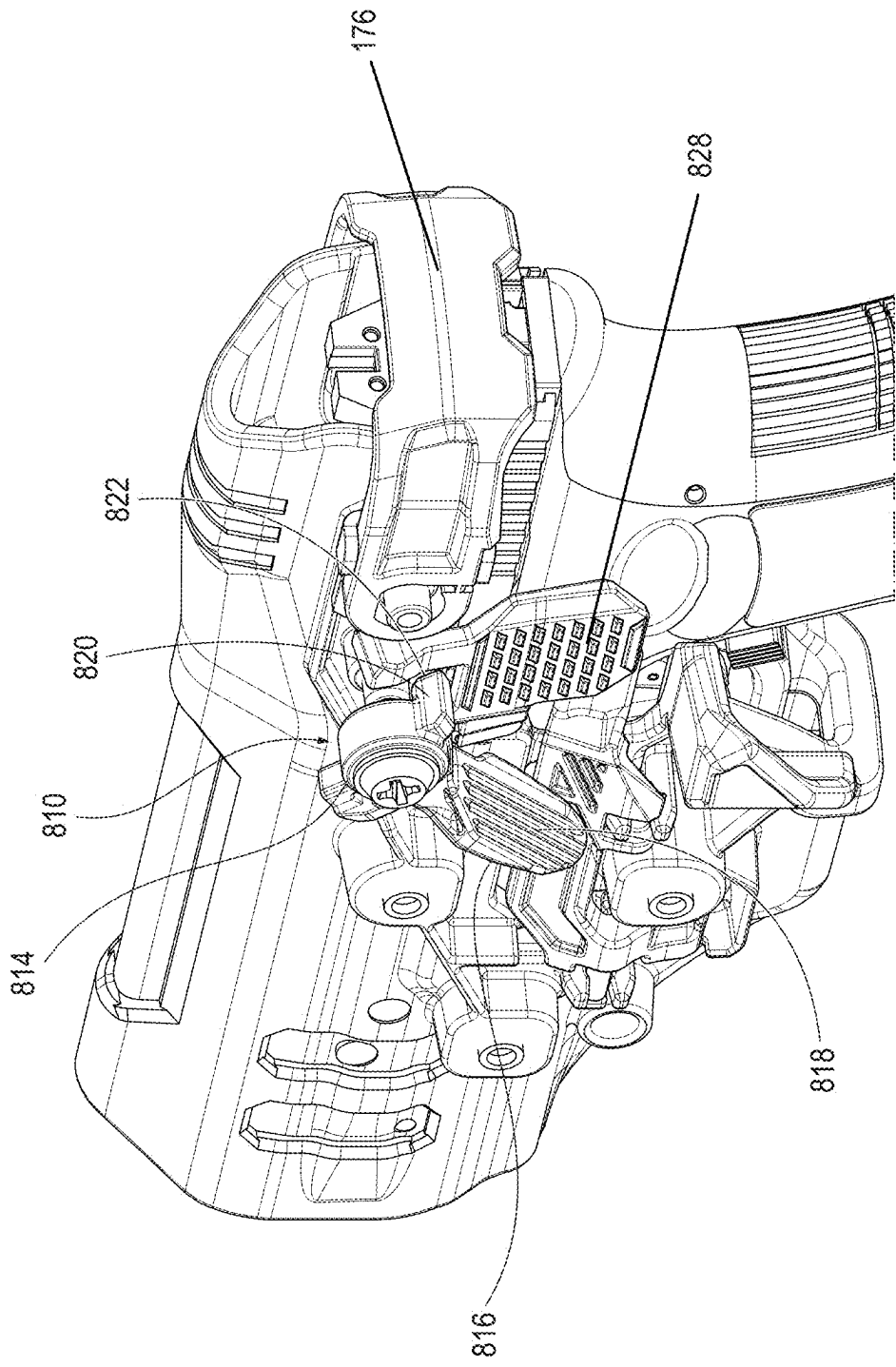


FIG. 63

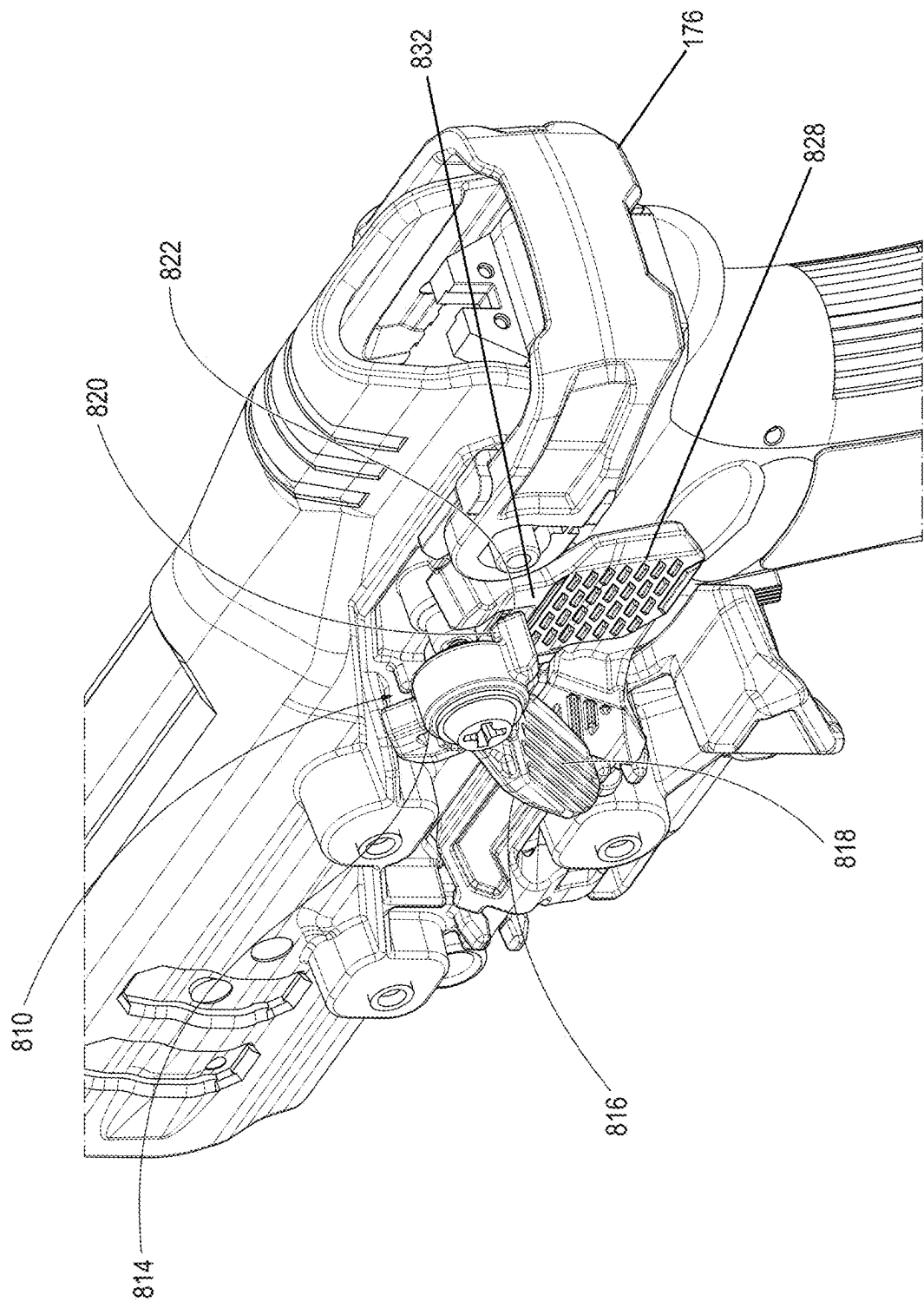


FIG. 64

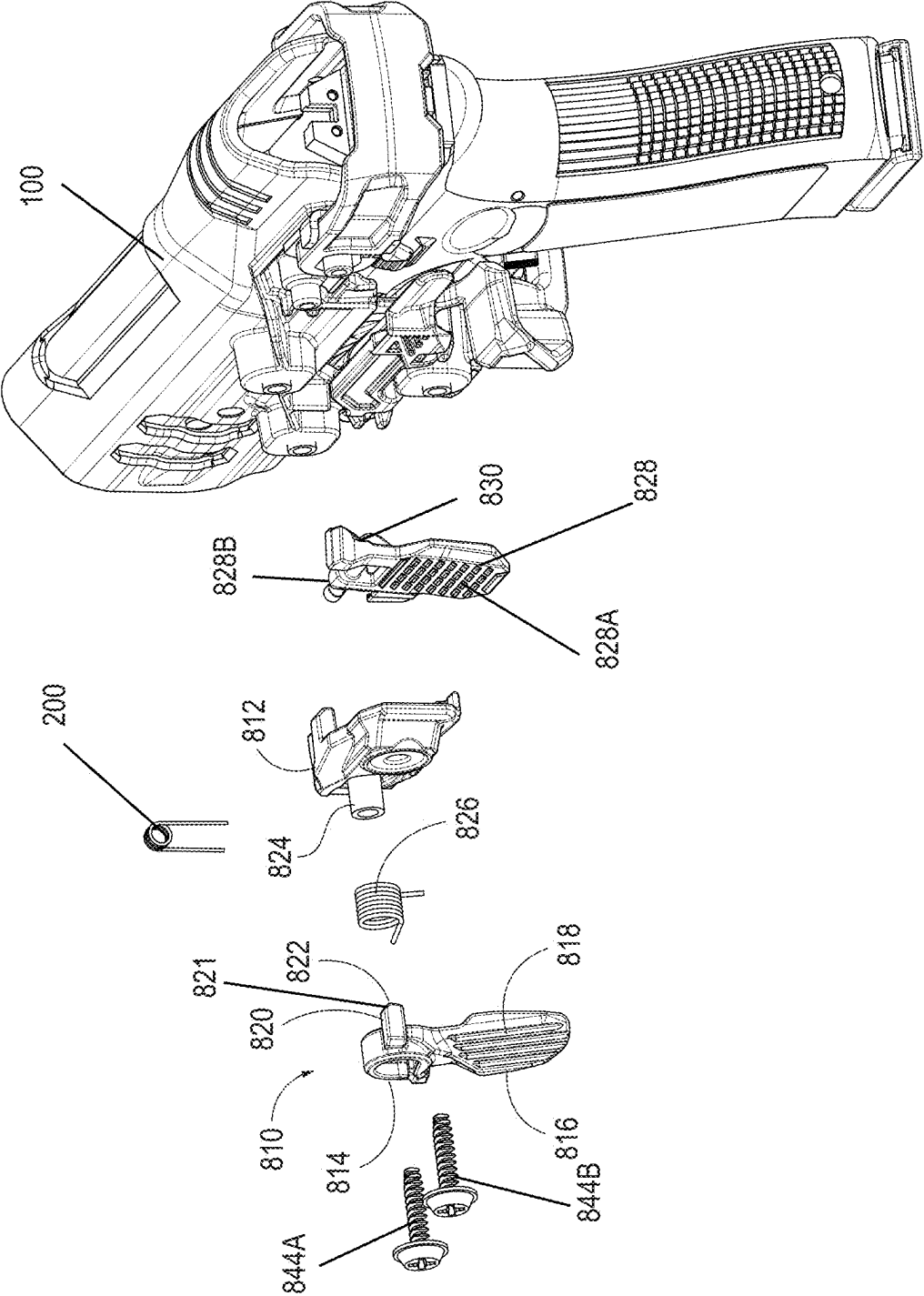


FIG. 65

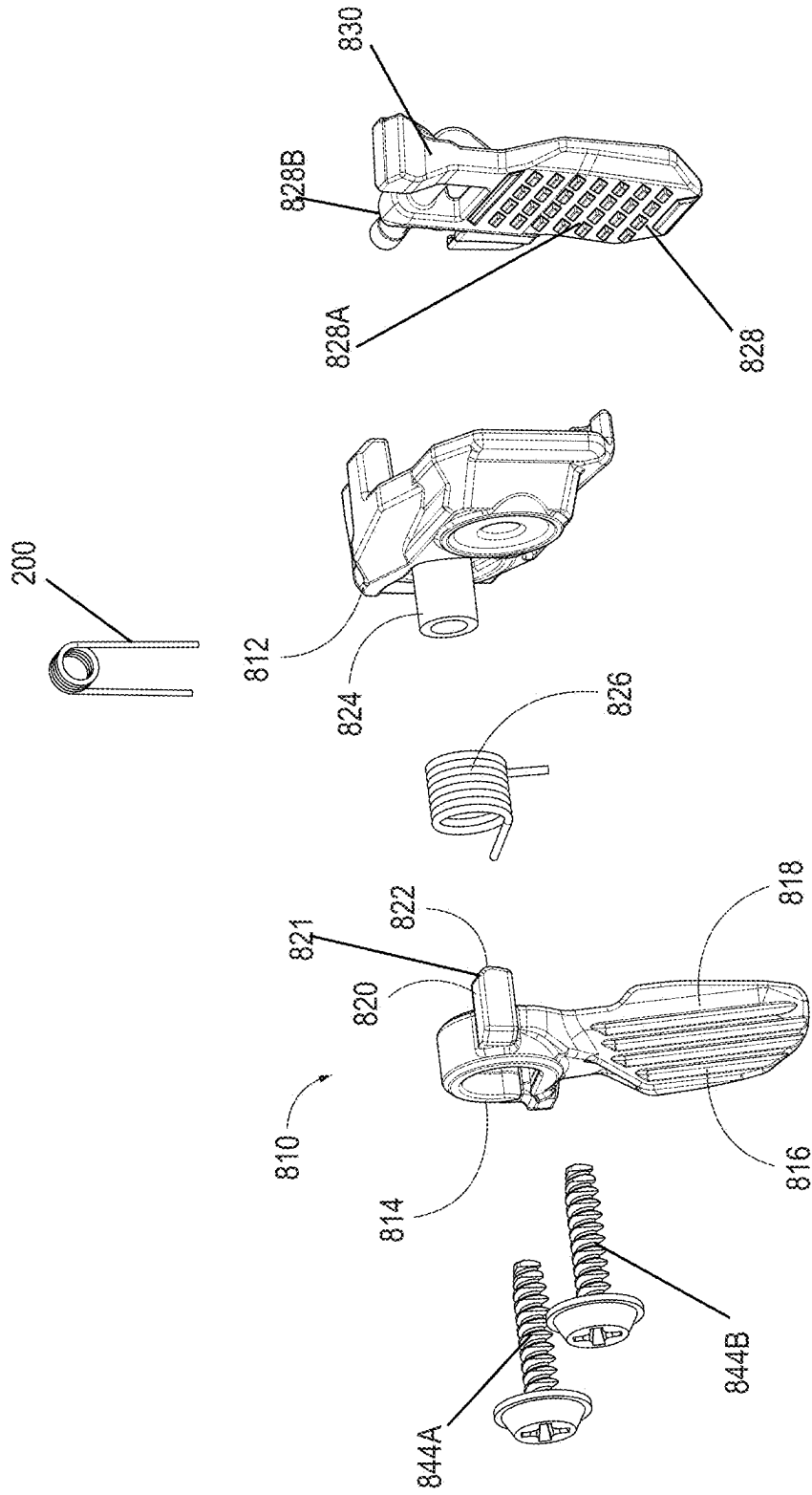


FIG. 66

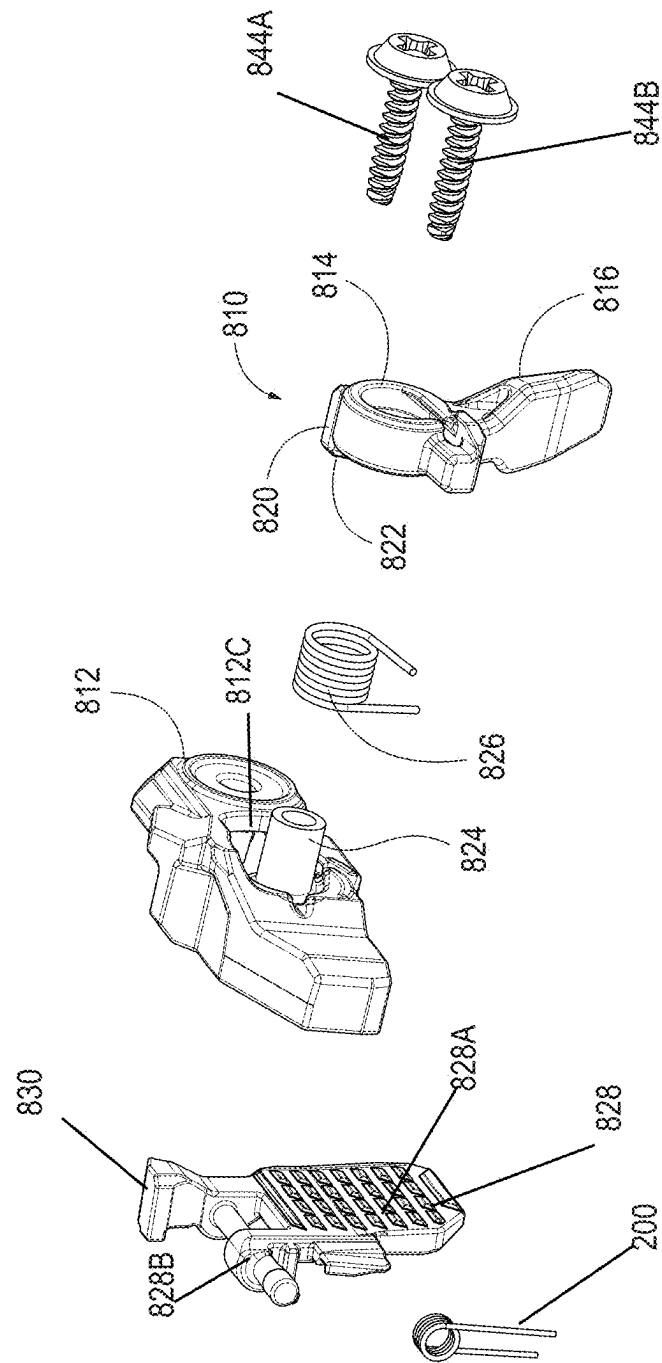


FIG. 67

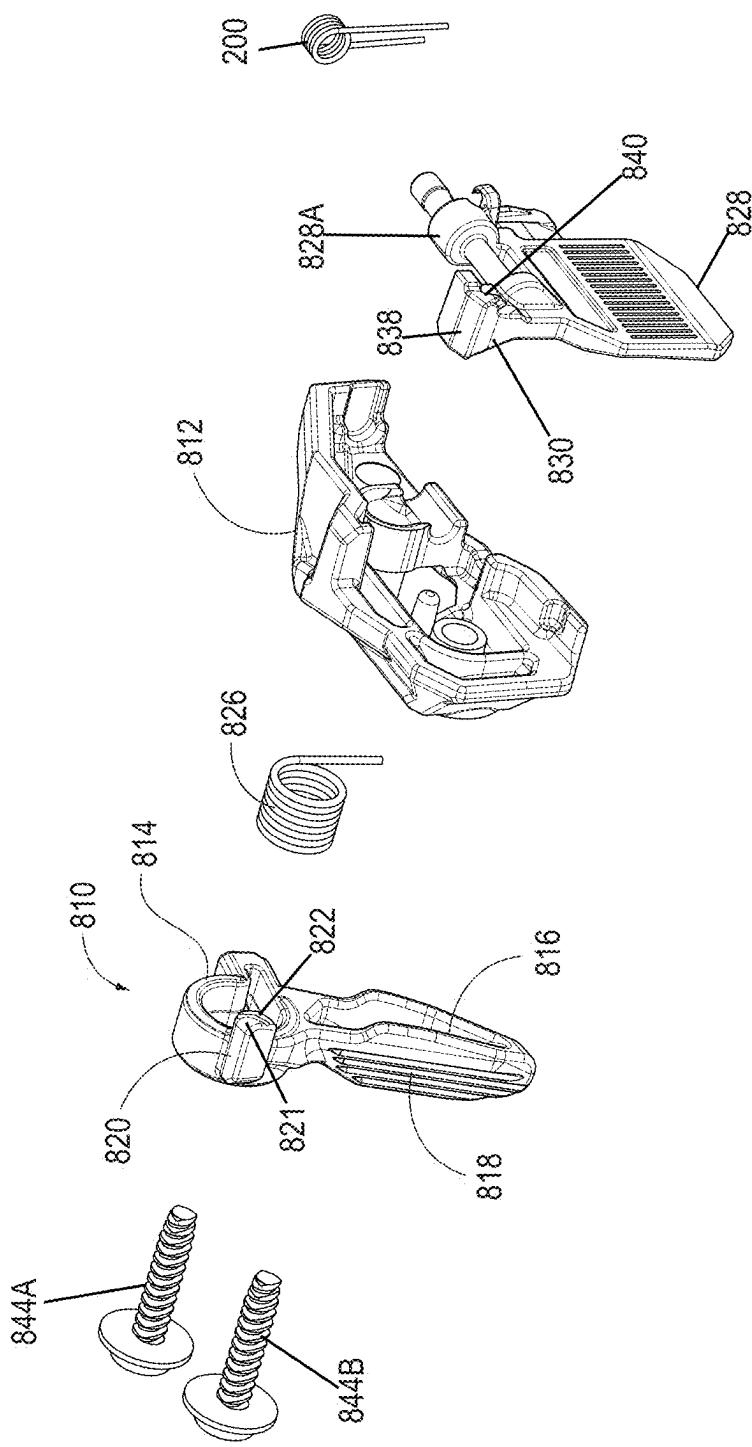


FIG. 68

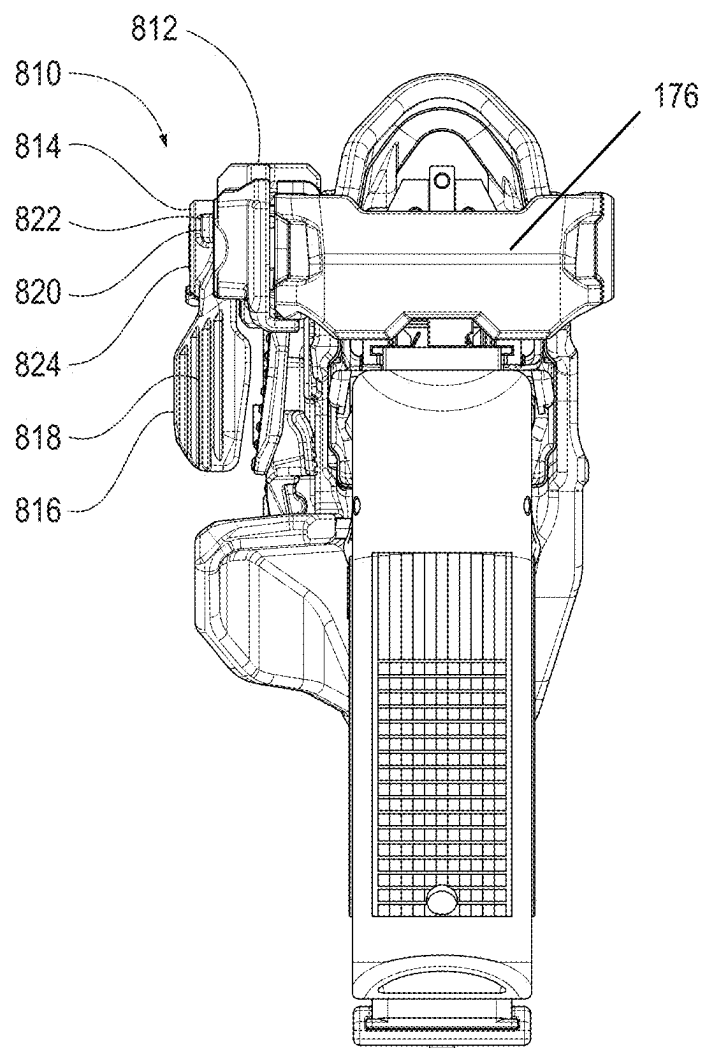


FIG. 69

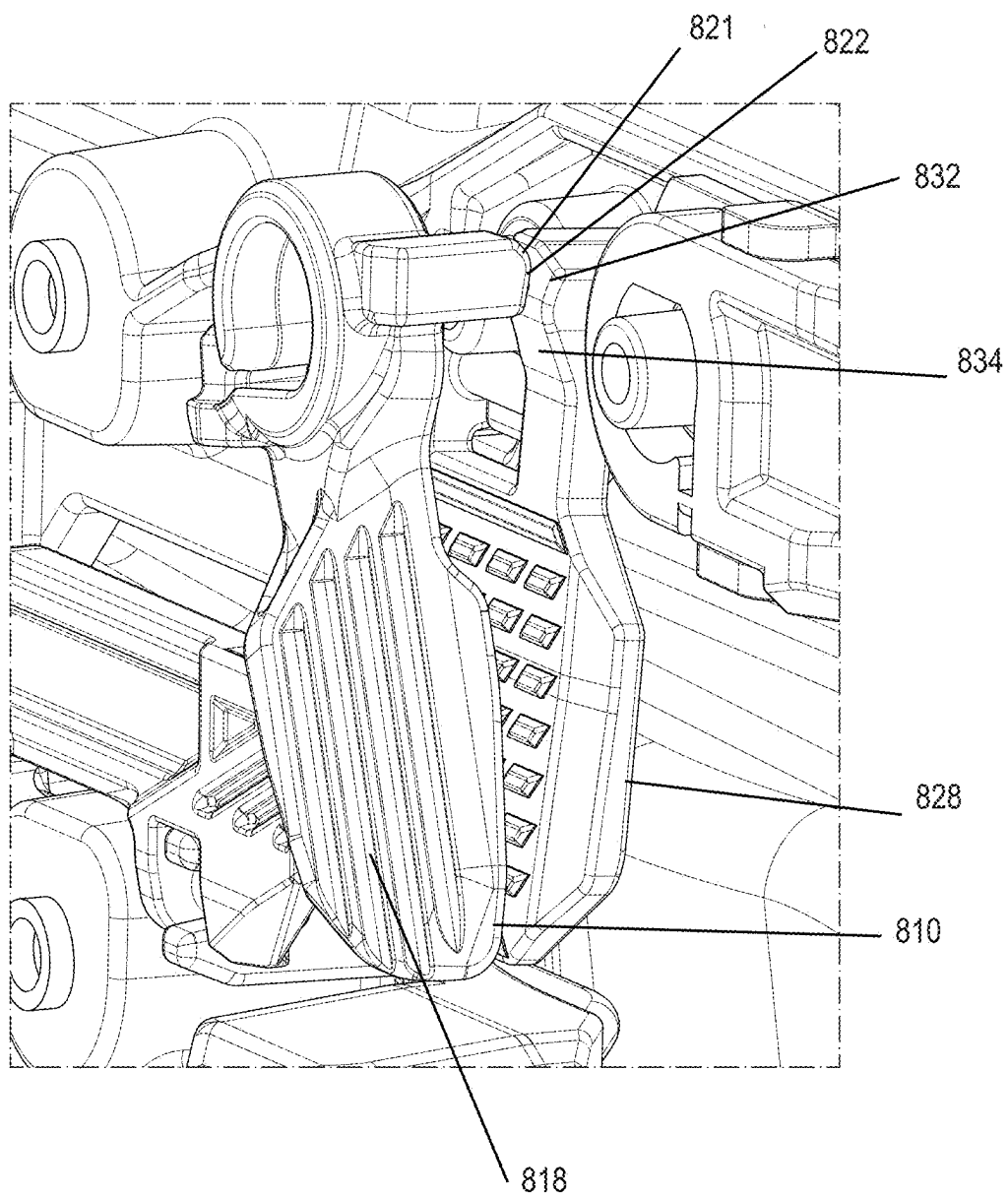


FIG. 70

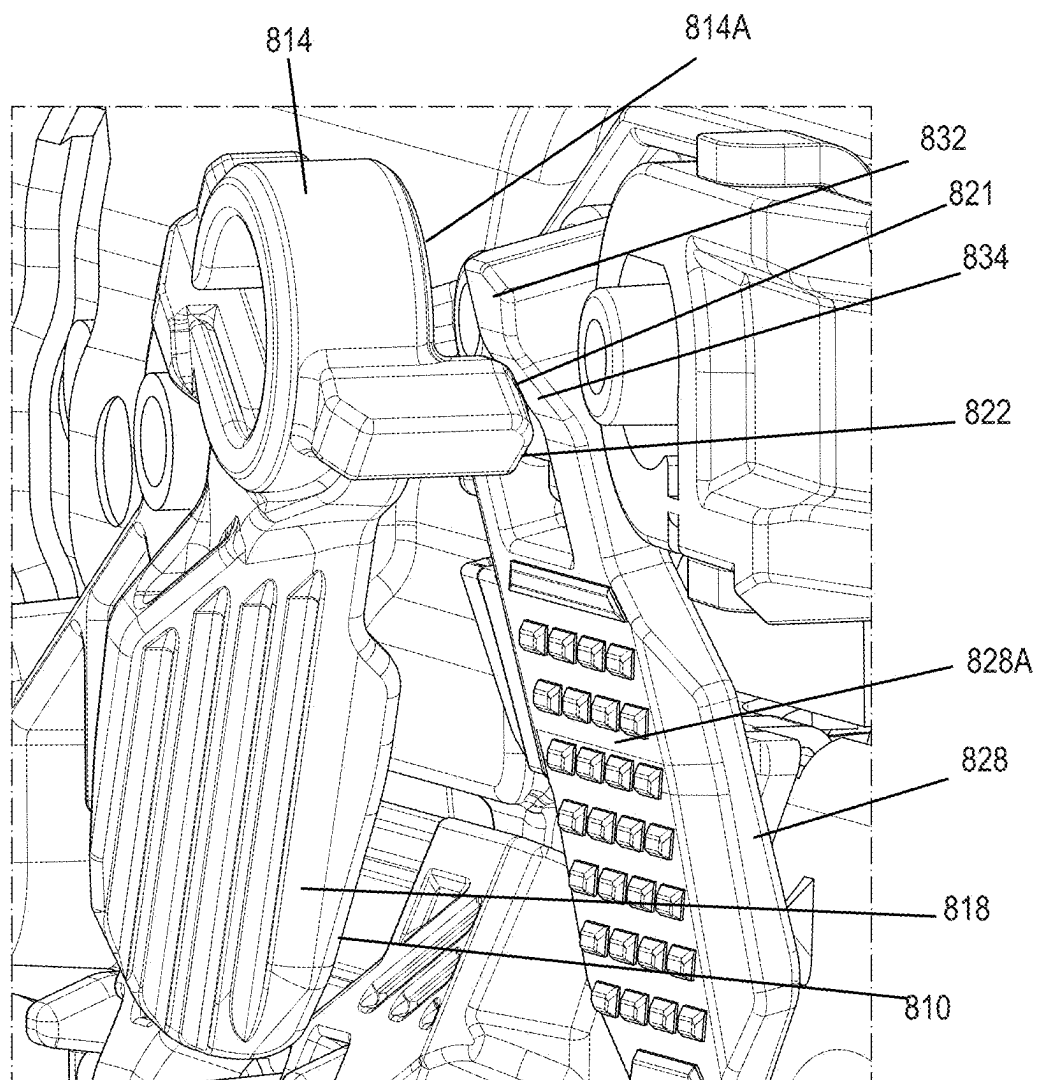


FIG. 71

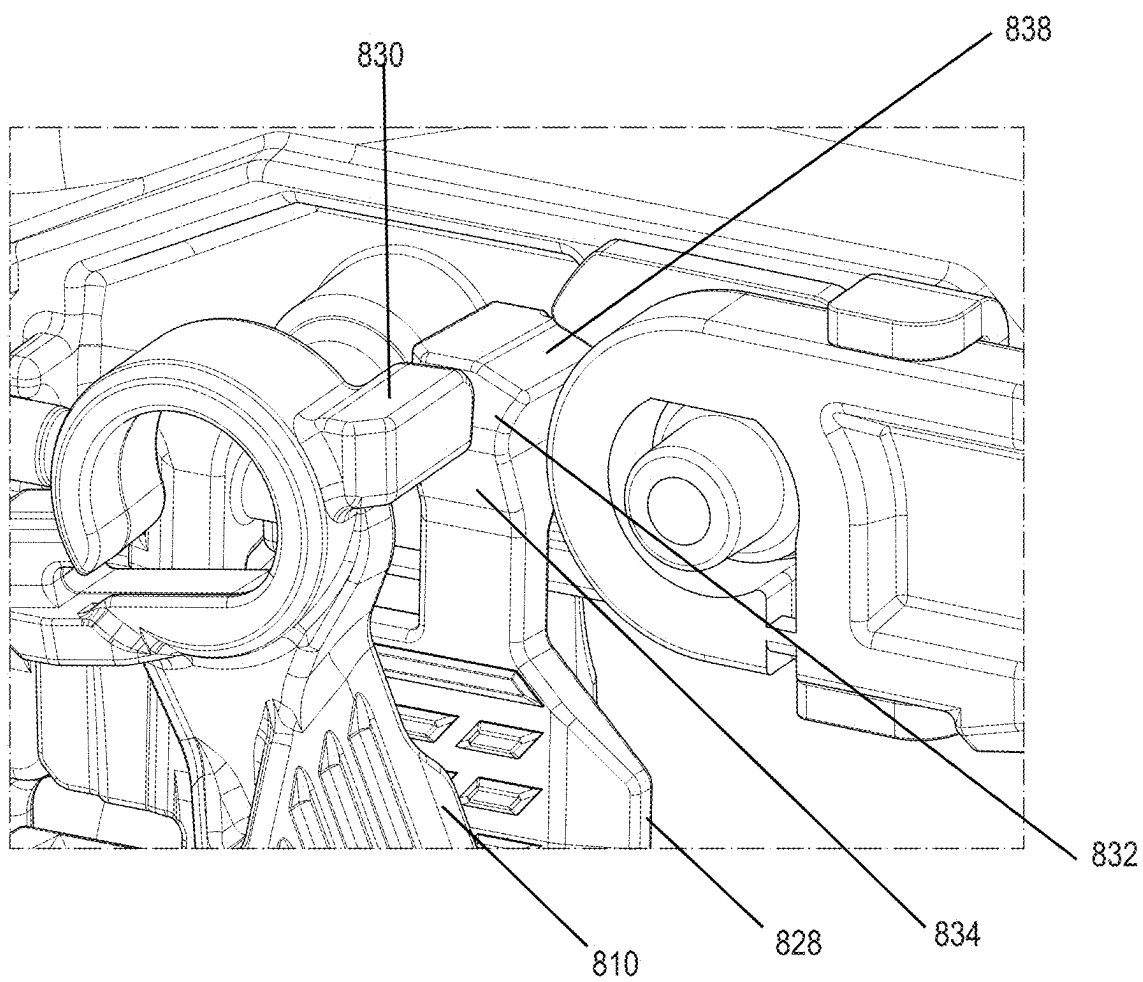


FIG. 72

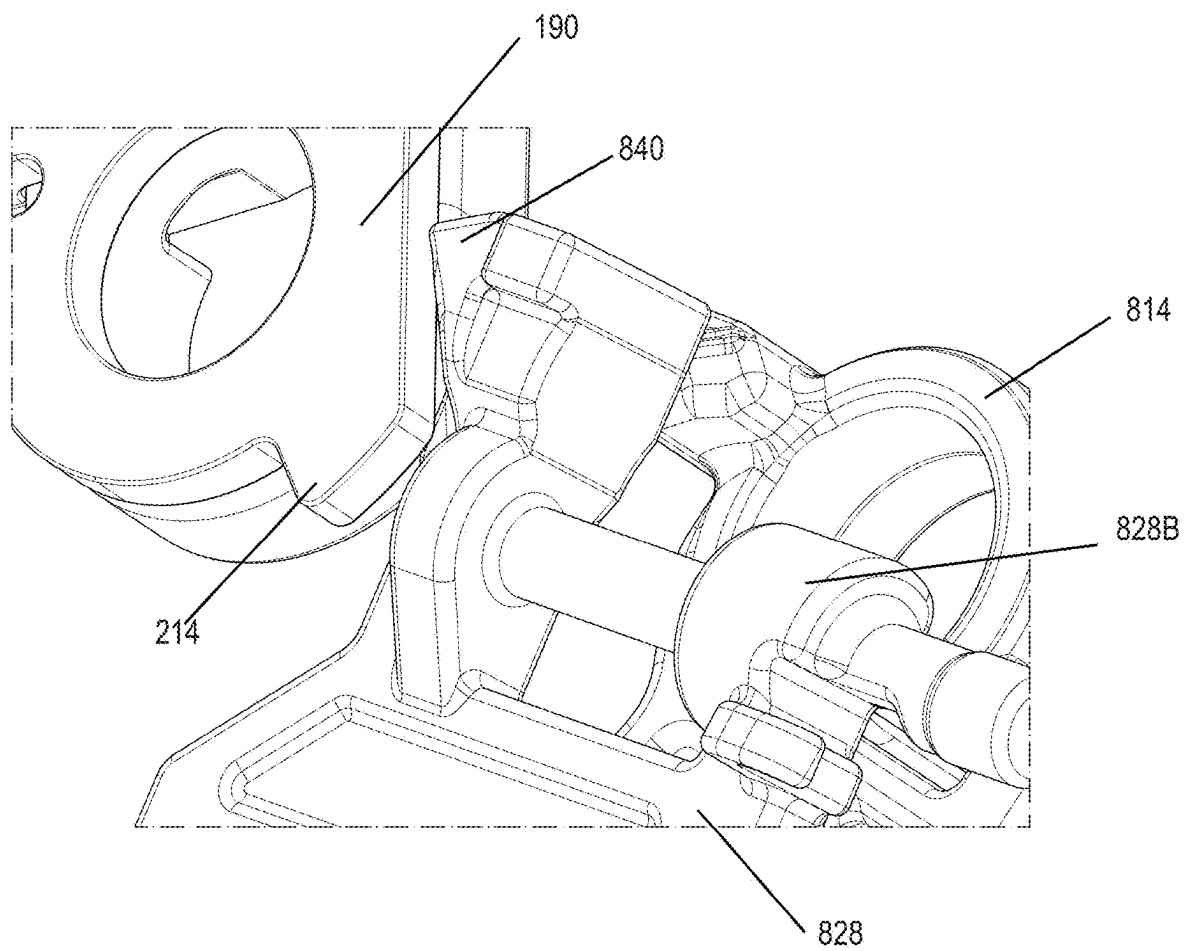


FIG. 73

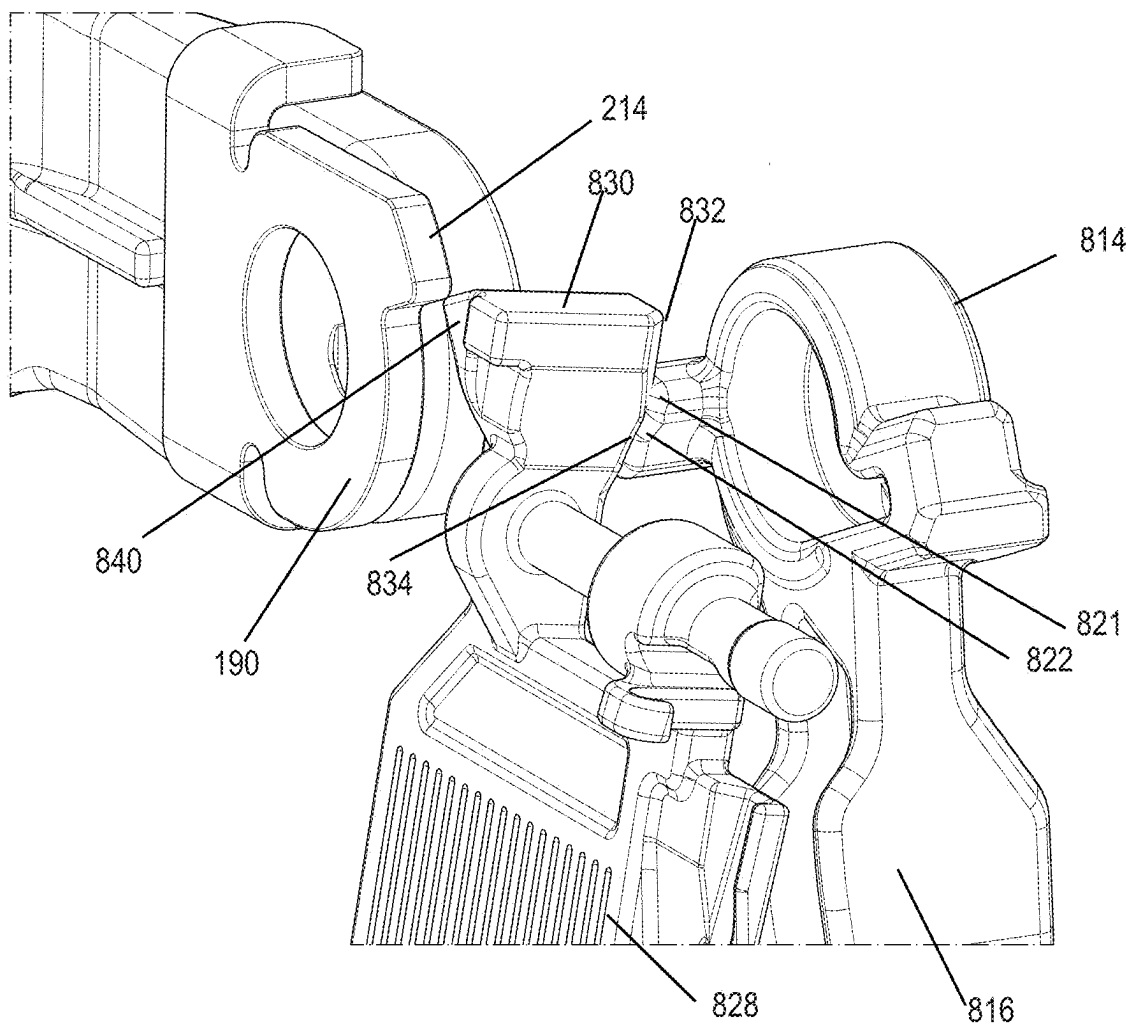


FIG. 74

ROTATING LOCK LEVER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/116,806 filed Mar. 2, 2023 which claims priority to U.S. Provisional Application No. 63/315,971 filed Mar. 2, 2022, the disclosures of which are incorporated by reference herein in their entirety.

BACKGROUND OF THE DISCLOSURE

[0002] In many countries (e.g., Germany, Poland and the United States) law enforcement officers often carry a handgun in an openly visible manner while on duty. The handgun is typically carried in a holster designed to protect the handgun and hold it securely. The holster may include a belt loop that allows it to be hung from a wide belt, for example, a police officer's service belt. The holster may also include a locking mechanism to prevent an assailant from drawing the holster user's weapon and prevent the handgun from inadvertently falling out of the holster, for example, when the holster user is running, climbing over a fence, etc. It is important that the retention mechanisms are secure in such challenging situation.

[0003] In addition to having the handgun be secure in the holster in challenging situations, ease of withdrawal when urgently needed by the user is an important feature. That is, easy actuation of release mechanisms in stressful situations.

[0004] A holster that securely retains a handgun and that provides improvement in ease of use when the handgun is urgently needed would be welcomed by the industry. Accessories for such a holster would also be welcomed by the industry.

[0005] To have holsters be commercially viable, they need to be priced to meet expectations of the consuming populace. So any improvement of assembly and cost efficiencies by minimizing the number of parts and minimizing the number of assembly steps would also be welcomed by the industry.

SUMMARY

[0006] In various embodiments of the disclosure, a holster includes a holster body configured to receive a handgun, the holster body comprising, a handgun receiving and withdrawal axis; a user proximal wall portion; a cavity; a rearward opening of the cavity, a first user actuatable handgun retention mechanism located at the user proximal wall portion for selectively preventing the handgun, when in the cavity, from being withdrawn from the cavity, the first actuatable handgun retention mechanism comprising a trigger guard blocking portion movable from a trigger guard blocking position to a non-blocking position; a second user actuatable handgun retention mechanism, the second user actuatable handgun retention mechanism comprising a housing and a hood assembly, the hood assembly comprising: a shroud that is pivotally supported by the holster body so that the shroud selectively pivots between a handgun retaining position in which the shroud extends across a portion of the rearward opening of the cavity of the holster body and a release position in which the shroud is displaced from the portion of the rearward opening so a withdrawal path of the handgun is unobstructed by the shroud; a bracket comprising a lug portion; and a thumb button, the thumb button com-

prising a thumb receiving tab configured for receiving a user's thumb and a pivot portion pivotally attached to the holster and comprising a protrusion portion comprising an angled locking surface configured to engage the lug portion of the bracket when the shroud is in the handgun retaining position, wherein the pivot portion of the thumb button is configured to pivot the thumb button about an axis parallel to the handgun receiving and withdrawal axis between an undepressed position and a depressed position depressed inwardly towards the user proximal wall portion, the protrusion portion further comprising an outwardly facing blocking surface and a ramp surface sloped downwardly and inwardly from the blocking surface; a rotating lock lever assembly comprising: a housing coupled to the user proximal wall portion and comprising a fastener boss extending from the housing; a pivot portion pivotally mounted to the fastener boss and comprising an axis of rotation transverse to the handgun receiving and withdrawal axis, a surface of the pivot portion facing inwardly towards the user proximal wall portion and defining a first plane; an elongate member extending from the pivot portion of the rotating lock lever assembly, the elongate member comprising a lever button portion; and a blocking projection extending from the pivot portion of the rotating lock lever assembly inwardly towards the user proximal wall portion beyond the first plane, the blocking projection comprising an engagement surface, wherein the engagement surface defines a second plane spaced inwardly towards the user proximal wall portion from the first plane, the blocking projection further comprising a ramp surface sloping upwardly and outwardly from the engagement surface, wherein the rotating lock lever assembly is configured for movement of the rotating lock lever assembly between a protrusion portion blocking position, an intermediate handshake position, and a protrusion portion non-blocking position, wherein the engagement surface of the blocking projection is configured to engage with the blocking surface of the protrusion portion when in the protrusion portion blocking position to block rotation of the protrusion portion of the thumb button when in the blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism to be in the retaining position and the shroud to be in the retaining position, and wherein the blocking projection is configured to allow rotation of the protrusion portion of the thumb button when in the non-blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism to be in the trigger guard non-blocking position and the shroud to be in the release position; wherein the elongate member is configured to block user access to the thumb receiving tab of the thumb button by covering at least part of the thumb button when the rotating lock lever assembly is in the protrusion portion blocking position; and wherein the elongate member is configured to be pivoted by actuation of the lever button portion by the user about the pivot portion of the thumb button such that the thumb receiving tab is accessible by the thumb of a user extending forwardly in the direction of the handgun receiving and withdrawal axis; wherein the rotating lock lever assembly further comprises a coil spring on the fastener boss, and wherein the pivot portion of the rotating lock lever is pivotally mounted to the fastener boss such that the coil spring biases the rotating lock lever assembly to the protrusion blocking position; wherein the elongate member comprising the lever button portion is pivotable about the

pivot portion of the rotating lock lever assembly by a user's thumb moving forwardly and parallel to the handgun receiving and withdrawal axis engaging the lever button portion of the elongate member, wherein the elongate member rotates from a position blocking at least a portion of the thumb receiving tab of the thumb button about a first angular range of rotary motion to a position not blocking the thumb receiving tab of the thumb button and allowing the user's forwardly extending thumb to be positioned over the thumb receiving tab of the thumb button and to be able to press the thumb button inwardly; and wherein the ramp surface of the blocking projection is configured to engage with the ramp surface of the protrusion portion when the rotating lock lever assembly is in the intermediate handshake position, wherein the ramp surface of the protrusion portion is pivoted outwardly away from the user proximal wall portion beyond the first plane defined by the inwardly facing surface of the pivot portion of the rotating lock lever assembly when the thumb button is depressed inwardly; wherein when the rotating lock lever assembly is moved to the protrusion portion non-blocking position, the elongate member is not able to return to the protrusion portion blocking position until the thumb button pivots back to the undepressed position, such that until the thumb button pivots back to the undepressed position, the ramp surface of the protrusion portion blocks the ramp surface of the blocking projection of the rotating lock lever assembly, and therefore blocks the return of the rotating lock lever assembly to the blocking position.

[0007] In embodiments, the rotating lock lever assembly housing further comprises a fastener extending through the fastener boss attaching the housing to the holster body.

[0008] In embodiments, the coil spring is configured to return the rotating lock lever assembly to the protrusion blocking position when force is removed by the user's thumb on the lever button portion, and wherein the thumb button allows the return of the rotating lock lever assembly to the protrusion blocking position when the thumb button is in the undepressed position.

[0009] In embodiments, the engagement of the ramp surface of the protrusion portion and the ramp surface of the blocking projection of the rotating lock lever assembly assists the elongate member in moving towards the non-blocking position when the thumb button is depressed by the user.

[0010] In embodiments, the shroud includes a U-shaped member.

[0011] In embodiments, a hood spring biases the U-shaped member to pivot toward the release position.

[0012] In embodiments, the hood spring biases the lug portion of the hood assembly against the protrusion portion of the thumb button while the U-shaped member is in the handgun retaining position.

[0013] In embodiments, the angled locking surface of the protrusion portion is configured for engagement with the lug portion when the hood assembly is closed, wherein rotation of the thumb button through a first range of rotary motion causes the angled locking surface of the protrusion portion to disengage from the lug portion of the hood assembly, wherein the rotating lock lever assembly is configured to block the thumb button from rotation of the thumb button through the first range of rotary motion when the rotating lock lever assembly is in the protrusion portion blocking

position, wherein motion of the hood assembly is blocked when the rotating lock lever assembly is in the protrusion portion blocking position.

[0014] In embodiments, in order for the hood assembly to close, the thumb button is configured to rotate to allow the lug portion of the hood assembly to travel past the back of the thumb button, wherein when the rotating lock lever assembly is at rest in the blocked position, it blocks all motion of the thumb button and does not allow the lug portion to rotate past the backside of the thumb button causing the need for the rotating lock lever assembly to move to the non-blocking position prior to closing the hood assembly.

[0015] In embodiments, the protrusion portion further includes an engagement surface on a rear surface thereof facing the holster body; wherein moving the hood assembly from an open position to a closed position causes the lug portion to engage with the engagement surface and be blocked by the engagement surface until the user rotates the thumb button inwardly towards the proximal wall portion to unblock the lug portion from the engagement surface; wherein, once the lug portion has moved past the engagement surface, lug portion rests on the angled locking surface corresponding to the hood assembly being in the retaining position.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.

[0017] FIG. 1 is a perspective view showing a handgun and a holster in accordance with the detailed description.

[0018] FIG. 2 is a perspective view showing a handgun and a holster in accordance with the detailed description.

[0019] FIG. 3A is an exploded perspective view showing a holster body and a lever.

[0020] FIG. 3B is an exploded perspective view showing a belt receiving member, a holster body and a lever.

[0021] FIG. 3C is an exploded perspective view showing a belt receiving member, a holster body and a lever.

[0022] FIG. 4 is an exploded perspective view showing a holster body and a lever.

[0023] FIG. 5 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 5.

[0024] FIG. 6 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 6.

[0025] FIG. 7 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 7.

[0026] FIGS. 8A and 8B are plan views of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIGS. 8A and 8B.

[0027] FIG. 9 is a cross-sectioned perspective view of an assembly including a holster and a handgun having a trigger

guard. The holster and the handgun are cross-sectioned for purposes of illustration in FIG. 9.

[0028] FIG. 10 is a plan view of an assembly including a holster and a handgun having a trigger guard. The holster and the handgun are cross-sectioned for purposes of illustration in FIG. 10.

[0029] FIGS. 11A-11C are stylized cross-sectional views each showing a pin and a pin receiving portion.

[0030] FIG. 12 is a stylized cross-sectional view showing a pin and a pin receiving portion.

[0031] FIG. 13 is a perspective view showing a pin and a pin receiving portion.

[0032] FIG. 14 is a perspective view showing a handgun and a holster in accordance with the detailed description.

[0033] FIG. 15 is an exploded perspective view showing a holster body and two active handgun retention mechanisms operated by a single thumb button.

[0034] FIG. 16 is a perspective view showing retention mechanisms seen in the exploded perspective view of FIG. 15.

[0035] FIG. 17 is an exploded perspective view further illustrating some of the elements seen in the sub-assembly of FIG. 16.

[0036] FIG. 18A is a perspective view of components of handgun retention mechanisms.

[0037] FIG. 18B is an exploded view of the components of FIG. 18A.

[0038] FIG. 19 is a further exploded view of components of the handgun retention mechanisms including a housing for securing the pivoting thumb button.

[0039] FIG. 20 is a further exploded view of components of the handgun retention mechanisms taken from the side opposite that of FIG. 19.

[0040] FIG. 21 is an exploded view of an embodiment of a holster with a thumb button operating a single retention mechanism and a thumb button guard.

[0041] FIG. 22 is a plan view of a holster with a holstered handgun and a user's hand about the grasp the handgun.

[0042] FIG. 23 is a plan view of a holster with a holstered handgun and a user's hand engaging the handgun.

[0043] FIG. 24 is a plan view of a holster with a holstered handgun and a user's hand engaging the handgun.

[0044] FIG. 25 is a perspective view showing a holster including a hood spring, a lever spring, and a tab spring. For purposes of illustration and explanation, the hood spring, the lever spring, and the tab spring are each illustrated a second time at a location spaced away from the holster.

[0045] FIG. 26 is a perspective view showing an assembly including a lever, a lever spring, a tab, and a tab spring. For purposes of illustration and explanation, the lever spring and the tab spring are each illustrated a second time at a location spaced away from the assembly.

[0046] FIG. 27 is a perspective view of an assembly including a lever, a lever spring, a tab, and a tab spring. For purposes of illustration and explanation, the lever spring and the tab spring are each illustrated a second time at a location spaced away from the assembly.

[0047] FIGS. 28A and 28B are perspective views showing an assembly including a hood assembly and a thumb receiving tab.

[0048] FIGS. 29A and 28B are perspective views showing an assembly including a hood assembly and a thumb receiving tab.

[0049] FIG. 30A is a plan view showing a hood assembly part having a lug portion and a thumb receiving tab having a protrusion portion that selectively engages the lug portion of the hood assembly part. FIG. 31A is a perspective view of the assembly shown in FIG. 30A. In the embodiments of FIGS. 30A and 31A, the hood assembly part is in a handgun retaining position.

[0050] FIG. 30B is a plan view showing a hood assembly part having a lug portion and a thumb receiving tab having a protrusion portion that selectively engages the lug portion of the hood assembly part. FIG. 31B is a perspective view of the assembly shown in FIG. 30B. In the embodiments of FIGS. 30B and 31B, the hood assembly part is in a handgun releasing position.

[0051] FIG. 32A is a perspective view showing a holster in accordance with the detailed description.

[0052] FIG. 32B is a perspective view showing selected parts from the holster shown in FIG. 32A. The parts shown in FIG. 32B include a U-shaped shroud, a bracket and a finger receiving tab.

[0053] FIG. 32C is a perspective view further illustrating the parts shown in FIG. 32B. In FIG. 32C, the U-shaped shroud, the bracket and the finger receiving tab are shown from a different viewpoint.

[0054] FIGS. 33A and 33B are two perspective views showing a holster including a U-shaped shroud. In the embodiment of FIG. 33A, the U-shaped shroud is in a handgun retaining position. In the embodiment of FIG. 33B, the U-shaped shroud is in a release position.

[0055] FIGS. 34A and 34B are perspective views of a belt engaging member.

[0056] FIG. 35A is a front view of the belt engaging member shown in FIGS. 34A and 34B.

[0057] FIG. 35B is a right side view of the belt engaging member shown in FIGS. 34A and 34B.

[0058] FIG. 35C is a top view of the belt engaging member shown in FIGS. 34A and 34B.

[0059] FIG. 35D is a rear view of the belt engaging member shown in FIGS. 34A and 34B.

[0060] FIG. 35E is a left side view of the belt engaging member shown in FIGS. 34A and 34B.

[0061] FIG. 35F is a bottom view of the belt engaging member shown in FIGS. 34A and 34B.

[0062] FIGS. 36A and 36B are two perspective views showing a holster having a first handgun retention mechanism including a pivotable lever and a second handgun retention mechanism including a U-shaped shroud. In the embodiment of FIGS. 36A and 36B, the U-shaped shroud is in a handgun retaining position.

[0063] FIG. 37A is a front view of the holster shown in FIGS. 36A and 36B.

[0064] FIG. 37B is a right side view of the holster shown in FIGS. 36A and 36B.

[0065] FIG. 37C is a top view of the holster shown in FIGS. 36A and 36B.

[0066] FIG. 37D is a rear view of the holster shown in FIGS. 36A and 36B.

[0067] FIG. 37E is a left side view of the holster shown in FIGS. 36A and 36B.

[0068] FIG. 37F is a bottom view of the holster shown in FIGS. 36A and 36B.

[0069] FIGS. 38A and 38B are two perspective views showing a holster having a handgun retention mechanism including a pivotable lever.

[0070] FIG. 39A is a front view of the holster shown in FIGS. 38A and 38B.

[0071] FIG. 39B is a right side view of the holster shown in FIGS. 38A and 38B.

[0072] FIG. 39C is a top view of the holster shown in FIGS. 38A and 38B.

[0073] FIG. 39D is a rear view of the holster shown in FIGS. 38A and 38B.

[0074] FIG. 39E is a left side view of the holster shown in FIGS. 38A and 38B.

[0075] FIG. 39F is a bottom view of the holster shown in FIGS. 38A and 38B.

[0076] FIG. 40 is an exploded perspective view showing a belt receiving member, a holster body and a lever.

[0077] FIG. 41 is an exploded perspective view showing a holster body and a lever.

[0078] FIG. 42 is an exploded perspective view showing a belt receiving member, a holster body and two locking mechanisms.

[0079] FIG. 43 is an exploded perspective view showing a holster body and two locking mechanisms.

[0080] FIG. 44 is a perspective view showing the belt receiving member, the holster body and the lever shown in FIG. 40.

[0081] FIG. 45 is a top view showing the belt receiving member, the holster body and the lever shown in FIG. 40.

[0082] FIG. 46 is a perspective view showing the belt receiving member, the holster body and the locking mechanisms shown in FIG. 42.

[0083] FIG. 47 is a top view showing the belt receiving member, the holster body and the locking mechanisms shown in FIG. 42.

[0084] FIG. 48A is a view of an exemplary embodiment of a level-2 holster with a RDS dust cover from an upward end of the holster.

[0085] FIG. 48B is a view of the exemplary embodiment of the holster from FIG. 48A from the downward end of the holster.

[0086] FIG. 48C is a view from the user distal side of the exemplary embodiment of the holster from FIG. 48A.

[0087] FIG. 48D is a view from the user proximal side of the exemplary embodiment of the holster from FIG. 48A.

[0088] FIG. 48E is a view from the user distal side of the exemplary embodiment of the holster from FIG. 48A with the dust cover in the first, open position.

[0089] FIG. 48F is a view from the user proximal side of the exemplary embodiment of the holster from FIG. 48A with the dust cover in the second, closed position.

[0090] FIG. 49 is a view of the exemplary dust cover of FIG. 48A.

[0091] FIG. 50A is a close up view of an exemplary configuration of the followers and two-position detent.

[0092] FIG. 50B is a close up view of the exemplary configuration of the followers and two-position detent of FIG. 50A.

[0093] FIG. 51A is a view of an exemplary embodiment of a level-3 holster with a RDS dust cover from an upward end of the holster

[0094] FIG. 51B is a view of the exemplary embodiment of the holster from FIG. 51A from the downward end of the holster.

[0095] FIG. 51C is a perspective view of the exemplary embodiment of the holster from FIG. 51A from the downward end of the holster.

[0096] FIG. 51D is a view from the user proximal side of the exemplary embodiment of the holster from FIG. 51A.

[0097] FIG. 51E is a view from the user distal side of the exemplary embodiment of the holster from FIG. 51A.

[0098] FIG. 51F is a view of the shroud from the exemplary embodiment of FIG. 51A.

[0099] FIG. 51G is a view from the user proximal side of the exemplary embodiment of the holster from FIG. 51A with the shroud in the release position and the dust cover in the second, closed position

[0100] FIG. 51H is a view from the user proximal side of the exemplary embodiment of the holster from FIG. 51A with the shroud in the release position and the dust cover in the first, open position.

[0101] FIG. 52A is an interior side view of an exemplary embodiment of a lockout button assembly and a thumb receiving tab with the lockout button assembly in a protrusion portion blocking position.

[0102] FIG. 52B is an interior side view of the exemplary embodiment of FIG. 52A with the lockout button assembly in a protrusion portion non-blocking position.

[0103] FIG. 52C is an interior side view similar to FIG. 52A with the housing removed.

[0104] FIG. 52D is an interior side view similar to FIG. 52B with the housing removed.

[0105] FIG. 53A is a front view of the exemplary embodiment of FIG. 52A with the lockout button assembly in a protrusion portion non-blocking position.

[0106] FIG. 53B is a front view similar to FIG. 53A with the housing removed.

[0107] FIG. 54A is an exterior side view of the exemplary embodiment of FIG. 52A with the lockout button assembly in a protrusion portion non-blocking position.

[0108] FIG. 54B is an exterior side view similar to FIG. 54A with the housing removed.

[0109] FIG. 55 is a user proximal side of an exemplary embodiment of a holster with the exemplary embodiment of the lockout button assembly of FIG. 52A with the lockout button assembly in a protrusion portion blocking position.

[0110] FIG. 56 is a user proximal side of an exemplary embodiment of a holster with the exemplary embodiment of the lockout button assembly of FIG. 52A with the lockout button assembly in a protrusion portion non-blocking position and a handgun in a withdrawn position.

[0111] FIG. 57 is an exploded perspective view of an exemplary embodiment of a holster with the exemplary embodiment of the lockout button assembly.

[0112] FIG. 58A is a cross-sectional view of an exemplary embodiment of the lockout button assembly in a blocking position.

[0113] FIG. 58B is a cross-sectional view of an exemplary embodiment of the lockout button assembly in a non-blocking position.

[0114] FIG. 59 is an exterior side view of a handgun and a holster having a rotating lock lever assembly in a blocking position in accordance with an exemplary embodiment of the disclosure;

[0115] FIG. 60 is a perspective view of the handgun and holster of FIG. 59;

[0116] FIG. 61 shows the handgun and holster of FIG. 60 with rotating lock lever assembly housing removed;

[0117] FIG. 62 is an exterior side view of the handgun and holster of FIG. 59 showing the rotating lock lever assembly in a non-blocking position with rotating lock lever assembly housing removed;

[0118] FIG. 63 is a perspective view of the handgun and holster of FIG. 62 showing the rotating lock lever assembly in a non-blocking position with rotating lock lever assembly housing removed;

[0119] FIG. 64 is a perspective view of the handgun and holster of FIG. 62 showing the rotating lock lever assembly in a non-blocking position with rotating lock lever assembly housing removed.

[0120] FIG. 65 is a perspective, exploded view of the handgun and holster of FIG. 59;

[0121] FIG. 66 is a perspective, exploded view of the rotating lock lever assembly and thumb button of FIG. 65;

[0122] FIG. 67 is a perspective, exploded view of the rotating lock lever assembly and

[0123] thumb button of FIG. 65;

[0124] FIG. 68 is a perspective, exploded view of the rotating lock lever assembly and thumb button of FIG. 65;

[0125] FIG. 69 is a rear view of the handgun and holster of FIG. 59 showing the rotating lock lever assembly in a blocking position;

[0126] FIG. 70 is a detail view of the rotating lock lever assembly in a blocking position;

[0127] FIG. 71 is a detail view of the rotating lock lever assembly when the rotating lock lever assembly is moved forward from the blocking position but before the non-blocking position in a handshake position;

[0128] FIG. 72 is a detail view of the rotating lock lever assembly in a blocking position;

[0129] FIG. 73 is a detail view of the rotating lock lever assembly in a blocking position; and

[0130] FIG. 74 is a detail view of the rotating lock lever assembly in the handshake position.

[0131] While the embodiments of the disclosure are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

[0132] Referring to FIGS. 1-4, perspective views of a handgun 80 and a holster 100 are shown. FIG. 1 shows the handgun 80 withdrawn from the holster 100 and FIG. 2 shows the handgun inserted into the holster 100. The handgun being conventional and having a forward end 82, a handgun body 84, a grip 86 at a rearward end 87 of the handgun 80, a trigger guard 90, a back strap 92 with a backstrap recess 94, a slide 98 positioned above the handgun body, and a rail 99 positioned below the slide. In embodiments, the holster 100 for receiving and withdrawing the handgun having a trigger guard 90 comprises a polymer holster body 102, an elongate polymer pivoting lever 104 that is part of a first handgun retention mechanism 106 that is thumb actuated. The pivoting lever has an actuation tab 107 configured as by a thumb receiving button that is disposed in an opening 105 in the holster body, actuation of the thumb receiving button moves a trigger guard block

portion 128 of the lever between retention and release positions. The retention mechanism 106 has two distinct separated pivoting connections 109 each formed from a pair of cooperating connector pivot portions 110, one of each pair unitary with the holster body and the other of each pair unitary with the lever. In embodiments, the pivot portions 110 of the lever 104 and pivot portions 110 of the holster body 102 are coupled using a simple assembly process with no separate hinge pin.

[0133] The holster body has three unitary bosses 111 on a proximal wall portion 113 for attachment to a plate portion of a user attachment means, such as a belt engaging member. See FIGS. 12 and 13. The holster having a proximal side toward and holster body having a proximal

[0134] Continuing to referring to FIGS. 1-4, in embodiments, the holster body 102 has a plurality of holster wall portions defining a cavity 108 extending along a handgun receiving and withdrawal axis 110H. In embodiments, the retention mechanism 106 comprises a lever 104 pivotally supported by the holster body 102 and retained by holster attachment plate portion 165 or by other means. The lever 104 may comprise an elongate central portion 124 integrally formed with a first forward pivot portion 112 and the holster body 102 may comprises a second forward pivot portion 114 integrally formed with one of the holster wall portions. The first forward pivot portion 112 may mate with the second forward pivot portion 114 to form a forward pivoting connection 109. In embodiments, the lever 104 further comprises a first rearward pivot portion 116 integrally formed with the elongate central portion 124 of the lever 104 and the holster body 102 comprises a second rearward pivot portion 118 integrally formed with one of the holster wall portions. The first rearward pivot portion 116 may mate with the second rearward pivot portion 118 to form a rearward pivoting connection 109. In embodiments all pivot portions of the retention mechanism are axially aligned and co-axial when assembled.

[0135] Continuing to referring to FIGS. 1-4, in embodiments, is a retention mechanism having a lever that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is contacting the a thumb button portion of the lever. In embodiments, the retaining mechanism comprises a lever having a thumb button portion, and the lever pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the retaining mechanism comprises a thumb-actuated button that pivots about a first axis and a lever that pivots about a second axis that is parallel to the first axis.

[0136] Referring to FIGS. 5-8B, in embodiments, the lever 104 of the retention mechanism 106 has an elongate central portion 124, a thumb receiving portion 107, configured as a thumb button, and a blocking portion 128. The thumb receiving portion of the lever comprising a first arm 130 extending away from the elongate central portion 124 in a first direction, the elongate central portion 124 and the first arm 130 cooperating to form a first L-shaped portion 132. The blocking portion 128 of the lever 104 comprises a second arm 134 extending away from the elongate central portion 124 in a second direction opposite the first direction, the elongate central portion 124 and the second arm 134 cooperating to form a second L-shaped portion 136.

[0137] Referring to FIGS. 6 and 7, the lever 104 is elongate in the direction of the axis X and has a lever length

L1. The pivot portions **112**, **116** have a separation distance D1 and have a pivoting connection length L2. The pivoting connection length L2 to the overall lever length L1 is in embodiments greater than 0.70 or 70%; in embodiments, greater than 0.60 or 60%; in embodiments greater than 0.80 or 80%.

[0138] Referring to FIGS. 1 and 2, a forward or handgun insertion direction Z and a rearward or handgun withdrawal direction -Z are illustrated using arrows labeled “Z” and “-Z,” respectively. An upward direction Y and a downward direction -Y are illustrated using arrows labeled “Y” and “-Y,” respectively. A direction X extending away from the user’s body and the user attachment side of the holster is illustrated using an arrow labeled “X.” A direction -X extending toward the user’s body and the user attachment side of the holster is illustrated using an arrow labeled “-X.” The directions illustrated using these arrows may be conceptualized, by way of example and not limitation, from the point of view of a user who is wearing a holster hung from a service belt and inserting a handgun into the holster. The directions illustrated using these arrows may also be conceptualized, by way of example and not limitation, from the point of view of a user holding a handgun in a normal firing position and viewing the gunsights of the handgun. The directions illustrated using these arrows may be applied to the apparatus shown and discussed throughout this application. In embodiments, the Z direction and the -Z direction are both generally orthogonal to the XY plane defined by the X direction and the Y direction. In embodiments, the X direction and the -X direction are both generally orthogonal to the ZY plane defined by the Z direction and the -Z direction. In embodiments, the Y direction and the -Y direction are both generally orthogonal to the ZX plane defined by the Z direction and the -Z direction. Various direction-indicating terms are used herein as a convenient way to discuss the objects shown in the figures. It will be appreciated that many direction indicating terms are related to the instant orientation of the object being described. It will also be appreciated that the objects described herein may assume various orientations without deviating from the spirit and scope of this detailed description. Accordingly, direction-indicating terms such as “upwardly,” “downwardly,” “forwardly,” “rearwardly,” etc. should not be interpreted to limit the scope of the invention recited in the attached claims.

[0139] Referring to FIG. 9, a cross-sectioned perspective view of a holster body **102** is shown. In the example embodiment of FIG. 9, the holster body **102** is formed from two polymers, one overmolded onto the other, providing an interior layer **138** and an exterior layer **140**. In embodiments, the interior layer **138** is softer than the exterior layer **140** minimizing any wear, damage or markings on the exterior surface of the handgun that is received in the cavity **108** defined by the holster. In embodiments, the interior layer **138** and the exterior layer **140** are combined using an assembly process. In embodiments, the interior layer **138** comprises a thermoplastic elastomer (TPE) and the exterior layer **140** a thermoplastic material. In embodiments, the interior layer **138** comprises Hytrel® polymer, available from DuPont, and the exterior layer **140** comprises nylon. The exterior layer being harder and stiffer than the interior layer.

[0140] Referring to FIG. 10, a cross-sectional view of a holster body **102** is shown. In embodiments, the holster body **102** is formed to provide a handgun fitting function that can

be adjusted by rotating a tensioning screw **142**. The shape of the forward portion of the handgun can be conceptualized as a four cornered geometric figure, roughly a rectangle, and the shape of the cavity **108** defined by the holster body **102** can be configured to make point contact with the four corners of the figure or rectangle. In embodiments, the shape of the cavity **108** defined by the holster body **102** can be conceptualized as two opposing V-shapes, an upper V-shape **148** and a lower V-shape **146**. The upper V-shape **148** and the lower V-shape **146** are shown using dashed lines in FIG. 10. The two legs of the upper V-shape **148** and the two legs of the lower V-shape **146** may each contact one corner of the figure or rectangle. In embodiments, the lower V-shape **146** defines a slot **144** near the lower end of the lower V-shape **146**. A tensioning screw **142** is positioned to selectively decrease the angle between the two legs of the lower V-shape **146**. As the angle between the two legs of the lower V-shape **146** decreases, the two legs of the lower V-shape **146** apply upwardly directed component forces to two lower corners **150** of the rectangle. The forces applied to the lower corners **150** of the rectangle urge the two upper corners **152** of the rectangle against the two legs of the upper V-shape **148**.

[0141] The components herein may be formed of thermoplastic polymers using an injection molding process.

[0142] Referring to FIGS. 3, 4 and 11A-11C, a retention mechanism **106** for a holster **100** may include a lever pivotally supported by the holster body **102**. The lever **104** may comprise an elongate central portion **124** integrally formed with a first forward pivot portion **112**, and the holster body **102** may comprise a second forward pivot portion **114** integrally formed with a wall portion of the holster body **102**. The first forward pivot portion **112** may be mated with the second forward pivot portion so that the lever **104** is pivotally supported by the holster body **102**. In embodiments, one of the forward pivot portions **112**, **114** is a pin portion **154** and the other of the forward pivot portions **112**, **114** is a C-shaped pin receiving portion **156**. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span greater than 180 degrees; in embodiments greater than 185°; and in embodiments greater than 190°. In embodiments, one of the forward pivot portions **112**, **114** is a pin portion **154** and the other of the forward pivot portions **112**, **114** is a U-shaped pin receiving portion **156**.

[0143] With continuing reference to FIGS. 3, 4 and 11A-11C, the lever **104** may further comprise a first rearward pivot portion **116** integrally formed with the elongate central portion **124** and the holster body **102** may comprise a second rearward pivot portion **118** integrally formed with a wall portion of the holster body **102**. The first rearward pivot portion **116** may be mated with the second rearward pivot portion so that the lever **104** is pivotally supported by the holster body **102**. In embodiments, one of the rearward pivot portions **116**, **118** is a pin portion **154** and the other of the rearward pivot portions **116**, **118** is a C-shaped pin receiving portion **156**. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span greater than 180 degrees; in embodiments greater than 185°; and in embodiments greater than 190°. In embodiments, one of the rear-

ward pivot portions **116**, **118** is a pin portion **154** and the other of the rearward pivot portions **116**, **118** is a U-shaped pin receiving portion **156**.

[0144] Referring to FIGS. 3B and 3C, a holster **100** in accordance with this detailed description may comprise a holster attachment plate portion **165** that is part of or all of a user attachment means for the holster which may be a belt engaging member. The plate portion **165** may retain the lever **104** in a pivoting connection arrangement with the holster body **102**. The lever **104** may be pivotally supported by the holster body **102** and may be moveable between a handgun trigger guard capture position and a handgun trigger guard release position. In embodiments, the plate portion **165** is fixed to the holster body **102** by a plurality of screws **162**. The lever **104**, may be, for example, captured between the belt engaging member **160**, or other user attachment means, and the holster body **102**. In embodiments, the lever **104** can be freely separated from the holster body **102** after the belt engaging member **160** is removed from the holster body **102**. In embodiments, the belt engaging member **160** comprises a first belt loop portion **164** defining a first passageway **166** to receive a belt and a second belt loop portion **168** defining a second passageway **170** for receiving the belt.

[0145] The user attachment means **160** may comprise various holster supporting devices without deviating from the spirit and scope of this detailed description. Examples of holster supporting devices that may be suitable in some applications are disclosed in the following United States patents all of which are hereby incorporated by reference herein: U.S. D653848, U.S. D567707, U.S. D508318, U.S. Pat. Nos. 9,134,093, 8,783,532, 8,517,234, 8,469,245, 8,297,562, 8,251,266, 8,235,263, 7,866,515, 7,320,420, 9,423,210, 9,664,480, 9,841,255, 9,222,751, 8,544,706, 8,215,525, 8,100,304, 7,971,762, 7,922,050, 7,690,541, 6,478,202, 6,189,751, and 5,467,909.

[0146] Referring to FIGS. 14-17, in embodiments, a holster **100** has a first handgun retention mechanism **106'** that engages the trigger guard as previously described with respect to FIGS. 1-13, and a second handgun retention mechanism **174**. A dual actuation member **179** with a thumb button **180** actuates both the first handgun retention mechanism **106'** and the second handgun retention mechanism **174**. In embodiments, each handgun retention mechanism selectively prevents the handgun **80** from being withdrawn from the cavity **108** defined by the wall portions **109** of the holster body **102**.

[0147] The second handgun retention mechanism **106'** comprises a slide retention member configured as a U-shaped shroud **176** that is pivotally supported by the holster body **102** so that the U-shaped shroud **176** selectively pivots between a handgun retaining position in which the U-shaped shroud **176** extends across a portion of a rearward opening **178** of the cavity **108** and a release position in which the U-shaped shroud **176** extends forward of the rearward opening **178** so that a withdrawal path of the handgun **80** is unobstructed. In embodiments, the U-shaped shroud **176** pivots about a shroud pivot axis AC that extends laterally and is perpendicular to the handgun insertion and withdrawal axis AA.

[0148] Referring to FIGS. 15-20, in embodiments, the first handgun retention mechanism **106'** comprises a lever **104'** pivotally supported by the holster body **102**. The lever **104'** may be similarly configured to the lever **106** of FIGS. 1-13.

In embodiments, the lever **104'** has an elongate central portion **124**, a depressible actuation portion **186**, and a trigger guard blocking portion **128**. The depressible actuation portion **186** of the lever **104** comprises a first arm **130** extending away from the elongate central portion **124** in a first direction so that the elongate central portion **124** and the first arm **130** cooperate to form a first L-shape **182**. The trigger guard blocking portion **128** of the lever **104'** comprises a second arm **134** extending away from the elongate central portion **124** in a second direction opposite the first direction so that the elongate central portion **124** and the second arm **134** cooperate to form a second L-shape **184**. In embodiments, the lever **104'** is pivotally supported by the holster body so that the lever **104'** pivots about a lever axis **110** that extends in upward and downward directions as the holster is worn or forward and rearward with respect to the handgun. In embodiments, the lever pivots between a handgun trigger guard blocking or capture position and a handgun trigger guard non-blocking or release position. The trigger guard blocking portion **128** of the lever **104** extends into a trigger guard portion of the holster body **102** when the lever **104'** is disposed in the handgun trigger guard capture position.

[0149] Continuing to referring to FIGS. 15-20, a feature and benefit of embodiments is a retention mechanism having a lever that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is used to rotate the lever. In embodiments, the retaining mechanism comprises a thumb-actuated button that pivots about a first axis and a lever that pivots about a second axis that is parallel to the first axis. In embodiments, the lever pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the thumb-actuated button pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster.

[0150] Referring to FIGS. 16-20, the second handgun retention mechanism is illustrated. The slide blocking member configured as a shroud **176** is part of a hood assembly **188**. Fasteners **197**, **198** pivotally secure the U-shaped shroud, second handgun retention mechanism housing **201**, bracket **190** with slot **144**, to the holster body. Fastener **199** secures the opposite side of the shroud to the holster body. The second handgun retention mechanism **174** utilizes the retention mechanism housing **201** to secure components of the retention mechanism in place. The housing has two cooperating components, and inward member **202** and an outward member **204** that may be arranged in a clamshell-like fashion that captures the leaf spring **194** and a pivoting shaft **206** of the dual actuation member **179** within the housing **201**. In embodiments, the holster includes a spring **196** that biases the U-shaped shroud **176** to pivot toward the release position. In embodiments, the holster **100** includes the leaf spring **194** that is received in the slot **144** of the bracket **190** when the U-shaped shroud **176** is in the handgun retaining position. The base **195** of the leaf spring **194** is fixed with respect to the holster body and the tip **196** of the leaf spring **194** can flex inwardly and outwardly. The bracket **190** is attached to the U-shaped shroud **176** such that as the leaf spring **194** keeps the bracket from rotating forwardly it also keeps the shroud from rotating forwardly. In embodiments, rotation of the thumb receiving tab or button **180** through a first range of rotary motion causes deflection of the leaf spring **194** so that a distal portion of the leaf spring **194**

moves outside of the slot 144 allowing the bracket 190 and also the shroud 176 to rotate forwardly. In embodiments, rotation of the thumb button 180 through a second range of rotary motion causes the thumb button 180 to engage the depressible actuation portion 186 of the lever 104 causing at least part of the trigger guard blocking portion 128 of the lever to be withdrawn from the trigger guard receiving portion of the holster body 102. In this embodiment, the thumb receiving portion of the lever 104 of the embodiment of FIGS. 1-4 has been replaced with an actuation portion or depressible portion that is depressed by the inner side of the thumb button 180.

[0151] The first and second active handgun retention mechanisms may be arranged such that the second and first mechanisms can actuate sequentially or simultaneously, in embodiments.

[0152] Referring to FIG. 21, another embodiment of a holster 300 comprising a holster body 306 similar to that of FIGS. 14 and 15 with a top or forward wall portion that mostly covers the slide of a handgun holstered therein, a pair of bosses 310 that can receive components of a second active handgun retention mechanism or, as illustrated in FIG. 21, a removable thumb button guard 312 secured to the bosses with fasteners 316. The lever 104 may be configured as discussed previously with respect to FIGS. 1-9 and has a thumb button 180. The plate portion 165 of a belt engaging member may capture the pivoting lever onto the holster body.

[0153] Referring to FIGS. 22-24, the sequence of gripping and drawing a handgun 80 from a holster 100 having an optimally positioned thumb button 107 is illustrated. The handgrip 330 of the handgun 80 has a backstrap 322 region with a backstrap recess 324 that receives the user's hand 340 initially by the webbing 342 of the hand contacting the recess 324. The user will then wrap his middle, ring, and pinky fingers 350, 351, 352 around the grip as shown in FIG. 23 and will have his forefinger 353 in a longitudinal recess for said finger on the side of the holster opposite the thumb button. The thumb 354 will naturally be positioned at the thumb button 107 such that a normal grasping action of the grip, consistent with the "master grip" will depress the thumb button 107 and actuate the retention mechanism 104. Referring to FIGS. 22-25, 36A, 36B, 38A, 38B and 40-43, the master grip and ergonomics are further facilitated in embodiments where the thumb button 180 is positioned with no holster body directly rearward of the rearward margin of the thumb button 180, as well as no holster body portions below the thumb button 180. In use, as the handgun 80 is gripped prior to actuation of the thumb button 180, the user's thumb is in the natural gripping position at the surface of the thumb button 180, the thumb button 180 is depressed, coincident with grasping the handgun 80, and as the handgun 80 is withdrawn, the user's thumb slides off of the thumb button 180 into immediate contact with the handgun 80. That is, there is no engagement with the exterior surface of the holster body 102 by the user's thumb either during actuation of the release of the retention mechanism or as the handgun 80 is withdrawn. In that the thumb button 180 is positioned in close proximity to the surface of the handgun 80, the thumb movement inward as the thumb transitions from engagement with the thumb button 180 to engagement with the body of the handgun 80 is minimal.

[0154] Referring to FIGS. 25-27C, in embodiments, the holster includes the hood spring 196, a lever spring 198 and

a tab spring 200. In embodiments, each spring comprises a length of wire, the wire of the spring forming a first leg 208, a second leg 210 and a coil 212 disposed between the first leg 208 and the second leg 210. In embodiments, the first leg 208 of the hood spring 196 is fixed relative to the holster body 102 and the second leg 210 of the hood spring 196 is seated against the hood assembly 188 so that the U-shaped shroud 176 is biased to pivot toward the release position. In embodiments, the lever spring 198 is positioned and adapted to bias the lever 104 to rotate toward the handgun trigger guard capture position. In embodiments, the first leg 208 of the lever spring 198 is seated against to the holster body 102 and the second leg 210 of the lever spring 198 is seated against the lever 104. In embodiments, the tab spring 200 is positioned and adapted to bias the thumb button 180 to rotate in a direction that moves a distal end of the thumb button tab 180 away from the holster body 102. In embodiments, the first leg 208 of the tab spring 200 is seated against to the holster body 102 and the second leg 210 of the tab spring 200 is seated against the thumb receiving tab 180.

[0155] Referring to FIGS. 28A-31B, in embodiments, the U-shaped shroud 176 is part of a hood assembly 188. In embodiments, a bracket 190 of the hood assembly 188 includes a lug portion 214 and the thumb button 180 includes a protrusion portion 216 that engages the lug portion 214 of the hood assembly 188 while the U-shaped shroud 176 is in the handgun retaining position. In embodiments, the holster 100 further includes a hood spring 196 that biases the U-shaped shroud 176 to pivot toward the release position. In embodiments, the hood spring 196 biases the lug portion 214 of the hood assembly 188 against the protrusion portion 216 of the thumb button 180 while the U-shaped shroud 176 is in the handgun retaining position. In embodiments, rotation of the thumb button 180 through a first range of rotary motion causes the protrusion portion 216 of the thumb button 180 to disengage from the lug portion 214 of the hood assembly 188. In embodiments, rotation of the thumb button 180 through a second range of rotary motion causes the thumb button 180 to engage the tab receiving portion of the lever 104 causing at least a portion of the trigger guard retaining portion of the lever 104 to be withdrawn from the trigger guard receiving portion of the holster body 102.

[0156] Referring to FIGS. 33A and 33B, a feature and benefit of embodiments is a retention mechanism having a lever 104' that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is contacting a thumb button 180. In embodiments, the retaining mechanism comprises a thumb button 180 that pivots about a first axis and a lever 104' that pivots about a second axis that is parallel to the first axis. In embodiments, the lever 104' pivots about an axis that is parallel to a handgun insertion and withdrawal axis 110H of the holster 100. In embodiments, the thumb button 180 pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster 100. FIGS. 33A and 33B are two perspective views showing a holster including a U-shaped shroud. In the embodiment of FIG. 33A, the U-shaped shroud is in a handgun retaining position. In the embodiment of FIG. 33B, the U-shaped shroud is in a release position.

[0157] Referring to FIGS. 1-47, in embodiments, an exteriorly worn holster 100 for receiving a handgun 80 has a holster body 102 having a plurality of unitary holster wall

portions defining a handgun receiving cavity 108 and a rearward opening 88. In embodiments, the handgun has a handgun body 84, a handgrip 86, a slide 98, and a trigger guard 90. In embodiments, the holster body 102 has a handgun receiving and withdrawal axis 110H. In embodiments, the plurality of holster wall portions comprise a user proximal wall portion 113 and a user distal wall portion 115, the proximal wall portion 113 having thumb button recess 218 with a thumb button recess edge portion 220 defining the thumb button recess 218. In embodiments, the thumb button recess 218 extends forwardly from the rearward opening 88.

[0158] In embodiments, a first user actuatable handgun retention mechanism 106 of the holster 100 is located at the proximal wall portion 113 for selectively preventing the handgun 80, when in the cavity 108, from being withdrawn from the cavity 108. In embodiments, the first handgun retention mechanism 106 comprises a first lever 104 pivotally mounted on the proximal wall portion 113 of the holster body 102 with a first axis of rotation. In embodiments, the first lever 104 has an actuation tab 107 rearwardly positioned on the lever 104 and positioned within the thumb button recess 218. In embodiments, the actuation tab 107 has an upper margin 222, a rearward margin 224, and a lower margin 226. In embodiments, the lever 104 further has a trigger guard blocking portion 128 forwardly positioned on the lever 104, the lever 104 being configured and positioned such that depression of the actuation tab 107 rotates the lever 104 about its respective axis of rotation to move the trigger guard blocking portion 128 from a trigger guard blocking position to a non-blocking position.

[0159] In embodiments, an upper portion 242 of the thumb button recess edge portion 220 extends rearwardly of the actuation tab 107 along the upper margin 222 of the actuation tab 107 and a lower portion 246 of the thumb button recess edge portion 220 extends rearwardly of the actuation tab 107 at the lower margin 226 of the actuation tab 107.

[0160] In embodiments, the holster 100 has only a single user actuatable handgun retention mechanism 106 and the actuation tab is configured as an actuation tab 107. In embodiments, the holster 100 comprises a pair of thumb guides 230 positioned above and below the actuation tab 107, the thumb guides 230 projecting laterally outward from the holster body proximal wall portion 113 beyond an undepressed position of the actuation tab 107. In embodiments, the thumb guides 230 each extend laterally outward from an outward surface of the proximal wall portion 113 a distance of at least 0.35 inches.

[0161] In embodiments, the holster 100 has a second handgun retention mechanism 174 including a pivoting shroud 176 positioned rearwardly of the slide 98 of the handgun 80 holstered in the holster 100. In embodiments, the second handgun retention mechanism 174 comprises a second lever 104', the second lever 104' having an actuation tab configured as a thumb button 180, the actuation tab of the first lever position behind the thumb button 180 whereby depression of the thumb button 180 also depresses the actuation tab of the first lever.

[0162] In embodiments, the holster 100 further comprises a pair of thumb guides 230 positioned above and below the thumb button 180, the thumb guides 230 projecting laterally outward from the holster body 102 proximal wall portion 113 beyond an undepressed position of the thumb button 180.

[0163] In embodiments, one of the thumb guides 230 is positioned above the thumb button 180 and comprises a housing for the second handgun retention mechanism 174.

[0164] In embodiments, a belt engaging member 160 is attached to the holster body 102 with a plurality of threaded fasteners 162 extending into the holster body 102.

[0165] In embodiments, one of said threaded fasteners 162 is positioned above the first lever 104, one of the plurality of threaded fasteners 162 is positioned below the first lever 104, and one of the plurality of threaded fasteners 162 is positioned forward of the first lever 104, wherein the first lever 104 is captured between the holster body 102 and the belt engaging member 160.

[0166] In embodiments, a column portion 236 of the belt engaging member 160 extends rearwardly, wherein when the handgun 80 is holstered. In embodiments, the thumb button 180 is positioned between the handgun body 84 and the belt engaging member 160, wherein the belt engaging member 160 has a pair of ribs 232 having their elongate dimension extending forwardly and rearwardly and positioned to confront the thumb guides 230 whereby access to the thumb button 180 is restricted on four sides of the thumb button 180 and an access path 252 for the thumb is provided rearwardly of the thumb button 180. Ribs 232 and bottom rib 254 define a cavity 256 or recess with three sides in the column portion 236 of belt engaging member 160. By providing a cavity 256 in belt engaging member 160, an access path 252 is provided, while allowing the holster 100 to be positioned closer to the body of the user than if cavity 256 were not present and the fourth wall of the cavity 256 was instead provided by a surface coplanar with the rest of the column portion 236 of belt engaging member 160. The ribs 232 may provide structural strengthening of the lower plate portion 258 as well as providing access inhibiting structure to prevent access by others from front of the holster 100 or the back side of the holster 100, as the holster 100 is worn. The recess or cavity 256 opening upwardly providing access to the holster wearer as well as guide structure to easily and non-visually guide the user's thumb to the proper location both to actuate a thumb button 180 and for the master grip.

[0167] In embodiments, the belt engaging member 160 comprises a jacket slot belt loop 228. In embodiments, the jacket slot belt loop 228 has a column portion 236 including a lower plate portion 258 having a holster attachment region 260, a mid level portion 302, and an upper portion 264 with a pair of clamping belt loop portions 234. The lower plate portion 258 having apertures 266 for receiving threaded fasteners 162 for attachment of the holster body 102. The mid level portion 302 may have a distal side 268 that cooperates with the holster body 102 to capture components of a retention mechanism 106 between the holster body 102 and the mid level portion 302 and thereabove a pair of upright ribs 232 protruding outwardly from the distal side 268 of mid level portion 302 defining recess or cavity 256 therebetween that extends upwardly to the upper portion 264 and is open upwardly.

[0168] In embodiments, the belt engaging member 160 comprises a jacket slot belt loop 228 having a pair of clamping belt loop portions 234 extending from the column portion 236, each of the clamping belt loop portions 234 being adjustable with respect to the column portion 236 by a respective threaded fastener 262.

[0169] The upper portion 264 having a central column 272. A pair of belt loop portions 234 are displaced distally inward of the central column 272 and displaced laterally from the central column 272. The belt loop portions 234 connect to the central column 272 at upper connecting portions 274. In certain embodiments, the central column 272, the upper connection portions 274 and the belt portions 234 all unitary with each other. Each belt loop portion 234 having a lower end 276 not unitarily joined to the central column 272 but having a closable spacing therefrom. The central column 272 may have protruding portions configured as bosses 280 to cooperate with the lower ends 276 of the belt loop portions 234. Threaded fasteners 262 may be utilized to adjust the spacing between each belt loop portions 234 and the central column 272 effecting a clamping action onto a belt, such as a duty belt, whereby the jacket slot belt loop 228 is secured to a user's belt. Optional elastomeric bushings (not shown) may be utilized between the lower ends 276 of the belt loop portions 234 and bosses 280 of the central column 272 wherein the bushings comprise a bore in a central axis of bosses 280 for receiving the threaded fasteners 262. A feature and advantage of such embodiments is that the level of clamping of the jacket slot belt loop 228 and holster 100 are readily adjustable. Moreover, the lateral offset of the belt loop portions 234 from the central column 272 provides stability for the mounting system and holster 100 as the length of the engagement of the jacket slot belt loop 228 with the holster 100 is extended.

[0170] In certain embodiments, threaded fasteners 262 can be unscrewed from lower ends 276 of the belt loop portions 234 such that there is a gap 284 between lower ends 276 of the belt loop portions 234 and bosses 280. This gap 284 allows the jacket slot belt loop 228 to receive a belt of the user without the user needing to weave the belt through belt slots 286 formed by the openings in between belt loop portions 234 and central column 272. This allows user to receive a belt in the belt slots 286 without removing other accessories already attached to the belt. Alternatively, a user can weave a belt through belt slots 286 without fully unscrewing threaded fasteners 262. Once a belt is received in the belt slots 286, the threaded fasteners 262 can be re-engaged with the lower ends 276 of clamping belt loop portions 234. The belt can be more tightly secured by tightening threaded fasteners 262. In certain embodiments, if a belt is of a narrower width than belt slots 286, the action of tightening threaded fasteners 262 may force the belt up in the belt slots 286 so that the belt is biased against the upper margin 288 of upper portion 292 of belt slots 286 (FIGS. 35A and 35D) 23. In this manner, belt slots 286 can accommodate a belt that has a width less than that of belt slots 286. In certain embodiments, the distance between the lower ends 290 of the belt slots 286 near the bosses 280 may be greater than the distance between the upper portion 292 of the belt slots 286 near upper margin 288.

[0171] In some embodiments, the outside face 294 of the clamping belt loop portions 234 may have protruding features 296, such as ribs or studs, in order to receive a belt more securely (FIG. 35B). On other embodiments, the back side 298 of column portion 236 of belt engaging member 160 may also have protruding features 300 such as ribs or studs, in order to receive belt more securely.

[0172] In embodiments, the thumb guides 230 each extend laterally outward from an outward surface of the proximal wall portion 113 a distance of at least 0.35 inches. In

embodiments, the thumb guides 230 each extend laterally outward from an outward surface of the proximal wall portion 113 a distance of at least 0.35 inches and the ribs 232 extend outwardly from a surface of the column portion 236 a distance of at least 0.15 inches.

[0173] In embodiments, the first lever 104 is seated in a lever receiving region 238 on the proximal wall portion 113 of the holster body and the proximal wall portion 113 has three threaded bosses 240 dispersed around the lever receiving region 238 for receiving a belt engaging member 160. In embodiments, the proximal wall portion 113 further has a plurality of lever region containment wall portions 244 extending from the outer surface of the proximal wall portion 113 and each of the plurality of lever containment wall portions 244 connect to at least one of the three threaded bosses 240. In embodiments, the three threaded bosses 240 are positioned for receiving a belt engaging member 160 utilizing a plurality of threaded fasteners 162.

[0174] In embodiments, the first lever 104 is seated in a lever receiving region 238 on the proximal wall portion 113 and the proximal wall portion 113 has three threaded bosses 240 dispersed around the lever receiving region 238 for receiving a belt engaging member 160 and, when the belt engaging member 160 is attached, the first lever 104 is captured within the lever receiving region 238. In embodiments, the first lever 104 may be removed from an engagement with the proximal wall portion 113 when the belt engaging member is not attached to the proximal wall portion 113. In embodiments, the first lever 104 may be removed without tools when the belt engaging member is not attached. In embodiments, the first lever 104 may be removed by simply prying the first lever 104 outward when the belt engaging member 160 is not attached.

[0175] In embodiments, the holster body 102 has an upper wall portion having a pair of inclined wall portions defining a joint, a lower wall portion with a slit 250 extending from a forward opening to proximate a trigger guard receiving portion of the holster body defining a pair of forward clamping wall portions 248. In embodiments, a threaded fastener 142 is positioned so as to extend between the pair of forward clamping wall portions 248 for adjusting the forward clamping wall portions 248 about a forward portion of the handgun 80.

[0176] Referring to FIGS. 1-47, in embodiments, a holster 100 is provided for receiving a handgun 80 having a handgun body 84, a slide 98, and a trigger guard 90. In embodiments, the holster 100 comprises a holster body 102 having a plurality of holster wall portions defining a rearward opening cavity 108 extending along a handgun receiving and withdrawal axis 110H for receiving and holding the handgun 80. In embodiments, the plurality of holster wall portions comprise a user proximal wall portion 113 and a user distal wall portion 115. In embodiments, the holster includes a first handgun retention mechanism 106' and a second handgun retention mechanism 174, disposed at the proximal wall portion 113, each handgun retention mechanism selectively preventing the handgun 80, when in the cavity 108, from being withdrawn from the cavity 108. In embodiments, the first and second handgun retention mechanisms have a respective first lever 104' having a first axis of rotation and a second lever with an axis of rotation, each lever having a respective actuation tab portion on one end of the lever where depression of the actuation tab rotates the lever about its respective axis of rotation, one of the two

actuation tabs configured as a thumb receiving button with an outer thumb receiving surface, the other of the two actuation tabs positioned behind the thumb receiving button such that depression of the thumb receiving button effects a depression of the other of the two actuation tab portions, whereby depression of the thumb receiving button actuates both the first handgun retention mechanism **106'** and the second handgun retention mechanism **174**. In embodiments, the first lever **104'** of the first handgun retention mechanism **106'** is pivotally supported by the holster body **102**. In embodiments, the first lever **104'** comprises an elongate central portion **124** and a blocking portion **128** at an end opposite the respective actuation tab, the blocking portion **128** movable in and out of a handgun blocking position. In embodiments, the second handgun retention mechanism **174** comprising a U-shaped member **176** that is pivotally supported by the holster body **102** so that the U-shaped member **176** selectively pivots between a handgun retaining position in which the U-shaped member extends across a portion of a rearward opening of the cavity **108** and a release position in which the U-shaped member **176** is displaced from the portion of the rearward opening so that a withdrawal path of the handgun **80** is unobstructed by U-shaped member **176**.

Red Dot Sight Dust Cover

[0177] The present disclosure also provides a red dot sight (RDS) dust cover that can be integrated with any holster, including those of the present disclosure, that is further configured to accommodate a handgun with a RDS or any holster capable of accommodating a handgun with a RDS. Holsters of the present disclosure can be modified to accommodate a handgun with a RDS by extending a portion of an upward wall of the holster body to form a pocket for receiving the RDS. The RDS dust cover can be used with either level-2 or level-3 configuration.

[0178] In any of the foregoing embodiments, the holster can further be configured to receive a handgun with a red dot sight (RDS) and further include a dust cover configured to cover at least a portion of the RDS, the dust cover being pivotally connected to a portion of the holster body, such as the upward wall portion of the holster body. Thus, the dust cover can be pivoted from a first position, where it does not cover the RDS, and a second position, where it covers at least a portion of the RDS. In some embodiments, the holster body further includes a two-position detent positioned to engage two followers on the dust cover, where the two-position detent includes a first, open position detent and a second, closed position detent on each of a first side of the two-position detent and a second, opposite side of the two-position detent, the first, open position detents biasing the dust cover in the first position and the second, closed position detents biasing the dust cover in the second position. In some embodiments, the first, open position detent and the second, closed position detent on each side of the two-position detent are separated by a ridge having a peak such that the respective follower encounters resistance in moving from the first, open position detent to the second, closed position detent and vice versa, thus biasing the dust cover in the open or closed position depending on the relative position of the followers, i.e. in the first, open position detents or the second, closed position detents. In embodiments, the followers can be made of a flexible material such that they can deflect when passing over the ridge without breaking, but are of sufficient rigidity that they

can engage the first, open position detent or second, closed position detent to bias the dust cover in the first, open position or the second, closed position, respectively. The ridge can include a peak and thus have a sloped shape on each side of the ridge in the direction of each of the first, open position detent and the second, closed position detent, respectively, such that the ridge provides resistance while the sloped portions aid in deflecting the follower away from or toward the detents. The followers can be disposed at an end of the dust cover proximal to the upward portion of the holster body, one positioned on each of the first and second sides of the two-position detent, extending toward the two-position detent and configured to engage the first detents in the first position or the second detents in the second position. In some embodiments, the two-position detent can be positioned on a central rib of the upward portion of the holster body. In such instances, the followers on the dust cover can be positioned with one on each side of a central, vertical axis of the dust cover, the prongs disposed pointing toward the central, vertical axis, and positioned to engage the two-position detent. In operation, the dust cover can be displaced from the second position to the first position by the draw of the gun from the holster which displaces the followers from the second, closed detents to the first, open detents which biases the dust cover in the open position. Once the gun is returned to the holster, the dust cover can be moved manually by the user from the first position to the second position, moving the followers from the first detents to the second detents of the two-position detent, thereby biasing the dust cover in the second position and covering at least a portion of the RDS.

[0179] In some embodiments, the dust cover is configured to cover all portions of the RDS not covered by the holster body. In some embodiments, the dust cover is hingeably connected to the upward portion of the holster body. By way of example, but not limitation, the dust cover can further include two pegs, one that extends from a user proximal side of the dust cover toward the and through an aperture in the user proximal wall portion of the holster body and one that extends from a user distal side of the dust cover that is opposite of the user proximal side away from the user proximal wall and through an aperture in the user distal wall portion of the holster body, the pegs and apertures configured to permit hingeable operation of the dust cover. By way of further example, but not limitation, rather than the apertures being in the user proximal wall portion and the user distal wall portion, respectively, the apertures can be in a side of the upward portion of the holster body proximate to the user proximate wall portion of the holster body and in a side of the upward portion of the holster body proximate to the user distal wall portion. Thus, it should be understood that the pivotable nature of the dust cover can be effected by various designs that permit pivoting of the dust cover from the first position to the second position. Thus, the pivoting of the dust cover can be around an axis orthogonal to an axis between upward and downward portions of the holster body. For example, the pivoting can be from a downward direction to an upward direction, and vice versa. In this way, the dust cover can be tilted toward the gun when it is holstered to cover at least a portion of the RDS or away from the gun so that it can be drawn. It should be understood that the foregoing embodiments described with respect to the dust cover can be for a holster that includes a level-2 or level-3

configuration as described herein or to any other holster that can accommodate a RDS-equipped gun.

[0180] In some embodiments, where the holster is in a level-3 configuration, the dust cover can further include a foot portion that extends in the direction of the user proximal wall portion or the user distal wall portion from a user proximal side of the dust cover or a user distal side of the dust cover, respectively. The foot portion can be positioned near an end distal from the followers. For example, the foot portion can be positioned at the “top” of the dust cover. The shroud of the holster further includes a channel configured and positioned to catch the foot portion of the dust cover such that, if the dust cover is in the first position, when the shroud is moved back, the channel engages the foot portion and pulls the dust cover from its first position to its second position. Thus, the channel can be positioned on a user proximal side of the shroud or a user distal side of the shroud depending on whether the foot portion is on the user proximal side of the dust cover or the user distal side of the dust cover, respectively. In operation, for a holster with a level-3 configuration, when the shroud is in the release position, the dust cover is maintained in its second position. The dust cover can then be moved from its second position to its first position and vice versa as described above, either manually by the user, or pushed from second position to first position by the gun as it is withdrawn from the holster. When the shroud is in the release position and the dust cover is in the first position, the dust cover can be moved to the second position manually by the user. In the alternative, when the shroud is moved from the release position to the handgun retaining position, if the dust cover is in the first position, the channel of the shroud will catch the foot portion and move the dust cover from the first position to the second position. It should be understood that the foot portion can be on either or both the user proximal side of the user distal side of the dust cover and that the channel(s) in the shroud can be positioned to correspond to the position(s) of the foot portion(s).

[0181] Referring now to FIGS. 48A-51H, exemplary embodiments of the holster configured to accommodate a handgun with a RDS and a dust cover of the present disclosure are provided. FIGS. 48A-48F depict an exemplary level-2 configuration while FIGS. 51A-51H depict an exemplary level-3 configuration and aspects thereof. FIGS. 49 and 50 depict an exemplary dust cover of the present disclosure and a close up view of an exemplary pair of the followers and the two-position detent, respectively.

[0182] As shown in FIGS. 48A-48F, an exemplary embodiment of a level-2 holster configuration with a dust cover of the present disclosure is shown. Referring to FIG. 48A, the holster is shown from the upward end and includes the holster 100 having a holster body 102 which includes a user proximal wall portion 113, a user distal wall portion 115 and an upward wall portion 117. The holster is configured to accommodate a handgun having a RDS as evidenced by the projection 119 in the upward wall portion 117. The dust cover 500 is hingeably attached to the upper wall portion 117 in a position to cover at least a portion of the RDS when the handgun is in the holster 100 and the dust cover 500 is in the second, closed position as shown in FIG. 48A. As depicted, the dust cover 500 is hingeably attached to the holster 100 by two protrusions or pegs 501 (only one is shown) that extend through two corresponding apertures 502 (only one is shown) in the upper wall portion 117 on the user proximal

and user distal sides (503 and 504, respectively) of the dust cover 500. FIG. 48B depicts the same exemplary embodiment as in FIG. 48A from the downward end of the holster 100. FIG. 48B shows the two followers 505 at the end of the dust cover proximal to the upward portion of the holster body 102. It can be seen that the followers 505 extend on either side toward a central rib 506 of the upward portion of the holster body toward a two-position detent 507. Turning to FIGS. 48C-48D, the exemplary embodiment of FIG. 48A is shown from the user distal side 504 (FIG. 48C) and from the user proximal side 503 (FIG. 48D). Referring to FIGS. 48E and 48F, the holster 100 is shown with the dust cover 500 in the first, open position from the user distal side 504 (FIG. 48E) and the user proximal side 503 (FIG. 48F). When the dust cover 500 is in the first position, the handgun can be inserted or withdrawn without contacting the dust cover 500. As discussed above, when the dust cover 500 is in the second position as in FIGS. 48A-48D, it can be displaced to the open position by the draw of the handgun which moves the followers from the second, closed position detent 510 to the first, open position detent 509 of the two-position detent 507 to bias the dust cover 500 in the open position and permit the continued draw of the handgun.

[0183] In FIGS. 48A-48B, 48D and 48F, the dust cover 500 can include a foot portion 508 as shown, however, it should be understood that in embodiments where the holster 100 does not include a level-3 configuration with a shroud, the foot portion can be omitted.

[0184] As shown in FIG. 49, an exemplary embodiment of the dust cover 500 of the present disclosure can have a user distal side 504 and a user proximal side 503, two followers 505 at an end proximal to the upward portion of the holster, and a foot portion 508 at an end distal from the followers 505 that extends in the direction of the user proximal wall portion a user proximal side of the dust cover. As noted above, it should be understood that the foot portion is not required if the dust cover is not being used in a level-3 configuration and, even in a level-3 configuration, may not require the foot portion if the shroud is not configured with a channel to catch the dust cover. As shown in FIG. 50A, the two followers 505 can be positioned on either side of a two-position detent 507 which includes a first, open position detent 509 and a second, closed position detent 510 on each of a first side of the two-position detent 507 and a second, opposite side of the two-position detent 507 where the first, open position detent 509 and the second, closed position detent 510 are separated by a ridge 512, the first, open position detent 509 biasing the dust cover 500 in the first position and the second, closed position detents 510 biasing the dust cover 500 in the second position. As depicted the two-position detent 507 can be located on a central rib 506 of an upward portion of the holster 100. FIG. 50B shows an extreme close-up of the two-position detent and the followers of the dust cover of FIG. 50A, with the dust cover displaced from the two-position detent, which shows that the ridge includes a peak and sloped sides, sloping toward each of the first, open position detent 509 and the second, closed position detent 510.

[0185] As shown in FIGS. 51A-51H, an exemplary embodiment of a level-3 holster configuration with a dust cover 500 of the present disclosure is shown. Referring to FIGS. 51A-51H, a holster 100 with a level-3 configuration is shown which includes a U-shaped shroud 176 and the dust cover 500, where the U-shaped shroud 176 is capable of

engaging the foot portion **508** of the dust cover **500** to bias it in the second position to cover the RDS. As shown in FIGS. **51B-51C**, the dust cover can include two followers **505** at an end of the dust cover proximal to the upward portion of the holster body which can be positioned on each side of a two-position detent **507** that is disposed on a central rib **506** of an upward portion of the holster **100** and which can bias the dust cover **500** in the second, closed position as shown in FIGS. **51A-51E** and **51G** or in the first, open position as shown in FIG. **51H**. The U-shaped shroud **176** also includes a channel **511** which can engage the foot portion **508** of the dust cover **500** such that it can bias the dust cover **500** in the second, closed position and, if the dust cover **500** is in the first, open position, catch the foot portion **508** to displace the dust cover to the second, closed position. FIGS. **51D-51E** depict the holster **100** of FIGS. **51A-51C** from the user proximal side and the user distal side, respectively. As noted above, FIGS. **51G-51H** depict the holster **100** of FIGS. **51A-51E** with the dust cover **500** in the closed position and the shroud **176** is in the release position (FIG. **51G**) or the dust cover **500** is in the first, open position and the shroud **176** is in the release position (FIG. **51H**). It should be understood that in embodiments with a level-3 configuration holster, the dust cover is not required to include the foot portion unless that shroud with the channel as described is to be used to engage the foot portion. For example, the dust cover can be manually operable by the user independent of the shroud, as in the level-2 configuration.

[0186] FIG. **51F** depicts an exemplary U-shaped shroud **176** that includes the channel **511** for engaging the foot portion **508** of the dust cover **500**.

[0187] It should be understood that in any of the foregoing embodiment, the foot portion and channel can be disposed on either a user proximal side or a user distal side of the dust cover.

[0188] It should also be understood that the dust cover and its configuration can be applied to any holster, preferably a holster of the present disclosure.

Lockout Button Assembly

[0189] In certain embodiments the holster includes a lockout button assembly which allows for an additional security feature. The lockout button assembly can be used in combination with any of the foregoing embodiments. Unless expressly stated otherwise, features of the holster are the same or similar to features of the foregoing embodiments.

[0190] In some embodiments, the lockout button assembly adds an extra security feature by preventing rotation of the thumb receiving tab, for example thumb button **180** of prior embodiments, thus locking out the other retention mechanisms.

[0191] In some embodiments, the holster having the lockout button assembly includes a holster body configured to receive a handgun, that may optionally be equipped with a red dot sight (RDS), said holster body comprising a plurality of unitary holster wall portions defining a handgun receiving cavity and a rearward opening, the holster body having a handgun receiving and withdrawal axis, the plurality of holster wall portions comprising a user proximal wall portion, a user distal wall portion and, in optional embodiments, an upward wall portion having a projection to accommodate at least a portion of the RDS and a dust cover hingeably connected to the holster body and positioned and configured

to cover at least a portion of the RDS not covered by the holster body when the dust cover is in a second, closed position and to not cover the RDS when in a first, open position; a first user actuatable handgun retention mechanism at the user proximal wall portion for selectively preventing the handgun, when in the cavity, from being withdrawn from the cavity, the first actuatable handgun retention mechanism comprising a trigger guard blocking portion movable from a trigger guard blocking position to a non-blocking position; a second user actuatable handgun retention mechanism, the second user actuatable handgun retention mechanism comprising a shroud that is pivotally supported by the holster body so that the shroud selectively pivots between a handgun retaining position in which the shroud extends across a portion of a rearward opening of a cavity of the holster body and a release position in which the shroud is displaced from the portion of the rearward opening so a withdrawal path of the handgun is unobstructed by the shroud. The shroud includes a U-shaped member and, is included as part of a hood assembly in embodiments including a hood assembly. A bracket of the hood assembly includes a lug portion. A thumb receiving tab of the hood assembly includes a protrusion portion that engages the lug portion of the hood assembly while the shroud is in the handgun retaining position. further including a hood spring that biases the U-shaped member to pivot toward the release position.

[0192] Referring to FIGS. on. **52A-58**, exemplary embodiments of a lockout button assembly **700** are shown. As shown in FIGS. **52A-52D**, a lockout button assembly **700** includes an elongated member **702** having a lockout button **704** positioned on a proximal end **706** of the elongated member **702** and a blocking portion **708** at a distal end **710** of the elongated member **702**. The lockout button assembly **700** is configured to be slidably mounted to the housing **201'** for the second user actuatable handgun retention mechanism. The lockout button assembly **700** may be manually moved by user movement of the lockout button **704** to move the lockout button assembly **700** between a protrusion portion blocking position (FIGS. **52A**, **52C** and **55**) and a protrusion portion non-blocking position (FIGS. **52B**, **52D**, **53A**, **53B**, **54A**, **54B** and **56**). In the protrusion portion blocking position, the blocking portion **708** is configured to block rotation of the protrusion portion **216** of the thumb receiving tab or thumb button **180** allowing the first user actuatable handgun retention mechanism to be in the trigger guard blocking position and the U-shaped member of the second user actuatable handgun retention mechanism to be in the retaining position. In the protrusion portion non-blocking position, the blocking portion **708** is configured to allow rotation of the protrusion portion **216** of the thumb button **180** allowing the first user actuatable handgun retention mechanism to be in the trigger guard non-blocking position and U-shaped member **176** of the second user actuatable handgun retention mechanism to be in the release position. The prior embodiments described herein do not have a lockout button assembly **700** including a blocking portion **708** configured to block the rotation of the protrusion portion **216** of the thumb button **180**.

[0193] In certain embodiments, the housing **201'** for the second user actuatable handgun retention mechanism includes a slot **712**. The lockout button portion **704** comprises a tab **714** configured to be manually moved within the slot **712** to allow movement of the lockout button assembly **700** between the protrusion portion blocking position and the

protrusion portion non-blocking position. The lockout button assembly 700 may be moved by the user by pressing either the tab 714 or the lockout button portion 704.

[0194] In certain embodiments, the blocking portion 708 of the lockout button assembly 700 includes a laterally projecting engagement portion 716 configured to engage with the protrusion portion 216 of the thumb button 180 when the lockout button assembly 700 is in the protrusion portion blocking position. When the lockout button portion 704 is pushed, the protrusion portion 216 can rotate freely in the space behind the blocking portion 708.

[0195] In certain embodiments, referring to FIGS. 52A, 58A and 58B, the lockout button assembly 700 further has a detent feature which temporarily locks or makes it harder to move the lockout assembly within the housing 201' when the lockout assembly 700 is in the protrusion portion non-blocking position. In some embodiments, the detent feature is a detent ridge 720 on an outer side of the elongated member 702 which is configured to seat temporarily within a corresponding detent groove 722B within housing 201' when the lockout button assembly 700 is in the protrusion portion non-blocking position (FIG. 58B) such that the detent ridge 720 is configured to be seated within the detent groove 722 upon manual movement of the lockout button portion 704 when user moves the lockout button assembly 700 from the protrusion portion blocking position to the protrusion portion non-blocking position. In some embodiments, the detent feature additionally or alternatively includes a second detent groove 722A within housing 201' spaced proximally from detent groove 722B configured to temporarily seat detent ridge 720 when the lockout button assembly 700 is in the protrusion portion blocking position (FIG. 58A).

[0196] In some embodiments, as discussed in embodiments above, a hood spring 196 biases the U-shaped member 176 to pivot toward the release position, wherein the hood spring 196 biases the lug portion 214 of the hood assembly 188 against the protrusion portion of the thumb button 180 while the U-shaped member 176 is in the handgun retaining position. Rotation of the thumb button 180 through a first range of rotary motion causes the protrusion portion 216 of the thumb button 180 to disengage the lug portion 214 of the hood assembly. Rotation of the thumb button 180 is blocked when the lockout button assembly 700 is in the blocking position.

[0197] In some embodiments, rotation of the thumb button 180 through a second range of rotary motion causes the thumb button 180 to engage the tab receiving portion of the lever 104" causing at least a portion of the trigger guard retaining portion of the lever 104" to be withdrawn from the trigger guard receiving portion of the holster body 102. The lockout button assembly 700 prevents the rotary motion of the thumb button 180 by blocking the protrusion portion 216 of the thumb button 180 when in the blocking position, which prevents engagement of the tab receiving portion of the lever 104" and prevents the trigger guard retaining portion of the lever 104" to be withdrawn from the trigger guard receiving portion of the holster body 102.

[0198] FIG. 55 shows an exemplary embodiment of a holster 100' with an exemplary embodiment of the lockout button assembly 700 in a protrusion portion blocking position, the lever 104" in the trigger guard blocking position, the U-shaped member 176 in the handgun retaining position, and the dust cover 500 in the closed position.

[0199] FIG. 56 shows an exemplary embodiment of a holster 100' with an exemplary embodiment of the lockout button assembly 700 in a protrusion portion non-blocking position, the lever 104 in the trigger guard non-blocking position, the U-shaped member 176 in the handgun release position, the dust cover 500 in the open position, and the handgun 80 in a withdrawn position.

Rotating Lock Lever Assembly

[0200] Referring to FIGS. 59-74, in certain embodiments, the holster 100 includes a rotating lock lever assembly 810 configured as an additional security feature, as an alternative embodiment to the lockout button assembly discussed above. Rotating lock lever assembly 810 may be useable in combination with a first user actuable handgun retention mechanism 106' and a second user actuable handgun retention mechanism 174. Unless expressly stated otherwise, features of the holster and first and second user actuable handgun retention mechanisms 106', 174 are the same or similar to features of the foregoing embodiments and common reference numerals are used to designate like or similar elements.

[0201] In some embodiments, the rotating lock lever assembly 810 adds an extra security feature by preventing rotation of the thumb button 828, for example thumb button 828 is similar to thumb button 180 of prior embodiments, thus locking out the other retention mechanisms 106', 174.

[0202] In some embodiments, the holster 100 having the rotating lock lever assembly 810 may include the first user actuable handgun retention mechanism 106' at the user proximal wall portion 113, as discussed above, for selectively preventing the handgun 80, when in the holster cavity discussed above, from being withdrawn from the cavity, the first actuable handgun retention mechanism 106' comprising a trigger guard blocking portion movable from a trigger guard blocking position to a non-blocking position. The holster 100 may further include the second user actuable handgun retention mechanism 174, as discussed above, the second user actuable handgun retention mechanism 174 may include a hood assembly 188 including a U-shaped shroud or member 176 that is pivotally supported by the holster body between handgun retaining and release positions. The U-shaped member 176 may be included as part of a hood assembly 188. A bracket 190 (FIGS. 73-74) of the hood assembly 188 may include a lug portion 214. A thumb button 828 of the hood assembly may include a thumb receiving tab 828A configured for receiving a user's thumb and a pivot portion 828B disposed above thumb receiving tab 828A and pivotally attached to the holster. The pivot portion 828B may include a protrusion portion 830 that engages the lug portion 214 of the hood assembly 188 while the U-shaped member 176 is in the handgun retaining position. A hood spring 196 (shown in previous embodiments) may bias the U-shaped member 176 rearwardly to pivot toward the release position. In embodiments, a tab spring 200 is positioned on the pivot portion 828B and is adapted to bias the thumb button 828 to rotate in a direction that moves a distal end of the thumb receiving tab 828A away from the proximal wall portion 113 of holster 100, in the undepressed or non-actuated position.

[0203] In embodiments, the rotating lock lever assembly 810 is positionable between a protrusion portion blocking position (FIGS. 59, 60, 61, 69, 70, 72), an intermediate dual

ramp interaction or handshake position (FIGS. 71, 74) and a protrusion portion non-blocking position (FIGS. 62, 63, 64).

[0204] In certain embodiments, the rotating lock lever assembly 810 may have a rotating lock lever assembly housing 812 coupled to the user proximal wall portion 113 of the holster body. A pivot portion 814, which may include a pivot bore, may be pivotally mounted to the housing 812 and may have an axis of rotation transverse to the handgun receiving and withdrawal axis, and also transverse to the user proximal wall portion 113 of the holster body.

[0205] In certain embodiments, the rotating lock lever assembly housing 812 may further include a fastener hole 842 and a fastener boss 824 configured to receive a pair of fasteners 844A and 844B for mounting rotating lock lever assembly housing 812 to holster 100. A coil spring 826 may be mounted on the exterior of fastener boss 824. The pivot portion 814 may be pivotally mounted to the fastener boss 824 with the coil spring 826 between the fastener boss 824 and the pivot portion 814 such that the coil spring 826 biases the rotating lock lever assembly 810 rearwardly to the protrusion blocking position.

[0206] In some embodiments, rotating lock lever assembly 810 may have an elongate member 816 extending from pivot portion 814. The axis of rotation of the pivot portion 814 may be disposed above the elongate member 816, and above thumb receiving tab 828A of thumb button 828. When the rotating lock member 810 is at rest, in the blocking position, elongate member 816 may extend at an angle downwardly and rearwardly from pivot portion 814. When the rotating lock member 810 is in the intermediate dual ramp interaction, handshake position, elongate member 816 may extend substantially downwardly from pivot portion 814. When the rotating lock member 810 is in the non-blocking position, elongate member 816 may extend at an angle downwardly and forwardly from pivot portion 814. In certain embodiments, an inner surface 814A (FIG. 71) of the pivot portion 814 facing inwardly towards the proximal wall portion 113 defines a first plane.

[0207] Elongate member 816 may have a lever button portion 818 on a rearwardly facing face thereof engageable with a user's thumb extending forwardly in the direction of the handgun receiving and withdrawal axis from the rear of the holster 100. Rotating lock lever assembly 810 may further have a blocking projection 820 which may extend rearwardly from pivot portion 814 and also project inwardly forming an inwardly facing engagement surface 822. In some embodiments, the engagement surface 822 defines a second plane spaced inwardly towards the proximal wall portion 113 from the first plane defined by the inner surface 814A of pivot portion 814 such that the blocking projection 820 projects inwardly towards the protrusion portion 830 of thumb button 828 past inner surface 814A. The blocking projection 820 may further include a ramp surface 821 sloping upwardly, outwardly and rearwardly from the engagement surface 822.

[0208] In certain embodiments, when the elongate member 816 is in the protrusion portion blocking position, elongate member 816, which includes the lever button portion 818, is configured to be actuated by a user's thumb moving parallel to the handgun receiving and withdrawal axis against lever button portion 818 and against the biasing force of coil spring 826 through a first range of rotary motion. As such, the elongate member 816 rotates forwardly

from a position blocking at least a portion of the thumb receiving tab 828A of the thumb button 828 to a downwardly extending position allowing the user's forwardly extending thumb to be positioned over the thumb receiving tab 828A of the thumb button 828 and in position to depress the thumb button 828 inwardly.

[0209] In some embodiments, when in the blocking position, rotating lock lever assembly 810 is biased to be at rest by coil spring 826 and blocks access to thumb button 828 thereby preventing the thumb button 828 from rotating from an undepressed position (FIG. 70) to a depressed position (FIG. 71), which prevents the first actuable handgun retention mechanism 106' and the second actuable handgun retention mechanism 174 from moving to trigger guard non-blocking and handgun release positions, as discussed in previous embodiments. In the blocking position, the rotating lock lever assembly 810 at least partially covers the thumb button 828 and prevents access by the thumb of a user extending forwardly in the direction of the handgun receiving and withdrawal axis, and mechanically inhibits the thumb button 828 from being moved, even if it is pressed, until the rotating lock lever assembly 810 is moved away from the blocking position, as discussed below. The rotating lock lever assembly 810 also blocks access from outside proximal wall portion 113 perpendicular to the withdrawal axis. Thus, the user cannot go through the rotating lock lever assembly 810, or depress thumb button 828 otherwise, until elongate member 816 of the rotating lock lever assembly 810 is rotated out of the way.

[0210] The rotating lock lever assembly housing 812 may further include a slot opening 812C surrounding fastener boss 824 and defining upper and lower limit surfaces 812A, 812B. When in the blocking position, blocking projection 820 makes contact with an upper limit surface 812A which blocks further biased rotation by coil spring 826 (FIG. 59). When in the non-blocking position, blocking projection 820 makes contact with a lower limit surface 812B which blocks further biased rotation by the user's thumb.

[0211] Referring to FIGS. 66 and 70, protrusion portion 830 of thumb button 828 may have a blocking surface 832 and a ramp surface 834 sloped downwardly and inwardly from the blocking surface 832. The blocking surface 832 of the protrusion portion 830 may engage the engagement surface 822 of the blocking portion 820 of the rotating lock lever assembly 810 at the second plane when the elongate member 816 is in the protrusion portion blocking position and the lever button portion 818 is covering the thumb button 828. The interaction between engagement surface 822 and blocking surface 832 at the second plane mechanically inhibits the thumb button 828 from being moved, even if it is pressed, as shown in FIGS. 61 and 70, because blocking surface 832 is unable to rotate past the second plane, even when a user presses thumb receiving tab 828a, due to engagement surface 822 blocking such rotation or movement.

[0212] The rotating lock lever assembly 810 is activated by the user moving a thumb forwardly and parallel to the holster withdrawal axis and over the thumb receiving tab 828A by pushing the rotating lock lever assembly 810 forwardly so that the thumb directly covers thumb receiving tab 828A. When the user has moved the rotating lock lever assembly 810 forwardly about the first range of rotary motion so that the lever button portion 818 extends downwardly, the user's thumb is positioned directly over the

thumb receiving tab **828A** of the thumb button **828**. The user is then able to depress the thumb receiving tab **828A** by moving the thumb inwardly against the thumb receiving tab **828A** of the thumb button **828** so that thumb button **828** rotates about a first range of rotary motion. This is possible because blocking surface **832** is now able to rotate past the second plane, because engagement surface **822** is no longer blocking such rotation or movement. Referring to FIG. 71, while the thumb receiving tab **828A** of thumb button **828** is depressed inwardly toward the user proximal wall portion **113** the rotating lock lever **810** is in a handshake position. In the handshake position, the part of the protrusion portion **830** engaging the blocking projection **820** moves downwardly from the blocking surface **832** to the ramp surface **834** of the protrusion portion **830** of thumb button **828**. Ramp surface **834** of protrusion portion **830** now engages with and blocks the ramp surface **821** of blocking projection **820** of rotating lock lever assembly **810** from returning rearwardly to the blocking position by creating a dual ramp interaction or handshake between the thumb button **828** and the rotating lock lever assembly **810** preventing the rotating lock lever assembly **810** from returning to the blocking position, as discussed below.

[0213] In between the blocking position and non-blocking position, the handshake between the thumb button **828** and the rotating lock lever assembly **810** may provide an assist to the movement of the rotating lock lever assembly **810** forwardly to the non-blocking position by the ramp surface **834** of the protrusion portion **830** of thumb button **828** placing outward pressure on the ramp surface **821** of blocking projection **820** of rotating lock lever assembly **810** causing the elongate member **816** to move forwardly about a second range of rotary motion to the non-blocking position. Alternatively, the user's thumb may assist in the movement of the rotating lock lever assembly **810** forwardly about the second range of rotary motion to the non-blocking position by continuing to move the thumb forward.

[0214] The elongate member **816** is not able to return to the protrusion portion blocking position until the thumb receiving tab **828A** of thumb button **828** is rotated outwardly to an undepressed position. This is because, until the thumb button **828** is rotated back outwardly to an undepressed position by the biasing of tab spring **200** and the user releasing pressure by the thumb on the thumb receiving tab **828A**, the ramp surface **834** of the protrusion portion **830** of the thumb button **828** is configured to block the engagement surface **822** of the blocking projection **820** of the rotating lock lever assembly **810**, and therefore block the return of the rotating lock lever assembly **810** to the blocking position.

[0215] In certain embodiments, in the non-blocking position there is substantially no interaction between the blocking projection **820** of the rotating lock lever assembly **810** and the protrusion portion **830** of the thumb button **828**.

[0216] In some embodiments, the handshake between blocking projection **820** and protrusion portion **830** also works in reverse such that when only the thumb button **828** is depressed and not the rotating lock lever assembly **810**, and rotating lock lever **810** is in the non-blocking position, when the thumb button is released, elongate member **816** will be biased rearwardly to the blocking position from the non-blocking position such that the ramp surface **821** of blocking projection **820** will ride up the ramp surface **834** of the protrusion portion **830** of the thumb button **828** to the

blocking surface **832** of the protrusion portion **830** of the thumb button **828**. Or in, the alternative, releasing only the rotating lock lever assembly **810** with the user's thumb will cause the thumb button **828** to return to the undepressed position such that the ramp surface **834** of the protrusion portion **830** of the thumb button **828** rides up ramp surface **821** of blocking projection **820**.

[0217] Depressing the thumb button **828** inwardly towards the proximal wall portion **113** about an axis parallel to the handgun receiving and withdrawal axis (FIG. 71), allows the first actuatable handgun retention mechanism **106'** and the second actuatable handgun retention mechanism **174** to move to trigger guard non-blocking and handgun release positions. Once the hood assembly **188** of the second actuatable handgun retention mechanism **174** is moved to an open, release position, the hood assembly **188** cannot be returned to a closed, blocking position without moving the rotating lock lever assembly **810** to the non-blocking position, and moving thumb button **828** to a depressed position.

[0218] The rotating lock lever assembly **810** may be configured for manual movement of the rotating lock lever assembly **810** between a protrusion portion blocking position and a protrusion portion non-blocking position, wherein the blocking projection **820** is configured to block rotation of the protrusion portion **830** of the thumb button **828** when in the blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism **106'** to be in the trigger guard blocking position and the shroud **176** to be in the retaining position, and wherein the blocking projection **820** is configured to allow rotation of the protrusion portion **830** of the thumb button **828** outwardly when in the non-blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism **106'** to be in the trigger guard non-blocking position and the shroud **176** to be in the release position. As discussed above, hood spring **196** biases the U-shaped member **176** to pivot toward the release position. The hood spring **196** biases the lug portion **214** of the hood assembly **188** against the protrusion portion **830** of the thumb button **828** while the U-shaped member **176** is in the handgun retaining position. Protrusion portion **830** of thumb button **828** may further include an angled locking surface **838** configured for engagement with the lug portion **214** of hood assembly **188** when the hood assembly **188** is in the closed retaining position, wherein rotation of the thumb button **828** through a first range of rotary motion causes the angled locking surface **838** of the protrusion portion **830** to disengage from the lug portion **214** of the hood assembly. The rotating lock lever assembly **810** is configured to block the thumb button **828** from rotation of the thumb button **828** through the first range of rotary motion when the rotating lock lever assembly **810** is in the protrusion portion blocking position. Motion of the hood assembly **188** is blocked when the rotating lock lever assembly **810** is in the protrusion portion blocking position by engagement of lug portion **214** and the protrusion portion **820**.

[0219] In order for the hood assembly **188** to close, the thumb button **828** is configured to rotate about pivot portion **828B** to allow the lug portion **214** of the hood assembly to travel past the inwardly facing backside of the thumb button **828**, wherein when the rotating lock lever assembly **810** is at rest in the blocked position (FIG. 72), it blocks all motion of the thumb button **828** and does not allow the lug portion **214** to rotate past the inwardly facing backside (FIG. 73) of

the thumb button **828** causing the need for the rotating lock lever assembly **810** to move to the non-blocking position, and thumb button **828** to depressed position, prior to closing the hood assembly **188**.

[0220] Referring to FIG. **68**, protrusion portion **830** of thumb button **828** may further include an engagement surface **840** on a rear surface thereof. Moving the hood assembly **188** from an open position (FIG. **73**) to a closed position (FIG. **74**) causes the lug portion **214** to engage with the engagement surface **840** and be blocked by engagement surface **840** until the user rotates the thumb button **828** inwardly towards the proximal wall portion **113** of holster **100** to unblock the lug portion **214** from engagement surface **840**. Once the lug portion **214** has moved past the engagement surface **840** it can rest once again on angled locking surface **838** corresponding to the hood assembly **188** being in the closed and retaining position.

[0221] Referring to FIG. **74**, the rotating lock lever assembly **810** may be configured to rotate out of the blocked position to allow the thumb button **828** to freely move once the lug portion **214** of the hood assembly **188** provides pressure on the engagement surface **840** on the rear surface of the protrusion portion **830**, and further wherein the hood assembly and corresponding lug portion **214** cannot move to the closed and blocking position when the rotating lock lever assembly **810** is in the blocked position. Alternatively, as discussed above, thumb button **828** must be depressed in order to rotate angled locking surface **838** out of the way of lug portion **214**.

[0222] The following United States patents and applications are hereby incorporated by reference herein: U.S. Pat. Nos. 5,048,735, 5,100,036, 5,129,562, 5,275,317, 5,284,281, 5,372,288, 5,395,021, 5,419,474, 5,449,103, 5,509,591, 5,573,157, 5,810,221, 5,810,221, 5,918,784, 5,918,784, 6,112,962, 6,189,751, 6,230,946, 6,267,279, 6,276,581, 6,533,149, 6,547,111, 6,547,111, 6,634,527, 6,641,009, 6,641,009, 6,752,300, 6,752,300, 6,769,582, 6,799,392, 6,854,626, 7,200,965, 7,434,712, 7,461,765, 7,461,765, 7,530,456, 7,530,456, 7,556,181, 7,556,181, 7,841,497, 7,841,497, 7,922,050, 7,922,050, 8,141,758, 8,141,758, 8,177,108, 8,235,263, 8,474,670, 8,517,235, 8,602,276, 8,602,276, 8,631,981, 8,631,981, 8,646,665, 8,720,753, 8,720,753, 8,720,754, 8,720,755, 8,851,344, 8,985,412, 9,022,262, 9,022,262, 9,057,579, 9,057,580, 9,134,093, 9,134,093, 9,175,925, 9,175,925, 9,228,802, 9,267,760, 9,347,741, 9,383,165, 9,410,767, 9,500,426, 9,777,986, 9,835,400, 10,619,974, U.S. Ser. No. 16/747,986 filed Jan. 21, 2020 and U.S. Ser. No. 16/748,151 filed Jan. 21, 2020. Components illustrated in such patents may be utilized with embodiments herein. Incorporation by reference is discussed, for example, in MPEP section 2163.07 (B).

[0223] The patents and other references mentioned above in all sections of this application are herein incorporated by reference in their entirety for all purposes.

[0224] All of the features disclosed in this specification (including the references incorporated by reference, including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive.

[0225] Each feature disclosed in this specification (including references incorporated by reference, any accompanying claims, abstract and drawings) may be replaced by alterna-

tive features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

[0226] The invention is not restricted to the details of the foregoing embodiment(s). The invention extends to any novel one, or any novel combination, of the features disclosed in this specification (including any incorporated by reference references, any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed. The above references in all sections of this application are herein incorporated by references in their entirety for all purposes.

[0227] Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that any arrangement calculated to achieve the same purpose could be substituted for the specific examples shown. This application is intended to cover adaptations or variations of the present subject matter. Therefore, it is intended that the invention be defined by the attached claims and their legal equivalents, as well as the following illustrative aspects. The above described aspects embodiments of the invention are merely descriptive of its principles and are not to be considered limiting. Further modifications of the invention herein disclosed will occur to those skilled in the respective arts and all such modifications are deemed to be within the scope of the invention.

What is claimed is:

1. A holster, comprising:

- a holster body configured to receive a handgun, the holster body comprising,
 - a handgun receiving and withdrawal axis;
 - a user proximal wall portion;
 - a cavity;
 - a rearward opening of the cavity
- a first user actuatable handgun retention mechanism located at the user proximal wall portion for selectively preventing the handgun, when in the cavity, from being withdrawn from the cavity, the first actuatable handgun retention mechanism comprising a trigger guard blocking portion movable from a trigger guard blocking position to a non-blocking position;
- a second user actuatable handgun retention mechanism, the second user actuatable handgun retention mechanism comprising a housing and a hood assembly, the hood assembly comprising:
 - a shroud that is pivotally supported by the holster body so that the shroud selectively pivots between a handgun retaining position in which the shroud extends across a portion of the rearward opening of the cavity of the holster body and a release position in which the shroud is displaced from the portion of the rearward opening so a withdrawal path of the handgun is unobstructed by the shroud;
 - a bracket comprising a lug portion; and
 - a thumb button, the thumb button comprising a thumb receiving tab configured for receiving a user's thumb and a pivot portion pivotally attached to the holster and comprising a protrusion portion comprising an angled locking surface configured to engage the lug portion of the bracket when the shroud is in the handgun retaining position, wherein the pivot portion of the thumb button is configured to pivot the thumb button about an axis parallel to the handgun receiving and withdrawal axis

between an undepressed position and a depressed position depressed inwardly towards the user proximal wall portion, the protrusion portion further comprising an outwardly facing blocking surface and a ramp surface sloped downwardly and inwardly from the blocking surface;

a rotating lock lever assembly comprising:

a housing coupled to the user proximal wall portion and comprising a fastener boss extending from the housing;

a pivot portion pivotally mounted to the fastener boss and comprising an axis of rotation transverse to the handgun receiving and withdrawal axis, a surface of the pivot portion facing inwardly towards the user proximal wall portion and defining a first plane;

an elongate member extending from the pivot portion of the rotating lock lever assembly, the elongate member comprising a lever button portion; and

a blocking projection extending from the pivot portion of the rotating lock lever assembly inwardly towards the user proximal wall portion beyond the first plane, the blocking projection comprising an engagement surface, wherein the engagement surface defines a second plane spaced inwardly towards the user proximal wall portion from the first plane, the blocking projection further comprising a ramp surface sloping upwardly and outwardly from the engagement surface;

wherein the rotating lock lever assembly is configured for movement of the rotating lock lever assembly between a protrusion portion blocking position, an intermediate handshake position, and a protrusion portion non-blocking position, wherein the engagement surface of the blocking projection is configured to engage with the blocking surface of the protrusion portion when in the protrusion portion blocking position to block rotation of the protrusion portion of the thumb button when in the blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism to be in the trigger guard blocking position and the shroud to be in the retaining position, and wherein the blocking projection is configured to allow rotation of the protrusion portion of the thumb button when in the non-blocking position allowing the trigger guard blocking portion of the first user actuatable handgun retention mechanism to be in the trigger guard non-blocking position and the shroud to be in the release position;

wherein the elongate member is configured to block user access to the thumb receiving tab of the thumb button by covering at least part of the thumb button when the rotating lock lever assembly is in the protrusion portion blocking position; and wherein the elongate member is configured to be pivoted by actuation of the lever button portion by the user about the pivot portion of the thumb button such that the thumb receiving tab is accessible by the thumb of a user extending forwardly in the direction of the handgun receiving and withdrawal axis;

wherein the rotating lock lever assembly further comprises a coil spring on the fastener boss, and wherein the pivot portion of the rotating lock lever is pivotally

mounted to the fastener boss such that the coil spring biases the rotating lock lever assembly to the protrusion blocking position;

wherein the elongate member comprising the lever button portion is pivotable about the pivot portion of the rotating lock lever assembly by a user's thumb moving forwardly and parallel to the handgun receiving and withdrawal axis engaging the lever button portion of the elongate member, wherein the elongate member rotates from a position blocking at least a portion of the thumb receiving tab of the thumb button about a first angular range of rotary motion to a position not blocking the thumb receiving tab of the thumb button and allowing the user's forwardly extending thumb to be positioned over the thumb receiving tab of the thumb button and to be able to press the thumb button inwardly; and

wherein the ramp surface of the blocking projection is configured to engage with the ramp surface of the protrusion portion when the rotating lock lever assembly is in the intermediate handshake position, wherein the ramp surface of the protrusion portion is pivoted outwardly away from the user proximal wall portion beyond the first plane defined by the inwardly facing surface of the pivot portion of the rotating lock lever assembly when the thumb button is depressed inwardly;

wherein when the rotating lock lever assembly is moved to the protrusion portion non-blocking position, the elongate member is not able to return to the protrusion portion blocking position until the thumb button pivots back to the undepressed position, such that until the thumb button pivots back to the undepressed position, the ramp surface of the protrusion portion blocks the ramp surface of the blocking projection of the rotating lock lever assembly, and therefore blocks the return of the rotating lock lever assembly to the blocking position.

2. The holster of claim 1, wherein the rotating lock lever assembly housing further comprises a fastener extending through the fastener boss attaching the housing to the holster body.

3. The holster of claim 2, wherein the coil spring is configured to return the rotating lock lever assembly to the protrusion blocking position when force is removed by the user's thumb on the lever button portion, and wherein the thumb button allows the return of the rotating lock lever assembly to the protrusion blocking position when the thumb button is in the undepressed position.

4. The holster of claim 3, wherein the engagement of the ramp surface of the protrusion portion and the ramp surface of the blocking projection of the rotating lock lever assembly assists the elongate member in moving towards the non-blocking position when the thumb button is depressed by the user.

5. The holster of claim 1, wherein the shroud comprises a U-shaped member.

6. The holster of claim 5, further comprising a hood spring that biases the U-shaped member to pivot toward the release position.

7. The holster of claim 6, wherein the hood spring biases the lug portion of the hood assembly against the protrusion portion of the thumb button while the U-shaped member is in the handgun retaining position.

8. The holster of claim **7**, wherein the angled locking surface of the protrusion portion is configured for engagement with the lug portion when the hood assembly is closed, wherein rotation of the thumb button through a first range of rotary motion causes the angled locking surface of the protrusion portion to disengage from the lug portion of the hood assembly, wherein the rotating lock lever assembly is configured to block the thumb button from rotation of the thumb button through the first range of rotary motion when the rotating lock lever assembly is in the protrusion portion blocking position, wherein motion of the hood assembly is blocked when the rotating lock lever assembly is in the protrusion portion blocking position.

9. The holster of claim **8**, wherein, in order for the hood assembly to close, the thumb button is configured to rotate to allow the lug portion of the hood assembly to travel past the back of the thumb button, wherein when the rotating lock lever assembly is at rest in the blocked position, it blocks all

motion of the thumb button and does not allow the lug portion to rotate past the backside of the thumb button causing the need for the rotating lock lever assembly to move to the non-blocking position prior to closing the hood assembly.

10. The holster of claim **9**, wherein the protrusion portion further comprises an engagement surface on a rear surface thereof facing the holster body; wherein moving the hood assembly from an open position to a closed position causes the lug portion to engage with the engagement surface and be blocked by the engagement surface until the user rotates the thumb button inwardly towards the proximal wall portion to unblock the lug portion from the engagement surface; wherein, once the lug portion has moved past the engagement surface, lug portion rests on the angled locking surface corresponding to the hood assembly being in the retaining position.

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